

Fundamental Limits of Network Inference and Matching

by

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Dissertation

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Abstract

Modern networked systems, from social media platforms and protein–protein interactomes to peer-to-peer blockchains, share a common mathematical skeleton: agents interact pairwise, and both the resulting network and the observations arising from it are shaped by randomness. The theory of random graphs provides a unifying abstraction in which operational questions about real networks become statistical questions about latent structure. This thesis studies a suite of inference problems on such random-graph models, in idealized settings and under deviations from them, and characterizes fundamental limits of what can be inferred together with algorithms that attain them.

The first part studies inference from noisy observations of multiple unlabeled networks and investigates how to optimally combine information across them. A central message is that multiple correlated views of the same latent structure are fundamentally more informative than any single pair of views: there are regimes in which an inferential task is impossible from two observations but becomes possible once more than two networks are observed. This phenomenon is quantified for the recovery of a latent vertex correspondence across correlated inhomogeneous random graphs, and for the hypothesis testing problem of detecting a latent correlation among Gaussian-weighted graphs. A parallel inferential question is formulated for circular data, where nearly matching necessary and sufficient conditions are derived for detecting a planted phase-coherent subset or community.

The second part revisits inference and consensus when the idealized random-graph assumptions are perturbed under adversarial corruption. Two problems are studied: the first is the graph matching problem of the first part, now in the presence of two kinds of adversaries: a *weak* adversary whose corrupted nodes interact randomly with the network, and a *strong* adversary whose corrupted nodes interact adversarially. Achievability and impossibility results are obtained in both cases for the problems of matching the uncorrupted nodes and identifying the corrupted nodes. The second problem is a longest-chain blockchain consensus protocol under a communication model that allows point-to-point messages between honest peers to be lost in transit. Bounds are derived on the fraction of adversarial computational power the system can tolerate while remaining secure.

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I am also very lucky to have had a fantastic set of collaborators, Sophie and Flore, from whom I have learned a great deal, and I am deeply grateful for their support and encouragement throughout our collaborations. I look forward to many more in the years ahead. I am also grateful to the colleagues and friends with whom I have had rich research discussions over the years — Ani, Mehrdad, Neeladri, Saptarshi, Souvik, Sushil, Vijay, Weina, and Yashaswini, among many others. Research can be isolating, and these conversations were a reliable reminder that it does not have to be.

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My family has been a source of love and support throughout this journey, even when they had little frame of reference for what my work actually looks like from the inside. My parents wanted the best for me at every step, and their belief in me and patience through the uncertainty meant a great deal. Even at a distance, they have been present in all the ways that count. I was also very lucky to have my sister and her family nearby, and on many occasions, driving over for a weekend was exactly what I needed to reset. Some of my best weekends of the past five years were spent at her place, doing not much at all, and I will miss being a short drive away. I am also grateful to my extended family for the warmth, check-ins, and, of course, the food.

One unexpected gift of academic life is the travel it demands. Conferences sent me to corners of the country I might never have visited otherwise, and along the way I discovered a genuine hobby of exploration. The road trips, national parks, city museums, and weekend escapes were made immeasurably richer by the people I shared them with: Yasha, Neeladri, Ayesha, Saptarshi, Ameya, Hao, Ashwin, Viraj, Uzma, Charul, Tulasi, Fereshteh, and others. I often think back to hiking in Zion or to the spontaneous drive out to Niagara Falls, and find something clarifying in those memories. Just as much, I find myself drawn to the more chaotic stretches of the road. A recent winter trip through the Southwest with Neeladri is the kind of thing I will remember for a long time, with its long drives, improvised stops at national parks, and the particular thrill of just keeping going. I hope there is more of that ahead.

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Having grown up in large cities and warm climates, I came to Champaign-Urbana with some apprehension. It turns out I was wrong to worry. The town won me over quickly, and somewhere along the way it stopped feeling like a place I was merely passing through. Its quiet charm brings with it a pace that made more room for thought than I knew I needed. After six years here, ordinary places have become tied to particular seasons, routines, conversations, and memories. The thought that they will no longer be a short walk away is harder to sit with than I expected.

I owe a quiet but significant debt to all the teachers I have had over the years. They are too many to list, but each one left a mark. Watching the different ways people approach the same kinds of problems shaped how I think about my own work as much as anything else has. The teacher I owe the most to, naturally, is my advisor. From Bruce I have learned far more than probability and networks. I have learned the value of clarity, patience, and precision: how to sit with a problem long enough to understand what it is really asking, and how rigor and intuition can coexist in a thought. If there is anything in this thesis that reflects those lessons, it is because I have been fortunate to learn from teachers who took their work seriously, treated ideas with care, and taught me to do the same.

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Chapter 1

Introduction

Modern networked systems, from social media platforms and protein–protein interactomes to peer-to-peer blockchains, share a common mathematical skeleton: agents interact pairwise, and both the resulting network and the observations arising from it are shaped by randomness. A unifying abstraction for such systems is the theory of *random graphs*, in which nodes and edges are generated by a stochastic model. Under this abstraction, operational questions about real networks become statistical questions about latent structure, and the study of these questions combines tools from probability and statistical inference.

This thesis is concerned with statistical inference on random networks (graphs). It considers a suite of inference problems, some in idealized settings and others under deviations from those settings, organized around two related questions.

- (a) (Inference). Given noisy, partial, or unlabeled observations of one or more random graphs, what latent structure can be recovered or tested for? When is the task possible in an information theoretic sense, and when is it fundamentally impossible?
- (b) (Robustness). How do these inferential limits and other guarantees change when a fraction of the graph is adversarially corrupted, or when interactions between the components is lossy?

The thesis answers concrete instances of each of these questions. In each case, the problem is posed on a suitable stochastic model, and its solution takes the form of a pair of results: a *converse* identifying the parameter regime in which the task is impossible, and an *achievability* result exhibiting an algorithm that succeeds in the complementary regime. When the two bounds match, a fundamental limit is identified: a sharp threshold separating the parameter regimes in which the task is solvable from those in which it is provably impossible. Such limits are useful because they provide a benchmark against which any deployed algorithm or protocol can be measured. A method whose performance approaches the threshold is quantifiably near-optimal, while a method that falls short reveals room for algorithmic improvement.

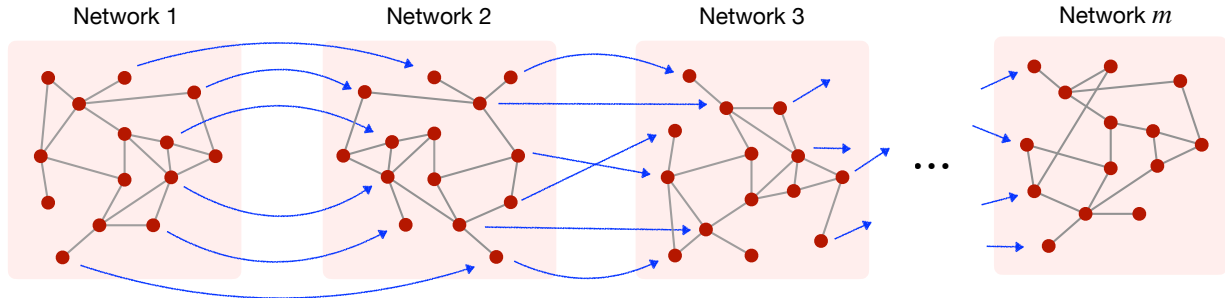


Figure 1.1: Example of a matching (alignment) between multiple networks

The remainder of this chapter motivates each of the problems treated, describes the methodological thread that runs through the thesis, and lays out the organization of the chapters that follow.

1.1 Statistical inference on networks

The first three chapters of the thesis treat inference questions on random graphs and on directional data: given noisy or incomplete observations, what can be said about the latent structure that produced them? Three concrete instances are studied, each drawn from somewhat different application contexts but united by the same statistical structure. Each is motivated below.

1.1.1 Network alignment

Network alignment, also known as graph matching, is the problem of identifying which nodes in one network correspond to which nodes in another. The motivating premise is simple: the same underlying entities (people, proteins, physical landmarks) often appear in multiple networks under different labels, and the structural information carried by the networks (who connects with whom) is enough to recover the correspondence even when the labels themselves are missing, scrambled, or deliberately anonymized. Figure 1.1 presents an illustration.

The applications cut across very different domains. In computational biology, the protein–protein interactomes (PPI) of two species share many homologous interactions, and aligning the yeast and fruit fly networks allows researchers to identify functional modules conserved across the two species, with downstream consequences for both basic science and drug discovery [SXB08, KHGPM16, BSI06]. In the study of the human brain, the connectome is reasonably stable across healthy individuals, and aligning an individual’s connectome to a reference is useful for detecting abnormalities [STK05, CPPDG24]. In privacy research, the matching algorithm itself becomes the threat: Narayanan and Shmatikov showed that anonymized Twitter activity could be linked to public Flickr profiles to recover the identities of about a third of co-active users [NS09], and that anonymized Netflix viewing histories could be linked to public IMDb activity to reveal individual viewing histories [NS08]. In autonomous systems, robots exploring overlapping regions of an unknown environment build local maps that they merge by aligning the corresponding scene graphs [YDF⁺22].

Further natural settings include matching dependency parses for textual entailment in natural language processing [HNM05] and feature-based image matching for object detection and tracking in computer vision [SS05, YYL⁺16].

The algorithms used in practice for these tasks are largely heuristic. Spectral methods, message-passing relaxations, and degree- or neighborhood-profile-based comparisons are well understood empirically but rarely come with provable performance guarantees on the inputs they are most likely to encounter. Information-theoretic limits play a complementary role here. A converse identifies the regime in which *no* algorithm, regardless of its sophistication or computational budget, can succeed. Such results are operationally useful, since they tell a privacy practitioner what level of perturbation is sufficient to defeat any adversary, including ones not yet invented. A matching achievability result, which exhibits an algorithm that succeeds in the complementary regime, closes the gap and identifies the inherent difficulty of the problem. The size of the gap between the information-theoretic threshold and the performance of a deployed heuristic, measures how much room remains for algorithmic improvement on the same data. Chapters 3 and 6 of this thesis carry out this characterization in two specific extensions of the basic setup: when more than two correlated graphs are available, and when the underlying random graph model is corrupted at the node level.

1.1.2 Correlation detection

Given two or more unlabeled networks, the question is whether their edge structures are correlated under some latent permutation, or whether the networks are statistically independent. Unlike the alignment problem, no correspondence is recovered here: only a yes/no certificate. Because no correspondence is required, correlation detection is at least as easy as matching: a successful matching implicitly answers detection, but successful detection does not entail a recoverable correspondence. The two thresholds therefore can, and in some settings do, separate.

This separation is what makes detection an interesting problem in its own right. As an upstream screen, detection answers a worthwhile yes/no question before any matching pipeline is invested in: is there enough shared structure between the networks to make alignment worth attempting at all? In the PPI example motivated above from biology, a positive detection certifies homology between two species’ interactomes even when the alignment itself is unstable. In auditing data privacy, detection answers whether two datasets still leak information about a common latent object. In neuroscience and signal processing, detection certifies that two streams of observations are driven by a shared latent process, even when the latent variables themselves are unobservable.

Chapter 4 considers correlation detection in the setting where the observations are multiple noisy copies of Gaussian-weighted graphs.

1.1.3 Phase-coherent structure in circular data

A different family of inference questions in this thesis takes observations on the unit circle. Circular or directional data arise whenever the natural sample space is cyclic: angles in degrees, times of day, headings of moving objects, or phases of an oscillator. The corresponding detection question in this thesis asks whether all observations are unstructured (uniform on the circle), or whether a small subset of the observed angles is concentrated around an unknown common phase.

These models have natural concrete applications. In event-stream analytics, the time-stamps of events such as financial transactions, equipment failures, or cyber-attacks inherently live on a 24-hour or annual cycle, and a small population of events driven by a diurnal or seasonal mechanism shows up as a planted concentration around an unknown phase [MJ09]. For example, in neuroscience, the instantaneous phase of a neural oscillator is approximately uniform when the population is desynchronized. Conversely, a subpopulation driven by a common rhythm becomes localized around a shared direction, and detecting this localization is foundational to the study of phase-locked oscillations. In multivariate time-series analysis, the cross-spectral coherences between pairs of signals at a given frequency are themselves angular quantities, and a community of signals sharing a common driver appears as a planted phase-coherent community in the resulting angular network on the edges of a complete graph.

Fundamental limits of detection in this setting play the same role as their counterparts in graph-based detection: they identify the signal-to-noise floor below which no test, regardless of complexity, can reliably distinguish the structured alternative from a uniform null. For an applied scientist with limited samples, such a threshold is concrete operational guidance: it says how strongly the planted angles must be concentrated, as a function of the population size N and the size K of the planted subset, before any test stands a chance of succeeding. Chapter 5 derives such thresholds, near-tight up to constants and occasional logarithmic factors, under two different noise models, and for both planted-subset and planted-community structure.

1.2 Robust inference and consensus on networks

The random-graph models that make the inferential thresholds above extractable are deliberately stylized. Real networks do not, in general, fit them. For example, in real networks, degree distributions are heavy-tailed, edges are clustered, communities are themselves learned rather than known, and a non-trivial fraction of nodes may behave in ways the generative model does not capture. A natural and operationally important question is what happens when the modelling assumptions partially break. Which guarantees survive? Which break catastrophically? At what rate does performance degrade as the corruption grows? This is the robustness question, and the second part of the thesis takes it up in two complementary settings.

The first concerns network alignment with corrupt nodes. The standard correlated random-graph model assumes that the two observed networks are subsamples of a common parent and that the only

source of disagreement between them is independent edge noise. In real applications this is wrong at the node level. A user’s social-network account may be hacked and rewired arbitrarily by an attacker [NS09]; a protein in a PPI network may interact essentially at random because of expression failure, toxicity, or because the protein is itself an unknown transcription factor [Fio19, KPA⁺12]. The edges incident on corrupted nodes are typically unreliable, and the corruption is spatially clustered around those nodes rather than spread independently across the graph. Because the worst-case corruption is qualitatively different from independent edge noise, the question of how a matching algorithm degrades under node corruption is open even in the basic homogeneous setting. Chapter 6 addresses this question for two explicit corruption models.

The second setting is the security of distributed consensus protocols under realistic communication. Existing security analyses of blockchain protocols typically assume *synchronous* communication: a known finite upper bound on the delay of any point-to-point message between honest peers [PSS17, Ren19, GKL20, DKT⁺20, GKR20]. The assumption is comfortable for analysis but unrealistic in deployment. Wireless and mobile peer-to-peer networks lose messages routinely; partitions and eclipse attacks deny service in bursts; the actual delay distribution between honest peers has no known finite upper bound. A security analysis that does not explicitly account for message loss can yield parameter recommendations that overstate what a deployed system actually guarantees, with material consequences for any user who relies on the advertised security parameters. Chapter 7 introduces a communication model in which a fixed fraction of messages are permanently lost in transit, and asks how much adversarial computational power the proof-of-work longest-chain protocol can tolerate while still achieving consensus.

1.3 Other work not featured in the thesis

A separate line of work, pursued through external collaboration during the course of this Ph.D., concerns the design of *flexibility structures* in two-sided matching markets. The setting is a bipartite graph in which one side represents demand and the other represents supply, and the central question is how to design the compatibility structure so as to maximize the expected size of a maximum matching in the resulting random graph. This question motivates platforms such as ride-hailing services and on-demand labor markets, where the platform shapes flexibility through contracts, service-area assignments, and incentives. Two companion papers explore the question in complementary models.

- (i) *Spatial matching*. Here the underlying compatibility graph is a bipartite random geometric graph: each supply node connects with all demand nodes that fall within its service region. The question is how to choose the service regions, subject to a sum constraint on their total volume, so as to maximize the expected matching size. The paper [ASY26b] establishes a *uniformity principle*: more uniform allocations of the service-region volumes across supply nodes (in the sense of majorization) yield larger expected matchings.

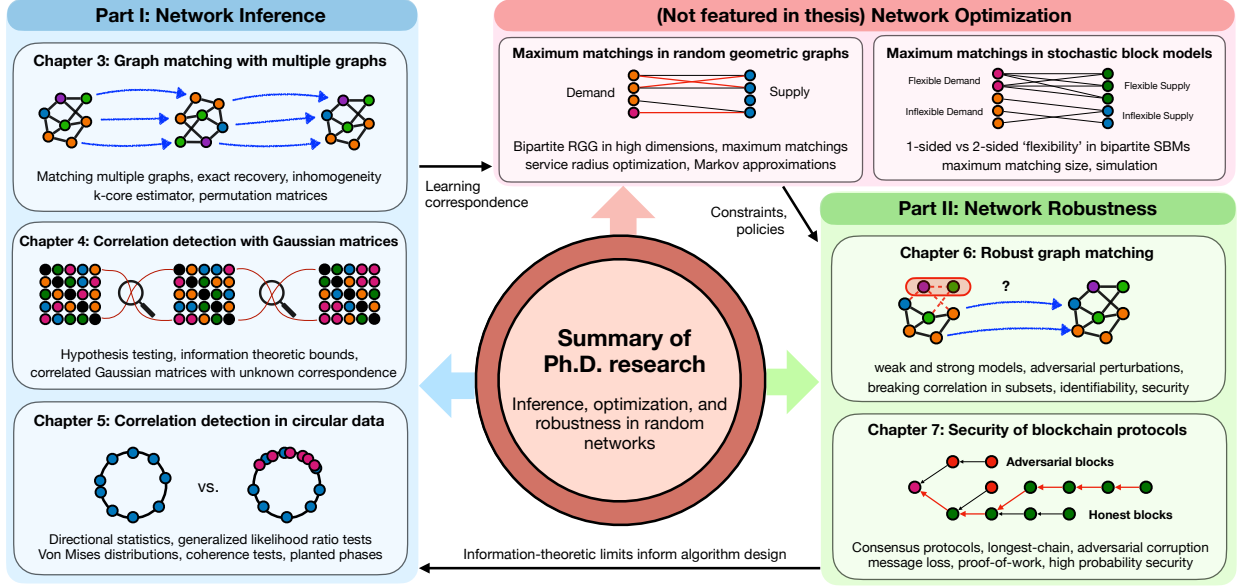


Figure 1.2: Summary of research.

- (ii) *Block models*. Here the compatibility graph is a 2×2 stochastic block model, in which agents on either side of the market are either *regular* or *flexible*, with regular–regular pairs least likely to connect, regular–flexible pairs more likely, and flexible–flexible pairs most likely. For this model, the paper derives an exact variational formula for the asymptotic size of a maximum matching, by extending the local weak convergence framework of [BLS11] to multi-type unimodular Galton–Watson trees [ASY26a]. The formula is then used to compare two flexibility-allocation strategies: one in which a service platform incentivizes flexibility on a single side of the market, and one in which it splits the budget equally between the two sides. This approach sharpens and extends the comparisons of [FMZ26].

Due to space constraints, these works are not included in the present thesis; the interested reader is referred to the available preprints [ASY26b, ASY26a].

1.4 Organization of the thesis

The chapters are grouped into two parts. Part I concerns inference in idealized settings, and Part II concerns deviations from those settings. Figure 1.2 summarizes the contributions of each of the chapters. Background material common to several chapters is collected in Chapter 2.

Part I: Statistical Inference on Networks

Part I considers inference from noisy or unlabeled observations of one or more random graphs, with the aim of identifying sharp information-theoretic thresholds for recovery and detection.

- Chapter 3 establishes fundamental limits for recovering the underlying node correspondence

(i.e. node level matchings) among *multiple* correlated graphs. In the setting of inhomogeneous random graphs, a matching algorithm is presented and analyzed: graphs are first partially matched pairwise, and the partial matchings are then combined by transitivity. The analysis yields a sufficient condition on the problem parameters for exactly matching all nodes across all the graphs. In the setting of homogeneous (Erdős–Rényi) graphs, this condition is shown to be also necessary, so that the algorithm operates down to the information-theoretic threshold. This uncovers a regime in which exact matching between two graphs alone is impossible but becomes possible with $m > 2$ graphs. Converse results are also established in the inhomogeneous setting, where transitivity again plays a role. Along the way, independent results on the k -core of inhomogeneous random graphs are derived. The chapter is based on the papers [AH25b] and [AH24a].

- Chapter 4 addresses the hypothesis testing problem of deciding whether $m \geq 2$ complete weighted graphs with Gaussian edge weights are mutually correlated after unknown relabelings of their vertices. Under the null model all edge weights are independent standard Gaussians, whereas under the planted model the graphs share a latent vertex alignment and each pair of corresponding edge weights has correlation ρ . For fixed m , we identify the sharp information-theoretic threshold for detection. Above the threshold, a generalized likelihood-ratio test achieves strong detection, whereas even weak detection is impossible below the threshold. The result extends the two-graph detection threshold of Wu, Xu, and Yu [WXY23] to any fixed number of graphs, exhibits a side-information regime in which two graphs alone are insufficient but multiple graphs enable detection, and, together with the recovery threshold of Vassaux and Massoulié [VM25], shows that this Gaussian multi-graph model has no detection–recovery gap. The chapter is based on the paper [AH25a].
- Chapter 5 introduces hypothesis testing problems for circular data, where observations take values on the unit circle and may contain a hidden, phase-coherent structure. Under the null hypothesis, the data are independent and uniform on the unit circle; under the alternative, either (i) a planted subset of size K concentrates around an unknown phase (the flat setting), or (ii) a planted community of size k induces coherence among the edges of a complete graph (the community setting). In each setting, two models are considered: a hard-cluster distribution, in which correlated planted observations lie in an arc of known length and unknown location, and a von Mises distribution, in which correlated planted observations follow a von Mises distribution with a common unknown location parameter. For each of the four resulting models, nearly matching necessary and sufficient conditions are derived, up to constants and occasional logarithmic factors, thereby establishing phase transitions for detectability. The chapter is based on the paper [AH26].

Part II: Robust Inference and Consensus on Networks

Part II returns to inference and consensus, but under deviations from the idealized model.

- Chapter 6 illustrates two models for graph matching in the presence of corrupt nodes. The *weak* model, motivated by biological networks, allows one or both networks to have a positive fraction of molecular entities interact *randomly* with their network. For this model, no estimator can correctly recover a positive fraction of the corrupt nodes. Necessary conditions for any estimator to correctly identify and match all the uncorrupt nodes are derived, and these conditions are shown to also be sufficient for the k -core estimator. The *strong* model, motivated by social networks, permits one or both networks to have a positive fraction of users connect *arbitrarily*. For this model, detection of corrupt nodes is impossible; nevertheless, when only one of the networks is compromised, we can correctly match a positive fraction of nodes under appropriate conditions, albeit without identifying them. The chapter is based on the paper [AH24b].
- Chapter 7 details the proof-of-work longest-chain consensus protocol in the presence of message losses. Security analyses of blockchain consensus have primarily focused on the synchronous model, where point-to-point communication delays are upper bounded by a known finite constant. This assumption is unrealistic in noisy settings where messages may be lost (i.e., incur infinite delay). A new communication model, the $0\text{-}\infty$ model, is introduced to capture such losses, and a region of tolerable adversarial power under which the consensus protocol is secure is derived. The guarantees take the form of a simple bound on the probability that a transaction violates desired security properties, and this violation probability decays almost exponentially in the security parameter. The analysis proceeds by constructing combinatorial objects from blocktrees and identifying random variables associated with them that are amenable to analysis. The resulting bound improves on the state of the art and extends the known regime for the tolerable adversarial threshold in settings where messages may be lost. The chapter is based on the paper [ASH22].

Chapter 2

Background

The introduction staked out two themes: inference and robustness, that animate this thesis. In each case, the object under study is a random graph, and the machinery of analysis is a blend of combinatorics, probability, and statistics. This chapter gathers the background material needed for the rest of the thesis. The material below is standard, and presented here mainly to fix notation and to establish a shared vocabulary across subsequent chapters.

2.1 Graphs and graph models

A *graph* is a pair $G = (V, E)$ consisting of a vertex set V and an edge set $E \subseteq \binom{V}{2}$, where $\binom{V}{2}$ denotes the set of unordered pairs of distinct vertices. When $|V| = n$, we typically identify V with $[n] \triangleq \{1, 2, \dots, n\}$. Two vertices $u, v \in V$ are *adjacent* if $\{u, v\} \in E$, in which case we write $u \sim v$. The *degree* of a vertex v , denoted $\deg(v)$, is the number of vertices adjacent to it. A graph is *simple* if it has no self-loops and no multiple edges; all graphs in this thesis are simple unless stated otherwise. A *directed graph* or *digraph* replaces unordered pairs with ordered pairs: $E \subseteq V \times V$. A *weighted graph* attaches a real-valued weight to each edge.

It is convenient to represent a simple graph on n vertices by its *adjacency matrix* $A \in \{0, 1\}^{n \times n}$, where $A_{ij} = 1$ iff $\{i, j\} \in E$. For simple undirected graphs, A is symmetric with zero diagonal, so the graph is fully specified by the strictly upper-triangular entries.

A *random graph model* is a probability distribution over graphs on a common vertex set. We introduce the four models that recur throughout the thesis, in roughly increasing order of structure.

2.1.1 The Erdős–Rényi model

The simplest and most studied random graph model is the Erdős–Rényi model $G(n, p)$ [ER59]. A graph on n labelled vertices is sampled by including each of the $\binom{n}{2}$ possible edges independently with probability $p \in [0, 1]$. Equivalently, the strictly upper-triangular entries of the adjacency matrix

are i.i.d. $\text{Bern}(p)$. Despite its simplicity, $G(n, p)$ already exhibits a rich set of phase transitions: connectivity, the emergence of a giant component, and the appearance of small subgraphs each undergo sharp thresholds at specific scalings of p with n . The Erdős–Rényi model serves as the null model for most of the graph matching and correlation detection work in this thesis.

2.1.2 Stochastic block models

Real networks are rarely homogeneous; vertices often belong to communities or types, and the probability of an edge depends on the types of its endpoints. The *stochastic block model* (SBM) captures this in the simplest way. Each of n vertices is assigned a type in $[K]$ (either deterministically or by i.i.d. sampling from a distribution over types), and for each pair i, j with types a, b , an edge is placed independently with probability B_{ab} , where $B \in [0, 1]^{K \times K}$ is a symmetric matrix of between-type edge probabilities. The special case $K = 1$ recovers $G(n, p)$. SBMs are the workhorse model for community detection and for any analysis that involves vertex heterogeneity.

2.1.3 Inhomogeneous random graphs

Inhomogeneous random graphs generalize the Erdős–Rényi model and the SBM by letting the edge probability depend on the pair of endpoints. Let $\mathbf{P} = (p_{ij})_{i, j \in [n]}$ be a symmetric matrix with zero diagonal and entries in $[0, 1]$, called the *parameter matrix*. A graph G is sampled from the inhomogeneous random graph distribution $\text{RG}(n, \mathbf{P})$ if each edge $\{i, j\}$ with $i < j$ is included independently with probability p_{ij} . The Erdős–Rényi model is the special case $p_{ij} = p$, and the SBM is the special case in which p_{ij} takes finitely many values determined by the types of i and j . Inhomogeneous random graphs are the natural model for networks with arbitrary degree heterogeneity.

2.1.4 Random geometric graphs

A *random geometric graph* (RGG) embeds each vertex as a point in a metric space and places an edge between two vertices whenever their distance falls below a threshold. Formally, let $X_1, \dots, X_n$ be i.i.d. samples from a distribution on a metric space $(\mathcal{X}, d)$ and fix a *connection radius* $r > 0$. The RGG $G(n, r)$ has an edge $\{i, j\}$ iff $d(X_i, X_j) \leq r$. A soft version of the model has the edge present with probability p if the locality condition is met, and absent otherwise. Unlike the previous three models, the edges of an RGG are *not* conditionally independent given a low-dimensional label: geometric constraints (triangle inequality, clustering) make its analysis qualitatively different.

2.1.5 Correlated random graphs

Several chapters study not a single random graph but a *pair* (or even a tuple) of graphs on a common vertex set whose edges are statistically dependent. The canonical construction is the *correlated Erdős–Rényi* model with parameters (n, p, s) :

- (i) Sample a parent graph $G_0 \sim G(n, p)$.

- (ii) Obtain G_1 and G_2 by independently retaining each edge of G_0 with probability s (“subsampling”).
- (iii) Apply a latent permutation $\pi^*: [n] \rightarrow [n]$ to relabel the vertices of G_2 .

The result is a pair of graphs (G_1, G_2) whose adjacency matrices agree on the vertex pairs whose underlying edge in G_0 was retained in both subsamples, with one copy of the graph scrambled by the unknown permutation π^* . Two families of problems arise from this construction:

- *Graph matching*: recover π^* given (G_1, G_2) .
- *Correlation detection*: decide whether (G_1, G_2) was generated from a correlated pair ($s > 0$) or from two independent Erdős–Rényi graphs ($s = 0$).

Chapters 3 and 6 study the former; Chapter 4 studies the latter, in a Gaussian relaxation where the Bernoulli edge indicators are replaced by real-valued weights.

2.2 Matchings between graphs

Given two graphs G_1 and G_2 on vertex sets V_1 and V_2 of equal size, a (*full*) *matching* is a bijection $\pi: V_1 \rightarrow V_2$. When $|V_1| = |V_2| = n$, such a matching is equivalent to a permutation in the symmetric group $\mathcal{S}_n$, and we use the two terms interchangeably. In matrix form, a matching corresponds to an $n \times n$ permutation matrix P with $P_{ij} = \mathbf{1}\{\pi(i) = j\}$.

Many applications require the flexibility of a *partial matching*: an injection π whose domain is a subset $\text{dom}(\pi) \subseteq V_1$. Partial matchings naturally arise when some vertices are “unrecoverable” or are declared ambiguous by an estimator. Throughout the thesis, we measure the quality of an estimated matching $\hat{\pi}$ against the ground truth π^* by the *overlap* or *error* on $\text{dom}(\hat{\pi})$:

$$\text{ov}(\hat{\pi}, \pi^*) \triangleq \sum_{i \in \text{dom}(\hat{\pi})} \mathbf{1}\{\hat{\pi}(i) = \pi^*(i)\}, \quad \text{err}(\hat{\pi}, \pi^*) \triangleq |\text{dom}(\hat{\pi})| - \text{ov}(\hat{\pi}, \pi^*).$$

Three recovery regimes are commonly distinguished: *exact recovery* ($\hat{\pi} = \pi^*$ with probability $1 - o(1)$), *almost-exact recovery* ($\text{ov}/n \rightarrow 1$), and *partial recovery* ($\text{ov}/n \rightarrow \alpha$ for some $\alpha > 0$).

2.3 Asymptotic notation

All asymptotics in this thesis are taken as $n \rightarrow \infty$, where n is typically the number of vertices of the graph under study. For two sequences of positive reals a_n, b_n , we use standard asymptotic notation:

- $a_n = O(b_n)$ if $\limsup_{n \rightarrow \infty} a_n/b_n < \infty$.
- $a_n = \Omega(b_n)$ if $b_n = O(a_n)$.
- $a_n = \Theta(b_n)$ if $a_n = O(b_n)$ and $a_n = \Omega(b_n)$.

- $a_n = o(b_n)$ if $\lim_{n \rightarrow \infty} a_n/b_n = 0$.
- $a_n = \omega(b_n)$ if $b_n = o(a_n)$.

2.4 Statistical tools

In this section, we visit some tools from statistics that are used throughout the thesis.

2.4.1 Parameter estimation: maximum likelihood and maximum a posteriori

Let $\mathcal{P} = \{p_\theta : \theta \in \Theta\}$ be a parametric family of distributions indexed by a parameter space Θ , and let X be an observation drawn from p_{θ^*} for some unknown $\theta^* \in \Theta$. An *estimator* is any measurable function $\hat{\theta}(X)$ that maps observations to Θ (or to a larger set). The quality of an estimator is typically measured by a loss $\ell(\hat{\theta}, \theta^*)$ and evaluated in expectation (*risk*) or with high probability.

Two natural choices of estimator recur throughout the thesis. The *maximum likelihood estimator* (MLE) is the parameter that maximizes the likelihood of the observed data,

$$\hat{\theta}^{\text{MLE}}(X) \triangleq \arg \max_{\theta \in \Theta} p_\theta(X).$$

When θ is itself modeled as a random variable with a prior distribution π on Θ , the corresponding Bayesian estimator is the *maximum a posteriori* (MAP) estimator, obtained by maximizing the posterior $\pi(\theta | X) \propto p_\theta(X) \pi(\theta)$:

$$\hat{\theta}^{\text{MAP}}(X) \triangleq \arg \max_{\theta \in \Theta} p_\theta(X) \pi(\theta).$$

When the prior is uniform on Θ , the MAP and MLE estimators coincide. In particular, the latent permutations in the graph matching problems studied in this thesis are sampled uniformly at random from the symmetric group, so the MLE and MAP analyses there are equivalent.

2.4.2 Hypothesis testing

A *hypothesis test* chooses between two candidate distributions p_0 (the null) and p_1 (the alternative) based on an observation X . A test is a measurable function $\phi(X) \in \{0, 1\}$, with $\phi = 0$ meaning “accept the null.” Its two error probabilities are the *false-alarm* (or Type I) probability

$$p_{\text{fa}}(\phi) \triangleq \mathbb{P}_0(\phi(X) = 1)$$

and the *miss* (or Type II) probability

$$p_{\text{miss}}(\phi) \triangleq \mathbb{P}_1(\phi(X) = 0).$$

The *Neyman–Pearson lemma* states that, among all tests with $p_{\text{fa}} \leq \alpha$, the one that minimizes p_{miss} is a thresholded likelihood ratio $\phi(X) = \mathbf{1}\{p_1(X)/p_0(X) \geq \tau\}$.

Hypothesis testing underpins several chapters. Chapter 4 asks whether two Gaussian-edge-weighted graphs share hidden correlation or are independent; Chapter 5 addresses an analogous question for paired samples of circular data. In both cases, the fundamental quantities are precisely p_{fa} and p_{miss} , and the interest is in the rate at which both can be driven to zero.

2.4.3 Information-theoretic limits and phase transitions

An *information-theoretic lower bound* is a statement of the form: below some threshold on the signal-to-noise ratio, no estimator, regardless of its runtime, can succeed with probability bounded away from zero. Such bounds are typically proved via Fano’s inequality or by direct computation of total variation distance

$$\text{TV}(p_0, p_1) \triangleq \frac{1}{2} \int |p_0 - p_1|$$

or Kullback–Leibler divergence

$$\text{D}_{\text{KL}}(p_0 \| p_1) \triangleq \int p_0 \log(p_0/p_1).$$

The matching upper bound is often delivered by a concrete algorithm achieving the bound, such as the MLE or by a relaxation of it. When upper and lower bounds coincide, the result is a *phase transition*: a sharp threshold above which recovery is possible and below which it is impossible.

2.4.4 Concentration of measure

Many proofs in this thesis utilize that the empirical value of a random quantity does not deviate much from its mean. Three tools appear repeatedly:

- *Markov’s* and *Chebyshev’s* inequalities, for crude first-moment and second-moment bounds;
- the *first- and second-moment methods*, which bound $\mathbb{P}(X = 0)$ and $\mathbb{P}(X > 0)$ respectively for nonnegative integer-valued random variables;
- *Chernoff-type* bounds and their extensions (e.g. Hoeffding) for sums of independent random variables.

For Erdős–Rényi and related models, these bounds deliver exponential control on degrees, subgraph counts, and related quantities.

2.5 Circular data

Chapter 5 is concerned with a class of data that does not fit the linear framework of the previous section: *circular* or *directional* data, i.e., observations taking values on the unit circle $S^1 = [0, 2\pi)$.

Circular data arise naturally whenever the sample space has a cyclic structure, e.g. compass headings, time-of-day, phase of an oscillator, orientation in a plane.

The simplest non-trivial distribution on S^1 is the *von Mises* distribution with mean direction $\mu \in S^1$ and concentration parameter $\kappa \geq 0$:

$$f(\theta; \mu, \kappa) = \frac{1}{2\pi I_0(\kappa)} \exp(\kappa \cos(\theta - \mu)),$$

where I_0 is the modified Bessel function of the first kind. Von Mises is the circular analog of the Gaussian: it is the maximum-entropy distribution on S^1 with a prescribed first moment, and it plays the role of the natural “noise” model in the hypothesis-testing question of Chapter 5.

Part I

Statistical Inference on Networks

Chapter 3

Network alignment with multiple inhomogeneous graphs

3.1 Introduction

The graph alignment problem is to recover the underlying latent node correspondence between correlated networks, i.e. to match the nodes across graphs. The problem derives its importance from the fact that nodes in networks are often *unlabeled* or *scrambled* – for example, user identities in social networks are anonymized for privacy, and proteins in interactome assays are not natively aligned across species. Applications include:

- (a) Social networks: Matching two social networks amounts to recovering the hidden node identities of users in the anonymized network. [NS09] established the efficacy of this approach: A third of the users active on both Twitter and Flickr could be identified with only a 12% error rate, simply by matching the anonymous Twitter graph to the Flickr graph. In a related work [NS08], it is shown that anonymous movie ratings of Netflix users can be recovered by matching to the public Internet Movie Database (IMDb).
- (b) Biological networks: The protein-protein interactomes of two different species are often correlated networks. [SXB08] showed that matching the PPI network of yeast to that of fruit flies allowed identification of conserved functional components between the two species. This transfer of knowledge between species is a useful tool in biology and genomics [KHGPM16, BSI06]. Similarly, the brain connectomes of healthy humans are correlated [STK05], and their alignment is useful in detecting abnormalities [CPPDG24].

- (c) Natural language processing: Given a passage of text and a hypothesis, consider the ob-

This chapter is based on joint work with Bruce Hajek. Part of the material was published as [AH25b]. © 2025 IEEE. Reprinted, with permission, from T. Ameen and B. Hajek, “Exact random graph matching with multiple graphs,” in *2025 IEEE International Symposium on Information Theory (ISIT)*, 2025. In accordance with IEEE policy, the authors retain the right to reuse this material in a dissertation.

jective of determining whether the hypothesis is entailed by the passage. Haghighi and co-authors [HNM05] presented an approach to solving this, by first constructing dependency trees of the text and hypothesis respectively, and then matching them to conclude with a yes/no answer after thresholding a score of the output matching.

- (d) Computer vision: Graph matching is employed extensively for vision tasks. One example is object detection: [SS05] showed that objects in images can be detected by matching graph representations of the test image to a feature image that contains the object. Yet another example in pattern recognition is the problem of tracking the movement of objects in a sequence of frames [YYL⁺16].
- (e) Autonomous agents: Robots and self-driving cars rely on a multitude of sensors to collect data about their environment. Combining information from various sensors increases data fidelity. For example, [YDF⁺22] showed that graph alignment is useful for scene-map construction when multiple robots explore an unknown environment.

There is a long history of utilizing domain knowledge to implement graph alignment algorithms for these tasks – a survey can be found for example in [YYL⁺16, VF13]. Even so, the theoretical study of graph matching algorithms and their performance guarantees is a relatively recent undertaking. The majority of work has focused on pairs of random graphs, particularly Erdős-Rényi graphs. This chapter studies thresholds for exact recovery with multiple inhomogeneous random graphs.

3.1.1 Related work

The theoretical study of graph matching algorithms and their performance guarantees has focused on the homogeneous setting of Erdős-Rényi (ER) graphs. Pedarsani and Grossglauser [PG11] introduced the subsampling model to generate two such correlated graphs: the model entails twice subsampling each edge independently from a parent ER graph to obtain two sibling graphs, both of which are marginally ER graphs themselves. The goal is then to match nodes between the two graphs to recover the underlying latent correspondence. This has been the framework of choice for many works that study graph matching. For example, Cullina and Kiyavash [CK16, CK17] studied the problem of *exactly matching* two ER graphs, where the objective is to match *all* vertices correctly. They identified a threshold phenomenon for this task: exact recovery is possible if the problem parameters are above a threshold, and impossible otherwise. Subsequently, threshold phenomena were also identified for *partial* graph matching between ER graphs, where the objective is to match only a positive fraction of nodes [GML21, HM23, WXY22, DD23b, Du25]. The case of almost-exact recovery – where the objective is to match all but a negligible fraction of nodes – was studied by Cullina and co-authors [CKMP19]: a necessary condition for almost exact recovery was identified, and the same condition was shown to also sufficient be for the *k*-core estimator; the estimator is described in Section 3.3. This estimator proved useful for graph matching in other contexts such as the stochastic block model [GRS22] and inhomogeneous random graphs [RS23]. Ameen and Hajek [AH24b] showed some robustness properties of the *k*-core estimator in the context

of matching ER graphs under node corruption. The estimator also plays an important role in this chapter.

Other works have studied *computational* aspects of graph alignment: Various *efficient* algorithms have been proposed, including algorithms based on the spectrum of the graph adjacency matrices [FMWX22], and node degree and neighborhood based algorithms [DCKG19, DMWX21, MRT23], as well as algorithms based on iterative methods [DL23] and counting subgraphs [MWXY23, BCL⁺19, CR24]. While all these works study Erdős-Rényi graphs, there is recent interest in computational aspects of graph alignment in other models such as the stochastic block model [CR24], as well as inhomogeneous random graphs [DFW23]. Still other works have studied the graph matching problem under side information – this includes settings where a subset of nodes are matched a priori and provided as seeds [PG11, MX20, LS18], the setting where node attributes are present [WZWW24, ZWWW24, YC24], and the setting where correlation is localized to a subset of nodes [HSY24].

Incorporating information from *multiple* graphs to match them has been recognized as an important research direction, for instance in the work of Gaudio and co-authors [GRS22]. Several studies have approached this from an empirical perspective [LLB⁺09, VM17, ZZG24]. To our knowledge, a limited number of works study the theory of matching multiple graphs; we review them below. The works [JLK21] and [RS21] note that it is possible to exactly match m graphs whenever it is possible to exactly match any two graphs by pairwise matching all the graphs exactly. In contrast, we show that under appropriate conditions, it is possible to exactly match m graphs even when no two graphs can be pairwise matched exactly.

In independent and concurrent work, Rácz and Zhang [RZ24] study the problem of exact community recovery and graph matching from multiple correlated stochastic block models with two balanced communities. Their Theorems 3 and 4 determine the sharp threshold for exact graph matching with m correlated SBMs with 2 balanced communities, where $p \triangleq a \log n/n$ and $q \triangleq b \log n/n$ are the intra and inter-community edge probabilities. Their result for exact graph matching is that it is possible iff $s(1 - (1 - s)^{m-1})(a + b)/2 > 1$. Our achievability result when specialized to SBMs (Corollary 1) agrees with this. In the homogeneous case $a = b$ (i.e., Erdős-Rényi graphs with $p_{ij} = C \log n/n$), this reduces to $Cs(1 - (1 - s)^{m-1}) > 1$, which also agrees with the threshold established in our Theorems 1 and 3. Beyond graph matching, [RZ24] determine the sharp threshold for exact community recovery from m correlated SBMs (their Theorems 1 and 2), revealing a regime where community recovery is possible even when exact graph matching is not. Our work, in contrast, studies only the graph matching problem, in the broader class of inhomogeneous random graphs with general parameter matrices $\mathbf{P}$, which subsumes the SBM and ER model as a special case.

Ameen and Hajek [AH25b] determined the sharp threshold for exactly matching m correlated Erdős-Rényi graphs, showing that exact recovery is possible if and only if $Cs(1 - (1 - s)^{m-1}) > 1$. This chapter substantially generalizes these results in several directions: we treat the full class of inhomogeneous random graphs with arbitrary parameter matrices $\mathbf{P}$, which subsumes Erdős-Rényi graphs,

stochastic block models, random geometric graphs, and Chung–Lu graphs as special cases. Beyond recovering the sharp ER threshold as a corollary (Theorem 3.3), we establish sufficient conditions for exact matching of inhomogeneous graphs (Theorem 3.1), converse results in the heterogeneous setting (Theorem 3.4), and a separation between these bounds demonstrating that the parameter matrix $\mathbf{P}$ itself can serve as side information for matching (Theorem 3.2 and Corollary 3.5). We also derive results on the k -core of inhomogeneous random graphs (Section 3.4) that may be of independent interest.

The recent work of Vassaux and Massoulié [VM25] studies the information-theoretic limit for exact recovery in the *Gaussian* setting, where the observations are multiple correlated Wigner matrices, and also the limit for partial recovery in the homogeneous Erdős–Rényi setting. Also related is the recent work [EG25], which studies statistical-computational gaps and the effect of infinitely many graphs in the Gaussian setting, and [AH25a], which studies the problem of detecting whether m graphs are correlated, also in the Gaussian setting. Closest to our analysis of matching inhomogeneous random graphs is the work [RS23], where sufficient conditions for exact and partial graph matching between *two* inhomogeneous random graphs are studied.

3.1.2 Contributions

This chapter studies fundamental limits of exact recovery with multiple inhomogeneous random graphs. These graphs allow for heterogeneity by having each edge $\{i, j\}$ in the graph appear independently with probability p_{ij} , and subsume other models such as ER graphs and stochastic block models, and are closely related to random geometric graphs.

We consider the generalization of the matching problem to m graphs and propose an algorithm to combine information among the graphs. A key idea is that of *transitive closure* – our algorithm works by pairwise matching graphs partially and then boosting the matchings by transitivity. Our analysis yields a sufficient condition for exact recovery of multiple inhomogeneous graphs. In the homogeneous case where $p_{ij} = p$, we show that this condition is also necessary, thereby characterizing the sharp reconstruction threshold for exactly matching multiple Erdős–Rényi graphs. In the heterogeneous setting, we also provide converse results and transitive closure plays a role here as well. We present simulation results to corroborate the utility of transitive closure as a black-box, efficient and optimal bridge between the binary and m -ary matching problems.

3.2 Model

Let $\mathbf{P} = (p_{ij})_{i,j \in [n]}$ with $0 \leq p_{ij} = p_{ji} \leq 1$ denote a symmetric matrix with zero diagonals. A graph G is an inhomogeneous random graph on the vertex set $[n] \triangleq \{1, 2, \dots, n\}$ with parameter matrix $\mathbf{P}$, denoted by $G \sim \text{RG}(n, \mathbf{P})$, if each edge $\{i, j\}$ with $i < j$ exists independently with probability p_{ij} . Consider the natural extension of the subsampling model to m graphs. Let $s \in (0, 1]$, and obtain m correlated inhomogeneous graphs $G'_1, \dots, G'_m$ by subsampling each edge of a parent graph $G \sim$

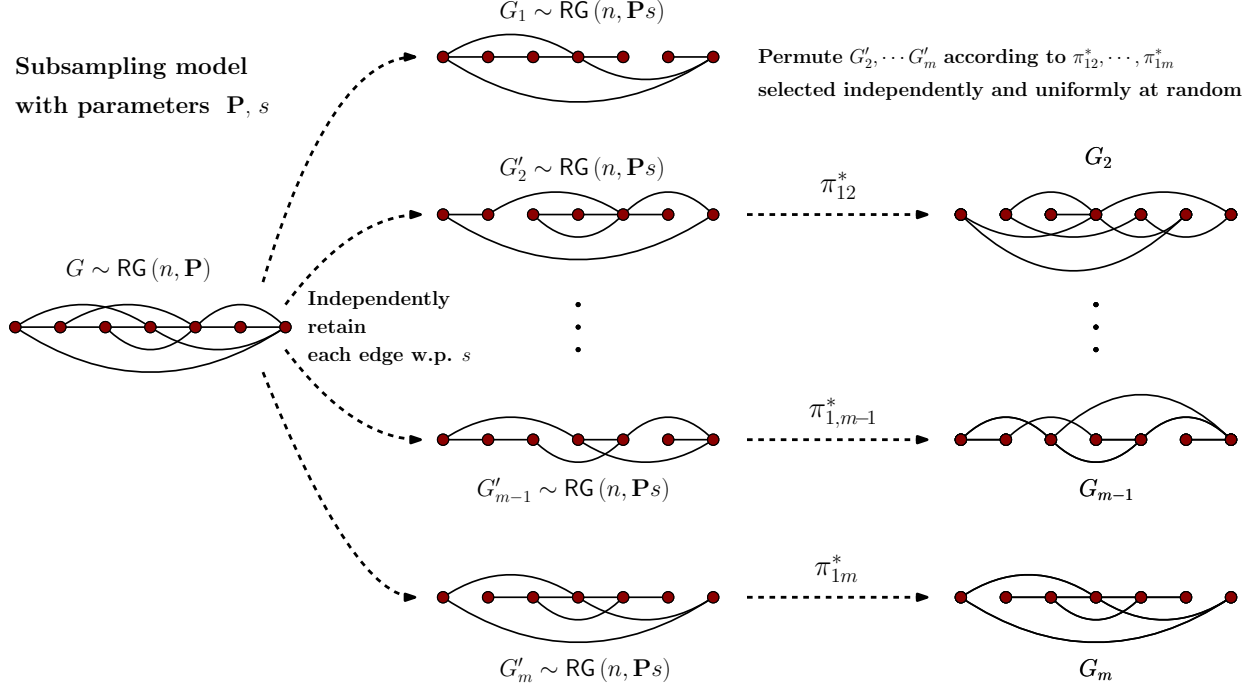


Figure 3.1: Obtaining m correlated inhomogeneous graphs via the subsampling model.

$\text{RG}(n, \mathbf{P})$ independently with probability s . These sibling graphs are themselves inhomogeneous random graphs with parameter matrix $\mathbf{P}s$. Let $\pi_{12}^*, \dots, \pi_{1m}^*$ denote independent permutations that are each drawn uniformly at random from the set of all permutations on $[n]$. Obtain the graphs $G_2, \dots, G_m$ by permuting the nodes of G'_ℓ according to $\pi_{1\ell}^*$, for each $\ell = 2, \dots, m$. For example, $\{i, j\}$ is an edge in G'_ℓ if and only if $\{\pi_{1\ell}^*(i), \pi_{1\ell}^*(j)\}$ is an edge in G_ℓ . An illustration of the subsampling model is provided in Figure 3.1.

We will use the tuple $(\pi_{12}, \dots, \pi_{1m})$ of permutations to identify the correspondence between all m graphs. Stated thus, the graph G_1 is used as a reference and so we will interchangeably use G_1 and G'_1 . Note that the underlying correspondence between all the graphs is fixed upon fixing $\pi_{12}^*, \dots, \pi_{1m}^*$: for any two graphs G_k and G_ℓ , their correspondence is given by $\pi_{k\ell}^* \triangleq \pi_{1\ell}^* \circ (\pi_{1k}^*)^{-1}$.

In this chapter, we are concerned with *exact recovery*. The goal is to determine necessary and sufficient conditions on the parameters $\mathbf{P}, s$ and m so that given correlated inhomogeneous graphs $G_1, \dots, G_m$ from the subsampling model, it is possible to exactly recover the underlying correspondences $\pi_{12}^*, \dots, \pi_{1m}^*$ with probability $1 - o(1)$.

Formally, a *matching* $(\mu_{12}, \dots, \mu_{1m})$ is a collection of injective functions with domain $\text{dom}(\mu_{1i}) \subseteq V$ for each i , and co-domain V . An *estimator* is simply a mechanism to map any collection of graphs $(G_1, \dots, G_m)$ to a matching. We say that an estimator *completely* matches the graphs if the output mappings $\mu_{12}, \dots, \mu_{1m}$ are all complete, i.e. they are all permutations on $[n]$. If the matching $(\mu_{12}, \dots, \mu_{1m})$ satisfies $\mu_{1j} = \pi_{1j}^*$ for all $2 \leq j \leq m$, then all nodes are correctly matched across all graphs, i.e. the graphs are exactly matched.

Some special cases of inhomogeneous random graphs The inhomogeneous random graph model captures a variety of generative models. Following [RS23], we introduce these models conditioned on latent variables intrinsic to them for convenience.

- (a) Erdős-Rényi graph: Given a parameter $p \in [0, 1]$, the homogeneous case of ER graphs sets $p_{ij} = p$ for all distinct i and j .
- (b) Stochastic block model (SBM) [Abb18, HLL83]: Given a partition $V_1, \dots, V_M$ of the node set $[n]$ into M communities and edge probabilities $(q_{ab})_{1 \leq a \leq b \leq M}$, the SBM sets $p_{ij} = q_{ab}$ for all distinct i and j such that $i \in V_a$ and $j \in V_b$.
- (c) Random geometric graph (RGG) [LR23]: Given a collection $x_1, \dots, x_n$ of points in a set $S \subseteq \mathbb{R}^d$, and parameters $p \in [0, 1]$ and $r > 0$, the RGG sets $p_{ij} = p \cdot \mathbf{1}_{\{\|x_i - x_j\|_2 \leq r\}}$ for distinct i and j .
- (d) Chung-Lu graph (CLG) [CL02]: Given a collection $(w_i)_{1 \leq i \leq n}$ of positive weights satisfying $w_i \leq \sqrt{\sum_{j \in [n]} w_j}$ for all i , the CLG model sets $p_{ij} = \frac{w_i w_j}{\sum_{k \in [n]} w_k}$ for distinct i and j .

Correlated graphs from any of the above models are obtained via the subsampling model.

Notation Throughout this work, $p_{\max}$ denotes the largest entry in the matrix $\mathbf{P}$, i.e. $p_{\max} = \max_{i,j} p_{ij}$. For a scalar s , the matrix $\mathbf{P}s$ is obtained by multiplying each entry of $\mathbf{P}$ with s . For a graph G and node v in G , let $\delta_G(v)$ denote the degree of v . Given a matching μ , we denote by G^μ the graph on node set given by the image of μ obtained by relabeling the nodes of G according to μ , i.e. for $i, j \in \text{dom}(\mu)$:

$$\{i, j\} \in E(G) \iff \{\mu(i), \mu(j)\} \in E(G^\mu).$$

For any two graphs H_1 and H_2 on the same vertex set V , denote by $H_1 \vee H_2$ their *union graph* and by $H_1 \wedge H_2$ their *intersection graph*. An edge $\{i, j\}$ is present in $H_1 \vee H_2$ if it is present in either H_1 or H_2 . Similarly, the edge is present in $H_1 \wedge H_2$ if it is present in both H_1 and H_2 . For a random variable X , denote by $M_X(t) \triangleq \mathbb{E}[e^{tX}]$ the moment generating function of X . Standard asymptotic notation ($O(\cdot)$, $o(\cdot)$, $\Omega(\cdot)$, $\dots$) is used throughout, and it is implicit that $n \rightarrow \infty$.

3.3 Main results and algorithm

3.3.1 Exact recovery threshold: positive results

Our first result is a sufficient condition for exactly matching all nodes across all the graphs.

Theorem 3.1. *Let $d_i = \sum_{j \neq i} p_{ij}$. Suppose that there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p_{\max} =$*

$o(n^{\alpha/2-\varepsilon-1})$ and

$$\sum_{i=1}^n e^{-d_i s(1-(1-s)^{m-1})} = o(n^{-\alpha}). \quad (3.1)$$

There exists an estimator, namely Algorithm 1 described in the next subsection, whose output $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ satisfies

$$\mathbb{P}(\hat{\pi}_{12} = \pi_{12}^*, \dots, \hat{\pi}_{1m} = \pi_{1m}^*) = 1 - o(1).$$

A few remarks are in order.

- The left hand side of (3.1) is approximately the expected number of isolated nodes in the graph $H \triangleq G_1 \wedge (G'_2 \vee \dots \vee G'_m)$ which is distributed as $\text{RG}(n, \mathbf{P}s(1 - (1-s)^{m-1}))$. In the Poisson limit, each node independently has degree $\text{Pois}(d_i s(1 - (1-s)^{m-1}))$ in H , so the node is isolated in H with probability $e^{-d_i s(1-(1-s)^{m-1})}$.
- The sufficient condition (3.1) of Theorem 3.1 can be slightly improved to the condition

$$\sum_{i=1}^n e^{-d_i s(1-(1-s)^{m-1})} - \max_{j \in [n]} e^{-d_j s(1-(1-s)^{m-1})} = o(n^{-\alpha}). \quad (3.2)$$

Let $v^* \in \arg \min_{i \in [n]} d_i$. It can be shown that Algorithm 1 applied to the graphs G_k yields that all nodes except v^* can be exactly matched. The node v^* is then the only unmatched node and can be manually matched without ambiguity.

- [RS23] studied the graph alignment problem when $m = 2$. Setting $m = 2$ in (3.1) yields $\sum_{i=1}^n e^{-d_i s^2} = o(n^{-\alpha})$, which is equivalent to their sufficient condition.
- Algorithm 1 does not require access to the parameter matrix $\mathbf{P}$.
- Algorithm 1 is stated for deterministic matrices $\mathbf{P}$. However, if the matrices $\mathbf{P}$ are random such that $\mathbb{E}[p_{\max}] = o(n^{\alpha/2-\varepsilon-1})$ and

$$\mathbb{E} \left[\sum_{i=1}^n e^{-d_i s(1-(1-s)^{m-1})} \right] = o(n^{-\alpha}) \quad (3.3)$$

then the conclusion of the theorem still holds. This follows by the Markov inequality. For example, the condition $\mathbb{E}[p_{\max}] = o(n^{\alpha/2-\varepsilon-1})$ means there exists a sequence $\delta_n \rightarrow 0$ so that for all large n , $\mathbb{E}[p_{\max}] \leq \delta_n n^{\alpha/2-\varepsilon-1}$. Therefore by Markov's inequality,

$$\mathbb{P}(p_{\max} \geq \sqrt{\delta_n} n^{\alpha/2-\varepsilon-1}) \leq \sqrt{\delta_n}.$$

A similar statement can be made about the sum in (3.1). Therefore, with probability at least $1 - 2\sqrt{\delta_n}$, the matrix $\mathbf{P}$ satisfies $p_{\max} \leq \sqrt{\delta_n} n^{\alpha/2-\varepsilon-1}$ and $\sum_{i=1}^n e^{-d_i s(1-(1-s)^{m-1})} \leq \sqrt{\delta_n} n^{-\alpha}$.

The proof of Theorem 3.1 thus goes through to give the same conclusion as before.

The sufficient condition (3.1) can be specialized to various models that are captured by inhomogeneous random graphs. Some of these are presented next.

Corollary 3.1 (Correlated SBMs). *Let $G_1, \dots, G_m$ be correlated SBMs obtained from the subsampling model. Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $q_{ab} = o(n^{\alpha/2-\varepsilon-1})$ for all a and b . Suppose that*

$$\sum_{a=1}^M |V_a| \cdot \exp \left(-s \left(1 - (1-s)^{m-1} \right) \cdot \sum_{b=1}^m |V_b| q_{ab} \right) = o(n^{-\alpha}).$$

There is an estimator whose output $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ satisfies $\mathbb{P}(\hat{\pi}_{12} = \pi_{12}^, \dots, \hat{\pi}_{1m} = \pi_{1m}^*) = 1 - o(1)$.*

In the balanced 2-community model with $|V_1| = |V_2| = n/2$ and connectivity parameters $q_{11} = q_{22} = a \log(n)/n$ and $q_{12} = b \log(n)/n$, the sufficient condition becomes

$$\sum_{a=1}^M |V_a| \cdot \exp \left(-s \left(1 - (1-s)^{m-1} \right) \cdot \sum_{b=1}^m |V_b| q_{ab} \right) = n^{1-s(1-(1-s)^{m-1})(a+b)/2}.$$

The right hand side is $o(n^{-\alpha})$ for some $\alpha > 0$ whenever $s(1-(1-s)^{m-1})(a+b)/2 > 1$, which agrees with the sufficient condition for exact recovery in [RZ24].

Corollary 3.2 (Correlated RGGs). *Let $G_1, \dots, G_m$ be correlated RGGs obtained from the subsampling model. Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p = o(n^{\alpha/2-\varepsilon-1})$. For a node i , let $N_r(i)$ denote the number of nodes in its r -neighborhood, i.e. $N_r(i) \triangleq |\{j : \|x_i - x_j\|_2 \leq r\}|$. Suppose that*

$$\sum_{i=1}^n e^{-ps(1-(1-s)^{m-1}) \cdot N_r(i)} = o(n^{-\alpha}).$$

There is an estimator whose output $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ satisfies $\mathbb{P}(\hat{\pi}_{12} = \pi_{12}^, \dots, \hat{\pi}_{1m} = \pi_{1m}^*) = 1 - o(1)$.*

The following is a version of Corollary 3.2 assuming a random placement of the nodes.

Corollary 3.3 (Correlated RGGs with random locations). *Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p = o(n^{\alpha/2-\varepsilon-1})$. Let G be a random RGG with parameters p and r and points $X_1, \dots, X_n$ independently and uniformly distributed on the unit torus in $\mathbb{R}^2$. Let $G_1, \dots, G_m$ be obtained from the subsampling model with parameter s . Suppose that*

$$(n-1)\pi r^2 p s(1-(1-s)^{m-1}) - (1+\epsilon)(1+\alpha) \log n \rightarrow \infty \quad (3.4)$$

There is an estimator whose output $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ satisfies $\mathbb{P}(\hat{\pi}_{12} = \pi_{12}^, \dots, \hat{\pi}_{1m} = \pi_{1m}^*) = 1 - o(1)$. (Here $(n-1)\pi r^2 p s(1-(1-s)^{m-1})$ is the mean number of neighbors of a node in the intersection of G_1 with the union of graphs G'_2 through G'_m .)*

Proof. For each $i \in [n]$, $N_r(i)$ has the binomial distribution with parameters $n-1$ and πr^2 , which is

second order less spread out than the Poisson distribution with mean $(n-1)\pi r^2$. Therefore, using the fact $\mathbb{E}[e^{-\theta X}] = e^{-\lambda(1-e^{-\theta})}$ if $X \sim \text{Poisson}(\lambda)$, (3.3) holds if and only if

$$(n-1)\pi r^2[1 - \exp(-ps(1 - (1-s)^{m-1}))] - (1+\alpha)\log n \rightarrow \infty$$

Since $1 - \exp(-u) = u + o(u)$ as $u \rightarrow 0$, a sufficient condition for that is given by (3.4). $\square$

Corollary 3.4 (Correlated CLGs). *Let $G_1, \dots, G_m$ be correlated CLGs obtained from the subsampling model. Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $w_i = o(n^{\alpha/2-\varepsilon-1})\sqrt{\sum_{k \in [n]} w_k}$ for all i . Suppose that*

$$\sum_{i=1}^n e^{-w_i s(1-(1-s)^{m-1})} = o(n^{-\alpha}).$$

There is an estimator whose output $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ satisfies $\mathbb{P}(\hat{\pi}_{12} = \pi_{12}^, \dots, \hat{\pi}_{1m} = \pi_{1m}^*) = 1 - o(1)$.*

Next, we show that the sufficient condition (3.2) can be improved for a class of parameter matrices if the algorithm has access to $\mathbf{P}$.

Theorem 3.2. *Suppose ε and s are fixed with $\varepsilon > 0$ and $0 < s \leq 1$. If d varies with n such that:*

$$ds \geq (1 + \varepsilon) \log n, \tag{3.5}$$

then there exists a parameter matrix $\mathbf{P}$ with $d_i \triangleq \sum_{j \neq i} p_{ij} \leq d$ for $i \in [n]$ and $p_{\max} \triangleq \max_{i,j} p_{ij} = \log^2(n)/n$ such that exact recovery is possible, i.e. m graphs $G_1, G_2, \dots, G_m$ obtained by subsampling each edge independently with probability s from a parent graph $G \sim \text{RG}(n, \mathbf{P})$ can be exactly matched with probability $1 - o(1)$ by some algorithm that has access to $\mathbf{P}$.

The proof of Theorem 3.2 is in Section 3.5.4; it uses randomization and does not construct a specific matrix $\mathbf{P}$. Instead, we view $\mathbf{P}$ as an instance of the random matrix $p_{\max} \cdot A$, where A is the adjacency matrix of an auxiliary random graph $G_{-1} \sim \text{ER}(n, q)$ (with $p_{\max}$ and q chosen appropriately). This matrix $\mathbf{P}$ is deliberately heterogeneous, with entries taking values in $\{0, p_{\max}\}$, and the idea is that the heterogeneity in $\mathbf{P}$ can itself be exploited for matching ($\mathbf{P}$ reveals the structure of G_{-1} , which acts as an additional graph). We also have the following corollary.

Corollary 3.5. *There exists a parameter matrix $\mathbf{P}$ satisfying*

$$\sum_{i=1}^n e^{-d_i s(1-2\varepsilon)} - \max_{j \in [n]} e^{-d_j s(1-2\varepsilon)} = \Omega(n^\varepsilon), \tag{3.6}$$

such that exact recovery is possible with probability $1 - o(1)$ by some algorithm that has access to $\mathbf{P}$.

Proof. Take $ds = (1 + \varepsilon) \log n$ in Theorem 2. Then there exists $\mathbf{P}$ with $d_i \leq d$ and $p_{\max} = \log^2(n)/n$ such that exact recovery is possible. We verify that $\mathbf{P}$ satisfies (4). Since $d_i s \leq (1 + \varepsilon) \log n$ for all

i :

$$\sum_{i=1}^n e^{-d_i s(1-2\varepsilon)} \geq n \cdot e^{-(1+\varepsilon)(1-2\varepsilon) \log n} = n^{1-(1+\varepsilon)(1-2\varepsilon)} = n^{\varepsilon+2\varepsilon^2} \geq n^{\varepsilon}.$$

Since the maximum term is at most 1, subtracting it preserves the $\Omega(n^{\varepsilon})$ lower bound. $\square$

Corollary 3.5 shows that if $(1-s)^{m-1} > 2\varepsilon$, then the sufficient condition (3.2) of Theorem 3.1 does not hold for some $\mathbf{P}$, but exact recovery is nevertheless possible.

3.3.2 Exact recovery threshold: negative results

While Corollary 3.5 shows that (3.2) is not a sharp sufficient condition for general parameter matrices, we prove that this condition is sharp in the homogeneous case of Erdős-Rényi graphs. This provides the sharp threshold for exactly matching m ER graphs.

Theorem 3.3. *Let $C > 0$ and $p_{ij} = C \log n/n$ for each $i < j$, and let $d_i = \sum_{j \neq i} p_{ij}$. Suppose that*

$$\sum_{i=1}^n e^{-d_i s(1-(1-s)^{m-1})} = \Omega(1), \quad (3.7)$$

or equivalently that $Cs(1-(1-s)^{m-1}) < 1$. Then the output $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ of any estimator satisfies

$$\mathbb{P}(\hat{\pi}_{12} = \pi_{12}^*, \dots, \hat{\pi}_{1m} = \pi_{1m}^*) = o(1).$$

In the setting of ER graphs with $p_{ij} = C \log(n)/n$, Theorems 3.1 and 3.3 establish that exact recovery is possible if and only if $Cs(1-(1-s)^{m-1}) > 1$. When $m = 2$, this condition reduces to $Cs^2 > 1$, recovering the reconstruction threshold of [CK17]. Thus, when $m > 2$, there is a regime given by

$$Cs(1-(1-s)^{m-1}) > 1 > Cs^2,$$

where exact recovery is impossible using only the two graphs but possible using all m graphs. This is illustrated in Figure 3.2.

We turn next to negative results in the heterogeneous case. To that end, consider the following lemma about isolated nodes in an inhomogeneous random graph. We state it generally for a matrix $\mathbf{Q}$ and later set $\mathbf{Q} = \mathbf{P}s$ or $\mathbf{Q} = \mathbf{P}s^2$ depending on the graph in consideration.

Lemma 3.1. *Let $H \sim \text{RG}(n, \mathbf{Q})$, and let $r_i = \sum_{j \neq i} q_{ij}$. Suppose that $q_{\max} = o(1/\sqrt{n})$ and*

$$\left(\sum_{i=1}^n e^{-r_i} \right) - \max_{j \in [n]} e^{-r_j} = \Omega(1). \quad (3.8)$$

With probability bounded away from 0, there are at least two isolated nodes in H .

Lemma 3.1 is proved in Appendix A.2.1. Applying it to $H = G_1$ yields a regime where there

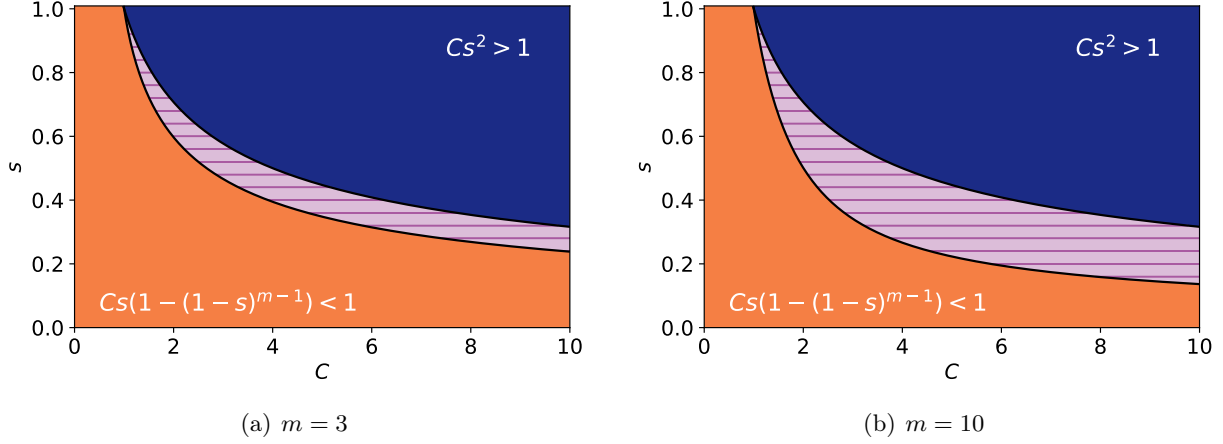


Figure 3.2: Regions in parameter space for the homogeneous setting. *Orange*: Exactly matching m graphs is impossible even with m graphs. *Blue*: Exactly matching 2 graphs is possible with 2 graphs. *Striped*: Impossible to match 2 graphs using only the 2 graphs, but possible using m graphs as side information.

are at least two isolated nodes u and v in G_1 , with probability bounded away from 0. Since it is impossible to distinguish between u and v in G_1 , it is impossible to match them with probability greater than $1/2$. Exactly matching the m graphs is therefore impossible in this regime. This implies the following converse to Corollary 3.5.

Theorem 3.4. *Consider the model of m correlated graphs with parameters $\mathbf{P}$ and s . Let $d_i = \sum_{j \neq i} p_{ij}$. If $p_{\max} = o(1/\sqrt{n})$ and*

$$\left(\sum_{i=1}^n e^{-d_i s} \right) - \max_{j \in [n]} e^{-d_j s} = \Omega(1), \quad (3.9)$$

then for any algorithm the probability of exactly matching all m graphs is $1 - \Omega(1)$.

We now present some remarks about the impossibility results.

- If we ask for a given $\mathbf{P}, s, m$ whether exact recovery is possible by some algorithm, then implicitly algorithms with access to $\mathbf{P}$ should be included because $\mathbf{P}$ is deterministic. In all cases, Theorem 3.1 provides a performance floor while Theorem 3.4 provides a performance ceiling. Even so, for certain structured matrices $\mathbf{P}$, the gap between Theorem 3.1 and 3.4 vanishes, yielding a sharp threshold. For example, Theorem 3.3 shows Theorem 3.1 is sharp for homogeneous $\mathbf{P}$ while Corollary 3.5 shows Theorem 3.4 is sharp for some choices of $\mathbf{P}$. A result of [RZ24] also provides another example of a sharp threshold (balanced 2 community SBM): since their Theorem 4 (converse) matches the achievability in our Corollary 3.1. The sharp thresholds are unclear for the RGG and CLG models, and it remains an open problem to give the sharp recovery threshold for a given general $\mathbf{P}, s, m$.

- Lemma 3.1 can also be used to give a weaker converse to the homogeneous setting (i.e. Erdős-Rényi graphs, Theorem 3.3) when $m = 2$. The maximum likelihood estimator (Appendix A.1) selects a matching to maximize the number of edges in the intersection graph. Setting $H = G_1 \wedge G'_2 \sim \text{RG}(n, \mathbf{P}s^2)$ in Lemma 3.1, we have that if $p_{\max} = o(1/\sqrt{n})$ and

$$\left(\sum_{i=1}^n e^{-d_i s^2} \right) - \max_{j \in [n]} e^{-d_j s^2} = \Omega(1), \quad (3.10)$$

then, with probability $\Omega(1)$, there are at least two nodes u and v that are isolated in $G_1 \wedge G'_2$, i.e. u and v both have disjoint sets of neighbors in G_1 and G'_2 . Thus, the MLE places equal probability on (i) the true matching and (ii) the matching which maps all other nodes correctly but swaps u and v , and so the estimator fails with probability $\Omega(1)$.

- In the heterogeneous setting, the existence of two isolated nodes in the intersection graph $G_1 \wedge G'_2$ does not imply impossibility of exact matching, because the MLE no longer has this simple form. Two isolated nodes u and v in $G_1 \wedge G'_2$ may be distinguishable via their differing connectivity patterns. To illustrate an extreme case, if u satisfies $p_{ui} = 0$ for all i , it is always isolated, while v might have disjoint sets of neighbors in G_1 and G'_2 , and this suffices to distinguish them. Effectively, the algorithm can match to the matrix $\mathbf{P}$ itself.

3.3.3 Estimation Algorithm

The algorithm that achieves the guarantee of Theorem 3.1 is based on the idea of *transitive closure*: First, the graphs are matched pairwise to obtain *partial* matchings $\hat{\mu}_{ij}$ between each pair of graphs. These matchings can then be boosted via transitivity. The idea is formalized in Algorithm 1, which runs in two steps: In step 1, the k -core estimator, for a suitable choice of k , is used to pairwise match all the graphs. For any i and j , the k -core estimator selects a permutation $\hat{\nu}_{ij}$ to maximize the size of the k -core of $G_i \wedge G_j^{\hat{\nu}_{ij}}$ (the k -core of a graph G is the largest subset of vertices $\text{core}_k(G)$ such that the induced subgraph has minimum degree at least k). It then outputs a matching $\hat{\mu}_{ij}$ by restricting the domain of $\hat{\nu}_{ij}$ to $\text{core}_k(G_i \wedge G_j^{\hat{\nu}_{ij}})$. Note that the matchings $\hat{\mu}_{ij}$ need not be complete matchings. In step 2, these partial matchings are *boosted* as follows: If a node v is unmatched between two graphs G_i and G_j , then search for a sequence of graphs $G_i, G_{k_1}, \dots, G_{k_\ell}, G_j$ such that v is matched between any two consecutive graphs in the sequence. If such a sequence exists, then extend $\hat{\mu}_{ij}$ to include v by transitively matching it from G_i to G_j . We emphasize the following characteristics of the algorithm.

- Algorithm 1 only ever makes *pairwise* comparisons between graphs. In contrast, the maximum likelihood estimator for the simpler homogeneous setting selects the permutations $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ simultaneously based on all the graphs. Despite this, Algorithm 1 works down to the threshold for exact recovery.
- The transitive closure sub-routine (step 2) has low computational complexity, $O(m^2 n)$: For

Algorithm 1: Matching through transitive closure

require: Graphs $G_1, G_2, \dots, G_m$ on a common vertex set $[n]$, Integer k

// Step 1: Pairwise matching using the k -core estimator

```
1 for  $\{i, j\}$  in  $\{1, \dots, m\}$  such that  $i < j$  do
2    $\hat{\nu}_{ij} \leftarrow \arg \max_{\pi} |\text{core}_k(G_i \wedge G_j^{\pi})|$ 
3    $\hat{\mu}_{ij} \leftarrow \hat{\nu}_{ij}$  with domain restricted to  $\text{core}_k(G_i \wedge G_j^{\hat{\nu}_{ij}})$ 
4 end
// Step 2: Boosting through transitive closure
5 for  $v \in [n]$  do
6   for  $j = 2, \dots, m$  do
7     if there is a sequence of indices  $1 = k_1, \dots, k_{\ell} = j$  in  $[m]$  such that
8        $\hat{\mu}_{k_{\ell-1}, j} \circ \dots \circ \hat{\mu}_{k_2, k_3} \circ \hat{\mu}_{1, k_2}(v) = v'$  for some  $v' \in [n]$  then
9       Set  $\hat{\pi}_{1j}(v) = v'$ 
10    end
11 end
12 return  $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ 
```

each node $v \in G_1$, the $m - 1$ corresponding vertices in graphs $G_2, \dots, G_m$ can be found by breadth first search using at most one evaluation of $\hat{\mu}_{k, \ell}(\cdot)$ for each k, ℓ pair with $1 \leq k \neq \ell \leq m$. Thus, the complexity is $O(m^2)$ per node in G_1 and $O(m^2n)$ overall.

- The transitive closure sub-routine can be used to boost the pairwise matchings obtained by *any* estimator in step 1, i.e. it can be used as a black-box bridge between binary graph matching and m -ary graph matching whenever partial pairwise matchings are available. Our choice of the k -core estimator for step 1 is motivated by its precision, i.e. the k -core estimator does not make mistakes with high probability [CKMP19, RS23]. This is crucial to avoid compounding errors during the boosting step. This precision is quantified formally later in the analysis (see Section 3.5.1).

3.4 Preliminaries: On the k -core of Inhomogeneous Graphs

This section presents useful preliminaries and lemmas for the proof of Theorem 3.1. The results are stated for general inhomogeneous graphs and may be of interest more generally beyond the scope of the graph alignment problem. The main objective of this section is to establish that under appropriate conditions, all nodes with sufficiently large degree in an inhomogeneous graph are in its k -core with high probability.

Lemma 3.2. *Let $G \sim \text{RG}(n, \mathbf{P})$ and $d_i = \sum_{j \neq i} p_{ij}$. Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such*

that $p_{\max} = o(n^{\alpha/2-\varepsilon-1})$ and

$$\sum_{i=1}^n e^{-d_i} = o(n^{-\alpha}).$$

Then, for any positive integer k and any $v \in [n]$,

$$\mathbb{P}(\{v \notin \text{core}_k(G)\} \cap \{\delta_G(v) \geq k + 2/\alpha\}) = o(1/n).$$

We build up to the proof of this lemma by establishing some results about the number of low-degree nodes (in Section 3.4.1), and about the size of the k -core in inhomogeneous random graphs (in Section 3.4.2). These results are then used to prove Lemma 3.2 in Section 3.4.3.

3.4.1 On low degree nodes in an inhomogeneous graph

Lemma 3.3. *Let $p \in [0, 1]$. Consider independent random variables X, Y and Z with distributions*

$$X \sim \text{Bern}(p), \quad Y \sim \text{Pois}(p), \quad Z \sim 2 \cdot \text{Pois}(p/2).$$

Then, for all t , the moment generating functions of X, Y and Z are ordered as

$$M_X(t) \leq M_Y(t) \leq M_Z(t).$$

Proof. We have

$$M_X(t) = 1 + p(e^t - 1) \stackrel{(a)}{\leq} \exp(p(e^t - 1)) = M_Y(t) \stackrel{(b)}{\leq} \exp\left(p\left(\frac{e^{2t} - 1}{2}\right)\right) = M_Z(t),$$

where (a) uses $1 + x \leq e^x$ and (b) uses that $k \mapsto (e^{kt} - 1)/k$ is increasing in k for all t . $\square$

Lemma 3.4. *Let $n > 0$ and let $X_1, \dots, X_n$ be independent random variables with $X_i \sim \text{Bern}(p_i)$. Let m be an integer so that $0 \leq m \leq n$, and define*

$$X \triangleq \sum_{i \leq m} X_i + \sum_{j > m} 2X_j.$$

Then, for any $t < \mathbb{E}X$,

$$\mathbb{P}(X \leq t) \leq \exp\left(-\frac{\mathbb{E}X}{2} + \frac{t}{2} \log\left(\frac{e \cdot \mathbb{E}X}{t}\right)\right)$$

Proof. Let $Y_i \sim \text{Pois}(p_i)$ and $Z_i \sim 2 \cdot \text{Pois}(p_i/2)$. It follows that

$$\begin{aligned} M_X(t) &= \prod_{i=1}^m M_{X_i}(t) \cdot \prod_{j=m+1}^n M_{X_j}(2t) \\ &\leq \prod_{i=1}^m M_{Z_i}(t) \cdot \prod_{j=m+1}^n M_{Y_j}(2t) \\ &= \prod_{i=1}^m M_{Z_i}(t) \cdot \prod_{j=m+1}^n M_{2Y_j}(t) = M_Z(t), \end{aligned}$$

where $Z \triangleq \sum_{i=1}^m Z_i + \sum_{j=m+1}^n 2Y_j$ is distributed as $Z \sim 2 \cdot \text{Pois}(\mathbb{E}X/2)$.

Therefore, Chernoff tail bounds that hold for Z also hold for X . In particular, for any $t < \mathbb{E}X$,

$$\mathbb{P}(X \leq t) \leq \inf_{\theta \geq 0} e^{\theta t} M_X(-\theta) \leq \inf_{\theta \geq 0} e^{\theta t} M_Z(-\theta) = \inf_{\theta \geq 0} \exp\left(\theta t + \frac{\mathbb{E}X}{2} (e^{-2\theta} - 1)\right).$$

Setting $\theta = \frac{1}{2} \log(\mathbb{E}X/t)$ gives the desired result. $\square$

Lemma 3.5. *Let $G \sim \text{RG}(n, \mathbf{P})$, and let $d_i = \sum_{j \neq i} p_{ij}$. Let $r > 0$ be an integer and let Z_r denote the set of nodes of G with degree less than or equal to r . Suppose that for some $\alpha > 0$,*

$$\sum_{i=1}^n e^{-d_i} = o(n^{-\alpha}). \quad (3.11)$$

– For any $\delta > 1 - \alpha/2$,

$$\mathbb{P}(|Z_r| > n^\delta) = o(1/n). \quad (3.12)$$

– If $p_{\max} = o(n^{\alpha/2-1})$, then (3.12) is true for all $\delta > 1 - \alpha$.

Proof. If strictly more than n^δ nodes all have degree less than or equal to r , then a subset of exactly n^δ nodes has average degree less than r , so

$$\begin{aligned} \mathbb{P}(|Z_r| > n^\delta) &= \mathbb{P}\left(\exists T \subseteq [n]: \left\{|T| \geq n^\delta\right\} \cap \left\{\max_{u \in T} \delta_G(u) \leq r\right\}\right) \\ &= \mathbb{P}\left(\exists T \subseteq [n]: \left\{|T| = n^\delta\right\} \cap \left\{\max_{u \in T} \delta_G(u) \leq r\right\}\right) \\ &\leq \mathbb{P}\left(\exists S \subseteq [n]: \left\{|S| = n^\delta\right\} \cap \left\{\sum_{u \in S} \delta_G(u) \leq r|S|\right\}\right). \end{aligned} \quad (3.13)$$

Let G_{uv} denote the indicator random variable for the presence of edge $\{u, v\}$ in G . For any set

$S \subseteq [n]$, define

$$\mu_S \triangleq \sum_{i \in S} \mathbb{E}[\delta_G(i)] = \sum_{i \in S} d_i.$$

Since for any $i \in [n]$, we have that $e^{-d_i} \leq \sum_{j=1}^n e^{-d_j} \leq n^{-\alpha}$, it follows that $d_i \geq \alpha \log n$. Therefore, for any $S \subseteq [n]$, we have that $\mu_S \geq \alpha|S| \log n$. Hence,

$$\begin{aligned} \mathbb{P} \left(\sum_{u \in S} \delta_G(u) \leq r|S| \right) &= \mathbb{P} \left(\sum_{\substack{u \in S \\ v \in [n] \setminus S}} G_{uv} + \sum_{\substack{u, v \in S \\ u < v}} 2G_{uv} \leq r|S| \right) \stackrel{(a)}{\leq} \exp \left(-\frac{\mu_S}{2} + \frac{r|S|}{2} \log \left(\frac{e \cdot \mu_S}{r|S|} \right) \right) \\ &\stackrel{(b)}{\leq} \exp \left(-\frac{\alpha|S|}{2} \log(n) + \frac{r|S|}{2} \log \left(\frac{e \cdot \alpha \log n}{r} \right) \right) \end{aligned} \quad (3.14)$$

where (a) uses Lemma 3.4 and (b) uses that the function on the right hand side of (b) is decreasing in μ_S on the interval $(r|S|, \infty)$, and that $r|S| \leq \alpha|S| \log n \leq \mu_S$ for all sufficiently large n . Finally, combining (3.13) and (3.14), and using that $|S| = n^\delta$ yields

$$\begin{aligned} \mathbb{P}(|Z_r| > n^\delta) &\leq \sum_{S \subseteq [n]: |S|=n^\delta} \mathbb{P} \left(\sum_{u \in S} \delta_G(u) \leq r|S| \right) \leq \binom{n}{n^\delta} \exp \left(-\frac{\alpha n^\delta}{2} \log n + \frac{r n^\delta}{2} \log \left(\frac{e \cdot \alpha \log n}{r} \right) \right) \\ &\stackrel{(a)}{\leq} \exp \left(\left(1 - \delta - \frac{\alpha}{2} \right) n^\delta \log n + \frac{r n^\delta}{2} \log \left(\frac{e \cdot \alpha \log n}{r} \right) \right) = o(1/n), \end{aligned}$$

whenever $\delta > 1 - \alpha/2$. Here, (a) uses the fact that $\binom{n}{n^\delta} \leq \exp((1 - \delta)n^\delta \log n)$. Hence, (3.12) follows.

Suppose that $p_{\max} = o(n^{\alpha/2-1})$. Notice that

$$\sum_{u \in S} \deg_G(u) = \sum_{u \in S} \sum_{\substack{v \in [n] \\ v \neq u}} G_{uv} \geq \sum_{u \in S} \sum_{v \in [n] \setminus S} G_{uv} \triangleq Y_S,$$

and so we have that

$$\mathbb{P} \left(\sum_{u \in S} \deg_G(u) \leq r|S| \right) \leq \mathbb{P}(Y_S \leq r|S|) \stackrel{(a)}{\leq} e^{-\mathbb{E}Y_S} \cdot \left(\frac{e \cdot \mathbb{E}Y_S}{r|S|} \right)^{r|S|}, \quad (3.15)$$

where (a) uses that Y_S is a sum of independent Bernoulli random variables, and so the Chernoff tail bound (3.21) can be applied. On the other hand,

$$\mathbb{E}Y_S = \sum_{u \in S} d_u - \sum_{u \in S} \sum_{\substack{v \in S \\ v \neq u}} 2p_{uv} \stackrel{(a)}{\geq} \sum_{u \in S} d_u - \binom{|S|}{2} \cdot (2p_{\max}) \stackrel{(b)}{\geq} \alpha|S| \log(n) - |S|^2 \cdot p_{\max} \triangleq \tilde{\mu}_S,$$

where (a) uses that $p_{uv} \leq p_{\max}$ and (b) uses that $\mu_S \geq \alpha|S| \log n$. Continuing from (3.15), we have

$$\mathbb{P}(Y_S \leq r|S|) \leq e^{-\mathbb{E}Y_S} \cdot \left(\frac{e \cdot \mathbb{E}Y_S}{r|S|} \right)^{r|S|} \leq e^{-\tilde{\mu}_S} \cdot \left(\frac{e \cdot \tilde{\mu}_S}{r|S|} \right)^{r|S|}, \quad (3.16)$$

where the last step uses that the function $x \mapsto e^{-x} \cdot \left(\frac{e \cdot x}{r|S|} \right)^{r|S|}$ is decreasing in x whenever $x \geq r|S|$. Thus, for any S such that $|S| = n^\delta$, we have from (3.16) that

$$\begin{aligned} \mathbb{P}(Y_S \leq rn^\delta) &\leq \exp\left(-n^\delta \left(\alpha \log(n) - r - n^\delta p_{\max}\right)\right) \cdot \left(\frac{\alpha \log(n) - n^\delta p_{\max}}{r}\right)^{rn^\delta} \\ &= \exp\left(-n^\delta \left(\alpha \log(n) - r - n^\delta p_{\max} - r \log\left(\frac{\alpha \log(n) - n^\delta p_{\max}}{r}\right)\right)\right). \end{aligned} \quad (3.17)$$

Combining this with (3.13), a union bound yields that

$$\begin{aligned} \mathbb{P}(|Z_r| > n^\delta) &\leq \mathbb{P}\left(\exists S \subseteq [n] : \{|S| = n^\delta\} \cap \left\{\sum_{u \in S} \deg_G(u) \leq r|S|\right\}\right) \leq \binom{n}{n^\delta} \mathbb{P}(Y_S \leq rn^\delta) \\ &\stackrel{(a)}{\leq} \exp\left(-n^\delta \left[(\alpha + \delta - 1) \log(n) - n^\delta p_{\max} - r \log\left(\frac{\alpha \log(n) - n^\delta p_{\max}}{r}\right) - r - 1\right]\right), \end{aligned} \quad (3.18)$$

where (a) uses that $\binom{n}{n^\delta} \leq \exp((1 - \delta)n^\delta \log n)$. Finally, note that the right-hand side of (3.18) is $o(1/n)$ whenever $\alpha + \delta > 1$ and $n^\delta p_{\max} = o(\log n)$. Both conditions are satisfied when $p_{\max} = o(n^{\alpha/2-1})$ by choosing δ such that $1 - \alpha < \delta < 1 - \alpha/2$. $\square$

3.4.2 On the size of the k -core of an inhomogeneous random graph

The following concentration inequalities for a sum of Bernoulli random variables are used frequently in the analysis.

Lemma 3.6. *Let X be a sum of n independent Bernoulli random variables, and let $\mathbb{E}[X] = np$. Then,*

(a) *For any $\delta > 0$,*

$$\mathbb{P}(X \geq (1 + \delta)np) \leq \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}}\right)^{np} \leq \left(\frac{e}{1 + \delta}\right)^{(1+\delta)np}. \quad (3.19)$$

(b) *For any $\delta > 5$,*

$$\mathbb{P}(X \geq (1 + \delta)np) \leq 2^{-(1+\delta)np}. \quad (3.20)$$

Algorithm 2: Łuczak expansion

require: Graph G , Set $U \subseteq V(G)$.

```
1  $U_0 \leftarrow U$ 
2 for  $i = 0, 1, 2, 3, \dots$  do
3   if there exists  $u \in V \setminus U_i$  such that  $u$  has 3 or more neighbors in  $U_i$  then
4      $U_{i+1} \leftarrow U_i \cup \{u\}$ 
5   else
6     return  $U_i$ 
7   end
8 end
```

(c) For any $\delta \in (0, 1)$,

$$\mathbb{P}(X \leq (1 - \delta)np) \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{1-\delta}} \right)^{np} = e^{-np} \cdot \left(\frac{e}{1 - \delta} \right)^{(1-\delta)np}. \quad (3.21)$$

Proof. All proofs follow from the Chernoff bound and can be found, or easily derived, from Theorems 4.4 and 4.5 of [MU17]. $\square$

Let $G \sim \text{RG}(n, \mathbf{P})$ and let $k > 0$. Let v be any node of G and consider the induced subgraph $G - v$ of G on the vertex set $[n] \setminus \{v\}$. The objective of this section is to show that the k -core of $G - v$ is sufficiently large with probability $1 - o(1/n)$, formalized next.

Lemma 3.7. *Let $G \sim \text{RG}(n, \mathbf{P})$, and let v be a node of G . Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p_{\max} = o(n^{\alpha/2 - \varepsilon - 1})$ and*

$$\sum_{i=1}^n e^{-d_i} = o(n^{-\alpha}).$$

Then for any $\delta > 1 - \alpha$, the size of the k -core of $G - v$ satisfies

$$\mathbb{P}\left(|\text{core}_k(G - v)| < n - 3n^\delta\right) = o(1/n).$$

Proof. To bound the size of the k -core, consider Algorithm 2 to identify a subset of the k -core, originally proposed by [Luc91].

Note that the **for** loop eventually terminates - the set $V \setminus U_i$ is empty, for example, when $i = n$ for any input set U . The key is to realize that the **for** loop terminates quickly when the input U is Z_{k+1} , i.e. the set of vertices of the input graph G whose degree is $k + 1$ or less. Furthermore, for that input, the complement of the set output by the algorithm is contained in the k -core. This is formally stated in Lemma 3.8 in what follows.

Lemma 3.8. *Let $G \sim \text{RG}(n, \mathbf{P})$ and let v be a node of G . Let U_f be the output of Algorithm 2 with input graph $G - v$ and $U = Z_{k+1}$. Then,*

(a) $U_f^c \subseteq \text{core}_k(G - v)$.

(b) If there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p_{\max} = o(n^{\alpha/2-\varepsilon-1})$ and $\sum_{i=1}^n e^{-d_i} = o(n^{-\alpha})$, then for any $\delta > 1 - \alpha$,

$$\mathbb{P}(|U_f| > 3n^\delta) = o(1/n).$$

Finally, Lemma 3.8 directly implies Lemma 3.7. By part (a), $\text{core}_k(G - v) \supseteq U_f^c$, so $|\text{core}_k(G - v)| \geq n - 1 - |U_f|$. By part (b), $|U_f| \leq 3n^\delta$ with probability $1 - o(1/n)$, giving $|\text{core}_k(G - v)| \geq n - 3n^\delta$ with the same probability. $\square$

Proof of Lemma 3.8

(a) The proof is by construction: Since U_f is obtained by adding exactly f nodes to U_0 , it follows that $U_f^c \subseteq U_0^c = Z_{k+1}^c$, so each node in U_f^c has degree $k + 2$ or more in $G - v$. Further, each node in U_f^c has at most 2 neighbors in U_f , else the **for** loop would not have terminated. Thus, the subgraph of $G - v$ induced on the set U_f^c has minimum degree at least k , and the result follows.

(b) Let δ' be such that $1 - \alpha < \delta' < \min\{\delta, 1 - \alpha + 2\varepsilon\}$. If $|U_f| > 3n^\delta$, then either $|U_0| > 3n^{\delta'}$ or there is some M in $\{0, 1, \dots, f\}$ for which $|U_M| = 3n^{\delta'}$. Therefore,

$$\begin{aligned} \mathbb{P}(|U_f| > 3n^\delta) &\leq \mathbb{P}(|U_0| > 3n^{\delta'}) + \mathbb{P}(\exists M \in \{0, 1, \dots, f\} \text{ s.t. } |U_M| = 3n^{\delta'}) \\ &= o(1/n) + \underbrace{\mathbb{P}(\exists M \in \{0, 1, \dots, f\} \text{ s.t. } |U_M| = 3n^{\delta'})}_{(\star)}, \end{aligned}$$

where the first term is $o(1/n)$ by Lemma 3.5. Note that each iteration $i = 0, 1, \dots, M - 1$ of the **for** loop adds exactly 1 vertex and at least 3 edges to the subgraph of $G - v$ induced on U_M . Therefore, the induced subgraph $G|_{U_M}$ has $3n^{\delta'}$ vertices and at least $3(|U_M| - |U_0|)$ edges.

Let us couple the inhomogeneous random graph G with an Erdős-Rényi random graph $\tilde{G}$ on n nodes with marginal distribution $\tilde{G} \sim \text{ER}(n, p_{\max})$, such that G is a subgraph of $\tilde{G}$. Thus,

$$\begin{aligned} (\star) &\leq \mathbb{P}(\exists \text{ subgraph } H = (W, F) \text{ of } G - v \text{ s.t. } |W| = 3n^{\delta'} \text{ and } |F| \geq 3(3n^{\delta'} - |U_0|)) \\ &\leq \mathbb{P}(|U_0| > n^{\delta'}) + \mathbb{P}(\exists \text{ subgraph } H = (W, F) \text{ of } G \text{ s.t. } |W| = 3n^{\delta'} \text{ and } |F| \geq 6n^{\delta'}) \\ &\leq \mathbb{P}(|U_0| > n^{\delta'}) + \mathbb{P}(\exists \text{ subgraph } \tilde{H} = (\tilde{W}, \tilde{F}) \text{ of } \tilde{G} \text{ s.t. } |\tilde{W}| = 3n^{\delta'} \text{ and } |\tilde{F}| \geq 6n^{\delta'}) \\ &\leq o(1/n) + \underbrace{\binom{n}{3n^{\delta'}} \cdot \mathbb{P}\left(\text{Binom}\left(\binom{3n^{\delta'}}{2}, p_{\max}\right) > 6n^{\delta'}\right)}_{(\star\star)}, \end{aligned}$$

where the last step uses Lemma 3.5 and a union bound over all possible choices of W . Finally, using

that $\binom{n}{k} \leq \left(\frac{ne}{k}\right)^k$ and the fact that $\text{Bin}(n, p) \preceq \text{Bin}(n', p')$ whenever $n' \geq n$ and $p' \geq p$ yields

$$\begin{aligned} (\star\star) &\leq \left(\frac{n^{1-\delta'}e}{3}\right)^{3n^{\delta'}} \mathbb{P}\left(\text{Binom}\left(\frac{9n^{2\delta'}}{2}, n^{\alpha/2-\varepsilon-1}\right) > 6n^{\delta'}\right) \stackrel{(a)}{\leq} (n^{1-\delta'})^{3n^{\delta'}} \left(\frac{3e}{4} \cdot n^{\alpha/2-\varepsilon-(1-\delta')}\right)^{6n^{\delta'}} \\ &= \left(\frac{3e}{4} \cdot n^{\alpha/2-\varepsilon-(1-\delta')/2}\right)^{6n^{\delta'}} = o(1/n), \end{aligned}$$

since $\alpha/2 - \varepsilon - (1 - \delta')/2 < 0$. Note (a) uses the concentration bound (3.19) from Lemma 3.6. This concludes the proof of Lemma 3.8.

3.4.3 Relating high degree nodes to the k -core: Proof of Lemma 3.2

Proof of Lemma 3.2. For any set $A \subseteq [n]$, let $N_v(A)$ denote the number of neighbors of v in A , i.e.

$$N_v(A) \triangleq |\{u \in A : \{u, v\} \in E(G)\}|.$$

Fix $v \in [n]$. Let $G - v$ denote the induced subgraph of G on the vertex set $[n] - \{v\}$. Let C_v denote the k -core of $G - v$. Since C_v is a subset of the k -core of G , it follows that

$$\{v \notin \text{core}_k(G)\} \subseteq \{N_v(\text{core}_k(G)) \leq k - 1\} \subseteq \{N_v(C_v) \leq k - 1\}.$$

Let $\delta \in (1 - \alpha, 1 - \alpha + 2\varepsilon)$. It follows that

$$\mathbb{P}(\{v \notin \text{core}_k(G)\} \cap \{\delta_G(v) \geq k + 2/\alpha\}) \leq q_1 + q_2, \quad (3.22)$$

where

$$q_1 \triangleq \mathbb{P}\left(|C_v| < n - 3n^\delta\right) \stackrel{(a)}{=} o(1/n),$$

where (a) follows from Lemma 3.7, and

$$\begin{aligned} q_2 &\triangleq \mathbb{P}\left(\{N_v(C_v) \leq k - 1\} \cap \{\delta_G(v) \geq k + 2/\alpha\} \cap \{|C_v| \geq n - 3n^{1-\alpha/2}\}\right) \\ &\leq \sum_{\substack{A \subseteq V \\ |A| \geq n - 3n^\delta}} \mathbb{P}(\{N_v(C_v) < k\} \cap \{\delta_G(v) \geq k + 2/\alpha\} \mid C_v = A) \cdot \mathbb{P}(C_v = A) \\ &\leq \max_{\substack{A \subseteq V \\ |A| \geq n - 3n^\delta}} \mathbb{P}(\{N_v(A) < k\} \cap \{\delta_G(v) \geq k + 2/\alpha\}) \\ &\leq \max_{\substack{A \subseteq V \\ |A| \geq n - 3n^\delta}} \mathbb{P}(N_v(A^c) \geq 1 + 2/\alpha) \end{aligned} \quad (3.23)$$

For all sets A of size larger than $n - 3n^\delta$, we have that

$$\mathbb{P}(N_v(A^c) \geq 1 + 2/\alpha) = \mathbb{P}\left(\sum_{i \in A^c} G_{vi} \geq 1 + 2/\alpha\right) \leq \mathbb{P}\left(\text{Bin}(3n^\delta, n^{\alpha/2-1}) \geq 1 + 2/\alpha\right), \quad (3.24)$$

where the last step uses that $\sum_{i \in A^c} G_{vi} \preceq \text{Bin}(|A^c|, p_{\max}) \preceq \text{Bin}(3n^\delta, n^{\alpha/2-1})$. Combining (3.23) and (3.24) yields

$$q_2 \leq \mathbb{P}\left(\text{Bin}(3n^\delta, n^{\alpha/2-1}) \geq 1 + 2/\alpha\right) \leq \left(\frac{3\alpha e}{2 + \alpha} \cdot n^{-\alpha/2}\right)^{1+2/\alpha} = \text{const.} \times n^{-1-\alpha/2} = o(1/n).$$

Since both q_1 and q_2 are $o(1/n)$, the result follows from (3.22). $\square$

3.5 Proofs

This section presents proofs for Theorems 3.1 and 3.3. Section 3.5.1 presents the intuition behind the analysis of Algorithm 1, collecting useful lemmas along the way. These lemmas are proved in Section 3.5.3, after the proof of Theorem 3.1 in Section 3.5.2. Finally, the negative result (Theorem 3.3) is proved in Section 3.5.5.

3.5.1 Analyzing Algorithm 1: Intuition behind the proof of Theorem 3.1

Algorithm 1 succeeds if both step 1 and step 2 succeed, i.e.

- (a) Each instance of pairwise matching using the k -core estimator is *precise*, i.e. it is correct on its domain with probability $1 - o(1)$, i.e. $\hat{\mu}_{ij}(v) = \pi_{ij}^*(v)$ for all $v \in \text{dom}(\hat{\mu}_{ij})$ for all i and j .
- (b) For each node v and any two graphs G_i and G_j , there is a sequence of graphs such that v can be transitively matched through those graphs between G_i and G_j .

On step 1

This falls back to the regime of analyzing the performance of the k -core estimator in the setting of two graphs. [CKMP19] showed that the k -core estimator is *precise*: For any two correlated Erdős-Rényi graphs G_i and G_j with $\mathbf{P}_{ij} = C \log(n)/n$ for all $1 \leq i < j \leq n$, and constant subsampling probability s , the k -core estimator with sufficiently large k correctly matches all nodes in $\text{core}_k(G'_i \wedge G'_j)$ with probability $1 - o(1)$. Building on this analysis, [RS23] showed that this is true more generally:

Lemma 3.9 (Lemma III.4, [RS23]). *Let G_1 and G_2 be correlated inhomogeneous graphs obtained by subsampling each edge independently with probability s from a parent graph $G \sim \text{RG}(n, \mathbf{P})$. Suppose that there exists some $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p_{\max} \leq o(n^{\alpha/2-\varepsilon-1})$, and let $k > \frac{12}{1-\alpha+2\varepsilon}$. Then, with probability $1 - o(1)$, it holds that the matching $\hat{\mu}_k$ output by the k -core estimator satisfies*

$$\text{dom}(\hat{\mu}_k) = \text{core}_k(G'_1 \wedge G'_2) \text{ and } \hat{\mu}_k(v) = \pi^*(v) \ \forall v \in \text{dom}(\hat{\mu}_k).$$

This precision of the k -core estimator is critical to the success of Algorithm 1 since it ensures that there are no conflicts during the boosting step. Applying Lemma 3.9 to the setting of m graphs yields that each instance of pairwise matching in step 1 of Algorithm 1 is indeed precise.

Corollary 3.6. *Let $\hat{\mu}_{ij}$ denote the matching output by the k -core estimator on graphs G_i and G_j . Under the conditions of Lemma 3.9,*

$$\mathbb{P}(\exists 1 \leq i < j \leq m, \text{ and } v \in \text{core}_k(G'_i \wedge G'_j) \text{ such that } \hat{\mu}_{ij}(v) \neq \pi_{ij}^*(v)) = o(1).$$

Proof. Since the number of instances of pairwise matchings is constant whenever m is constant, a union bound yields

$$\begin{aligned} & \mathbb{P}(\exists 1 \leq i < j \leq m, \text{ and } v \in \text{core}_k(G'_i \wedge G'_j) \text{ such that } \hat{\mu}_{ij}(v) \neq \pi_{ij}^*(v)) \\ & \leq \sum_{i=1}^m \sum_{j=1}^m \mathbb{P}(\hat{\mu}_{ij}(v) \neq \pi_{ij}^*(v) \text{ for some } v \in \text{core}_k(G'_i \wedge G'_j)) = o(1). \end{aligned}$$

This concludes the proof. $\square$

On step 2

The challenging part of the proof is to show that boosting through transitive closure matches all the nodes with probability $1 - o(1)$ under the sufficient condition. It is instructive to visualize this using *transitivity graphs*.

Definition 3.1 (Transitivity graph, $\mathcal{H}(v)$). For each node $v \in [n]$, let $\mathcal{H}(v)$ denote the graph on the vertex set $\{g_1, \dots, g_m\}$ such that an edge $\{g_i, g_j\}$ is present in $\mathcal{H}(v)$ if and only if $v \in \text{core}_k(G'_i \wedge G'_j)$.

On the event that each instance of pairwise matching using the k -core is correct, the edge $\{g_i, g_j\}$ is present in $\mathcal{H}(v)$ if and only if v is correctly matched using the k -core estimator between G_i and G_j , i.e. $\pi_{1i}^*(v)$ is matched to $\pi_{1j}^*(v)$. Thus, in order for step 2 to succeed (i.e. to exactly match all vertices across all graphs), it suffices that the graph $\mathcal{H}(v)$ be connected for each node $v \in [n]$. However, studying the connectivity of the transitivity graphs is challenging because in any graph $\mathcal{H}(v)$, no two edges are independent. This is because the k -cores of any two intersection graphs $G'_a \wedge G'_b$ and $G'_c \wedge G'_d$ are correlated, because all the graphs G_a, G_b, G_c and G_d are themselves correlated. To overcome this, we introduce another graph $\tilde{\mathcal{H}}(v)$ that relates to $\mathcal{H}(v)$ and is amenable to analysis. Let $\deg_G(v)$ denote the degree of a node v in graph G .

Definition 3.2. For each node $v \in [n]$, let $\tilde{\mathcal{H}}(v)$ denote a complete *weighted* graph on the vertex set $\{g_1, \dots, g_m\}$ such that the weight on any edge $\{g_i, g_j\}$ is $\tilde{c}_v(i, j) := \deg_{G'_i \wedge G'_j}(v)$.

The relationship between the graphs $\mathcal{H}(v)$ and $\tilde{\mathcal{H}}(v)$ stems from Lemma 3.2 which relates the degree of node v in $G'_i \wedge G'_j$ to the inclusion of v in $\text{core}_k(G'_i \wedge G'_j)$ for each i and j . Since the graph $G'_i \wedge G'_j \sim \text{RG}(n, \mathbf{P}s^2)$, Lemma 3.2 implies that with probability $1 - o(1/n)$, if a pair $\{g_i, g_j\}$

has edge weight $\tilde{c}_{ij} \geq k + 2/\alpha$ in $\tilde{\mathcal{H}}(v)$, then the corresponding edge $\{g_i, g_j\}$ is present in the transitivity graph $\mathcal{H}(v)$, i.e. v is correctly matched between G_i and G_j in the instance of pairwise k -core matching between them. The graph $\mathcal{H}(v)$ is not connected only if it contains a (non-empty) vertex cut $U \subset \{1, \dots, m\}$ with no edge crossing between U and U^c . For a cut U , let $c_v(U)$ denote the number of such crossing edges in $\mathcal{H}(v)$. Furthermore, define the *cost* of the cut U in $\tilde{\mathcal{H}}(v)$ as

$$\tilde{c}_v(U) \triangleq \sum_{i \in U} \sum_{j \in U^c} \tilde{c}_v(i, j).$$

Lemma 3.2 is a statement about a single graph, but it can be invoked to prove the following.

Lemma 3.10. *Let $G_1, \dots, G_m$ be correlated inhomogeneous random graphs from the subsampling model with parameters $\mathbf{P}$ and s . Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p_{\max} = o(n^{\alpha/2 - \varepsilon - 1})$. Let $v \in [n]$ and let U be a vertex cut of $\{1, \dots, m\}$ such that $|U| \leq \lfloor m/2 \rfloor$. Then,*

$$\mathbb{P} \left(\{c_v(U) = 0\} \cap \left\{ \tilde{c}_v(U) > \frac{m^2}{4} \left(k + \frac{2}{\alpha} \right) \right\} \right) = o(1/n). \quad (3.25)$$

The proof of Lemma 3.10 is deferred to Section 3.5.3. It suffices to analyze the probability that the graph $\tilde{\mathcal{H}}(v)$ has a cut U such that its cost $\tilde{c}_v(U)$ is too small. To that end, we show that the bottleneck arises from vertex cuts of small size. Formally,

Lemma 3.11. *Let $G_1, \dots, G_m$ be correlated inhomogeneous graphs obtained from the subsampling model. Let $v \in [n]$ and let Z_ℓ denote the set $\{1, \dots, \ell\}$ for ℓ in $\{1, \dots, \lfloor m/2 \rfloor\}$. For any vertex cut Z of $\{1, \dots, m\}$, let $\tilde{c}_v(Z)$ denote its cost in the graph $\tilde{\mathcal{H}}(v)$. The following stochastic ordering holds:*

$$\tilde{c}_v(Z_1) \preceq \tilde{c}_v(Z_2) \preceq \dots \preceq \tilde{c}_v(Z_{\lfloor m/2 \rfloor}).$$

The proof of Lemma 3.11 is deferred to Appendix A.3 due to space constraints. Lemmas 3.10 and 3.11 imply that the tightest bottleneck to the connectivity of $\mathcal{H}(v)$ is the event that $\tilde{c}_v(Z_1)$ is below the threshold $r \triangleq \frac{m^2}{4} \left(k + \frac{2}{\alpha} \right)$, i.e. the sum of degrees of v over the intersection graphs $(G_1 \wedge G'_j : j = 2, \dots, m)$ is less than r . This event occurs only if the degree of v is simultaneously less than r in each of the intersection graphs $(G_1 \wedge G'_j : j = 2, \dots, m)$. The last ingredient to the analysis is that under the condition $\sum_{i=1}^n e^{-d_i s(1 - (1-s)^{m-1})} = o(n^{-\alpha})$, this event occurs with probability $o(1/n)$.

Lemma 3.12. *Let $G_1, \dots, G_m$ be obtained from the subsampling model with parameters $\mathbf{P}$ and s . Let $d_i = \sum_{j \neq i} p_{ij}$ and $v^* \in \arg \min_{v \in V} d_v$. Let $k \geq \frac{12}{1 - \alpha + 2\varepsilon}$ be constant and let $r \triangleq \frac{m^2}{4} \left(k + \frac{2}{\alpha} \right)$. Suppose there exist $\alpha \in (0, 1]$ and $\varepsilon > 0$ such that $p_{\max} = o(n^{\alpha/2 - \varepsilon - 1})$ and*

$$\sum_{v \in V \setminus v^*} e^{-d_v s(1 - (1-s)^{m-1})} = o(n^{-\alpha}). \quad (3.26)$$

Then

$$(\star) \triangleq \mathbb{P} \left(\exists v \in V \setminus v^* : \{\deg_{G_1 \wedge G'_2}(v) \leq r\} \cap \cdots \cap \{\deg_{G_1 \wedge G'_m}(v) \leq r\} \right) = o(1).$$

Consequently, under the sufficient condition (3.1), the transitivity graph $\mathcal{H}(v)$ is connected for each node v . The above ideas are used to prove Theorem 3.1.

3.5.2 Piecing it together: proof of Theorem 3.1

Let $\hat{\pi}_{12}, \dots, \hat{\pi}_{1m}$ denote the output of Algorithm 1 with $k \geq \frac{12}{1-\alpha+2\varepsilon}$. Let E_1 denote the event that Algorithm 1 fails to match all m graphs exactly, i.e.

$$E_1 = \{\hat{\pi}_{12} \neq \pi_{12}^*\} \cup \cdots \cup \{\hat{\pi}_{1m} \neq \pi_{1m}^*\}.$$

First, we show that the output of Algorithm 1 is correct with probability $1 - o(1)$ whenever the sufficient condition holds. If the event E_1 occurs, then either step 1 failed, i.e. there is a k -core matching $\hat{\mu}_{ij}$ that is incorrect, or step 2 failed, i.e. at least one of the graphs $\mathcal{H}(v)$ is not connected. Therefore,

$$\mathbb{P}(E_1) \leq \mathbb{P} \left(\bigcup_{i,j} \bigcup_{v \in \text{core}_k(G'_i \wedge G'_j)} \{\hat{\mu}_{ij} \neq \pi_{ij}^*\} \right) + \mathbb{P} \left(\bigcup_{v \in [n]} \{\mathcal{H}(v) \text{ is not connected}\} \right) \leq o(1) + q,$$

where the last step uses Corollary 3.6, and q denotes the probability that a transitivity graph $\mathcal{H}(v)$ is not connected for some v in $[n]$. For each ℓ in the set $\{1, \dots, \lfloor m/2 \rfloor\}$, let U_ℓ denote the set

$\{1, \dots, \ell\}$. Then,

$$\begin{aligned}
q &= \mathbb{P} \left(\bigcup_{v \in [n]} \bigcup_{\ell=1}^{\lfloor m/2 \rfloor} \{ \exists U \subset \{1, \dots, m\} : |U| = \ell \text{ and } c_v(U) = 0 \} \right) \\
&\leq \sum_{\ell=1}^{\lfloor m/2 \rfloor} \binom{m}{\ell} \cdot \mathbb{P}(\exists v \in [n] : c_v(U_\ell) = 0) \\
&\leq \sum_{\ell=1}^{\lfloor m/2 \rfloor} m^\ell \left[\mathbb{P} \left(\exists v \in [n] : \tilde{c}_v(U_\ell) \leq \frac{m^2}{4}(k + 2/\alpha) \right) \right. \\
&\quad \left. + \mathbb{P} \left(\exists v \in [n] : \{c_v(U_\ell) = 0\} \cap \{\tilde{c}_v(U_\ell) > \frac{m^2}{4}(k + 2/\alpha)\} \right) \right] \\
&\stackrel{(a)}{\leq} \sum_{\ell=1}^{\lfloor m/2 \rfloor} m^\ell \left[\mathbb{P} \left(\exists v \in [n] : \tilde{c}_v(U_\ell) \leq \frac{m^2}{4}(k + 2/\alpha) \right) + o(1) \right] \\
&\stackrel{(b)}{\leq} o(1) + \sum_{\ell=1}^{\lfloor m/2 \rfloor} m^\ell \cdot \mathbb{P} \left(\exists v \in [n] : \tilde{c}_v(U_1) \leq \frac{m^2}{4}(k + 2/\alpha) \right) \\
&\stackrel{(c)}{\leq} o(1) + \sum_{\ell=1}^{\lfloor m/2 \rfloor} m^\ell \cdot \mathbb{P} \left(\exists v \in [n] : \{\deg_{G_1 \wedge G'_2}(v) \leq r\} \cap \dots \cap \{\deg_{G_1 \wedge G'_m}(v) \leq r\} \right) \stackrel{(d)}{\leq} o(1)
\end{aligned}$$

Here, (a) uses Lemma 3.10 and a union bound, and (b) uses the fact that for any $\ell \geq 2$, the random variable $\tilde{c}_v(U_\ell)$ stochastically dominates $\tilde{c}_v(U_1)$ (Lemma 3.11). Lastly, (c) uses the definition of $\tilde{c}_v(U_1)$ and (d) uses Lemma 3.12. Therefore, transitive closure successfully matches all nodes in $[n]$ with probability $1 - o(1)$. It follows that $\mathbb{P}(E_2) = o(1)$. This concludes the proof.

3.5.3 Proofs of Lemmas 3.10 and 3.12

Proof of Lemma 3.10.

For any vertex cut U ,

$$\begin{aligned}
\left\{ \tilde{c}_v(U) > \frac{m^2}{4} \left(k + \frac{2}{\alpha} \right) \right\} &\stackrel{(a)}{\subseteq} \left\{ \tilde{c}_v(U) > |U|(m - |U|) \left(k + \frac{2}{\alpha} \right) \right\} \\
&= \left\{ \sum_{i \in U} \sum_{j \in U^c} \deg_{G'_i \wedge G'_j}(v) > |U|(m - |U|) \left(k + \frac{2}{\alpha} \right) \right\} \\
&\subseteq \bigcup_{i \in U} \bigcup_{j \in U^c} \left\{ \deg_{G'_i \wedge G'_j}(v) > k + \frac{2}{\alpha} \right\},
\end{aligned}$$

where (a) uses the fact that the maximum of a set of numbers is greater than or equal to the average. On the other hand,

$$\{c_v(U) = 0\} = \bigcap_{i \in U} \bigcap_{j \in U^c} \{v \notin \text{core}_k(G'_i \wedge G'_j)\}.$$

Let p_1 denote the probability in the LHS of (3.25). It follows from the union bound that

$$p_1 \leq \sum_{i \in U} \sum_{j \in U^c} \mathbb{P} \left(\{v \notin \text{core}_k(G'_i \wedge G'_j)\} \cap \left\{ \deg_{G'_i \wedge G'_j}(v) > k + \frac{2}{\alpha} \right\} \right) = o(1/n),$$

since for any choice of i and j , the graph $G'_i \wedge G'_j \sim \text{RG}(n, \mathbf{P}s^2)$.

Proof of Lemma 3.12

If $s = 1$, then the graphs $G_1, \dots, G_m$ are all isomorphic, and

$$(\star) = \mathbb{P}(\exists v \in V \setminus v^* : \deg_{G_1}(v) \leq r) \leq \sum_{v \in V \setminus v^*} \mathbb{P}(\deg_{G_1}(v) = 0) \leq \sum_{i=1}^n e^{-d_i} = o(1),$$

and so the result follows. We therefore assume without loss of generality that $s < 1$. Observe that

$$\begin{aligned} (\star) &\leq \mathbb{P} \left(\bigcup_{r_2=0}^r \cdots \bigcup_{r_m=0}^r \left(\exists v \in V \setminus v^* : \{\deg_{G_1 \wedge G'_2}(v) = r_2\} \cap \cdots \cap \{\deg_{G_1 \wedge G'_m}(v) = r_m\} \right) \right) \\ &\leq \sum_{r_2=0}^r \cdots \sum_{r_m=0}^r \sum_{v \in V \setminus v^*} \mathbb{E}_{D_v} \left[\mathbb{P} \left(\{\deg_{G_1 \wedge G'_2}(v) = r_2\} \cap \cdots \cap \{\deg_{G_1 \wedge G'_m}(v) = r_m\} \mid \deg_{G_1}(v) = D_v \right) \right] \\ &\stackrel{(a)}{=} \sum_{r_2=0}^r \cdots \sum_{r_m=0}^r \sum_{v \in V \setminus v^*} \mathbb{E}_{D_v} \left[\prod_{\ell=2}^m \binom{D_v}{r_\ell} s^{r_\ell} (1-s)^{D_v - r_\ell} \cdot \mathbf{1}_{\{r_\ell \leq D_v\}} \right] \\ &\stackrel{(b)}{\leq} \sum_{r_2=0}^r \cdots \sum_{r_m=0}^r \left[\prod_{\ell=2}^m \left(\frac{e}{r_\ell} \cdot \frac{s}{1-s} \right)^{r_\ell} \right] \cdot \sum_{v \in V \setminus v^*} \mathbb{E}_{D_v} \left[\prod_{\ell=2}^m D_v^{r_\ell} (1-s)^{D_v} \right], \end{aligned} \quad (3.27)$$

where (a) uses that the degrees of node v in the graphs $G_1 \wedge G'_2, \dots, G_1 \wedge G'_m$ are conditionally independent of the degree of v in G_1 , and that each edge of G_1 is independently present in $G_1 \wedge G'_j$ with probability s . Further, (b) uses that $\binom{n}{k} \leq (ne/k)^k$ with the implicit definition $(1/0)^0 \triangleq 1$.

Note that

$$\begin{aligned}
\sum_{v \in V \setminus v^*} \mathbb{E}_{D_v} \left[\prod_{\ell=2}^m D_v^{r_\ell} (1-s)^{D_v} \right] &= \sum_{v \in V \setminus v^*} \left(\sum_{d=0}^{n-1} \mathbb{P}(D_v = d) \prod_{\ell=2}^m d^{r_\ell} (1-s)^d \right) \\
&= \sum_{d=0}^{n-1} \left(\sum_{v \in V \setminus v^*} \mathbb{P}(D_v = d) \cdot d^{\sum_{\ell=2}^m r_\ell} \cdot (1-s)^{(m-1)d} \right). \quad (3.28)
\end{aligned}$$

Proceed by splitting the outer summation in (3.28) over d at $d^* \triangleq n^{\frac{\alpha}{2(m-1)r}}$. We have

$$\begin{aligned}
\sum_{d=0}^{d^*} \sum_{v \in V \setminus v^*} \mathbb{P}(D_v = d) \cdot d^{\sum_{\ell=2}^m r_\ell} \cdot (1-s)^{(m-1)d} &\leq \sum_{v \in V \setminus v^*} (d^*)^{\sum_{\ell=2}^m r_\ell} \sum_{d=0}^{d^*} \mathbb{P}(D_v = d) \cdot (1-s)^{(m-1)d} \\
&\leq (d^*)^{\sum_{\ell=2}^m r_\ell} \sum_{v \in V \setminus v^*} \mathbb{E}_{D_v} \left[(1-s)^{(m-1)D_v} \right] \\
&\stackrel{(c)}{=} (d^*)^{\sum_{\ell=2}^m r_\ell} \sum_{v \in V \setminus v^*} \prod_{u \neq v} (1 - p_{uv}s + p_{uv}s(1-s)^{m-1}) \\
&\stackrel{(d)}{\leq} (d^*)^{\sum_{\ell=2}^m r_\ell} \sum_{v \in V \setminus v^*} \exp \left(- \sum_{u \neq v} p_{uv}s (1 - (1-s)^{m-1}) \right) \\
&\stackrel{(e)}{\leq} n^{\alpha/2} \cdot \sum_{v \in V \setminus v^*} e^{-d_v s (1 - (1-s)^{m-1})} = o(1). \quad (3.29)
\end{aligned}$$

Here, (c) uses the moment generating function of D_v , which is distributed as a sum of independent Bernoulli random variables with means p_{uv} respectively. Further, (d) uses that $1 - x \leq e^{-x}$ for any x , and finally (e) uses the definition of d^* and the condition (3.26). The other part of the split sum can be bounded as

$$\begin{aligned}
\sum_{d=d^*}^{n-1} \sum_{v \in V \setminus v^*} \mathbb{P}(D_v = d) \cdot d^{\sum_{\ell=2}^m r_\ell} \cdot (1-s)^{(m-1)d} &\leq \sum_{v \in V \setminus v^*} \left[\max_{d \in [d^*, n]} d^{\sum_{\ell=2}^m r_\ell} \cdot (1-s)^{(m-1)d} \right] \mathbb{P}(D_v \geq d^*) \\
&\stackrel{(f)}{\leq} (d^*)^{\sum_{\ell=2}^m r_\ell} (1-s)^{(m-1)d^*} \cdot \sum_{v \in V \setminus v^*} \mathbb{P}(D_v \geq d^*) \\
&\stackrel{(g)}{\leq} n^{1+\alpha/2} \cdot ((1-s)^{m-1})^{n^{\frac{\alpha}{2(m-1)r}}} = o(1), \quad (3.30)
\end{aligned}$$

whenever $s > 0$. Here, (f) uses that the function $d \mapsto d^{\sum_{\ell=2}^m r_\ell} (1-s)^{(m-1)d}$ is decreasing on the interval $[d^*, n]$ for all sufficiently large n . Finally, (g) uses the definition of d^* and that $\mathbb{P}(D_v \geq d^*) \leq 1$ for all v . Plugging (3.29) and (3.30) in (3.28) yields that $\sum_{v \in V \setminus v^*} \mathbb{E}_{D_v} [\prod_{\ell=2}^m D_v^{r_\ell} (1-s)^{D_v}] = o(1)$.

Therefore, continuing from (3.27),

$$\begin{aligned}
(3.27) &= \sum_{r_2=0}^r \cdots \sum_{r_m=0}^r \left[\prod_{\ell=2}^m \left(\frac{e}{r_\ell} \cdot \frac{s}{1-s} \right)^{r_\ell} \right] \cdot o(1) \\
&\leq (r+1)(m-1) \cdot \max_{1 \leq r' \leq r} \left(\frac{es}{r'(1-s)} \right)^{r'(m-1)} \cdot o(1) = o(1),
\end{aligned}$$

since r and m are both constants independent of n . This completes the proof.

3.5.4 Proof of Theorem 3.2

The proof is nonconstructive; the idea is to show that recovery is possible if the parameter matrix $\mathbf{P}$ is random with a suitable distribution, and then show that a set of deterministic choices for $\mathbf{P}$ satisfying the requirements of the proposition has positive probability (actually probability converging to one) under the distribution of $\mathbf{P}$. Let G_{-1} denote an Erdős-Rényi graph with parameters n and q , where q is to be determined later. Let $\mathbf{P}$ denote the random matrix equal to $p_{\max}$ times the incidence matrix of G_{-1} .

Given $\mathbf{P}$, let $G \sim \text{RG}(n, \mathbf{P})$, let $G'_1, \dots, G'_m$ denote the graphs obtained from G by subsampling each edge with probability s , and let G_k be obtained from G'_k by randomly uniformly permuting the labels, for $k \in [m]$. We will consider the algorithm that separately matches G_{-1} to G_k for each k , and then matches $G_1, \dots, G_m$ to each other via transitive closure through G_{-1} . For $\ell \in [m]$, G'_ℓ can be viewed as being obtained from G_{-1} by independently subsampling the edges of G_{-1} with probability $p_{\max}s$. Therefore, if $r_{ij} \triangleq \mathbb{P}(\mathbf{1}_{\{e \in E(G_{-1})\}} = i, \mathbf{1}_{\{e \in E(G_\ell)\}} = j)$, then

$$\begin{pmatrix} r_{00} & r_{01} \\ r_{10} & r_{11} \end{pmatrix} = \begin{pmatrix} 1-q & 0 \\ q(1-p_{\max}s) & qp_{\max}s \end{pmatrix}$$

It is known [CK17, WXY22] for correlated Erdős-Rényi graphs such as this that exact recovery is possible if there is some $\varepsilon' > 0$ such that

$$n(\sqrt{r_{00}r_{11}} - \sqrt{r_{01}r_{10}})^2 \geq (1 + \varepsilon') \log n.$$

Assume without loss of generality that $\epsilon < 1$. Let $\epsilon' = \epsilon/4$ and select q so that $qs = \frac{1+2\epsilon'}{\log n}$. With that choice we conclude that the algorithm described exactly matches G_ℓ to G_{-1} with probability $1 - o(1)$ for each ℓ , and hence matches $G_1, \dots, G_m$ to each other with probability $1 - o(1)$.

Let $\delta_{G_{-1}}(i)$ denote the degree of node i in graph G_{-1} . Note that $\delta_{G_{-1}}(i) \sim \text{Binom}(n-1, q) \preceq \text{Binom}(n, q)$ for all nodes i . Therefore by the union bound and Chernoff inequality [Theorem

4.4[MU17]],

$$\mathbb{P}\left(\bigcup_{i=1}^n \{\delta_{G_{-1}}(i) > (1 + \varepsilon')nq\}\right) \leq n \cdot \mathbb{P}(\text{Binom}(n, q) > (1 + \varepsilon')nq) \leq ne^{-nq(\varepsilon')^2/3} = o(1). \quad (3.31)$$

Let $D_i = p_{\max} \cdot \delta_{G_{-1}}(i)$ for $i \in [n]$, so that D_i is the random version of d_i . Then (3.5), (3.31), and the fact $(1 + \varepsilon')(1 + 2\varepsilon') < 1 + \epsilon$ imply that $\mathbb{P}(D_i \leq d \text{ for } i \in [n]) = 1 - o(1)$. In summary, the intersection of the following three events has probability $1 - o(1)$:

- (a) $\max_{i,j} p_{ij} = p_{\max} \leq (\log n)^2/n$.
- (b) $G_1, \dots, G_m$ are exactly matched by the transitive closure algorithm via G_{-1} .
- (c) $D_i \leq d$ for $i \in [n]$.

Therefore, sampling from the distribution of $\mathbf{P}$ generates with positive probability, a deterministic parameter matrix $\mathbf{P}$ satisfying 1-3 above, and for which exact graph matching is possible.

3.5.5 On necessary conditions: proofs of Theorem 3.3

In this section, we prove Theorem 3.3, i.e. our sufficient condition is also necessary for the homogeneous case. Theorem 3.3 has a simple proof following a genie-aided converse argument: reduce the problem to that of matching two graphs by providing extra information to the estimator.

Proof of Theorem 3.3. . If the correspondences $\pi_{12}^*, \dots, \pi_{1,m-1}^*$ were provided as extra information to an estimator, then the estimator must still match G_m with the union graph $G'_1 \vee G'_2 \vee \dots \vee G'_{m-1}$. This can be viewed as an instance of matching two inhomogeneous graphs obtained by *asymmetric* subsampling: the graph G_m is obtained from a parent graph $G \sim \text{RG}(n, \mathbf{P})$ by subsampling each edge independently with probability $s_1 \triangleq s$, and the graph $\tilde{G}_{m-1} \triangleq G'_1 \vee G'_2 \vee \dots \vee G'_{m-1}$ is obtained from G by subsampling each edge independently with probability $s_2 \triangleq 1 - (1 - s)^{m-1}$. [CK17] studied this model for matching two graphs: their Theorem 2 establishes that matching G_m and $\tilde{G}_{m-1}$ is impossible if $Cs_1s_2 < 1$, or equivalently if $Cs(1 - (1 - s)^{m-1}) < 1$. This is equivalent to the necessary condition (3.7). $\square$

3.6 Simulations

A result of [RS23] establishes that with high probability, the k -core estimator for two inhomogeneous graphs matches only the nodes in the k -core of their true intersection graph. This is true even if the matrix $\mathbf{P}$ has constant entries that do not depend on n . Although the k -core estimator itself is not efficient to implement, its output can be efficiently simulated by computing the k -core of the true intersection graph. This is possible only because the simulator has access to the ground truth. Under the assumption that the k -core estimator outputs the k -core of the true intersection graph, we can simulate the performance of Algorithm 1. Figure 3.3 shows the mean fraction of matched

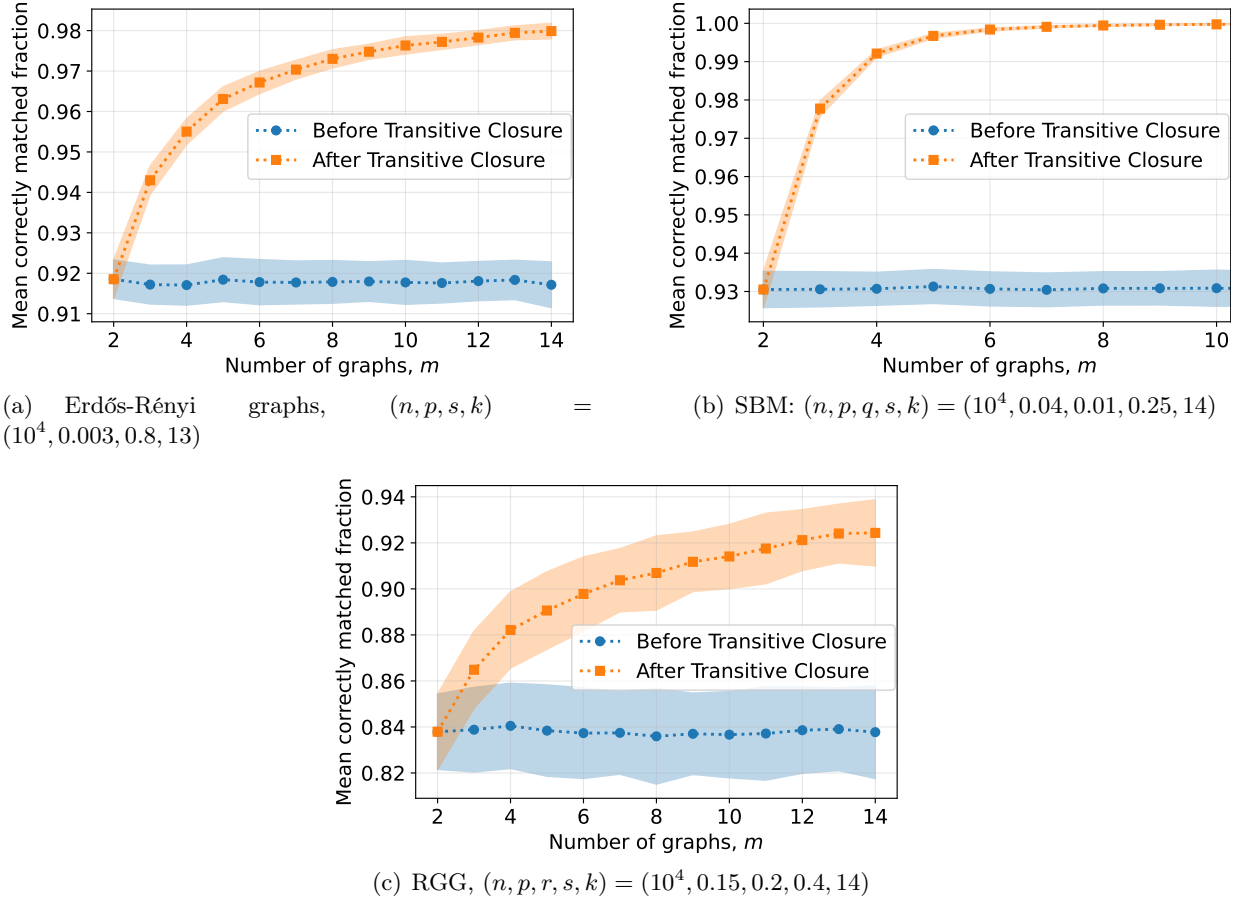


Figure 3.3: Mean fraction of correctly matched nodes for various graphs obtained by simulating Algorithm 1

nodes as a function of m before and after transitive closure, for a variety of graph models. For each value of m , the $\binom{m}{2}$ pairs of graphs are pairwise matched by simulating the k -core estimator, and the average fraction of matched nodes is recorded. The transitive closure subroutine is then applied to the partial matchings and the improvement in the fraction of matched nodes is recorded. In all figures, the shaded region represents the standard deviation for the fraction of matched nodes. Three settings are studied:

- (a) Erdős-Rényi graphs: Figure 3.3(a) studies ER graphs on 10^4 nodes. An edge is present in the parent graph with probability $p = 0.003$ and subsampled independently to the graphs G_1 and G_2 with probability 0.8.
- (b) Stochastic Block Model: Figure 3.3(b) similarly presents the fraction of matched nodes before and after transitive closure for a balanced 5-community stochastic block model on 10^4 nodes, with intra-community edge probability $p = 0.04$ and inter-community edge probability $q = 0.01$ respectively. The subsampling probability $s = 0.25$.
- (c) Random Geometric Graph: Figure 3.3(c) considers random geometric graphs generated as

follows: First, $n = 10^4$ points are sampled independently and uniformly on the unit square $[0, 1] \times [0, 1]$. The parent graph G is generated by connecting two points within Euclidean distance $r = 0.2$ with probability $p = 0.15$, and the graphs G_1 and G_2 are obtained by subsampling each edge independently with probability $s = 0.4$.

All data points are obtained by averaging across 50 independent runs of the algorithm for the ER and SBM models, and 100 runs for the RGG model. As Figure 3.3 shows, the transitive closure step boosts the fraction of matched nodes across any two graphs.

3.7 Discussion

In this chapter, we introduced and analyzed matching through transitive closure (Algorithm 1) – a black-box algorithm to combine information from multiple graphs to recover the underlying correspondence between them. Despite its simplicity, matching through transitive closure is optimal in the homogeneous (ER) setting, after the graphs are pairwise matched using the k -core estimator.

A limitation of Algorithm 1 is overall runtime – it does not run in polynomial time because the k -core estimator exhaustively searches the space of permutations. However, transitive closure (step 2) itself is *efficient* because it runs in polynomial time $O(m^2n)$. Therefore, a natural next step is to modify step 1 in our algorithm so that the pairwise matchings are done by an efficient algorithm. Optimality of such an algorithm is unclear: it is possible that the pairwise matchings output by the efficient algorithm are heavily correlated, so transitive closure is unable to boost them.

The analysis of Algorithm 1 yielded a sufficient condition (performance floor) to exactly match m inhomogeneous random graphs, and we also provide a necessary condition (performance ceiling). The structure of the $\mathbf{P}$ matrix plays a role in determining where the sharp recovery threshold falls between the floor and ceiling. Our work presents several directions for future research.

- **Sharpness of results for general inhomogeneous graphs.** Theorem 3.2 establishes that algorithms with access to the matrix $\mathbf{P}$ can exploit the structure within $\mathbf{P}$ to help with matching. Some matrices (such as the non-constructive $\mathbf{P}$ of Corollary 3.5) are ‘maximally informative’ in the sense that they allow matching all the way to the necessary condition (performance ceiling) of Theorem 3.4, while other matrices (such as the ER model) are ‘minimally informative’ because the sharp threshold remains at the sufficient condition (performance floor) of Theorem 3.1. Characterizing sharp thresholds as a function of $\mathbf{P}$, s and m is an open problem.
- **Relaxing restriction on $p_{\max}$.** The condition $p_{\max} = o(n^{\alpha/2-\varepsilon-1})$ is an intrinsic limiting artifact of the k -core analysis, shared by prior work employing this estimator [RS23]. On the achievability side, it arises because the Łuczak expansion bounds the k -core complement by coupling with $\text{ER}(n, p_{\max})$; on the converse side, the second moment method for pairs of isolated nodes (Lemma A.2 in Appendix A.2) introduces factors that require $np_{\max}^2 = o(1)$. In contrast, efficient degree-profile-based algorithms [DFW23] handle the denser regime $p_{ij} =$

$O(1)$ for two inhomogeneous graphs, but require correlation $1 - O(\log^{-2} n)$, far stronger than what our information-theoretic results need. Bridging these two regimes, i.e. extending our results to larger $p_{\max}$ while retaining the full range of correlation, is an open problem.

- **Polynomial-time algorithms.** Using a polynomial-time estimator in place of the k -core estimator in Step 1 of Algorithm 1 yields a polynomial-time algorithm to match m graphs. If this estimator is precise, i.e. it does not make mistakes, then it can be combined with the transitive closure step, yielding an algorithm to match m graphs. Can the performance guarantees of the k -core estimator be realized through polynomial time algorithms (such as the chandelier algorithm of [MWXY23]) that are precise?
- **Boosting for partial recovery.** This work focused on exact recovery, where the objective is to match all nodes across all graphs. It would be interesting to consider a regime where any instance of pairwise matching recovers at best a small fraction of nodes. Is it possible to quantify the extent to which transitive closure boosts the number of matched nodes?
- **Robustness.** Finally, how sensitive to perturbation is the transitive closure algorithm? Is it possible to quantify the extent to which an adversary may perturb edges in some of the graphs without losing the performance guarantees of the matching algorithm? Algorithms that perform well on models such as inhomogeneous graphs and are further generally robust are expected to also work well with real-world networks.

Chapter 4

Detecting correlation among multiple Gaussian networks

4.1 Introduction

The previous chapter studied *recovery* of the latent correspondence between m correlated graphs. A natural precursor to recovery is *detection*: given m unlabeled graphs, decide whether they are correlated under some latent permutation, or whether they are statistically independent. This is a hypothesis testing problem in which the observations are independent under the null model and correlated via a latent permutation under the alternative. The application domains are the same as those of graph matching — aligning protein-protein interactomes across species, comparing brain connectomes, ontology alignment, and the privacy-driven detection of cross-platform correlations between social networks — and so is the empirical fact that the number of available correlated views varies from two to many depending on the application. This chapter studies the fundamental limits of correlation detection for *networked* data when $m \geq 2$ graphs are available.

4.1.1 Related work

Wu, Xu and Yu [WXY23] studied the hypothesis testing problem to decide whether *two* unlabeled random graphs are correlated. In the setting where the graphs are Erdős-Rényi, they established the information-theoretic threshold for detection within a constant factor – this was later sharpened by Ding and Du [DD23a]. Computationally efficient tests for this problem have also been studied, for example by Barak and co-authors [BCL⁺19], Mao and co-authors [MWXY24] and Ding, Du and Li [DDL23]. The work of Wu, Xu and Yu [WXY23] also established the sharp information-theoretic threshold for detection in the setting of graphs with *Gaussian* weights. The case of general

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distributions was also studied recently by Oren-Loberman, Paslev and Huleihel [OLPH24].

The problem of detecting correlation between *two* graphs has been studied in other settings as well. For instance, necessary and sufficient conditions for correlation detection between unlabeled Galton-Watson trees were established by Ganassali, Lelarge and Massoulié [GLM24] and later by Ganassali, Massoulié and Semerjian [GMS24]. Another example is the work of Rácz and Sridhar [RS22], where a pair of graphs are either independent, or grow together until a time t^* and independently afterwards according to an appropriate evolution model.

The correlation detection problem is closely related to the *graph alignment* problem, where the unlabeled graphs are correlated and the objective is to recover the underlying correspondence between them. Here too, literature has broadly focused on two settings:

- (a) The Gaussian case, where two unlabeled complete graphs have Gaussian edge weights. The information-theoretic threshold for recovery for two graphs was independently established by Ganassali [Gan22] and by Yu, Wu, and Xu [WXY22]. Very recently, Vassaux and Massoulié [VM25] established the recovery threshold for m graphs for any $m \geq 2$.
- (b) The binary case, where the graphs are correlated Erdős-Rényi graphs. Here, the study of information theoretic limits for exact recovery, where the goal is to match all the nodes, was initiated by the works of Cullina and Kiyavash [CK16, CK17]. A flurry of works established thresholds for almost-exact recovery [CKMP19] and partial recovery [WXY22, DD23b, HM23, Du25]. Still other works studied information-theoretic thresholds under heterogeneity [RS23], perturbations [AH24b] and partial correlation [HSY24]. Variants of the recovery problem with *multiple* graphs have also been studied [JLK21, AH25b, RZ24].

Another closely related problem is *database alignment*. In this problem, the two observations are collections of high-dimensional feature *vectors* which are independent under the null hypothesis, and correlated via a latent permutation under the alternative hypothesis. The bulk of the literature on database alignment assumes that the feature vectors are Gaussian [Tam22, Tam23]. Information-theoretic limits for detection were investigated by Zeynep and Nazer in [ZN22], and later sharpened by Elimelech and Huleihel [EH23, EH24] and Jiao, Wu, and Xu [JWX25]. The case of general distributions was also studied by Paslev and Huleihel [PH24]. Finally, information-theoretic limits for recovering the underlying permutation have also been studied [CMK18, DCK19, BE21, BE24, DCK20, DCK23].

4.1.2 Contributions

To our knowledge, the problem of detecting correlation when there are *multiple* unlabeled observations is an open problem for all the aforementioned instances. This chapter considers the setting of $m \geq 2$ complete graphs on n unlabeled nodes with standard Gaussian edge weights, such that the correlation between corresponding edges in any pair of graphs is ρ .

- By analyzing the generalized likelihood ratio, we derive a sufficient condition $\rho^2 \geq \frac{8}{m} \frac{\log n}{n-1}$ for detection. The $1/m$ dependence uncovers an interval of ρ where detection is impossible with 2 graphs but possible with $m > 2$ graphs.
- By inductively leveraging the impossibility result of Wu, Xu and Yu [WXY23], we derive a necessary condition $\rho^2 \leq \left(\frac{4}{m-1} - \varepsilon\right) \frac{\log n}{n}$.
- By analyzing the second moment of the truncated likelihood ratio, we derive a sharp necessary condition that closes this gap. Specifically, we show weak detection is impossible if for some $\eta > 0$, $\rho^2 \leq \frac{(8-\eta)}{m} \frac{\log n}{n}$.

4.2 Preliminaries

Fix an integer $m \geq 2$ independent of n . For any finite set A , denote by $\binom{A}{2} := \{B \subset A : |B| = 2\}$ the collection of unordered pairs of distinct elements of A , and set

$$E_n = \binom{[n]}{2}, \quad N = |E_n| = \binom{n}{2}.$$

We observe m weighted complete graphs on the common vertex set $[n]$, encoded by their edge weights

$$X = (X_e^k)_{1 \leq k \leq m, e \in E_n} \in \mathbb{R}^{mN}.$$

Let $\mathfrak{S}_n$ denote the set of permutations of $[n]$. A permutation $\tau \in \mathfrak{S}_n$ induces a permutation on E_n , also denoted by τ , defined by $\tau(\{i, j\}) = \{\tau(i), \tau(j)\}$ for $\{i, j\} \in E_n$. An *alignment* π is an m -tuple $\pi = (\pi_1, \dots, \pi_m) \in \mathfrak{S}_n^m$ such that $\pi_1 = \text{Id}$. Under the *null law* $\mathbb{Q}$, all mN coordinates of X are independent standard Gaussians. Under the *planted law* $\mathbb{P}$, there is a latent alignment

$$\pi^* = (\pi_1^*, \dots, \pi_m^*), \quad \pi_1^* = \text{Id}, \quad \pi_2^*, \dots, \pi_m^* \stackrel{\text{iid}}{\sim} \text{Unif}(\mathfrak{S}_n),$$

Conditionally on π^* , the edge tuples

$$(X_{\pi_1^*(e)}^1, X_{\pi_2^*(e)}^2, \dots, X_{\pi_m^*(e)}^m), \quad e \in E_n,$$

are independent centered Gaussian vectors with common covariance matrix

$$\Sigma = (1 - \rho) I_m + \rho \mathbf{1} \mathbf{1}^\top, \tag{4.1}$$

where $\rho \in [0, 1)$ is the pairwise correlation between aligned edge weights. The data X is observed; the latent alignment π^* is not. The goal is to determine which hypothesis is true based on the observation.

Additional notation We use X^k as a shorthand for the collection of edge weights $(X_e^k)_{e \in E_n}$ in the k -th graph. Given an alignment $\pi = (\pi_1, \dots, \pi_m)$, write

$$\pi_{k\ell} := \pi_\ell \circ \pi_k^{-1},$$

for $k, \ell \in [m]$ so that $\pi_{k\ell}$ encodes the latent correspondence between the vertex labels of X^k and X^ℓ . Throughout, $\mathbb{Q}$ denotes the law of X under the null hypothesis H_0 , while $\mathbb{P}$ denotes the joint law of (X, π^*) under the planted hypothesis H_1 . With a slight abuse of notation, we write $\text{TV}(\mathbb{P}, \mathbb{Q})$ for the total variation distance between the marginal law of X under $\mathbb{P}$ and the null law $\mathbb{Q}$. Standard asymptotic notation ($O(\cdot), o(\cdot), \Omega(\cdot), \dots$) is used throughout, and it is implicit that $n \rightarrow \infty$. Further, we say $a_n \sim b_n$ if $a_n = (1 + o(1)) \cdot b_n$.

Remark 4.1. The above problem is equivalently a hypothesis testing problem between multiple *unlabeled* graphs – randomly labeling the nodes of these graphs is equivalent to applying uniformly random permutations to their labeled analogs.

The objective is to establish necessary and sufficient conditions on ρ for which it is possible to statistically distinguish between H_0 and H_1 . In light of Remark 4.1, such a test must rely solely on graph properties that are invariant with respect to relabeling the nodes. Let $\mathbb{Q}$ and $\mathbb{P}$ denote respectively the probability measures under H_0 and H_1 . Further, let $\mathbf{X}$ denote the collection $(X^1, \dots, X^m)$ of graphs. A test statistic $T(\mathbf{X})$ with threshold τ achieves

- *Strong detection* if the total error converges to 0:

$$\mathbb{P}(T(\mathbf{X}) < \tau) + \mathbb{Q}(T(\mathbf{X}) \geq \tau) = o(1).$$

- *Weak detection* if the test outperforms random guessing:

$$\mathbb{P}(T(\mathbf{X}) < \tau) + \mathbb{Q}(T(\mathbf{X}) \geq \tau) = 1 - \Omega(1).$$

Let $D_{\text{KL}}(\mathbb{P}, \mathbb{Q})$ denote the total variation distance between the two measures $\mathbb{P}$ and $\mathbb{Q}$. It is well known from detection theory that strong detection is possible iff $D_{\text{KL}}(\mathbb{P}, \mathbb{Q}) = 1 - o(1)$, whereas weak detection is possible iff $D_{\text{KL}}(\mathbb{P}, \mathbb{Q}) = \Omega(1)$.

By the Neyman-Pearson lemma, the optimal test statistic is the likelihood ratio

$$\frac{\mathbb{P}(\mathbf{X})}{\mathbb{Q}(\mathbf{X})} = \frac{1}{(n!)^{m-1}} \sum_{\pi_{12}, \dots, \pi_{1m} \in \mathfrak{S}_n} \frac{\mathbb{P}(\mathbf{X} | \pi_{12}, \dots, \pi_{1m})}{\mathbb{Q}(\mathbf{X})},$$

which is difficult to analyze due to the underlying dependence of each term on the corresponding permutation profile. Following the analysis in [WXY23] for the two-graph setting, we consider

instead the generalized likelihood ratio

$$\max_{\pi_{12}, \dots, \pi_{1m} \in \mathfrak{S}_n} \frac{\mathbb{P}(\mathbf{X} | \pi_{12}, \dots, \pi_{1m})}{\mathbb{Q}(\mathbf{X})}. \quad (4.2)$$

By analyzing the generalized likelihood ratio for two graphs, Wu, Xu, and Yu [WXY23] showed that strong detection is possible in this setting if $\rho^2 \geq 4 \log(n)/(n-1)$, and that weak detection is impossible if $\rho^2 \leq (4 - \varepsilon) \log(n)/n$ for any $\varepsilon > 0$.

4.3 Test statistic and main results

Our analysis is based on the test statistic

$$T(X) := \max_{\pi_2, \dots, \pi_m \in \mathfrak{S}_n} T_\pi(X), \quad (4.3)$$

which aggregates the aligned pairwise correlations over all graph pairs and over all edges, maximized over candidate alignments. The choice of T is motivated by the fact that thresholding T is approximately equivalent to the generalized likelihood-ratio test (GLRT) for distinguishing $\mathbb{Q}$ from $\mathbb{P}$; see Section B.4.

With that, the main results of this chapter are now presented.

Theorem 4.1. *Suppose that*

$$\rho^2 \geq \frac{8 \log n}{m n - 1}. \quad (4.4)$$

There exists a threshold τ for which the generalized likelihood ratio test based on T achieves strong detection, i.e. $\mathbb{P}(T < \tau) + \mathbb{Q}(T \geq \tau) = o(1)$.

Next, we present a necessary condition. This condition is not sharp, but can be obtained by a short proof using a genie aided converse.

Theorem 4.2 (Necessary condition via genie approach). *Suppose that for some $\varepsilon > 0$,*

$$\rho^2 \leq \left(\frac{4}{m-1} - \varepsilon \right) \frac{\log n}{n}. \quad (4.5)$$

Then weak detection is impossible, i.e. $\text{TV}(\mathbb{P}, \mathbb{Q}) = o(1)$.

The positive result in Theorem 4.1 establishes that for each $m > 2$, there is a region in parameter space where weak detection is impossible with 2 graphs alone, but using m graphs as side information allows strong detection. However, the thresholds in (4.4) and (4.5) differ by a multiplicative factor of $2(m-1)/m$. Vassaux and Massoulié [VM25] identified the threshold in (4.4) as the tight threshold for weak *recovery*, giving evidence that the threshold in Theorem 4.2 can be improved. Indeed, we

show by analyzing the second moment of a truncated likelihood ratio that there is no detection–recovery gap for this problem. Specifically, we prove the following theorem.

Theorem 4.3 (Necessary condition via the second moment method). *Fix $\eta \in (0, 1)$, and suppose that*

$$\rho^2 \leq \frac{8(1-\eta)}{m} \cdot \frac{\log n}{n}. \quad (4.6)$$

Then $\text{TV}(\mathbb{P}, \mathbb{Q}) \rightarrow 0$ as $n \rightarrow \infty$. In particular, weak detection is impossible in this regime.

4.4 A variation of Hanson-Wright inequality for Gaussians

The following proposition is a version of the Hanson-Wright inequality [HW71]. It is much less general than the Hanson-Wright inequality because it holds only for Gaussian random variables whereas Hanson-Wright applies to sub-Gaussian random variables. The proposition has the advantage of having a simple proof with explicit constants and it allows a tradeoff between the two matrix norms involved. Let $\|A\|$ denote the spectral norm and $\|A\|_F$ the Frobenius norm of a matrix A .

Proposition 4.1. *Let $Z = X^\top A X$, where A is an $n \times n$ matrix and X has the standard n -dimensional Gaussian distribution. For any constant γ with $0 < \gamma < 1$*

$$P(Z - \mathbb{E}[Z] \geq t) \leq \exp \left(-\frac{t^2}{4(\|A\|_F^2 + \|A\|t)} \right) \quad (4.7)$$

$$\leq \exp \left(-\frac{1}{4} \min \left\{ \frac{\gamma t^2}{\|A\|_F^2}, \frac{(1-\gamma)t}{\|A\|} \right\} \right). \quad (4.8)$$

Remark 4.2. If $\|A\|$ is relatively small we could select γ close to one. Since $\text{Var}(Z) = 2\|A\|_F^2$ the central limit theorem implies the coefficient of $\frac{t^2}{\|A\|_F^2}$ cannot have magnitude greater than $\frac{1}{4}$.

Taking $\gamma = \frac{1}{2}$ yields

$$P(Z - \mathbb{E}[Z] \geq t) \leq \exp \left(-\frac{1}{8} \min \left\{ \frac{t^2}{\|A\|_F^2}, \frac{t}{\|A\|} \right\} \right).$$

The following lemma will be used to apply the Hanson-Wright bound in the context of this chapter:

Lemma 4.1. *Let $m \geq 2$ and $0 \leq \rho < 1$. Suppose Y is a random Gaussian m vector with mean zero such that $\text{Var}(Y_k) = 1$ and $\text{Cov}(Y_k, Y_\ell) = \rho$ for $1 \leq k < \ell \leq m$. Let $Z = \sum_{1 \leq k < \ell \leq m} Y_k Y_\ell$. Then Z can be represented as in Proposition 4.1 for a matrix A with eigenvalue $\lambda_{\max} = \frac{m-1+(m-1)^2\rho}{2}$ with multiplicity one and eigenvalue $\lambda_{\min} = \frac{\rho-1}{2}$ with multiplicity $m-1$.*

Proof. The random vector Y can be represented as

$$Y = \begin{pmatrix} \sqrt{\rho} \mathbf{1}_{m \times 1} & \sqrt{1-\rho} I_{m \times m} \end{pmatrix} W$$

where W is a $\mathcal{N}(0_{(m+1) \times 1}, I_{(m+1) \times (m+1)})$ random vector and $Z = \frac{1}{2}Y^\top(E-I)Y$. Thus, $Z = W^\top \bar{A}W$ where

$$\bar{A} = \frac{1}{2} \begin{pmatrix} \sqrt{\rho} \mathbf{1}^\top \\ \sqrt{1-\rho} I \end{pmatrix} (E-I) \begin{pmatrix} \sqrt{\rho} \mathbf{1} & \sqrt{1-\rho} I \end{pmatrix}.$$

Note that $\lambda_{\max}$ is an eigenvalue of $\bar{A}$ with eigenvector $v_1 = \begin{pmatrix} m\sqrt{\frac{\rho}{1-\rho}} \\ \mathbf{1}_{m \times 1} \end{pmatrix}$, $\lambda_{\min}$ is an eigenvalue of $\bar{A}$ of multiplicity $m-1$ with eigenspace spanned by vectors of the form $\begin{pmatrix} 0 \\ v \end{pmatrix}$ such that $v \perp \mathbf{1}_{m \times 1}$, and 0 is an eigenvalue of $\bar{A}$ with eigenvector $v_0 = \begin{pmatrix} -\sqrt{\frac{1-\rho}{\rho}} \\ \mathbf{1}_{m \times 1} \end{pmatrix}$. Finally, by an orthonormal transformation of W , we can diagonalize $\bar{A}$ and reduce it to an $m \times m$ matrix A by deleting the row and column with the zero eigenvalue. $\square$

4.5 Achievable detection: proof of Theorem 4.1

Assume $\rho^2 \geq \frac{8}{m} \cdot \frac{\log n}{n-1}$ and take the threshold to be $\tau = \binom{n}{2} \binom{m}{2} \rho - n^c$ for a constant c with $1 < c < 1.5$. Without loss of generality, we may assume that the underlying permutations π_{1i}^* are all the identity permutation id. It then follows from the definition of T that $\mathbb{P}(T \leq \tau) \leq \mathbb{P}(T^* \leq \tau)$ where T^* is the log-likelihood for the identity permutation profile:

$$T^* = \sum_{1 \leq k < \ell \leq m} \sum_{1 \leq i < j \leq n} X_{ij}^k X_{ij}^\ell$$

Under $\mathbb{P}$, T^* is the sum of $\binom{n}{2}$ independent quadratic forms, each with the distribution of the random variable Z in Lemma 4.1. Hence T^* is also a quadratic form in jointly Gaussian random variables, and Proposition 4.1 yields

$$\begin{aligned} \mathbb{P}(T^* \leq \tau) &= \mathbb{P}(T^* - \mathbb{E}[T^*] \leq -n^c) \\ &\leq \exp \left(-\frac{1}{4} \frac{n^{2c}}{\binom{n}{2} (\lambda_{\max}^2 + (m-1)\lambda_{\min}^2) + \lambda_{\max} n^c} \right) \end{aligned}$$

Both $\lambda_{\max}$ and $\lambda_{\min}$ are bounded as $n \rightarrow \infty$, so $\mathbb{P}(T^* - \mathbb{E}[T^*] \leq -n^c) = o(1)$.

It remains to prove $\mathbb{Q}(T > \tau) = o(1)$. The distribution of T under $\mathbb{Q}$ does not depend on ρ , so $\mathbb{Q}(T > \tau)$ depends on ρ only through the value of τ . So without loss of generality for the remainder

of the proof assume $\rho^2 = \frac{8 \log n}{m(n-1)}$. By the union bound,

$$\begin{aligned} \mathbb{Q}(T > \tau) &= \mathbb{Q} \left(\max_{\pi_{12}, \dots, \pi_{1m}} \sum_{\substack{1 \leq i < j \leq n \\ 1 \leq k < \ell \leq m}} X_{\pi_{1k}(i)\pi_{1k}(j)}^k X_{\pi_{1\ell}(i), \pi_{1\ell}(j)}^\ell > \tau \right) \\ &\leq (n!)^{m-1} \cdot \mathbb{Q} \left(\sum_{\substack{1 \leq i < j \leq n \\ 1 \leq k < \ell \leq m}} X_{ij}^k X_{ij}^\ell > \tau \right) \end{aligned}$$

By Lemma 4.1 with $\rho = 0$, $\sum_{i < j} \sum_{1 \leq k < \ell \leq m} X_{ij}^k X_{ij}^\ell$ corresponds to a Gaussian quadratic form with eigenvalue $\frac{m-1}{2}$ having multiplicity $\binom{n}{2}$ and eigenvalue $-\frac{1}{2}$ with multiplicity $\binom{n}{2}(m-1)$. Hence, Proposition 4.1, and $n! \leq \exp((n + \frac{1}{2}) \log(n) - (n-1))$ yield:

$$\mathbb{Q}(T > \tau) \leq \exp \left((m-1)n \log n + \frac{m-1}{2} \log n - (m-1)(n-1) - \frac{\tau^2}{\left(\binom{n}{2}m(m-1) + 2(m-1)\tau\right)} \right).$$

Let $\mu = \binom{m}{2} \binom{n}{2} \rho$, so $\tau = \mu - n^c$. Note that

$$\begin{aligned} (m-1)n \log n &= \frac{\mu^2}{\binom{n}{2}m(m-1)}, \\ \frac{\tau^2}{\left(\binom{n}{2}m(m-1) + 2(m-1)\tau\right)} &\geq \frac{\tau^2}{\binom{n}{2}m(m-1)} \left(1 - \frac{2(m-1)\mu}{\binom{n}{2}m(m-1)} \right), \\ \tau^2 &\geq \mu^2 \left(1 - \frac{2n^c}{\mu} \right). \end{aligned}$$

Therefore,

$$\mathbb{Q}(T > \tau) \leq \exp(-(m-1)(n-1) + A_n + B_n)$$

where

$$\begin{aligned} A_n &= \frac{(m-1)n(\log n)2(m-1)\mu}{\binom{n}{2}m(m-1)} = \Theta(n^{1/2}(\log n)^{3/2}) \\ B_n &= \frac{(m-1)n(\log n)2n^c}{\mu} + \frac{m-1}{2} \log n \\ &= \Theta \left(n^c \left(\frac{\log n}{n} \right)^{1/2} \right). \end{aligned}$$

Thus, $\mathbb{Q}(T > \tau) = o(1)$ and the proof is complete.

4.6 Impossible detection via genie converse: proof of Theorem 4.2

The proof of Theorem 4.2 given here involves working directly with total variation distances. The idea in terms of a decision maker based with the hypothesis testing problem is the following. Even if there were a genie available that revealed to the decision maker how matrices X^2 through X^m were possibly aligned, the decision maker would not do better than guessing which hypothesis is true based on the information from the genie and the m matrices. A key idea is that in case the null hypothesis is true, the genie should reveal a plausible alignment of matrices X^2 through X^m . We begin by summarizing some well known properties of total variation distance in three lemmas.

Lemma 4.2. *Suppose P and Q are two joint probability distributions for random variables X, Y . Let P_X and Q_X denote the corresponding marginal probability distributions of X . If the conditional distribution of Y given X is the same for P and Q then $D_{\text{KL}}(P, Q) = D_{\text{KL}}(P_X, Q_X)$.*

Lemma 4.3. *For any probability measure P and event A , the distance between P and the conditional distribution of P given A satisfies: $D_{\text{KL}}(P, P(\cdot|A)) \leq P(A^c)$.*

Lemma 4.4. *Let P_X and Q_X be probability distributions for a random vector X . These distributions can be extended to joint distributions P and Q for random variables S, X such that:*

- $P(S \in \{0, 1\}) = Q(S \in \{0, 1\}) = 1$.
- $P(S = 0) = Q(S = 0) = D_{\text{KL}}(P_X, Q_X) = D_{\text{KL}}(P, Q)$
- The marginal distribution of X under P is P_X ,
- The marginal distribution of X under Q is Q_X
- $P_{X|S=1} = Q_{X|S=1}$ (equality of conditional distributions)

Proof of Theorem 4.2. The proof is by induction on m with the base case $m = 2$. The base case holds because it is the converse part of Theorem 1 in [WXY23]. For ease of notation we give here the proof for $m = 3$, and briefly explain at the end how the proof for general $m \geq 3$ can be given. So suppose $m = 3$ and suppose $\rho^2 \leq (2 - \varepsilon) \frac{\log n}{n}$.

To begin, we let $\mathbb{P}$ denote the joint distribution of X^1, X^2, X^3 and the unobserved random permutation profile $\pi^* = (\pi_{k\ell}^*)_{k, \ell \in [m]}$ under hypothesis H_1 and $\mathbb{Q}$ denote the joint distribution of X^1, X^2, X^3 under H_0 . We use $\otimes$ to denote product form distributions corresponding to independent random variables.

Using Lemmas 4.2 and 4.4 we shall consider extensions of $\mathbb{P}$ and $\mathbb{Q}$ to a larger ensemble of random objects and continue to use $\mathbb{P}$ and $\mathbb{Q}$ to denote the extensions. For a given subset of the objects we denote the marginal distribution of that set of objects under $\mathbb{P}$ or $\mathbb{Q}$ by using either $\mathbb{P}$ or $\mathbb{Q}$ with the objects as subscripts with no commas. For example, $\mathbb{P}_{X^2 X^3 \pi_{23}^*}$ denotes the marginal probability distribution of (X^2, X^3, π_{23}^*) under probability distribution $\mathbb{P}$.

Let $\varepsilon_n = D_{\text{KL}}(\mathbb{P}_{X^2 X^3}, \mathbb{Q}_{X^2 X^3})$. By the assumption on ρ and the result for $m - 1$, $\varepsilon_n = o(1)$ as $n \rightarrow \infty$. (This is the step that requires the proof by induction. It isn't the tightest part of the proof. For general $m \geq 3$ we require $\rho^2 \leq (\frac{4}{m-2} - \varepsilon) \frac{\log n}{n}$, which is implied by the assumption $\rho^2 \leq (\frac{4}{m-1} - \varepsilon) \frac{\log n}{n}$.)

Extend the probability distribution $\mathbb{Q}$ by adjoining a random permutation π_{23}^* such that the conditional distribution of π_{23}^* given (X^2, X^3) is the same under $\mathbb{Q}$ as under $\mathbb{P}$. As in the theory of multiple user information theory we can think of $\mathbb{P}$ defining a channel with input (X^2, X^3) and output π_{23}^* and we apply that same channel under probability distribution $\mathbb{Q}$. Since X_1 was independent of (X_2, X_3) under $\mathbb{Q}$ to begin with, X_1 is independent of (X^2, X^3, π_{23}^*) under $\mathbb{Q}$ extended to include π_{23}^* . By Lemma 4.2, $D_{\text{KL}}(\mathbb{P}_{X^2 X^3 \pi_{23}^*}, \mathbb{Q}_{X^2 X^3 \pi_{23}^*}) = \varepsilon_n$.

Focus further in this paragraph on the joint distributions of (X^2, X^3, π_{23}^*) . By Lemma 4.4 we can extend $\mathbb{P}_{X^2 X^3 \pi_{23}^*}$ and $\mathbb{Q}_{X^2 X^3 \pi_{23}^*}$ so that there is a binary random variable S jointly distributed with (X^2, X^3, π_{23}^*) so that (i) $\mathbb{P}(S = 0) = \mathbb{Q}(S = 0) = \varepsilon_n$ and (ii) $\mathbb{P}_{X^2 X^3 \pi_{23}^* | S=1} = \mathbb{Q}_{X^2 X^3 \pi_{23}^* | S=1}$. Equivalently, there exist choices of conditional probability distributions $\mathbb{P}_{S|X^2, X^3, \pi_{23}^*}$ and $\mathbb{Q}_{S|X^2, X^3, \pi_{23}^*}$ so that when $\mathbb{P}_{X^2 X^3 \pi_{23}^*}$ and $\mathbb{Q}_{X^2 X^3 \pi_{23}^*}$ are extended using those conditional distributions properties (i) and (ii) hold. Using those conditional probability distributions (again thinking of them as channels as in multiple user information theory), S can be adjoined to the larger ensemble $(X^1, X^2, X^3, \pi_{23}^*)$ under $\mathbb{P}$ and $\mathbb{Q}$ such that (i) and (ii) hold and under P : $X^1 - (X^2, X^3, \pi_{23}^*) - S$ is a Markov sequence. And under Q : X^1 is independent of $(X^2, X^3, \pi_{23}^*, S)$.

Let X^{23} denote the sum of X^2 and the version of X^3 that is aligned with X^2 using π_{23}^* : $X_{ij}^{23} \triangleq X_{ij}^2 + X_{\pi_{23}^*(i), \pi_{23}^*(j)}^3$ for $1 \leq i < j \leq n$. Note that X^{23} is a function of (X^2, X^3, π_{23}^*) . The conditional distribution $\mathbb{P}_{X^1 | X^2, X^3, \pi_{23}^*}$ can be described as follows. The permutation π_{21}^* is independent of (X^2, X^3, π_{23}^*) and uniformly distributed. Given $(X^2, X^3, \pi_{23}^*, \pi_{21}^*)$ the entries of X^1 are conditionally independent and

$$X_{\pi_{21}^*(i), \pi_{21}^*(j)}^1 \sim \mathcal{N}\left(\frac{\rho X_{ij}^{23}}{1 + \rho}, \frac{1 + \rho - 2\rho^2}{1 + \rho}\right)$$

In particular the conditional distribution $\mathbb{P}_{X^1 | X^2, X^3, \pi_{23}^*}$ depends on (X^2, X^3, π_{23}^*) only through X^{23} so that under $\mathbb{P}$: $X^1 - X^{23} - (X^2, X^3, \pi_{23}^*) - S$ is a Markov sequence. Also, since X^{23} is a function of (X^2, X^3, π_{23}^*) , under $\mathbb{Q}$: X^1 is independent of $(X^{23}, X^2, X^3, \pi_{23}^*, S)$.

By the choice of π_{23}^* and S and the fact that X^{23} is a function of (X^2, X^3, π_{23}^*) it follows that the conditional distribution $(X^{23}, \pi_{23}^*, X^2, X^3 | S = 1)$ is the same under $\mathbb{P}$ and $\mathbb{Q}$. Therefore, the conditional distribution $(\pi_{23}^*, X^2, X^3 | S = 1, X^{23})$ is also the same under $\mathbb{P}$ and $\mathbb{Q}$. By the Markov property (under $\mathbb{P}$) and independence property (under $\mathbb{Q}$) discussed in the previous paragraph, adding in conditioning on X^1 does not change the conditional distributions. In other words, the

conditional distribution $(\pi_{23}^*, X^2, X^3 | S = 1, X^{23}, X^1)$ is the same under $\mathbb{P}$ and $\mathbb{Q}$. That is:

$$\mathbb{P}_{\pi_{23}^*, X^2, X^3 | S=1, X^{23}, X^1} = \mathbb{Q}_{\pi_{23}^*, X^2, X^3 | S=1, X^{23}, X^1} \quad (4.9)$$

With the above preparations, we now have the string of inequalities:

$$\begin{aligned} D_{\text{KL}}(\mathbb{P}_{X^1 X^2 X^3}, \mathbb{Q}_{X^1 X^2 X^3}) &\stackrel{(a)}{\leq} D_{\text{KL}}(\mathbb{P}_{X^1 X^2 X^3 X^{23} \pi_{23}^*}, \mathbb{Q}_{X^1 X^2 X^3 X^{23} \pi_{23}^*}) \\ &\stackrel{(b)}{\leq} D_{\text{KL}}(\mathbb{P}_{X^1 X^2 X^3 X^{23} \pi_{23}^* | S=1}, \mathbb{Q}_{X^1 X^2 X^3 X^{23} \pi_{23}^* | S=1}) + 2\varepsilon_n \\ &\stackrel{(c)}{=} D_{\text{KL}}(\mathbb{P}_{X^1 X^{23} | S=1}, \mathbb{Q}_{X^1 X^{23} | S=1}) + 2\varepsilon_n \\ &\stackrel{(d)}{\leq} D_{\text{KL}}(\mathbb{P}_{X^1 X^{23} | S=1}, \mathbb{P}_{X^1 X^{23}}) + D_{\text{KL}}(\mathbb{P}_{X^1 X^{23}}, \mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23}}) \\ &\quad + D_{\text{KL}}(\mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23}}, \mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23} | S=1}) + 2\varepsilon_n \\ &\stackrel{(e)}{\leq} D_{\text{KL}}(\mathbb{P}_{X^1 X^{23}}, \mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23}}) + 4\varepsilon_n \end{aligned}$$

where (a) follows because including more variables cannot decrease variational distance, (b) follows by the triangle inequality of D_{KL} and two applications of Lemma 4.3, (c) follows from Lemma 4.2 and (4.9), (d) follows from the triangle inequality for variational distance and the fact $\mathbb{Q}_{X^1 X^{23} | S=1} = \mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23} | S=1}$, and (e) follows by applying Lemmas 4.2 and 4.3 to get: $D_{\text{KL}}(\mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23}}, \mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23} | S=1}) = D_{\text{KL}}(\mathbb{P}_{X^{23}}, \mathbb{P}_{X^{23} | S=1}) \leq \varepsilon_n$.

The term $D_{\text{KL}}(\mathbb{P}_{X^1 X^{23}}, \mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23}})$ is the variational distance for the detection problem for the two matrices X^1 and X^{23} . This problem is an instance of the Gaussian detection problem for two matrices and correlation coefficient ρ' given by $\rho' = \rho \sqrt{\frac{2}{1+\rho}}$. The assumption $\rho^2 \leq (2-\varepsilon) \frac{\log n}{n}$ implies $\rho'/\rho \rightarrow \sqrt{2}$ and $(\rho')^2 \leq (4-\frac{\varepsilon}{2}) \frac{\log n}{n}$ for all large n . Therefore $D_{\text{KL}}(\mathbb{P}_{X^1 X^{23}}, \mathbb{P}_{X^1} \otimes \mathbb{P}_{X^{23}}) = o(1)$ by the result for $m = 2$. Hence, tracing back through the string of inequalities, $D_{\text{KL}}(\mathbb{P}_{X^1 X^2 X^3}, \mathbb{Q}_{X^1 X^2 X^3}) = o(1)$. The proof for $m = 3$ is complete.

The proof for general $m \geq 3$ is similar. For all $m \geq 3$, matrices $X^2, \dots, X^m$ are essentially replaced by a single matrix $X^{2:m}$ where $X^{2:m}$ is the sum of matrices X^2 through X^m after X^3 through X^m are aligned with X^2 . Then the joint distribution of $X^1, X^{2:m}$ under $\mathbb{P}$ and $\mathbb{Q}$ after a scaling of $X^{2:m}$ by a constant is the same as the model for two matrices with parameter ρ' where $\rho' = \rho \sqrt{\frac{(m-1)}{1+(m-2)\rho}}$ so $\rho'/\rho \rightarrow \sqrt{m-1}$. $\square$

4.7 Impossible detection via second moment method: proof of Theorem 4.3

The proof of Theorem 4.3 is by the second moment method. Throughout this section, $L = L(X)$ denotes the likelihood ratio between the marginal planted law $\mathbb{P}_X$ and the null law $\mathbb{Q}$.

Since $\mathbb{P}_X \ll \mathbb{Q}$, Cauchy–Schwarz yields

$$\text{TV}(\mathbb{P}, \mathbb{Q}) = \frac{1}{2} \mathbb{E}_{\mathbb{Q}} |L - 1| \leq \frac{1}{2} \sqrt{\text{Var}_{\mathbb{Q}}(L)},$$

so that it suffices to show $\mathbb{E}_{\mathbb{Q}}[L^2] = 1 + o(1)$. As we discuss in Section 4.7.2, however, the naive second-moment computation is dominated by atypical values of L that, although rare under $\mathbb{P}$, contribute disproportionately to $\mathbb{E}_{\mathbb{Q}}[L^2]$. We therefore work with a truncated likelihood ratio that corresponds to an auxiliary measure $\mathbb{P}'$ close to $\mathbb{P}$ in total variation. The precise condition (4.6) on ρ^2 is only invoked near the end of the argument; weaker hypotheses on the rate at which $\rho \rightarrow 0$ are introduced as they become relevant.

The remainder of this section is organized as follows. Section 4.7.1 sets up the matrix and permutation notation needed for the analysis and writes the likelihood ratio as a Gaussian integral indexed by alignments. Section 4.7.2 introduces the truncated planted law $\mathbb{P}'$ and computes the leading-order contribution to the second moment. Section 4.7.3 introduces a class $\mathcal{H}$ of functions of permutations, that are exponentially negligible after averaging and absorbs the lower-order error terms in the second-moment expansion. Section 4.7.4 combines these ingredients to complete the proof.

4.7.1 The likelihood ratio

For a vertex permutation $\gamma \in \mathfrak{S}_n$, write $\gamma(e)$ for its induced action on an unordered edge $e \in E_n$. For an alignment $\pi = (\pi_1, \dots, \pi_m)$, define the symmetric $mN \times mN$ matrix K_π as follows. Its rows and columns are indexed by pairs $(k, e) \in [m] \times E_n$, and

$$(K_\pi)_{(k,e),(\ell,f)} = \mathbf{1}_{\{k \neq \ell\}} \cdot \mathbf{1}_{\{\pi_{k\ell}(e)=f\}}. \quad (4.10)$$

Equivalently, $(K_\pi)_{(k,e),(\ell,f)} = 1$ if $k \neq \ell$ and the edge f in graph ℓ corresponds to the same latent edge as e in graph k .

Under $\mathbb{P}$ the covariance matrix of X is Σ_{π^*} where

$$\Sigma_\pi = I_{mN} + \rho K_\pi,$$

where I_{mN} is the $mN \times mN$ identity matrix. We often write I when the dimension is clear. Therefore the conditional likelihood ratio

$$L_\pi(X) := \frac{d\mathbb{P}(X \mid \pi)}{d\mathbb{Q}(X)}$$

is given by

$$L_\pi(X) = \frac{1}{\sqrt{\det \Sigma_\pi}} \exp \left\{ -\frac{1}{2} X^\top (\Sigma_\pi^{-1} - I) X \right\}. \quad (4.11)$$

The full likelihood ratio is

$$L(X) = \mathbb{E}_\pi [L_\pi(X)].$$

Hence

$$\mathbb{E}_\mathbb{Q}[L(X)^2] = \mathbb{E}_{\pi \perp \tilde{\pi}} [\mathbb{E}_\mathbb{Q}[L_\pi(X)L_{\tilde{\pi}}(X)]] ,$$

where π and $\tilde{\pi}$ are independent alignments.

We close this section with some notations and properties of permutations. Given alignments π and $\tilde{\pi}$ let $d_{k\ell}(\pi, \tilde{\pi})$ denote the number of fixed points of $\pi_{\ell k} \circ \tilde{\pi}_{k\ell}$. It is equal to $|\{i \in [n] : \pi_{k\ell}(i) = \tilde{\pi}_{k\ell}(i)\}|$. Also let σ be the alignment such that $\sigma_1 = \text{Id}$ and $\sigma_k = \pi_k^{-1} \circ \tilde{\pi}_k$ for $2 \leq k \leq m$. Note that

$$\sigma_{k\ell} := \sigma_\ell \circ \sigma_k^{-1} = \pi_\ell^{-1} \circ \tilde{\pi}_\ell \circ \tilde{\pi}_k^{-1} \circ \pi_k = \pi_\ell^{-1} \circ \tilde{\pi}_{k\ell} \circ \pi_k.$$

Hence, i is a fixed point of $\sigma_{k\ell}$ if and only if any of the following equivalent conditions hold:

$$\begin{aligned} i = \pi_\ell^{-1} \circ \tilde{\pi}_{k\ell} \circ \pi_k(i) \quad \text{or} \quad \pi_\ell(i) = \tilde{\pi}_{k\ell} \circ \pi_k(i) \quad \text{or} \quad \pi_\ell \circ \pi_k^{-1} \circ \pi_k(i) = \tilde{\pi}_{k\ell} \circ \pi_k(i) \\ \text{or} \quad \pi_{k\ell} \circ \pi_k(i) = \tilde{\pi}_{k\ell} \circ \pi_k(i) \quad \text{or} \quad \pi_k(i) = \pi_{\ell k} \circ \tilde{\pi}_{k\ell} \circ \pi_k(i). \end{aligned}$$

In other words, i is a fixed point of $\sigma_{k\ell}$ if and only if $\pi_k(i)$ is a fixed point of $\pi_{\ell k} \circ \tilde{\pi}_{k\ell}$. Therefore, $d_{k\ell}(\pi, \tilde{\pi}) = d_{k\ell}(\sigma)$, where $d_{k\ell}(\sigma)$ is defined to be the number of fixed points of $\sigma_{k\ell}$. Therefore the following result from [VM25] is relevant and will play an important role near the end of the proof of Theorem 4.3.

Lemma 4.5 ([VM25]). *Let $(d_{k\ell})_{1 \leq k < \ell \leq m}$ be nonnegative integers. Set*

$$F((d_{k\ell})) = \left| \left\{ \sigma \in \mathcal{S}_n^{m-1} : d_{k\ell}(\sigma) \geq d_{k\ell} \text{ for all } 1 \leq k < \ell \leq m, \right\} \right|. \quad (4.12)$$

Then,¹

$$F((d_{k\ell})) \leq \frac{(n!)^{m-1}}{\left(\prod_{1 \leq k < \ell \leq m} (d_{k\ell}!) \right)^{2/m}}. \quad (4.13)$$

The same considerations as above hold for the edge permutations induced by the vertex permutations. (By definition, the edge permutation induced by a permutation $\tau \in \mathfrak{S}_n$, also called τ , is defined by $\tau(e) = f$ if and only if $e = \{i, j\}$ for some $1 \leq i < j \leq n$ and $f = \{\tau(i), \tau(j)\}$.) So similarly, let $D_{k\ell}(\pi, \tilde{\pi}) = |\{e \in \mathcal{E}_n : \pi_{k\ell}(e) = \tilde{\pi}_{k\ell}(e)\}|$. Then $D_{k\ell}(\pi, \tilde{\pi}) = D_{k\ell}(\sigma)$, where $D_{k\ell}(\sigma)$ is the number of fixed edges of the edge permutation induced by $\sigma_{k\ell}$. We write $D_{k\ell}(\pi, \tilde{\pi})$ or $D_{k\ell}(\sigma)$ interchangeably.

¹Here, the exponent is $2/m$ because the product is over $1 \leq k < \ell \leq m$, instead of $1 \leq k \neq \ell \leq m$ as in [VM25].

4.7.2 Truncating the planted measure

We require an upper bound on $\mathbb{E}_{\mathbb{Q}}[L_{\pi}(X)L_{\tilde{\pi}}(X)]$ that is uniform enough to average over the independent alignments π and $\tilde{\pi}$. The untruncated second moment is affected by rare values of $L_{\pi}(X)$ that are small-probability events under $\mathbb{P}$ but have a disproportionate effect under $\mathbb{Q}$. We control these contributions by replacing $\mathbb{P}$ with a probability measure $\mathbb{P}'$ obtained by conditioning on a high-probability event. Once we prove $\text{TV}(\mathbb{P}', \mathbb{Q}) = o(1)$, the desired conclusion follows from the triangle inequality. Specifically, $\mathbb{P}'$ is defined by setting

$$\mathbb{P}'(A) = \mathbb{P}(A|\mathcal{E}) = \frac{\mathbb{P}(A \cap \mathcal{E})}{\mathbb{P}(\mathcal{E})}$$

for all Borel measurable $A \subset \mathbb{R}^{m(n)} \times \mathfrak{S}_n^{m-1}$, where $\mathcal{E}$ is defined as follows:

$$\mathcal{E} \triangleq \bigcap_{1 \leq k < \ell \leq m} \bigcap_{S \subset [n]: |S| \geq n/2} \mathcal{E}_{k\ell, S},$$

where for $1 \leq k < \ell \leq m$ and $S \subset [n]$

$$\mathcal{E}_{k\ell, S} \triangleq \left\{ (X, \pi) : \sum_{e \in \binom{S}{2}} X_e^k X_{\pi_{k\ell}(e)}^\ell \leq \rho \binom{|S|}{2} + Cn^{3/2} \right\},$$

where $C > 0$ is a sufficiently large constant depending only on m .

It follows from [WXY23, Lemma 1] that

$$\mathbb{P} \left(\bigcap_{S \subset [n]: |S| \geq n/2} \mathcal{E}_{k\ell, S} \right) = 1 - e^{-\Omega(n)}.$$

for $1 \leq k < \ell \leq m$. Since $\mathcal{E}$ is the intersection of $\binom{m}{2}$ such events, the following lemma holds:

Lemma 4.6. $\mathbb{P}(\mathcal{E}) = 1 - e^{-\Omega(n)}$.

Recall for each (k, ℓ) with $1 \leq k < \ell \leq m$ that $d_{k\ell}(\pi, \tilde{\pi}) = d_{k\ell}(\sigma) = |\{i \in [n] : \pi_{k\ell}(i) = \tilde{\pi}_{k\ell}(i)\}|$. For each ordered pair $k, \ell \in [m]$ with $k \neq \ell$, let

$$O_{k\ell} = \begin{cases} \emptyset & \text{if } d_{k\ell}(\sigma) < n/2 \\ \{i \in [n] : \pi_{k\ell}(i) = \tilde{\pi}_{k\ell}(i)\} & \text{if } d_{k\ell}(\sigma) \geq n/2 \end{cases}.$$

Note that $|O_{k\ell}| = d_{k\ell}(\sigma) \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}}$.

We define a matrix K'_π related to K_π as follows:

$$(K'_\pi)_{(k,e),(\ell,f)} = \mathbf{1}_{\{k \neq \ell\}} \mathbf{1}_{\{\pi_{k\ell}(e)=f\}} \mathbf{1}_{\{e \in \binom{O_{k\ell}}{2}\}} \quad (4.14)$$

Note that K_π depends only on π while the sets $(O_{k\ell})$ and the matrix K'_π depend on both π and $\tilde{\pi}$. We collect here some matrix identities and trace expansions that will be used later. Their proofs are deferred to Chapter B. Recall that

$$D_{k\ell}(\pi, \tilde{\pi}) := |\{e \in E_n : \pi_{k\ell}(e) = \tilde{\pi}_{k\ell}(e)\}| = D_{k\ell}(\sigma).$$

Lemma 4.7. (a) If K'_π is defined by (4.14) with π replaced by $\tilde{\pi}$, then $K'_\pi = K'_{\tilde{\pi}}$.

(b) $K'_\pi \leq K_\pi$ pointwise and $K'_\pi \leq K_{\tilde{\pi}}$ pointwise.

(c) K_π and K'_π are symmetric matrices.

(d) $\text{Tr}(K_\pi K_{\tilde{\pi}}) = 2 \sum_{1 \leq k < \ell \leq m} D_{k\ell}(\sigma).$

(e)

$$\begin{aligned} \text{Tr}(K_\pi K'_\pi) &= \text{Tr}(K'_\pi K_\pi) = \text{Tr}(K'_\pi K_{\tilde{\pi}}) = \text{Tr}(K_{\tilde{\pi}} K'_\pi) \\ &= \text{Tr}((K'_\pi)^2) = 2 \sum_{1 \leq k < \ell \leq m} \binom{d_{k\ell}(\sigma)}{2} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}}. \end{aligned}$$

We also record three higher order trace expansions that will be useful later in the converse proof.

Lemma 4.8 (Trace expansions). For any two alignments π and $\tilde{\pi}$,

$$(a) \quad W_3(\pi, \tilde{\pi}) := \text{Tr}(K_\pi^2 K_{\tilde{\pi}}) = \sum_{\substack{k, \ell, k' \in [m] \\ k, \ell, k' \text{ distinct}}} \left| \{e \in E_n : (\tilde{\pi}_{k'k} \circ \pi_{\ell k'} \circ \pi_{k\ell})(e) = e\} \right|.$$

$$(b) \quad W_4(\pi, \tilde{\pi}) := \text{Tr}((K_\pi K_{\tilde{\pi}})^2) = \sum_{\substack{k, \ell, k', \ell' \in [m] \\ \{k, k'\} \cap \{\ell, \ell'\} = \emptyset}} \left| \{e \in E_n : (\tilde{\pi}_{\ell'k} \circ \pi_{k'\ell'} \circ \tilde{\pi}_{\ell k'} \circ \pi_{k\ell})(e) = e\} \right|.$$

$$(c) \quad \widetilde{W}_4(\pi, \tilde{\pi}) := \text{Tr}(K_\pi^3 K_{\tilde{\pi}}) = \sum_{\substack{k, \ell, k', \ell' \in [m] \\ \{k, k'\} \cap \{\ell, \ell'\} = \emptyset}} \left| \{e \in E_n : (\tilde{\pi}_{\ell'k} \circ \pi_{k'\ell'} \circ \pi_{\ell k'} \circ \pi_{k\ell})(e) = e\} \right|.$$

For $1 \leq k < \ell \leq m$, if $d_{k\ell}(\sigma) < n/2$ then $O_{k\ell} = \emptyset$, and if $d_{k\ell}(\sigma) \geq n/2$ then $\mathcal{E}_{k\ell, O_{k\ell}} \subset \mathcal{E}$. Therefore, on the event $(X, \pi) \in \mathcal{E}$,

$$\rho X^\top K'_\pi X = 2\rho \sum_{1 \leq k < \ell \leq m} \sum_{e \in \binom{O_{k\ell}}{2}} X_e^k X_{\pi_{k\ell}(e)}^\ell \leq \tau_\sigma$$

where

$$\tau_\sigma = 2 \sum_{1 \leq k < \ell \leq m} \left\{ \rho^2 \binom{d_{k\ell}(\sigma)}{2} + \rho C n^{3/2} \right\} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}}$$

Therefore, for every $r \geq 0$,

$$\mathbf{1}_{(X,\pi) \in \mathcal{E}} \leq \exp\{r(\tau_\sigma - \rho X^\top K'_\pi X)\}.$$

We will set $r = 1/2$ below; for the moment assume only that $0 \leq r \leq 1$. Consider the conditioned likelihood ratio,

$$L'(X) = \frac{1}{\mathbb{P}(\mathcal{E})} \mathbb{E}_\pi [L_\pi(X) \cdot \mathbf{1}_{(X,\pi) \in \mathcal{E}}].$$

Its second moment may be written as

$$\begin{aligned} \mathbb{E}_\mathbb{Q}[(L')^2] &= \frac{1}{\mathbb{P}(\mathcal{E})^2} \mathbb{E}_{\pi, \tilde{\pi}} \mathbb{E}_\mathbb{Q} [L_\pi(X) L_{\tilde{\pi}}(X) \cdot \mathbf{1}_{(X,\pi) \in \mathcal{E}} \mathbf{1}_{(X,\tilde{\pi}) \in \mathcal{E}}] \\ &\leq \frac{1}{\mathbb{P}(\mathcal{E})^2} \mathbb{E}_{\pi, \tilde{\pi}} \mathbb{E}_\mathbb{Q} [L_\pi(X) L_{\tilde{\pi}}(X) \cdot \mathbf{1}_{(X,\pi) \in \mathcal{E}}] \\ &\leq \frac{1}{\mathbb{P}(\mathcal{E})^2} \mathbb{E}_{\pi, \tilde{\pi}} \mathbb{E}_\mathbb{Q} \left[L_\pi(X) L_{\tilde{\pi}}(X) \exp\{r(\tau_\sigma - \rho X^\top K'_\pi X)\} \right]. \end{aligned} \quad (4.15)$$

Since $\mathbb{P}(\mathcal{E}) = 1 - o(1)$, the prefactor $\mathbb{P}(\mathcal{E})^{-2}$ is $1 + o(1)$. Hence, a sufficient condition for $\text{TV}(\mathbb{P}', \mathbb{Q}) = o(1)$ is $\mathbb{E}_{\pi, \tilde{\pi}} \mathbb{E}_\mathbb{Q} [L_\pi(X) L_{\tilde{\pi}}(X) \exp\{r(\tau_\sigma - \rho X^\top K'_\pi X)\}] = 1 + o(1)$. The following proposition will be used to bound the inner expectation in (4.15).

Proposition 4.2. *Suppose that $\rho^2 = o(1)$. Given two alignments π and $\tilde{\pi}$ and $r \in [0, 1]$, for sufficiently large n ,*

$$\log \mathbb{E}_\mathbb{Q} \left[L_\pi(X) L_{\tilde{\pi}}(X) \exp\{r(\tau_\sigma - \rho X^\top K'_\pi X)\} \right] = r\tau_\sigma + \frac{1}{2} \sum_{s \geq 1} \frac{1}{s} \text{Tr} [A^s]. \quad (4.16)$$

where

$$A = \rho^2 K_\pi K_{\tilde{\pi}} - 2r\rho K'_\pi - 2r\rho^2 (K_\pi K'_\pi + K'_\pi K_{\tilde{\pi}}) - 2r\rho^3 K_\pi K'_\pi K_{\tilde{\pi}}.$$

Proof. Using (4.11),

$$\begin{aligned} &\mathbb{E}_\mathbb{Q} [L_\pi(X) L_{\tilde{\pi}}(X) \exp\{r(\tau_\sigma - \rho X^\top K'_\pi X)\}] \\ &= \frac{1}{\sqrt{\det \Sigma_\pi} \sqrt{\det \Sigma_{\tilde{\pi}}}} \mathbb{E}_\mathbb{Q} \left[\exp \left\{ r\tau_\sigma - \frac{1}{2} X^\top (\Sigma_\pi^{-1} + \Sigma_{\tilde{\pi}}^{-1} - 2I + 2r\rho K'_\pi) X \right\} \right] \\ &\stackrel{(a)}{=} \frac{e^{r\tau_\sigma}}{\sqrt{\det \Sigma_\pi} \sqrt{\det \Sigma_{\tilde{\pi}}} \sqrt{\det (\Sigma_\pi^{-1} + \Sigma_{\tilde{\pi}}^{-1} - I + 2r\rho K'_\pi)}} \\ &\stackrel{(b)}{=} e^{r\tau_\sigma} \det (\Sigma_\pi + \Sigma_{\tilde{\pi}} - \Sigma_\pi \Sigma_{\tilde{\pi}} + 2r\rho \Sigma_\pi K'_\pi \Sigma_{\tilde{\pi}})^{-1/2} \\ &\stackrel{(c)}{=} e^{r\tau_\sigma} \det (I - A)^{-1/2}, \end{aligned}$$

where step (a) uses the Gaussian quadratic-form identity

$$\mathbb{E}[e^{-\frac{1}{2}x^\top Mx}] = \det(I + M)^{-1/2},$$

valid for $x \sim N(0, I)$ whenever $I + M$ is positive definite. The identity is applicable here because $\|\Sigma_\pi^{-1} + \Sigma_{\tilde{\pi}}^{-1} - 2I + 2r\rho K'_\pi\|_{\text{op}} = O(\rho)$ (which follows from (B.1) in Chapter B, and the fact that $\|K'_\pi\|_{\text{op}} \leq m - 1$ as explained below). Step (b) follows by multiplying the matrix inside the determinant on the left by Σ_π and on the right by $\Sigma_{\tilde{\pi}}$, and using multiplicativity of the determinant. Step (c) follows from $\Sigma_\pi = I + \rho K_\pi$ and $\Sigma_{\tilde{\pi}} = I + \rho K_{\tilde{\pi}}$. Since each row of $K_\pi, K_{\tilde{\pi}}$ and K'_π has at most $m - 1$ nonzero entries and they are all one, Gershgorin's circle theorem implies $\max\{\|K_\pi\|_{\text{op}}, \|K_{\tilde{\pi}}\|_{\text{op}}, \|K'_\pi\|_{\text{op}}\} \leq m - 1$. Therefore

$$\|A\|_{\text{op}} = O(\rho) = o(1) \quad (4.17)$$

so for all sufficiently large n , $\|A\|_{\text{op}} < 1$, so $-\log(I - A) = \sum_{s \geq 1} A^s/s$. Taking traces and using $\log \det B = \text{Tr}(\log B)$ yields

$$-\log \det(I - A) = \sum_{s \geq 1} \frac{\text{Tr}(A^s)}{s}. \quad \square$$

We next examine some of the terms appearing on the right-hand side of (4.16). Using the fact K'_π has zero diagonal and zero trace, Lemma 4.7(e), the fact $\text{Tr}(AB) = \text{Tr}(BA)$ for matrices A, B , and dropping negative terms yields

$$\begin{aligned} \frac{1}{2} \text{Tr}(A) &\leq \frac{1}{2} \rho^2 \text{Tr}(K_\pi K_{\tilde{\pi}} - 4r(K'_\pi)^2), \\ \frac{1}{4} \text{Tr}(A^2) &\leq r^2 \rho^2 \text{Tr}((K'_\pi)^2) + J_2 + o(\rho^5) \\ \text{Tr}(A^3) &\leq 3(2r)^2 \rho^4 \text{Tr}((K'_\pi)^2 K_\pi K_{\tilde{\pi}}) + o(\rho^5) \\ \text{Tr}(A^4) &\leq 16r^2 \rho^4 \text{Tr}((K'_\pi)^4) + o(\rho^5), \end{aligned}$$

where

$$\begin{aligned} J_2 &= 2r^2 \rho^3 \text{Tr}(K_\pi K'_\pi K'_\pi + K'_\pi K'_\pi K_{\tilde{\pi}}) + \frac{\rho^4}{4} \text{Tr}((K_\pi K_{\tilde{\pi}})^2) \\ &\quad + r^2 \rho^4 \text{Tr}((K_\pi K'_\pi + K'_\pi K_{\tilde{\pi}})^2) + 2r^2 \rho^4 \text{Tr}(K'_\pi K_\pi K'_\pi K_{\tilde{\pi}}). \end{aligned}$$

Lemma 4.9. *Suppose that $n^2 \rho^5 = o(1)$. With $W_3, W_4, \widetilde{W}_4$ as defined in Lemma 4.8, the following*

bounds hold:

$$\begin{aligned} J_2 &\leq 4\rho^3 W_3(\pi, \tilde{\pi}) + \frac{17\rho^4}{4} W_4(\pi, \tilde{\pi}) + 2\rho^4 \widetilde{W}_4(\pi, \tilde{\pi}) \\ \text{Tr}(A^3) &\leq 12\rho^4 W_4(\pi, \tilde{\pi}) + o(1) \\ \text{Tr}(A^4) &\leq 16\rho^4 W_4(\pi, \tilde{\pi}) + o(1) \end{aligned}$$

Also, the sum of all terms $s \geq 5$ in (4.16) is $O(N\rho^5) = o(1)$.

Proof. We use $r \leq 1$. Recall that $K'_\pi \leq K_\pi$ and $K'_\pi \leq K_{\tilde{\pi}}$ pointwise. Thus,

$$\text{Tr}(K_\pi K'_\pi K'_\pi) \leq \text{Tr}(K_\pi^2 K_{\tilde{\pi}}) = W_3(\pi, \tilde{\pi}), \quad \text{Tr}(K'_\pi K'_\pi K_{\tilde{\pi}}) \leq \text{Tr}(K_\pi^2 K_{\tilde{\pi}}) = W_3(\pi, \tilde{\pi}).$$

Similarly, $K_\pi K'_\pi + K'_\pi K_{\tilde{\pi}} \leq 2K_\pi K_{\tilde{\pi}}$ so $\text{Tr}((K_\pi K'_\pi + K'_\pi K_{\tilde{\pi}})^2) \leq 4 \text{Tr}((K_\pi K_{\tilde{\pi}})^2) = 4W_4(\pi, \tilde{\pi})$. Also, $K'_\pi K_\pi K'_\pi K_{\tilde{\pi}} \leq K_\pi^3 K_{\tilde{\pi}}$ so $\text{Tr}(K'_\pi K_\pi K'_\pi K_{\tilde{\pi}}) \leq \widetilde{W}_4(\pi, \tilde{\pi})$. Finally, $\text{Tr}((K'_\pi)^4) \leq \text{Tr}((K'_\pi)^2 K_\pi K_{\tilde{\pi}}) \leq W_4(\pi, \tilde{\pi})$. These yield the bounds in the centered equations in the lemma.

Recall that $\|A\|_{\text{op}} = O(\rho)$ uniformly in n . Hence $\text{Tr}(A^s) \leq Nm\|A\|_{\text{op}}^s \leq Nm(C\rho)^s$ for some constant C , implying the last statement of the lemma. $\square$

Assume for the remainder of this section that $n^2\rho^5 \rightarrow 0$ and $n\rho^2 \rightarrow \infty$. Applying Proposition 4.2, Lemma 4.7 and Lemma 4.9 and recalling

$$r\tau_\sigma = 2r \sum_{1 \leq k < \ell \leq m} \left\{ \rho^2 \binom{d_{k\ell}(\sigma)}{2} + \rho C n^{3/2} \right\} \cdot \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}}$$

we have for a constant C

$$\begin{aligned} \log \mathbb{E}_{\mathbb{Q}}[L_\pi(X) L_{\tilde{\pi}}(X) \mathbf{1}_{(X, \pi) \in \mathcal{E}}] &\leq \rho^2(2r - 4r + 2r^2) \sum_{1 \leq k < \ell \leq m} \binom{d_{k\ell}(\sigma)}{2} \cdot \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}} + \rho^2 \sum_{1 \leq k < \ell \leq m} D_{k\ell}(\sigma) \\ &\quad + 2\rho r C n^{3/2} \sum_{1 \leq k < \ell \leq m} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}} \\ &\quad + C\rho^3 W_3(\pi, \tilde{\pi}) + C\rho^4 (W_4(\pi, \tilde{\pi}) + \widetilde{W}_4(\pi, \tilde{\pi})) + o(1). \end{aligned}$$

Setting $r = 1/2$ then yields

$$\begin{aligned}
& \log \mathbb{E}_{\mathbb{Q}} [L_{\pi}(X) L_{\tilde{\pi}}(X) \mathbf{1}_{(X, \pi) \in \mathcal{E}}] \\
& \leq -\frac{1}{2} \rho^2 \sum_{1 \leq k < \ell \leq m} \binom{d_{k\ell}(\sigma)}{2} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}} + \rho^2 \sum_{1 \leq k < \ell \leq m} D_{k\ell}(\sigma) \\
& \quad + \rho C n^{3/2} \sum_{1 \leq k < \ell \leq m} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}} + C \rho^3 W_3(\pi, \tilde{\pi}) + C \rho^4 (W_4(\pi, \tilde{\pi}) + \widetilde{W}_4(\pi, \tilde{\pi})) + o(1) \\
& \stackrel{(a)}{=} \frac{1}{2} \sum_{1 \leq k < \ell \leq m} \rho^2 \binom{d_{k\ell}(\sigma)}{2} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}} + \rho^2 \sum_{1 \leq k < \ell \leq m} \binom{d_{k\ell}(\sigma)}{2} \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}} \\
& \quad + \rho C n^{3/2} \sum_{1 \leq k < \ell \leq m} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}} + Y(\pi, \tilde{\pi}) \\
& \stackrel{(b)}{\leq} (1 + o(1)) \rho^2 \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} + Y(\pi, \tilde{\pi}) \tag{4.18}
\end{aligned}$$

where

$$Y(\pi, \tilde{\pi}) = \rho^2 \sum_{1 \leq k < \ell \leq m} \left[D_{k\ell}(\sigma) - \binom{d_{k\ell}(\sigma)}{2} \right] + C \rho^3 W_3(\pi, \tilde{\pi}) + C \rho^4 (W_4(\pi, \tilde{\pi}) + \widetilde{W}_4(\pi, \tilde{\pi})) + o(1).$$

Here, (a) decomposes

$$D_{k\ell}(\sigma) = \binom{d_{k\ell}(\sigma)}{2} + \left[D_{k\ell}(\sigma) - \binom{d_{k\ell}(\sigma)}{2} \right]$$

and (b) uses that, on $\{d_{k\ell}(\sigma) \geq n/2\}$, $\rho C n^{3/2} = o\left(\rho^2 \binom{d_{k\ell}(\sigma)}{2}\right)$ by the assumption $n\rho^2 \rightarrow \infty$ (i.e. $\sqrt{n}\rho \rightarrow \infty$).

The next section shows that the term $Y(\pi, \tilde{\pi})$ can be neglected after the averaging over $(\pi, \tilde{\pi})$.

4.7.3 Exponentially negligible functions

The main result of this section is Lemma 4.12 below.

Definition 4.1. Let $\mathcal{H}$ denote the set of nonnegative functions $Y_n(\pi, \tilde{\pi})$ on $(\mathfrak{S}_n^{m-1})^2$ such that for every constant $\Gamma > 0$,

$$\mathbb{E}_{\pi \perp \tilde{\pi}} [\exp(\Gamma Y)] = 1 + o(1) \quad \text{as } n \rightarrow \infty.$$

Lemma 4.10. (a) If $Y_1, Y_2, \dots, Y_J \in \mathcal{H}$ for a constant number of terms J , then $Y_1 + \dots + Y_J \in \mathcal{H}$.

(b) If $Y = f(n)$ for some function f such that $f = o(1)$, then $Y \in \mathcal{H}$.

Proof. By the inequality $x_1 x_2 \dots x_J \leq (x_1^J + \dots + x_J^J)/J$,

$$\exp(\Gamma(Y_1 + \dots + Y_J)) \leq \frac{1}{J} (\exp(J\Gamma Y_1) + \dots + \exp(J\Gamma Y_J)),$$

which implies part (a). Part (b) follows from $\exp(\Gamma \cdot o(1)) = 1 + o(1)$. $\square$

The following lemma follows directly from classical results about permutations – the proof is in the appendix.

Lemma 4.11. *Fix an integer $j \geq 1$. Let $C_j = C_j^{(n)}$ denote the number of j -cycles in a uniformly random permutation of $\{1, \dots, n\}$.*

(a) *If $\theta_n = o(1)$ then $\mathbb{E}[e^{\theta_n C_j}] = 1 + o(1)$.*

(b) *If $\theta_n = o(\frac{\log n}{n})$ then $\mathbb{E}[e^{\theta_n C_j^2}] = 1 + o(1)$.*

Lemma 4.12. *Suppose $\frac{n\rho^4}{\log n} \rightarrow 0$ and $\rho^3 = o(\frac{\log n}{n})$. Then $\rho^2 \sum_{1 \leq k < \ell \leq m} [D_{k\ell} - \binom{d_{k\ell}(\sigma)}{2}]$, $\rho^3 W_3(\pi, \tilde{\pi})$, $\rho^4 W_4(\pi, \tilde{\pi})$, and $\rho^4 \widetilde{W}_4(\pi, \tilde{\pi})$ are in $\mathcal{H}$.*

Proof. By Lemma 4.10(a), to show $\rho^2 \sum_{1 \leq k < \ell \leq m} [D_{k\ell} - \binom{d_{k\ell}(\sigma)}{2}] \in \mathcal{H}$, it suffices to show that the individual terms $\rho^2 (D_{k\ell} - \binom{d_{k\ell}(\sigma)}{2}) \in \mathcal{H}$ for all k, ℓ with $1 \leq k < \ell \leq m$.

Now, $D_{k\ell}(\sigma) - \binom{d_{k\ell}(\sigma)}{2} = c_{k\ell}$ where

$$c_{k\ell} := \left| \left\{ \{i, j\} \subset [n] : \sigma_{k\ell}(i) = j, \sigma_{k\ell}(j) = i \right\} \right|.$$

In other words $c_{k\ell}$ is the number of 2-cycles (transpositions) in the cycle decomposition of the vertex permutation $\sigma_{k\ell}$. Since $\sigma_{k\ell}$ is uniformly distributed over $\mathfrak{S}_n$, $\rho^2 c_{k\ell} \in \mathcal{H}$ by Lemma 4.11.

Bounding $\rho^3 W_3(\pi, \tilde{\pi})$. By Lemma 4.8(a), it suffices by Lemma 4.10(a) to show that each summand is in $\mathcal{H}$ after multiplication by ρ^3 . Given distinct $k, \ell, k' \in [m]$, let

$$\tau = \tilde{\pi}_{k'k} \circ \pi_{\ell k'} \circ \pi_{k\ell}.$$

Since $\tilde{\pi}_{k'k}$ is independent of the other two permutations in τ and is uniformly distributed over $\mathfrak{S}_n$, τ is also uniformly distributed over $\mathfrak{S}_n$. Therefore,

$$\rho^3 \text{FixE}(\tau) = \rho^3 \binom{C_1(\tau)}{2} + \rho^3 C_2(\tau) \in \mathcal{H}, \quad (4.19)$$

where the first term uses $\rho^3 = o(\log n/n)$ and Lemma 4.11(b), while the second uses Lemma 4.11(a). Hence $\rho^3 W_3(\pi, \tilde{\pi}) \in \mathcal{H}$.

Bounding $\rho^4 W_4(\pi, \tilde{\pi})$. By Lemma 4.8(b), $W_4(\pi, \tilde{\pi})$ is the sum of

$$D_{k\ell k'\ell'} = \left| \left\{ e \in E_n : (\tilde{\pi}_{\ell'k'} \circ \pi_{k'\ell'} \circ \tilde{\pi}_{\ell k'} \circ \pi_{k\ell})(e) = e \right\} \right|$$

over $k, \ell, k', \ell' \in [m]$ such that $\{k, k'\} \cap \{\ell, \ell'\} = \emptyset$. By Lemma 4.10(a) it is sufficient to show that $\rho^4 D_{k\ell k'\ell'} \in \mathcal{H}$ for all such k, ℓ, k', ℓ' . We consider two cases.

First consider the case $k = k'$ and $\ell = \ell'$ corresponding to terms of the form $D_{k\ell k\ell}$. Note that $D_{k\ell k\ell}$ is the number of edges fixed by τ^2 where $\tau = \tilde{\pi}_{\ell k} \circ \pi_{k\ell}$ and τ is uniformly distributed over $\mathfrak{S}_n$. A vertex is fixed by τ^2 if it lies in a 1-cycle or a 2-cycle of τ . Therefore the number of vertices fixed by τ^2 is $C_1(\tau) + 2C_2(\tau)$. Edges between such vertices are fixed by τ^2 . In addition, each 4-cycle of τ contributes two transpositions to τ^2 , and hence contributes two more fixed edges. Thus

$$\begin{aligned} \rho^4 D_{k\ell k\ell} &= \rho^4 \text{FixE}(\tau^2) = \rho^4 \binom{C_1(\tau) + 2C_2(\tau)}{2} + 2\rho^4 C_4(\tau) \\ &\leq 2\rho^4 C_1^2(\tau) + 8\rho^4 C_2^2(\tau) + 2\rho^4 C_4(\tau) \end{aligned} \quad (4.20)$$

By Lemma 4.11 each term on the right-hand side of (4.20) is in $\mathcal{H}$ so by Lemma 4.10, $\rho^4 D_{k\ell k\ell} \in \mathcal{H}$.

Next, consider the second case, namely, that either $k \neq k'$ or $\ell \neq \ell'$ or both. We assume $k \neq k'$; the proof for $\ell \neq \ell'$ is similar. Either $k \neq 1$ or $k' \neq 1$. We assume $k \neq 1$; the case $k' \neq 1$ is similar. Let $\tau = \tilde{\pi}_{\ell'k} \circ \pi_{k'\ell'} \circ \tilde{\pi}_{\ell k'} \circ \pi_{k\ell}$. Using the definition $\pi_{k\ell} = \pi_\ell \pi_k^{-1}$ and similarly for $\tilde{\pi}_{k\ell}$, we find

$$\tau = \tilde{\pi}_k \circ \tilde{\pi}_{\ell'}^{-1} \circ \pi_{\ell'} \circ \pi_{k'}^{-1} \circ \tilde{\pi}_{k'} \circ \tilde{\pi}_{\ell}^{-1} \circ \pi_\ell \circ \pi_k^{-1}.$$

Note that $\tilde{\pi}_k$ appears only once and is uniformly distributed over $\mathfrak{S}_n$ and is independent of the other permutations defining τ . Hence τ is also uniformly distributed over $\mathfrak{S}_n$. Thus, recalling (4.19), $\rho^4 D_{k,\ell,k,\ell'} = \rho^4 \text{FixE}(\tau) \leq \rho^3 \text{FixE}(\tau) \in \mathcal{H}$.

The two cases cover all possibilities for k, ℓ, k', ℓ' , so $\rho^4 W_4(\pi, \tilde{\pi}) \in \mathcal{H}$.

Bounding $\rho^4 \widetilde{W}_4(\pi, \tilde{\pi})$. Note that $\widetilde{W}_4(\pi, \tilde{\pi})$ is the same as $W_4(\pi, \tilde{\pi})$ but with $D_{k\ell k'\ell'}$ being the number of fixed points of $\tilde{\pi}_{\ell'k} \circ \pi_{k'\ell'} \circ \pi_{\ell k'} \circ \pi_{k\ell}$. Since $\tilde{\pi}$ appears only once in each composition, all terms fall into the second case of the proof that $W_4(\pi, \tilde{\pi}) \in \mathcal{H}$ so that $\rho^4 \widetilde{W}_4(\pi, \tilde{\pi}) \in \mathcal{H}$. $\square$

4.7.4 Finishing the proof

Recall from (4.18) that if $n^2 \rho^5 \rightarrow 0$ and $n\rho^2 \rightarrow \infty$ then

$$\log \mathbb{E}_{\mathbb{Q}}[L_\pi(X) L_{\tilde{\pi}}(X) \mathbf{1}_{(X, \pi) \in \mathcal{E}}] \leq (1 + o(1)) \rho^2 \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} + Y(\pi, \tilde{\pi}),$$

where by Lemma 4.10 and Lemma 4.12, $Y \in \mathcal{H}$. Applying Holder's inequality for $p = 1 + \epsilon$ and $q = 1 + \frac{1}{\epsilon}$ (satisfying $\frac{1}{p} + \frac{1}{q} = 1$) yields that for any fixed $\epsilon > 0$:

$$\begin{aligned}
& \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\mathbb{E}_{\mathbb{Q}} [L_{\pi}(X) L_{\tilde{\pi}}(X) \mathbf{1}_{(X, \pi) \in \mathcal{E}}] \right] \\
& \leq \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left((1 + o(1)) \rho^2 \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} + Y(\pi, \tilde{\pi}) \right) \right] \\
& \leq \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left((1 + o(1)) \rho^2 (1 + \epsilon) \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} \right) \right]^{\frac{1}{1+\epsilon}} \\
& \quad \cdot \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left(\left(1 + \frac{1}{\epsilon}\right) Y(\pi, \tilde{\pi}) \right) \right]^{\frac{\epsilon}{1+\epsilon}} \\
& \leq \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left((1 + o(1)) \rho^2 (1 + \epsilon) \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} + o(1) \right) \right] \\
& \leq \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left(\rho^2 (1 + 2\epsilon) \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} + o(1) \right) \right]. \tag{4.21}
\end{aligned}$$

Remark 4.3 (Reduction to equality). It is enough to prove Theorem 4.3 when

$$\rho^2 = \frac{8(1-\eta) \log n}{m} \frac{n}{n}. \tag{4.22}$$

Indeed, if $0 \leq \rho_0 \leq \rho_1$, then a sample from the planted model with correlation ρ_1 can be passed through the coordinatewise Gaussian channel

$$Y_e^k = \sqrt{\rho_0/\rho_1} X_e^k + \sqrt{1 - \rho_0/\rho_1} Z_e^k, \quad Z_e^k \stackrel{\text{iid}}{\sim} N(0, 1),$$

with the same channel applied under the null. The null is mapped to the null, while the planted correlation is reduced from ρ_1 to ρ_0 . Since total variation cannot increase under a Markov kernel, indistinguishability at the boundary correlation implies indistinguishability for all smaller nonnegative correlations.

For the remainder of the proof of Theorem 4.3 we assume that $\rho^2 = \frac{8(1-\eta) \log n}{m} \frac{n}{n}$, and next show that the right-hand side of (4.21) is $1 + o(1)$. Select ϵ such that $(1 - \eta)(1 + 2\epsilon) = 1 - \eta/2$.

Let $\bar{\eta} = \eta/2$ and $\bar{\rho}^2 = \rho^2(1 + 2\epsilon)$ so that $\bar{\rho}^2 = \frac{8(1-\bar{\eta})}{m} \frac{\log n}{n}$. Then, (4.21) yields

$$\begin{aligned} & \mathbb{E}_{\pi \perp \tilde{\pi}} \mathbb{E}_{\mathbb{Q}} [L_{\pi}(X) L_{\tilde{\pi}}(X) \mathbf{1}_{(X, \pi) \in \mathcal{E}}] \\ & \leq \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left(\bar{\rho}^2 \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} + o(1) \right) \right] \\ & = (1 + o(1)) \cdot \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left(\bar{\rho}^2 \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} \right) \right]. \end{aligned} \quad (4.23)$$

Now, let

$$\mathcal{U} = \left\{ \sigma : 0 \leq d_{k\ell}(\sigma) \leq \frac{\sqrt{n}}{\log n} \text{ for } 1 \leq k < \ell \leq m \right\}.$$

Note that if $\sigma \in \mathcal{U}$, then

$$\frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} \leq \binom{d_{k\ell}(\sigma)}{2} \leq \frac{1}{2} \frac{n}{\log^2 n},$$

so that, using $\bar{\rho}^2 \leq \frac{8}{m} \frac{\log n}{n}$:

$$\begin{aligned} & \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left(\bar{\rho}^2 \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} \right) \mathbf{1}_{\{\sigma \in \mathcal{U}\}} \right] \\ & \leq \exp \left(\binom{m}{2} \frac{1}{2} \frac{n}{\log^2 n} \frac{8}{m} \frac{\log n}{n} \right) \\ & = \exp \left(\frac{2(m-1)}{\log n} \right). \end{aligned} \quad (4.24)$$

Lemma 4.13. *For integers d with $0 \leq d \leq n$, let*

$$f(d) := -\log(d!) + \frac{2(1 + \mathbf{1}_{\{d < \frac{n}{2}\}}) \cdot \binom{d}{2} \cdot (1 - \bar{\eta}) \log n}{n}.$$

Then, for all sufficiently large n , $f(d) \leq 0$ for $0 \leq d \leq n$, and $\max \left\{ f(d) : \frac{\sqrt{n}}{\log n} \leq d \leq n \right\} \sim -\frac{1}{2} \sqrt{n}$.

Proof. Let

$$g(d) \triangleq -\log(d!) + \frac{4 \binom{d}{2} (1 - \bar{\eta}) \log n}{n}$$

so that $f(d) = g(d)$ for $1 \leq d < n/2$. The function g satisfies

$$g(0) = g(1) = 0 \quad \text{and} \quad g(d) - g(d-1) = -\log d + \frac{(d-1)4(1-\bar{\eta}) \log n}{n}, \quad \text{for } d \geq 1.$$

Thus, for some d^* , g is decreasing over $0 \leq d \leq d^*$ and increasing over $d \geq d^*$. Since

$$g\left(\frac{n}{2}\right) \sim -\frac{\bar{\eta}}{2}n \log n < 0,$$

it follows that $f(d) = g(d) \leq 0$ for $0 \leq d < n/2$.

Next, by Stirling's formula, we have that $g\left(\frac{\sqrt{n}}{\log n}\right) \sim -\frac{1}{2}\sqrt{n}$ and $g\left(\frac{n}{2}\right) \sim -\frac{\bar{\eta}}{2}n \log n$. Since g is decreasing and then increasing, the maximum of g over $\sqrt{n}/\log n \leq d < n/2$ is attained at one of the endpoints. Therefore

$$\max \left\{ f(d) : \frac{\sqrt{n}}{\log n} \leq d < \frac{n}{2} \right\} = \max \left\{ g\left(\frac{\sqrt{n}}{\log n}\right), g\left(\frac{n}{2}\right) \right\} \sim -\frac{1}{2}\sqrt{n},$$

where we used that $\bar{\eta} > 0$ is fixed, so $\bar{\eta}n \log n \gg \sqrt{n}$.

Similarly, let

$$h(d) := -\log(d!) + \frac{2\binom{d}{2}(1-\bar{\eta})\log n}{n}$$

so that $f(d) = h(d)$ for $\frac{n}{2} \leq d \leq n$. The function h satisfies

$$h(d) - h(d-1) = -\log d + \frac{2(d-1)(1-\bar{\eta})\log n}{n} \quad \text{for } d \geq 1,$$

so, like g , it is decreasing for small d and then increasing. Hence the maximum of h over $n/2 \leq d \leq n$ is attained at one of the endpoints. Moreover, Stirling's formula gives

$$h(n) \sim -\bar{\eta}n \log n, \quad h\left(\frac{n}{2}\right) \sim -\frac{1+\bar{\eta}}{4}n \log n.$$

Therefore, for some constant $c_{\bar{\eta}} > 0$ depending only on $\bar{\eta}$,

$$\max \left\{ f(d) : \frac{n}{2} \leq d \leq n \right\} = \max \left\{ h(d) : \frac{n}{2} \leq d \leq n \right\} \leq -c_{\bar{\eta}}n \log n.$$

Combining the two ranges, and using again that $n \log n \gg \sqrt{n}$, we obtain as desired that

$$\max \left\{ f(d) : \frac{\sqrt{n}}{\log n} \leq d \leq n \right\} \sim -\frac{1}{2}\sqrt{n}.$$

□

Combining Lemma 4.5 and Lemma 4.13 for $\bar{\rho}^2 = \frac{8(1-\bar{\eta})}{m} \frac{\log n}{n}$ yields:

$$\begin{aligned}
& \mathbb{E}_{\pi \perp \tilde{\pi}} \left[\exp \left(\bar{\rho}^2 \sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell}(\sigma) < \frac{n}{2}\}}}{2} \binom{d_{k\ell}(\sigma)}{2} \right) \mathbf{1}_{\{\sigma \in \mathcal{U}^c\}} \right] \\
& \leq \frac{1}{(n!)^{m-1}} \sum_{(d_{k\ell}) : \max_{k\ell} d_{k\ell} \geq \frac{\sqrt{n}}{\log n}} F((d_{k\ell})) \exp \left(\left[\sum_{1 \leq k < \ell \leq m} \frac{1 + \mathbf{1}_{\{d_{k\ell} < \frac{n}{2}\}}}{2} \binom{d_{k\ell}}{2} \right] \bar{\rho}^2 \right) \\
& \stackrel{(a)}{\leq} \sum_{(d_{k\ell}) : \max_{k\ell} d_{k\ell} \geq \frac{\sqrt{n}}{\log n}} \exp \left(\sum_{1 \leq k < \ell \leq m} \left[-\frac{2}{m} \log(d_{k\ell}!) + \frac{1 + \mathbf{1}_{\{d_{k\ell} < \frac{n}{2}\}}}{2} \binom{d_{k\ell}}{2} \bar{\rho}^2 \right] \right) \\
& = \sum_{(d_{k\ell}) : \max_{k\ell} d_{k\ell} \geq \frac{\sqrt{n}}{\log n}} \exp \left(\frac{2}{m} \sum_{1 \leq k < \ell \leq m} f(d_{k\ell}) \right) \\
& \leq (n+1)^{\binom{m}{2}} \exp \left(-\frac{2(1+o(1))}{m} \times \frac{\sqrt{n}}{2} \right) = o(1).
\end{aligned} \tag{4.26}$$

Here, (a) uses the bound in Lemma 4.5. Combining (4.23), (4.24), and (4.26) yields

$$\mathbb{E}_{\pi \perp \tilde{\pi}} \mathbb{E}_{\mathbb{Q}} [L_{\pi}(X) L_{\tilde{\pi}}(X) \mathbf{1}_{(X, \pi) \in \mathcal{E}}] \leq 1 + o(1).$$

Returning to (4.15) and using $\mathbb{P}(\mathcal{E}) = 1 - o(1)$, we obtain

$$\mathbb{E}_{\mathbb{Q}}[(L')^2] \leq 1 + o(1).$$

Since $\mathbb{E}_{\mathbb{Q}}[L'] = 1$, this implies $\mathbb{E}_{\mathbb{Q}}[(L' - 1)^2] = o(1)$ and hence $\text{TV}(\mathbb{P}', \mathbb{Q}) = o(1)$. Finally, by the high-probability estimate for $\mathcal{E}$ and the standard bound between a measure and its conditioning on a high-probability event,

$$\text{TV}(\mathbb{P}, \mathbb{P}') \leq \mathbb{P}(\mathcal{E}^c) = e^{-\Omega(n)}.$$

The triangle inequality gives $\text{TV}(\mathbb{P}, \mathbb{Q}) = o(1)$, completing the proof of Theorem 4.3.

Chapter 5

Detecting planted correlation in circular data

5.1 Introduction

Many modern data sets are inherently directional, taking values on the unit circle rather than the real line. Examples include phases of oscillatory signals, times of day or year (viewed modulo 24 hours or one year), headings of moving objects, and phase differences between pairs of time series. In such settings, a natural statistical question is whether observed angles are unstructured (compatible with a uniform distribution on the circle), or whether there is a hidden subset that exhibits a preferred direction or phase.

This chapter studies circular data in the presence of planted structure. Under the null hypothesis, all observations are i.i.d. uniform on the circle, while under the alternative, there is a small unknown subset of indices (or community of vertices in a network) whose angles concentrate around an unknown phase. These hypothesis testing problems combine two classical themes: (i) circular statistics, where the von Mises distribution serves as a “Gaussian on the circle” [MJ09], and (ii) planted structure and phase transitions in high-dimensional detection, such as planted clique and community detection in stochastic block models.

Our goal is to characterize when detection is information-theoretically possible in directional planted models. We describe three application domains where such problems arise.

- (a) *Seasonal and diurnal effects in event data.* Consider time-stamped events (transactions, failures, attacks, messages) where $X_i \in [0, 2\pi)$ denotes the time of the i th event (modulo 24 hours or one year). Many applications exhibit background activity spread uniformly across all

This chapter is based on joint work with Bruce Hajek, available as the preprint [AH26]. To appear in *2026 IEEE International Symposium on Information Theory (ISIT)*, 2026. In accordance with IEEE policy, the authors retain the right to reuse this material in a dissertation.

times, alongside a smaller subset of events driven by a seasonal or diurnal mechanism (failures that concentrate during hot summer days, or cyber-attacks that concentrate at night). Under a natural null model, all event times are i.i.d. uniform on $[0, 2\pi)$; under the alternative, an unknown subpopulation of size K has times that concentrate in an unknown arc of length $2\pi\tau$ (for $\tau < 1$), or around an unknown preferred phase with concentration κ . Here τ or κ quantifies the strength of concentration, and K the size of the affected population.

- (b) *Phase-locked oscillators.* Suppose we measure the instantaneous phases $X_1, \dots, X_N \in [0, 2\pi)$ of N oscillators (neurons in a neural population, devices in a power grid, or Kuramoto oscillators modeling synchronization phenomena). When oscillators are fully desynchronized, their phases are close to uniform on the circle. However, when a subset of K oscillators is driven by a common source, their phases become localized near a common direction. The strength of this localization can be modeled by a concentration parameter κ , with the von Mises distribution providing a canonical alternative to uniformity.
- (c) *Phase-coherent communities in networks.* Consider a network of n time series (EEG channels, financial assets, or sensor readings). At a given frequency, the pairwise relationship between series i and j can be summarized by their cross-spectral coherence, a complex-valued quantity whose phase $X_{ij} \in [0, 2\pi)$ encodes the phase offset between the two signals. This yields a matrix $(X_{ij})_{1 \leq i < j \leq n}$ of angular measurements on the edges of the complete graph. Under unstructured interactions, these edge phases are approximately uniform. However, if a community C^* of k nodes is driven by a common oscillatory source (e.g. a shared rhythm in sensor networks), then the phases X_{ij} for $i, j \in C^*$ become aligned around some offset Θ^* and exhibit significantly greater coherence than background edges.

The task is to determine when a phase-coherent subset (or community) can be detected from the observed circular data, as a function of the population size (N or n), the subset size (K or k), and the concentration strength (τ or κ).

5.1.1 Models, tests, and objectives

A random variable X is said to have the von Mises distribution with location parameter $\Theta^* \in [0, 2\pi)$ and concentration parameter $\kappa > 0$, denoted $X \sim \text{vonMises}(\Theta^*, \kappa)$, if it has density

$$f(\theta; \Theta^*, \kappa) = \frac{1}{2\pi I_0(\kappa)} \exp(\kappa \cos(\theta - \Theta^*)) , \quad \theta \in [0, 2\pi) ,$$

where I_0 is the modified Bessel function of the first kind of order 0, $I_0(\kappa) = \frac{1}{2\pi} \int_0^{2\pi} \exp(\kappa \cos u) du$. Figure 5.1 shows an example of the von Mises distribution with different spread, depending on the value of κ .

We introduce four hypothesis testing problems below, with model parameters listed after the model names. In all models, the null hypothesis H_0 assumes directional homogeneity (i.i.d. uniform data

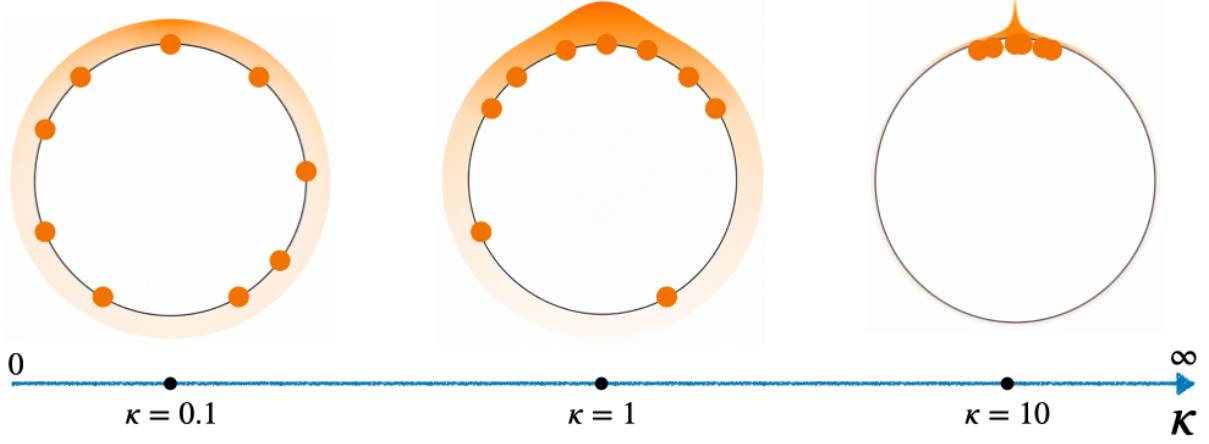


Figure 5.1: Example of the von Mises distribution for various κ

on the circle), while the alternative H_1 introduces a structured component.

- (a) **Hard-cluster flat model** (N, K, τ) . Angles $X_1, \dots, X_N \in [0, 2\pi)$ are observed. Under H_0 , the X_i are independent and uniform on $[0, 2\pi)$. Under H_1 , there is an unknown planted subset $S^* \subset [N]$ of size K and an unknown arc location $\Theta^* \in [0, 2\pi)$ such that

$$X_i \sim \text{Uniform}([\Theta^*, \Theta^* + 2\pi\tau]) \quad \text{for } i \in S^*, \quad X_i \sim \text{Uniform}([0, 2\pi]) \quad \text{for } i \notin S^*,$$

independently. This captures a hard cluster of events or oscillators localized in an unknown window of width $2\pi\tau$ on the circle.

- (b) **Von Mises flat model** (N, K, κ) . Angles $X_1, \dots, X_N \in [0, 2\pi)$ are observed, with the same null H_0 as in the hard-cluster flat model. Under H_1 , there is a planted subset S^* of size K and a preferred phase Θ^* such that

$$X_i \sim \text{vonMises}(\Theta^*, \kappa) \quad \text{for } i \in S^*, \quad X_i \sim \text{Uniform}([0, 2\pi]) \quad \text{for } i \notin S^*,$$

independently.

- (c) **Hard-cluster community model** (n, k, τ) . A symmetric matrix of angles $(X_{ij})_{1 \leq i, j \leq n}$ is observed, viewed as angular labels on the edges of the complete graph on n vertices. Under H_0 , the X_{ij} for $i < j$ are independent uniform on $[0, 2\pi)$. Under H_1 , there is an unknown community $C^* \subset [n]$ of size k and an unknown phase Θ^* such that

$$X_e \sim \text{Uniform}([\Theta^*, \Theta^* + 2\pi\tau]) \quad \text{for } e \in E(C^*),$$

where $E(C) \triangleq \{e = \{i, j\} : i, j \in C, i < j\}$ is the set of all edges with both end-points in C , and the angles for all edges with at least one endpoint outside C^* remain uniform on $[0, 2\pi)$.

- (d) **Von Mises community model** (n, k, κ) . A symmetric matrix of angles $(X_{ij})_{1 \leq i, j \leq n}$ is observed. Under H_0 the X_{ij} for $i < j$ are independent uniform on $[0, 2\pi)$. Under H_1 , there is an unknown community $C^* \subset [n]$ of size k and an unknown phase Θ^* such that

$$X_e \sim \text{vonMises}(\Theta^*, \kappa) \quad \text{for } e \in E(C^*),$$

whereas all other edges remain uniform on $[0, 2\pi)$.

In each model, under H_1 it is assumed that the planted set S^* (respectively C^*) is drawn uniformly at random among all subsets of size K (respectively k), and that the planted phase Θ^* is drawn independently from the uniform distribution on $[0, 2\pi)$. The law of the observations under H_1 , denoted by $\mathbb{P}$, is then the resulting mixture over (S^*, Θ^*) (respectively (C^*, Θ^*)), and $\mathbb{Q}$ denotes the corresponding law under H_0 .

The four models above are studied in parallel because they represent related manifestations of directional structure. The hard-cluster and von Mises models correspond to two canonical ways of modeling concentration on the circle, and in the regime of small window widths τ or large concentration parameters κ they are closely related, with the rough correspondence $\tau \approx \kappa^{-1/2}$. Likewise, the flat and community models differ primarily in their combinatorial structure, while sharing many of the same probabilistic mechanisms and bounding techniques. We study these models side by side to highlight which phenomena are specific to a particular geometry and which are intrinsic to directional planted structure more broadly.

We now formalize the detection problem. A *test* is a measurable function ϕ mapping data to $\{0, 1\}$, where $\phi(X) = 1$ means rejecting H_0 in favor of H_1 . The *false alarm* and *miss* probabilities are defined by

$$p_{\text{fa}}(\phi) \triangleq \mathbb{Q}\{\phi(X) = 1\}, \quad p_{\text{miss}}(\phi) \triangleq \mathbb{P}\{\phi(X) = 0\}.$$

We will be interested in the following detection regimes as n (or N) tends to infinity.

Definition 5.1 (Strong and weak detection).

- *Strong detection is possible* if there exists a sequence of tests (ϕ_n) such that $p_{\text{fa}}(\phi_n) \rightarrow 0$ and $p_{\text{miss}}(\phi_n) \rightarrow 0$.
- *Weak detection is possible* if there exists a sequence of tests (ϕ_n) and a constant $\delta > 0$ such that $p_{\text{fa}}(\phi_n) + p_{\text{miss}}(\phi_n) \leq 1 - \delta$ for all sufficiently large n .
- *Weak detection is impossible* if for every sequence of tests (ϕ_n) , $p_{\text{fa}}(\phi_n) + p_{\text{miss}}(\phi_n) \geq 1 - o(1)$. Equivalently, weak detection is impossible if the total variation distance between $\mathbb{P}$ and $\mathbb{Q}$ tends to zero.

Interval and coherence tests We propose and analyze natural testing procedures tailored to directional planted structure. For the flat models, we study an interval test with window length

$2\pi\tau$ and threshold γ . Given observations $X_1, \dots, X_N \in [0, 2\pi)$, define the statistic

$$T_{\text{int}}(X) \triangleq \sup_{\theta \in [0, 2\pi)} \sum_{i=1}^N \mathbf{1}\{X_i \in [\theta, \theta + 2\pi\tau]\}. \quad (5.1)$$

The interval test for the flat models with parameters (τ, γ) rejects H_0 if $T_{\text{int}}(X) \geq \gamma$.

Similarly, for the community models, we study an interval test with window length $2\pi\tau$. Given observations $(X_{ij})_{1 \leq i < j \leq n}$, consider the statistic

$$T_{\text{int}}^{\text{comm}}(X) \triangleq \sup_{\theta \in [0, 2\pi)} \sup_{C \subseteq [n]: |C|=k} \mathbf{1}\{X_e \in [\theta, \theta + 2\pi\tau] \text{ for all } e \in E(C)\}. \quad (5.2)$$

The community interval test with parameters (k, τ) rejects H_0 if $T_{\text{int}}^{\text{comm}}(X) = 1$.

For the von Mises community model, we also study a coherence test. For each edge $e = \{i, j\}$, let $Z_e = e^{iX_e}$ (with $i \triangleq \sqrt{-1}$), and define the coherence statistic

$$T_{\text{coh}}(X) \triangleq \max_{C \subseteq [n]: |C|=k} \left| \sum_{e \in E(C)} Z_e \right|, \quad (5.3)$$

The coherence test with parameters (k, β) rejects H_0 if $T_{\text{coh}}(X) \geq \beta$.

The interval and coherence tests correspond to generalized likelihood ratio tests for the hard-cluster and von Mises family of detection problems, respectively. A derivation is presented in Appendix C.2. The coherence test is related to the Rayleigh test in circular statistics, which compares the sum of Z_e over *all* edges e to a threshold. This statistic, although efficiently computable, performs worse than the coherence test at detecting planted communities; see Appendix C.3. Finally, the interval and coherence tests are themselves related; see Appendix C.4.

5.1.2 Related work

Directional statistics Directional statistics concerns data on circles or spheres (angles, phases), and has well-developed tools for modeling and inference [MJ09, Fis95]. A central problem in this area is testing circular uniformity: given i.i.d. angles $\theta_1, \dots, \theta_n \in [0, 2\pi)$, decide whether they are consistent with uniform($[0, 2\pi)$) or exhibit some systematic departure. Classical tests include the classical Rayleigh test [Ray80] (related to our coherence test), as well as Kuiper's test [Kui60] and Watson's test [Wat61] (comparing empirical and uniform CDFs), and Rao's spacing test [Rao76] (based on gaps between ordered angles). These methods (and their Bayesian counterparts [LRM18, MK21]) test goodness-of-fit in the homogeneous setting where all observations are drawn from a single common distribution, and the goal is to detect any global non-uniformity. A related contamination model is studied in [Ben06], which fits a mixture of von Mises and uniformly distributed data, and uses likelihood ratio and Watson-type goodness-of-fit procedures to choose between uniform, von Mises, and mixture fits. Here, each observation is drawn i.i.d. from the same mixture.

In contrast, our setting differs in two ways. First, under the alternative only a small, unknown subset of observations (or edges, in the community models) is structured, while the majority remain uniform. Second, we focus on high-dimensional regimes in which the problem size and planted subset size grow, and we ask for information-theoretic thresholds that separate detectability from impossibility. Classical uniformity tests typically have little power in such sparse planted regimes, where the global empirical distribution can remain near-uniform despite the coherent subset.

Beyond hypothesis testing, scan statistics have been studied in other contexts as well. In the circular setting, the scan statistic is the maximum number of observations contained in any arc of a given length as the arc sweeps around the circle. Foundational work of Cressie [Cre77, Cre80] analyzes the distribution of this statistic under circular uniformity, and broader developments and extensions are reviewed in the monograph [GNW05]. Scan statistics are also used in applications with periodic structure (e.g., seasonality), where one searches for a localized pulse of events by scanning a window around the year [WWG89, WNG93].

Planted structures in non-circular data From a high-dimensional testing viewpoint, our models belong to the broad class of problems with planted structure, where a low-complexity signal (a sparse subset, a community, or a low-rank perturbation) is embedded in a high-dimensional noise background. Canonical examples of such problems include planted clique [Jer92, AKS98, FK00], sparse PCA [dGJL04, JL09, DM14], and sparse submatrix detection [BI13]; see Wu and Xu [WX21] for a recent survey of information-theoretic and computational thresholds in such models. Particularly relevant in the graph setting are detectability and recovery thresholds for community detection in sparse stochastic block models, e.g. [MNS18, Mas14, BMNN16], and the survey [Abb18]. As a special case, the *planted dense subgraph* problem studies the detection of an anomalously dense vertex subset embedded in an otherwise homogeneous random graph, and has been analyzed from both statistical and computational perspectives [ACV14, VAC15, DM15, HWX17]. Our work can be viewed as a directional analog of these models: we replace scalar observations (Bernoulli or Gaussian) with circular valued measurements, and characterize the corresponding detection and impossibility regimes for planted clusters and communities. Our community models also differ from stochastic block models for a planted dense community in that for our models the marginal distributions of the edge labels are the same under either hypothesis.

A related line of work studies statistical–computational gaps in planted sparse structure problems, where detection is information-theoretically possible but conjecturally hard for polynomial time algorithms. For instance, Berthet and Rigollet [BR13] gives evidence for such gaps for sparse PCA via average case reductions from planted clique, and Brennan, Bresler, and Huleihel [BBH18] broadens this reduction-based program to obtain tight conditional lower bounds for several canonical planted models. Our focus here is on information-theoretic detectability and impossibility in directional planted models; extending these reduction frameworks to circular-valued observations is a natural direction for future work.

Table 5.1: Summary of necessary and sufficient conditions for detection

Model	Regime	Impossibility	Achievability
Hard-cluster flat (N, K, τ)	K constant	$\tau = \omega(N^{-1-\frac{1}{K-1}})$	$\tau = o(N^{-1-\frac{1}{K-1}})$
	$K = N^\alpha, \alpha \leq 1/2$	$\tau = \frac{K^2}{cN \log N}, c < 1 - 2\alpha$	$\tau = \frac{K^2}{cN \log N}, c > 2$
	$K = N^\alpha, \alpha > 1/2$	—	$\tau \in (0, 1 - \epsilon], \epsilon > 0$
Von Mises flat (N, K, κ)	$K = N^\alpha, \alpha \leq 1/2$	$\frac{1}{\sqrt{\kappa}} \geq \frac{K^2}{cN \log N} (c < \frac{1-2\alpha}{2\sqrt{\pi}})$	$\frac{1}{\sqrt{\kappa}} \leq \frac{K^2}{cN \log N} (c > 0.5057)$
	$K = N^\alpha, \alpha > 1/2$	—	$\kappa \geq \epsilon, \epsilon > 0$
Hard-cluster community (n, k, τ)	$3 \leq k \leq n$	—	$\tau \leq \left(\frac{k}{ne}\right)^{\frac{(1+\epsilon)k}{\binom{k}{2}-1}}$
	$k = c \log n$	$\tau > e^{-2/c}$	$\tau \leq e^{-2/c}$
	$k = o(\sqrt{n})$	$\tau \geq \exp\left(-\frac{(2-\epsilon)\log(n/k^2)}{k-1}\right)$	$\tau \leq \left(\frac{k}{ne}\right)^{\frac{(1+\epsilon)k}{\binom{k}{2}-1}}$
	$k = \omega(\log n)$	$\tau \geq 1 - \frac{2(1-\epsilon)}{k-1} \log\left(1 + \frac{\epsilon n}{k^2}\right)$	$\tau \leq 1 - \frac{2(1+\epsilon)}{k-1} \log \frac{ne}{k}$
Von Mises community (n, k, κ)	$3/\epsilon < k = o(\log n)$	$\kappa \leq \left(\frac{n}{k^2}\right)^{\frac{4-\epsilon}{k-1}}$	$\kappa \geq \left(\frac{4}{\pi^2} + \epsilon\right) \left(\frac{n}{k}\right)^{\frac{4(1+\epsilon)}{k-1}} \log k$
	$k = c \log n$	$c \log R(\kappa) < 2$	$c A^2(\kappa) > 2$ and $c > 2$, or $\kappa \geq \frac{2 \log k}{1 - \cos(\pi \exp(-2/c))}$
	$k = \omega(\log n)$	$\kappa \leq \sqrt{\frac{4(1-2c-\epsilon)\log n}{k-1}},$ $k \leq n^c$ and $c < 1/2$	$\kappa \geq (1 + \epsilon) \sqrt{\frac{8 \log n}{k-1}}$

5.1.3 Summary and discussion of main results

Information-theoretic detection thresholds For each of the four models, we derive sharp conditions (up to constants and occasional logarithmic factors) on the triplets (N, K, τ) , (N, K, κ) , (n, k, τ) , and (n, k, κ) that separate detectability from impossibility. Table 5.1 summarizes these results for specific regimes, with more general conditions for achievability and impossibility listed in the theorem statements. (Functions $A(\kappa)$ and $R(\kappa)$ are defined in Appendix C.1.) Our results assume that the decision maker does not know the planted phase Θ^* ; see Appendix C.5 for a discussion of how access to Θ^* can improve performance in the flat case, and Appendix C.6 for a discussion on the connection to stochastic block models in the community case.

A recurring theme in our results is a small K versus large K dichotomy. For the flat models, there is a qualitative change around $K = \sqrt{N}$: when $K = o(\sqrt{N})$, the planted subset is sufficiently sparse that detection requires the directions within the cluster to become increasingly concentrated (equivalently, $\kappa \rightarrow \infty$ in the von Mises model, or $\tau \rightarrow 0$ in the hard-cluster model), whereas when $K = \omega(\sqrt{N})$ the number of planted observations makes detection possible even at constant

concentration (e.g., $\kappa = \Theta(1)$) and a window width τ bounded away from 0 and 1). In contrast, for the community models the corresponding transition occurs for much smaller correlated sets, around $k = \log n$: if $k = o(\log n)$ then strong detection forces the edge-level concentration to strengthen with n (again $\kappa \rightarrow \infty$ or $\tau \rightarrow 0$), while if $k = \omega(\log n)$ the community contributes so many intra-community edges that detection can succeed with only bounded concentration.

In the small k regimes, the hard-cluster and von Mises distributions are closely aligned: For large κ the von Mises(Θ^*, κ) law is well approximated by the Gaussian distribution with mean Θ^* and standard deviation $\frac{1}{\sqrt{\kappa}}$ yielding the heuristic correspondence $\tau \approx \kappa^{-1/2}$ as $\kappa \rightarrow \infty$. This correspondence helps explain why the same scanning-based procedures and thresholds emerge in parallel across the hard-cluster and von Mises models.

The hard-cluster and von Mises distributions are also aligned in the large k regimes, where $\tau \rightarrow 1$ and $\kappa \rightarrow 0$, in the sense that both the uniform distribution on $[0, 2\pi\tau]$ and the von Mises distribution with spread parameter κ converge to the uniform distribution on $[0, 2\pi]$. Comparing the achievability conditions for the two community models suggests the correspondence $1 - \tau \approx \kappa^2/4$. The KL divergences in this regime satisfy

$$D_{\text{KL}}(\text{Unif}([0, 2\pi\tau]) || \text{Unif}([0, 2\pi])) = -\ln \tau = 1 - \tau + o(1 - \tau)$$

and

$$D_{\text{KL}}(\text{vonMises}(0, \kappa) || \text{Unif}([0, 2\pi])) = \kappa^2/4 + o(\kappa^2),$$

which suggests the same correspondence.

Methodology Our achievability results are obtained by analyzing the interval and coherence tests, combined with sharp tail bounds for the false alarm and miss probabilities. The converse results rely on second-moment methods and control of likelihood ratios, together with convex-order comparisons between hypergeometric and binomial overlap distributions. In the von Mises models, the analysis uses asymptotics for modified Bessel functions and a Gaussian approximation for the von Mises distribution in the large-concentration regime. The techniques illustrate how directional geometry and planted combinatorial structure can be analyzed in a unified way.

5.2 The hard-cluster flat model

In this section and the next three sections, the main result is presented as a theorem, followed by corollaries specializing the necessary and sufficient conditions to various regimes of interest, and then followed by subsections giving the proof of the theorem.

Theorem 5.1. *Consider the hard-cluster flat model with parameters (N, K, τ) .*

- (a) *(Achievability) Strong detection is achievable by the interval test with threshold γ in either of*

the following two cases:

(A1) $\gamma = K$ and $\frac{N^K \tau^{K-1}}{(K-1)!} \rightarrow 0$.

(A2) There exist sequences $c_N \rightarrow \infty$ and $\gamma \equiv \gamma_N$ with

$$\gamma_N \triangleq (N-K)\tau + K - c_N \sqrt{(N-K)\tau}, \quad \gamma_N \geq 1 + (N-1)\tau \text{ for all sufficiently large } N,$$

such that

$$\frac{(K-1)^2(1-\tau)^2}{(N-K)\tau} \rightarrow \infty, \quad (5.4)$$

$$\frac{(K(1-\tau) - c_N \sqrt{(N-K)\tau})^2}{2N\tau + K(1-\tau) - c_N \sqrt{(N-K)\tau}} - \log N \rightarrow \infty. \quad (5.5)$$

(b) (Converse) If $\tau \leq 1/2$ for sufficiently large N and

$$\frac{2N\tau^2}{K^2} \left(1 + \frac{K}{N\tau}\right)^{K+1} = o(1), \quad (5.6)$$

then weak detection is impossible.

Corollary 5.1 (Achievability regimes). *Under the hard-cluster flat model, strong detection is possible in each of the following regimes with the interval test.*

- $K \geq 2$ fixed and $\tau = o(N^{-1-\frac{1}{K-1}})$ [Take $\gamma = K$ so (A1) is satisfied.]
- $K = N^\alpha$ with $0 < \alpha \leq 1/2$, and $\tau = \frac{K^2}{(2+\epsilon)N \log N} = \frac{N^{2\alpha-1}}{(2+\epsilon) \log N}$ for some constant $\epsilon \in (0, 1/2)$. Here, the threshold γ_N is chosen so that $c_N \rightarrow \infty$ and $c_N = o(\sqrt{\log N})$.
- $K = N^\alpha$ with $1/2 < \alpha < 1$, and $\tau \in (0, 1-\epsilon]$ for some constant $\epsilon \in (0, 1)$. Here, the threshold γ_N is chosen so that $c_N \rightarrow \infty$ and $c_N = o(K/\sqrt{N\tau})$.

Corollary 5.2 (Impossibility regimes). *Under the hard-cluster flat model, weak detection is impossible in each of the following regimes.*

- $K \geq 2$ fixed and $\tau = \omega(N^{-1-\frac{1}{K-1}})$. [This scaling implies the condition (5.6) of Theorem 5.1(b).]
- $K = N^\alpha$ with $0 < \alpha < 1/2$, and $\tau = \frac{K^2}{\epsilon N \log N} = \frac{N^{2\alpha-1}}{\epsilon \log N}$ for any constant $\epsilon \in (0, 1-2\alpha)$. [For this choice of (K, τ) ,

$$\frac{2N\tau^2}{K^2} = \frac{2}{\epsilon^2 \log^2 N} N^{2\alpha-1}, \quad \frac{K(K+1)}{N\tau} = \epsilon \log N (1 + N^{-\alpha}),$$

hence

$$\frac{2N\tau^2}{K^2} \left(1 + \frac{K}{N\tau}\right)^{K+1} \leq \frac{2N\tau^2}{K^2} \exp\left\{\frac{K(K+1)}{N\tau}\right\} = \frac{2+o(1)}{\epsilon^2 \log^2 N} N^{2\alpha+\epsilon-1} \rightarrow 0,$$

since $\epsilon < 1 - 2\alpha$.]

5.2.1 Achievability for hard-cluster flat model – interval test

(a) Fix an integer threshold $\gamma \geq K$ and consider the interval test that thresholds the statistic (5.1), i.e., it decides H_1 if there exists an interval of length $2\pi\tau$ containing at least γ of the X_i 's, and decides H_0 otherwise. This decision rule is equivalent to the generalized maximum likelihood decision rule obtained by comparing the maximum over θ of the conditional likelihood given $\Theta^* = \theta$ to a threshold; see Appendix C.2.1.

Case (A1): $\gamma = K$. Here the probability of missed detection is zero: under H_1 the interval $[\Theta^*, \Theta^* + 2\pi\tau]$ contains all K points indexed by S^* , so the scan statistic is at least K and $p_{\text{miss}} = 0$. It remains to bound p_{fa} under H_0 .

We upper bound the probability of false alarm p_{fa} as follows. Let B_i be the event that the interval $[X_i, X_i + 2\pi\tau]$ contains at least γ points, including X_i itself. Then $p_{\text{fa}} = \mathbb{Q}(\cup_{i \in [N]} B_i)$, so by the union bound and the fact that $\mathbb{Q}(B_i) = \mathbb{Q}(B_1)$ for all i ,

$$p_{\text{fa}} \leq N \cdot \mathbb{Q}(B_1).$$

Under $\mathbb{Q}$, the cardinality of $\{j \in \{2, \dots, N\} : X_j \in [X_1, X_1 + 2\pi\tau]\}$ has the $\text{Binom}(N-1, \tau)$ distribution, so the total number of observations in $[X_1, X_1 + 2\pi\tau]$ is distributed as $1 + \text{Binom}(N-1, \tau)$. The event B_1 is the union of the events $\{X_j \in [X_1, X_1 + 2\pi\tau] \text{ for } j \in S\}$ over all subsets $S \subset \{2, \dots, N\}$ of cardinality $\gamma-1$. By another application of the union bound,

$$\mathbb{Q}(B_1) \leq \binom{N-1}{\gamma-1} \tau^{\gamma-1}.$$

Hence,

$$p_{\text{fa}} \leq N \binom{N-1}{\gamma-1} \tau^{\gamma-1} = \binom{N}{\gamma} \gamma \tau^{\gamma-1} \leq \frac{N^\gamma \tau^{\gamma-1}}{(\gamma-1)!}. \quad (5.7)$$

Specializing to $\gamma = K$, this shows that $p_{\text{fa}} \rightarrow 0$ whenever $\frac{N^K \tau^{K-1}}{(K-1)!} \rightarrow 0$. This concludes Case (A1).

Case (A2): $\gamma = \gamma_N$. If K increases with N then the critical value of τ could be large enough that under $\mathbb{P}$ the interval $[\Theta^*, \Theta^* + 2\pi\tau]$ contains not only the K points indexed by S^* but also, with high probability, a significant number of other points. This motivates the use of a threshold γ that is greater than K . The probability of false alarm depends only on N, τ , and γ and does not involve

K .

We first bound p_{fa} using a Chernoff bound, which is suitable when γ is at least the mean under $\mathbb{Q}$. As above, $p_{\text{fa}} \leq N \mathbb{Q}(B_1)$. Let $Y \triangleq 1 + \text{Binom}(N-1, \tau)$ denote the number of points falling in $[X_1, X_1 + 2\pi\tau]$ under $\mathbb{Q}$, with mean $\mu = 1 + (N-1)\tau$. A standard Chernoff bound yields $P\{Y \geq (1+\delta)\mu\} \leq \exp(-\mu\delta^2/(2+\delta))$, and therefore, for $\gamma \geq 1 + (N-1)\tau$,

$$\mathbb{Q}(B_1) = P\{1 + \text{Binom}(N-1, \tau) \geq \gamma\} \leq \exp\left\{-\frac{(\gamma - 1 - (N-1)\tau)^2}{1 + (N-1)\tau + \gamma}\right\}.$$

Hence, for $\gamma \geq 1 + (N-1)\tau$,

$$p_{\text{fa}} \leq N \cdot \exp\left\{-\frac{(\gamma - 1 - (N-1)\tau)^2}{1 + (N-1)\tau + \gamma}\right\}. \quad (5.8)$$

We next upper bound the probability of a miss. Suppose H_1 is true, so that the points $(X_i : i \in S^*)$ all lie in the interval $[\Theta^*, \Theta^* + 2\pi\tau]$. Detection will occur if at least $\gamma - K$ of the points $(X_i : i \in [N] \setminus S^*)$ also lie in this interval. The number of such points has the $\text{Binom}(N-K, \tau)$ distribution, so assuming $(N-K)\tau \geq \gamma - K$,

$$p_{\text{miss}} \leq P\{\text{Binom}(N-K, \tau) < \gamma - K\} \leq \exp\left\{-\frac{((N-K)\tau - \gamma + K)^2}{2(N-K)\tau}\right\}, \quad (5.9)$$

where we used the Chernoff bound $P\{\text{Binom}(\bar{n}, p) \leq (1-\delta)\bar{n}p\} \leq e^{-\bar{n}p\delta^2/2}$ with $\bar{n} = N-K$, $p = \tau$, and $\delta = \frac{(N-K)\tau - \gamma + K}{(N-K)\tau}$. Now suppose there exists a sequence $c_N \rightarrow \infty$ such that

$$\frac{(K-1)^2(1-\tau)^2}{(N-K)\tau} \rightarrow \infty, \quad (5.10)$$

$$\frac{(K(1-\tau) - c_N\sqrt{(N-K)\tau})^2}{2N\tau + K(1-\tau) - c_N\sqrt{(N-K)\tau}} - \log N \rightarrow \infty. \quad (5.11)$$

Select the threshold

$$\gamma = \gamma_N \triangleq (N-K)\tau + K - c_N\sqrt{(N-K)\tau},$$

and assume (as in the theorem statement) that $\gamma_N \geq 1 + (N-1)\tau$ for all sufficiently large N , so that (5.8) applies. With this choice of γ , we have $(N-K)\tau - \gamma + K = c_N\sqrt{(N-K)\tau}$, and (5.9) implies

$$p_{\text{miss}} \leq \exp(-c_N^2/2) \rightarrow 0.$$

Moreover, using $\gamma = \gamma_N$ in (5.8) and noting that $\gamma_N - 1 - (N-1)\tau = K(1-\tau) - c_N\sqrt{(N-K)\tau} - (1-\tau)$ and $1 + (N-1)\tau + \gamma_N = 2N\tau + K(1-\tau) - c_N\sqrt{(N-K)\tau} + O(1)$, condition (5.11) yields $p_{\text{fa}} \rightarrow 0$. This concludes Case (A2), and hence the proof of Theorem 5.1(a).

5.2.2 Converse for hard-cluster flat model

We shall employ a second moment method for identifying conditions such that $\text{TV}(\mathbb{P}, \mathbb{Q}) \rightarrow 0$, where $\text{TV}(\mathbb{P}, \mathbb{Q})$ is the total variation distance between $\mathbb{P}$ and $\mathbb{Q}$, thereby establishing the impossibility of weak detection. The method takes advantage of the fact that $\mathbb{P}$ is a mixture of distributions which each have a simple likelihood ratio with respect to $\mathbb{Q}$. The method is described in [WX21] and for the reader's convenience we explain it in detail here. The key is to bound $\text{Var}_{\mathbb{Q}}(L)$, where $L(X) = \frac{d\mathbb{P}}{d\mathbb{Q}}(X)$, which is also known as the χ^2 divergence between $\mathbb{P}$ and $\mathbb{Q}$ [Tsy09]. Since $\mathbb{P} \ll \mathbb{Q}$ it follows that

$$\text{TV}(\mathbb{P}, \mathbb{Q}) = \frac{1}{2} \mathbb{E}_{\mathbb{Q}}|L - 1| \leq \frac{1}{2} \sqrt{\text{Var}_{\mathbb{Q}}(L)}.$$

Hence, if $\text{Var}_{\mathbb{Q}}(L) \rightarrow 0$ as $N \rightarrow \infty$, then $\text{TV}(\mathbb{P}, \mathbb{Q}) \rightarrow 0$ and weak detection is impossible. Since $\mathbb{E}_{\mathbb{Q}}[L] = 1$ it follows that $\text{Var}_{\mathbb{Q}}(L) = \mathbb{E}_{\mathbb{Q}}[L^2] - 1$.

Averaging over the possible values of (S^*, Θ^*) , we may write

$$L(X) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{\binom{N}{K}} \sum_{S \subset [N]: |S|=K} L_{S,\theta}(X) d\theta, \quad (5.12)$$

where

$$L_{S,\theta}(X) \triangleq \tau^{-K} \mathbf{1}\{X_i \in [\theta, \theta + 2\pi\tau] \text{ for all } i \in S\}$$

is the conditional likelihood ratio given $S^* = S$ and $\Theta^* = \theta$. If we view S and Θ as random variables independent of X such that S is uniformly distributed over subsets of $[N]$ with $|S| = K$, Θ is uniformly distributed over $[0, 2\pi]$, and S and Θ are mutually independent, then (5.12) can compactly be rewritten as $L(X) = \mathbb{E}_{S,\Theta}[L_{S,\Theta}(X)]$. Moreover, if (S', Θ') represents an independent copy of (S, Θ) , then

$$L^2(X) = \mathbb{E}_{S,\Theta}[L_{S,\Theta}(X)] \mathbb{E}_{S',\Theta'}[L_{S',\Theta'}(X)] = \mathbb{E}_{S,\Theta,S',\Theta'}[L_{S,\Theta}(X) L_{S',\Theta'}(X)]$$

Then, taking expectation with respect to X having distribution $\mathbb{Q}$ and switching the order of integration yields:

$$\mathbb{E}_{\mathbb{Q}}[L^2] = \mathbb{E}_{S,\Theta,S',\Theta'} [\mathbb{E}_{\mathbb{Q}}[L_{S,\Theta}(X) L_{S',\Theta'}(X)]] . \quad (5.13)$$

Equation (5.13) is the key equation used in this and other sections in this chapter for calculating $\mathbb{E}_{\mathbb{Q}}[L^2]$ and $\text{Var}_{\mathbb{Q}}(L) = \mathbb{E}_{\mathbb{Q}}[L^2] - 1$.

Fix S, S', θ, θ' , and let $j = |S \cap S'|$. Let $2\pi\delta$ denote the length of the intersection of the two arcs $[\theta, \theta + 2\pi\tau]$ and $[\theta', \theta' + 2\pi\tau]$ (modulo 2π). Under $\mathbb{Q}$, the coordinates X_i are i.i.d. uniform on $[0, 2\pi)$,

and, therefore

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}}[L_{S,\theta}(X) \cdot L_{S',\theta'}(X)] &= \tau^{-2K} \cdot \mathbb{Q} \left\{ \begin{array}{ll} X_i \in [\theta, \theta + 2\pi\tau] \cap [\theta', \theta' + 2\pi\tau] & \text{for } i \in S \cap S', \\ X_i \in [\theta, \theta + 2\pi\tau] & \text{for } i \in S \setminus S', \\ X_i \in [\theta', \theta' + 2\pi\tau] & \text{for } i \in S' \setminus S \end{array} \right\} \\ &= \tau^{-2K} \delta^j \tau^{2(K-j)} = \delta^j \tau^{-2j}.\end{aligned}$$

We next average over θ, θ' . Let $u \in [0, 1/2]$ denote the circular distance between θ and θ' , normalized by 2π , i.e.

$$u \triangleq \frac{1}{2\pi} \min \{|\theta - \theta'|, 2\pi - |\theta - \theta'|\}, \quad u \sim \text{uniform} \left(\left[0, \frac{1}{2}\right] \right).$$

Therefore, the normalized overlap δ is a function $\delta = \delta_{\tau}(u)$ given by

$$\delta_{\tau}(u) = \begin{cases} (\tau - u)_{+}, & 0 < \tau \leq 1/2, \\ (2\tau - 1) + (1 - \tau - u)_{+}, & 1/2 < \tau < 1, \end{cases} \quad (5.14)$$

and thus, for each integer $j \geq 1$,

$$\mathbb{E}_{\theta, \theta'}[\delta^j] = 2 \int_0^{1/2} \delta_{\tau}(u)^j \, du = \begin{cases} \frac{2}{j+1} \tau^{j+1}, & 0 < \tau \leq \frac{1}{2} \\ \frac{2}{j+1} \left(\tau^{j+1} - (2\tau - 1)^{j+1} \right) + 2\left(\tau - \frac{1}{2}\right)(2\tau - 1)^j, & \frac{1}{2} < \tau < 1 \end{cases} \quad (5.15)$$

Now average over S, S' as well. Writing $J \triangleq |S \cap S'|$, we obtain

$$\mathbb{E}_{\mathbb{Q}}[L^2] = \mathbb{P}\{J = 0\} + \sum_{j=1}^K \mathbb{P}\{J = j\} \cdot \tau^{-2j} \cdot \mathbb{E}_{\theta, \theta'}[\delta^j], \quad (5.16)$$

and J has the hypergeometric distribution with parameters (N, K, K) . Using $\text{Var}_{\mathbb{Q}}(L) = \mathbb{E}_{\mathbb{Q}}[L^2] - 1$ yields

$$\text{Var}_{\mathbb{Q}}(L) = \sum_{j=1}^K \mathbb{P}\{J = j\} \left(\tau^{-2j} \cdot \mathbb{E}_{\theta, \theta'}[\delta^j] - 1 \right) \leq \sum_{j=1}^K \mathbb{P}\{J = j\} \cdot \tau^{-2j} \cdot \mathbb{E}_{\theta, \theta'}[\delta^j].$$

We specialize to the case $0 < \tau \leq 1/2$ for the remainder of the proof. Using (5.15) in (5.16),

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}}[L^2] &= \mathbb{P}\{J = 0\} + \sum_{j=1}^K \mathbb{P}\{J = j\} \left(\tau^{-2j} \cdot \frac{2}{j+1} \tau^{j+1} \right) \\ &= \mathbb{P}\{J = 0\} + \sum_{j=1}^K \mathbb{P}\{J = j\} \frac{2}{(j+1)\tau^{j-1}},\end{aligned}$$

so that

$$\text{Var}_{\mathbb{Q}}(L) \leq \sum_{j=1}^K \mathbb{P}\{J = j\} \frac{2}{(j+1)\tau^{j-1}}. \quad (5.17)$$

We now compare the overlap distribution to a binomial. The hypergeometric(N, K, K) distribution is second-order stochastically less spread out than $\text{Binom}(K, \frac{K}{N})$; see Hoeffding [Hoe63]. Since $y \mapsto z^y$ is convex for each fixed $z \geq 0$, this implies that for all $z \geq 0$,

$$\sum_{j=0}^K \mathbb{P}\{J = j\} \cdot z^j \leq \left(1 - \frac{K}{N} + \frac{Kz}{N}\right)^K.$$

Integrating over $z \in [0, r]$ for $r > 0$ yields

$$\begin{aligned} \sum_{j=0}^K \mathbb{P}\{J = j\} \frac{r^{j+1}}{j+1} &\leq \int_0^r \left(1 - \frac{K}{N} + \frac{Kz}{N}\right)^K dz = \frac{N}{K(K+1)} \left(1 - \frac{K}{N} + \frac{Kz}{N}\right)^{K+1} \Big|_{z=0}^r \\ &\leq \frac{N}{K^2} \left(1 + \frac{Kr}{N}\right)^{K+1}. \end{aligned}$$

Dropping the $j = 0$ term on the left, setting $r = 1/\tau$, multiplying both sides by $2\tau^2$, and using (5.17) gives

$$\text{Var}_{\mathbb{Q}}(L) \leq \sum_{j=1}^K \mathbb{P}\{J = j\} \frac{2}{(j+1)\tau^{j-1}} \leq \frac{2N\tau^2}{K^2} \left(1 + \frac{K}{N\tau}\right)^{K+1} \quad (5.18)$$

Consequently, $\text{TV}(\mathbb{P}, \mathbb{Q}) \leq \frac{1}{2} \sqrt{\text{Var}_{\mathbb{Q}}(L)} \rightarrow 0$ whenever the right-hand side of (5.18) tends to zero, and weak detection is impossible in that regime. This proves Theorem 5.1(b).

5.3 The von Mises flat model

Theorem 5.2. *Consider the von Mises flat model with parameters (N, K, κ) . For $\tau \in (0, 1)$, define*

$$p_{\kappa}(\tau) \triangleq \mathbb{P}\{\text{vonMises}(0, \kappa) \in [-\pi\tau, \pi\tau]\}, \quad g \triangleq K(p_{\kappa}(\tau) - \tau).$$

(a) *(Achievability) Suppose there exist sequences $c_N \rightarrow \infty$ and γ_N with*

$$\gamma_N \triangleq N\tau + g - c_N \sqrt{N\tau + g}, \quad \gamma_N \geq 1 + (N-1)\tau \text{ for all sufficiently large } N,$$

such that

$$\frac{g^2}{N\tau + g} \rightarrow \infty, \quad (5.19)$$

$$\frac{(g - c_N \sqrt{N\tau + g})^2}{2N\tau + g - c_N \sqrt{N\tau + g}} - \log N \rightarrow \infty. \quad (5.20)$$

Then the interval test with window length $2\pi\tau$ and threshold γ_N achieves strong detection.

(b) (Converse) Let $R(\kappa) \triangleq I_0(2\kappa)/I_0^2(\kappa)$, where I_0 is the modified Bessel function of the first kind. If

$$\frac{K^2}{N} (R(\kappa) - 1) - \log R(\kappa) \rightarrow -\infty, \quad (5.21)$$

then weak detection is impossible.

Corollary 5.3 (Achievability regimes). *Under the von Mises flat model, strong detection is possible in each of the following regimes with the interval test.*

- $K = N^\alpha$ with $0 < \alpha \leq 1/2$, and $\frac{1}{\sqrt{\kappa}} = \frac{K^2}{c_1 N \log N}$ for a constant $c_1 > c_0$, where

$$c_0 \triangleq \frac{2}{\pi} \min_{c_2 > 0} \frac{c_2}{(1 - 2Q(c_2))^2} \approx 0.5057, \quad c_2^* \approx 0.7518 \text{ (a minimizer),}$$

and Q is the complementary CDF of the standard Gaussian. Here, the test interval is chosen $\tau = c_2^*/(\pi\sqrt{\kappa})$, and the threshold γ_N is chosen so that $c_N \rightarrow \infty$ and $c_N = o(\sqrt{\log N})$. [For large κ , $\text{vonMises}(0, \kappa) \approx \mathcal{N}(0, 1/\kappa)$ so $p_\kappa(\tau) \rightarrow 1 - 2Q(c_2^*)$ and $g \sim K(1 - 2Q(c_2^*))$. With $\tau = c_2^*/(\pi\sqrt{\kappa})$ and $1/\sqrt{\kappa} = K^2/(c_1 N \log N)$, it follows that

$$\frac{g^2}{2N\tau \log N} \sim \frac{c_1(1 - 2Q(c_2^*))^2}{(2/\pi)c_2^*} > 1,$$

and, hence, (5.19)–(5.20) are satisfied.]

- $K = N^\alpha$ with $1/2 < \alpha < 1$, and $\kappa = \Omega(1)$. Here, the test interval is chosen $\tau \in [\epsilon, 1 - \epsilon]$ for some constant $\epsilon \in (0, 1/2)$, and the threshold γ_N is chosen so that $c_N \rightarrow \infty$ and $c_N = o(K/\sqrt{N})$. [Then $p_\kappa(\tau) - \tau$ is bounded away from 0, so $g = \Theta(K)$ and (5.19)–(5.20) hold.]

Corollary 5.4 (Impossibility regime). *Under the von Mises flat model, weak detection is impossible when $K = N^\alpha$ for some constant $0 < \alpha < 1/2$, and $\frac{1}{\sqrt{\kappa}} = \frac{K^2}{c N \log N}$ for some constant $c < \frac{1-2\alpha}{2\sqrt{\pi}}$. [In this scaling we have $\kappa \rightarrow \infty$, hence $R(\kappa) = I_0(2\kappa)/I_0^2(\kappa) \sim \sqrt{\pi\kappa}$ and $\log R(\kappa) \sim \frac{1}{2} \log \kappa$. Therefore the converse condition (5.21) is implied by*

$$\frac{K^2}{N} \sqrt{\pi\kappa} \leq \frac{1}{2} \log \kappa - \omega(1).$$

With $1/\sqrt{\kappa} = K^2/(cN \log N)$, the left-hand side equals $\sqrt{\pi} c^{-1} \log N$, while the right-hand side satisfies

$$\frac{1}{2} \log \kappa = \log(cN \log N / K^2) = (1 - 2\alpha) \log N + O(\log \log N).$$

Thus, if $c < (1 - 2\alpha)/(2\sqrt{\pi})$, the inequality holds with an $\omega(1)$ slack, and (5.21) follows.]

5.3.1 Achievability for von Mises flat model – interval test

Fix $\tau \in (0, 1)$ and write $p_\kappa(\tau)$ and g as in the theorem statement. We apply the interval test with window length $2\pi\tau$ and threshold $\gamma_N \triangleq N\tau + g - c_N\sqrt{N\tau + g}$.

False alarm. Under H_0 the observations are i.i.d. uniform on $[0, 2\pi)$, exactly as in the hard-cluster flat model. Hence the false alarm probability of the interval test is identical to that case. In particular, as in (5.8), whenever $\gamma_N \geq 1 + (N - 1)\tau$,

$$p_{\text{fa}} \leq N \cdot \exp \left\{ -\frac{(\gamma_N - 1 - (N - 1)\tau)^2}{1 + (N - 1)\tau + \gamma_N} \right\}. \quad (5.22)$$

Noting that

$$\gamma_N - 1 - (N - 1)\tau = g - c_N\sqrt{N\tau + g} - (1 - \tau), \quad 1 + (N - 1)\tau + \gamma_N = 2N\tau + g - c_N\sqrt{N\tau + g} + O(1),$$

we see that condition (5.20) implies that the exponent in (5.22) dominates $\log N$, and hence $p_{\text{fa}} \rightarrow 0$.

Missed detection. Under H_1 , consider the test interval of length $2\pi\tau$ centered at the planted phase Θ^* , namely $[\Theta^* - \pi\tau, \Theta^* + \pi\tau] \pmod{2\pi}$. Let Y be the number of observations falling in this window. Under H_1 , Y decomposes as

$$Y = Y_0 + Y_1, \quad Y_0 \sim \text{Binom}(K, p_\kappa(\tau)), \quad Y_1 \sim \text{Binom}(N - K, \tau),$$

with Y_0 and Y_1 independent. In particular,

$$\mu_1 \triangleq \mathbb{E}_{\mathbb{P}}[Y] = (N - K)\tau + Kp_\kappa(\tau) = N\tau + g.$$

Since the interval test rejects H_0 whenever $Y \geq \gamma_N$, we have

$$p_{\text{miss}} \leq \mathbb{P}\{Y < \gamma_N\} = \mathbb{P}\{Y_0 + Y_1 < \gamma_N\}.$$

Because $Y_0 + Y_1$ is a sum of independent Bernoulli random variables, we may apply the standard Chernoff lower-tail bound (Poisson-binomial bound). Writing $\gamma_N = (1 - \eta)\mu_1$ yields

$$p_{\text{miss}} \leq \exp \left(-\frac{\eta^2}{2} \mu_1 \right) = \exp \left(-\frac{(\mu_1 - \gamma_N)^2}{2\mu_1} \right).$$

Substituting $\mu_1 = N\tau + g$ and $\gamma_N = N\tau + g - c_N\sqrt{N\tau + g}$ gives

$$p_{\text{miss}} \leq \exp\left(-\frac{c_N^2}{2}\right) \rightarrow 0, \quad (5.23)$$

since $c_N \rightarrow \infty$. Combining with $p_{\text{fa}} \rightarrow 0$ proves strong detection. This proves part (a) of Theorem 5.2.

5.3.2 Converse for von Mises flat model

We bound $\text{Var}_{\mathbb{Q}}(L)$, where $L = \frac{d\mathbb{P}}{d\mathbb{Q}}$. Since $\mathbb{E}_{\mathbb{Q}}[L] = 1$ and $\text{TV}(\mathbb{P}, \mathbb{Q}) \leq \frac{1}{2}\sqrt{\text{Var}_{\mathbb{Q}}(L)}$, it suffices to show that $\mathbb{E}_{\mathbb{Q}}[L^2] \rightarrow 1$.

For fixed (S, θ) , the conditional likelihood ratio is

$$L_{S,\theta}(X) = I_0(\kappa)^{-K} \exp\left(\kappa \sum_{i \in S} \cos(X_i - \theta)\right). \quad (5.24)$$

Fix S, θ, S', θ' with $|S| = |S'| = K$ and let $j = |S \cap S'|$. Then

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[L_{S,\theta}(X)L_{S',\theta'}(X)] &= I_0(\kappa)^{-2K} \cdot \mathbb{E}_{\mathbb{Q}}\left[\exp\left(\kappa \sum_{i \in S} \cos(X_i - \theta) + \kappa \sum_{i \in S'} \cos(X_i - \theta')\right)\right] \\ &= I_0(\kappa)^{-2K} \cdot \mathbb{E}_{\mathbb{Q}}\left[\exp\left(\kappa \sum_{i \in S \cap S'} (\cos(X_i - \theta) + \cos(X_i - \theta'))\right)\right] \quad (5.25) \\ &\quad \times \mathbb{E}_{\mathbb{Q}}\left[\exp\left(\kappa \sum_{i \in S \setminus S'} \cos(X_i - \theta) + \kappa \sum_{i \in S' \setminus S} \cos(X_i - \theta')\right)\right] \\ &= I_0(\kappa)^{-2j} \cdot \mathbb{E}_{\mathbb{Q}}\left[\exp\left(\kappa (\cos(X_1 - \theta) + \cos(X_1 - \theta'))\right)\right]^j, \end{aligned}$$

where in (5.25) we used independence under $\mathbb{Q}$, and in the last step we used that the X_i are i.i.d. uniform on $[0, 2\pi)$, so each of the j intersection terms has the same expectation.

Using the identity $\cos a + \cos b = 2 \cos\left(\frac{a-b}{2}\right) \cos\left(\frac{a+b}{2}\right)$, we get

$$\cos(X_1 - \theta) + \cos(X_1 - \theta') = 2 \cos\left(\frac{\theta - \theta'}{2}\right) \cdot \cos\left(X_1 - \frac{\theta + \theta'}{2}\right),$$

and, therefore,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}\left[\exp\left(\kappa (\cos(X_1 - \theta) + \cos(X_1 - \theta'))\right)\right] &= \mathbb{E}_{\mathbb{Q}}\left[\exp\left(2\kappa \cos\left(\frac{\theta - \theta'}{2}\right) \cdot \cos\left(X_1 - \frac{\theta + \theta'}{2}\right)\right)\right] \\ &= I_0\left(2\kappa \cos\left(\frac{\theta - \theta'}{2}\right)\right), \end{aligned}$$

since $X_1 - (\theta + \theta')/2$ is uniform on $[0, 2\pi)$ under $\mathbb{Q}$. Hence

$$\mathbb{E}_{\mathbb{Q}}[L_{S,\theta}(X)L_{S',\theta'}(X)] = \left[\frac{I_0\left(2\kappa \cos\left(\frac{\theta-\theta'}{2}\right)\right)}{I_0^2(\kappa)} \right]^j. \quad (5.26)$$

To compute $\mathbb{E}_{\mathbb{Q}}[L^2]$, we view S, θ, S', θ' as independent and uniform over their domains. Letting $\vartheta \triangleq (\theta - \theta')/2$, we may write

$$\mathbb{E}_{\mathbb{Q}}[L^2] = \frac{1}{2\pi} \int_0^{2\pi} \sum_{j=0}^K \mathbb{P}\{|S \cap S'| = j\} \left[\frac{I_0(2\kappa \cos \vartheta)}{I_0^2(\kappa)} \right]^j d\vartheta. \quad (5.27)$$

We bound the sum over j using the fact that $|S \cap S'|$ is hypergeometric(N, K, K) and is second-order dominated by $\text{Binom}\left(K, \frac{K}{N}\right)$ (Hoeffding [Hoe63]). Since $z \mapsto z^j$ is convex for $z \geq 0$, this implies that for all $z \geq 0$,

$$\sum_{j=0}^K \mathbb{P}\{|S \cap S'| = j\} \cdot z^j \leq \sum_{j=0}^K \mathbb{P}\left\{\text{Binom}\left(K, \frac{K}{N}\right) = j\right\} \cdot z^j = \left(1 - \frac{K}{N} + \frac{Kz}{N}\right)^K.$$

Applying this with $z = \rho_{\kappa}(\vartheta)$, where

$$\rho_{\kappa}(\vartheta) \triangleq \frac{I_0(2\kappa \cos \vartheta)}{I_0^2(\kappa)},$$

and then using $1 + x \leq e^x$ yields

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[L^2] &\leq \frac{1}{2\pi} \int_0^{2\pi} \left(1 + \frac{K}{N}(\rho_{\kappa}(\vartheta) - 1)\right)^K d\vartheta \\ &\leq \frac{1}{2\pi} \int_0^{2\pi} \exp\left(\frac{K^2}{N}(\rho_{\kappa}(\vartheta) - 1)\right) d\vartheta = e^{-\frac{K^2}{N}} \cdot \frac{1}{2\pi} \int_0^{2\pi} \exp\left(\frac{K^2}{N}\rho_{\kappa}(\vartheta)\right) d\vartheta. \end{aligned} \quad (5.28)$$

Let $R(\kappa) \triangleq \max_{\vartheta} \rho_{\kappa}(\vartheta) = \rho_{\kappa}(0) = \frac{I_0(2\kappa)}{I_0^2(\kappa)}$. For $x \in [0, R(\kappa)]$ and $a > 0$, convexity of $x \mapsto e^{ax}$ on $[0, R(\kappa)]$ implies

$$e^{ax} \leq \left(1 - \frac{x}{R(\kappa)}\right)e^0 + \frac{x}{R(\kappa)}e^{aR(\kappa)} = 1 + \frac{e^{aR(\kappa)} - 1}{R(\kappa)}x.$$

Applying this with $a = K^2/N$ and $x = \rho_{\kappa}(\vartheta)$ in (5.28) gives

$$\mathbb{E}_{\mathbb{Q}}[L^2] \leq e^{-\frac{K^2}{N}} \cdot \frac{1}{2\pi} \int_0^{2\pi} \left(1 + \frac{e^{\frac{K^2}{N}R(\kappa)} - 1}{R(\kappa)}\rho_{\kappa}(\vartheta)\right) d\vartheta. \quad (5.29)$$

Lemma 5.1. *For every $\kappa > 0$,*

$$\frac{1}{2\pi} \int_0^{2\pi} \rho_{\kappa}(\vartheta) d\vartheta = 1.$$

Proof. Using $2 \cos(a) \cos(b) = \cos(a+b) + \cos(a-b)$, we have

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} I_0(2\kappa \cos \vartheta) \, d\vartheta &= \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \exp\left(2\kappa \cos \vartheta \cos(\varphi - \vartheta)\right) \, d\varphi \, d\vartheta \\ &= \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \exp\left(\kappa \cos \varphi + \kappa \cos(2\vartheta - \varphi)\right) \, d\vartheta \, d\varphi \\ &= I_0^2(\kappa), \end{aligned}$$

where in the last step we used the integral representation of $I_0(\kappa)$ twice. Dividing both sides by $I_0^2(\kappa)$ proves the claim. $\square$

Applying Lemma 5.1 to (5.29) yields

$$\mathbb{E}_{\mathbb{Q}}[L^2] \leq e^{-\frac{K^2}{N}} \left(1 + \frac{e^{\frac{K^2}{N} R(\kappa)} - 1}{R(\kappa)}\right) = \frac{e^{\frac{K^2}{N} (R(\kappa)-1)}}{R(\kappa)} + \left(1 - \frac{e^{-\frac{K^2}{N}}}{R(\kappa)}\right).$$

Therefore, a sufficient condition for $\mathbb{E}_{\mathbb{Q}}[L^2] \rightarrow 1$ (and, hence, $\text{Var}_{\mathbb{Q}}(L) \rightarrow 0$ and $\text{TV}(\mathbb{P}, \mathbb{Q}) \rightarrow 0$) is

$$\frac{K^2}{N} (R(\kappa) - 1) - \log R(\kappa) \rightarrow -\infty,$$

which is exactly (5.21). This proves part (b) of Theorem 5.2.

5.4 The hard-cluster community model

Theorem 5.3. *Consider the hard-cluster community model with parameters (n, k, τ) .*

(a) *(Achievability) The interval test with window length $2\pi\tau$ and threshold k yields $p_{\text{miss}} = 0$ and*

$$p_{\text{fa}} \leq \exp\left(\log \frac{\binom{k}{2}}{\tau} + k \left\{\log \frac{n}{k} + 1 + \frac{k-1}{2} \log \tau\right\}\right). \quad (5.30)$$

Thus, strong detection is achieved if the right hand side of (5.30) converges to zero.

(b) *(Converse) If*

$$\frac{k^2}{n} \left(\left(\frac{1}{\tau} \right)^{\frac{k-1}{2}} - 1 \right) \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad (5.31)$$

then weak detection is impossible.

Corollary 5.5 (Achievability regimes). *Under the hard-cluster community model, each of the following conditions implies strong detection by the interval test:*

- $k = c \log n$ for a constant $c > 0$ and $\tau \leq e^{-2/c}$.
- $3/\epsilon < k \leq n$ and $\tau \leq \left(\frac{k}{ne}\right)^{\frac{2+\epsilon}{k-1}}$ for a fixed $\epsilon \in (0, 1)$.
- $k = \omega(\log n)$ and $\tau \leq 1 - \frac{2(1+\epsilon)}{k-1} \log \frac{ne}{k}$
- $3 \leq k \leq n$ and $\tau = \left(\frac{k}{ne}\right)^{\frac{(1+\epsilon)k}{\binom{k}{2}-1}}$ for a fixed $\epsilon > 0$.
- $3/\epsilon < k = o(n)$ and $\tau = \tau_\epsilon \triangleq \left(\frac{k}{n}\right)^{\frac{2+\epsilon}{k-1}}$ for some $\epsilon \in (0, 1)$.

Proof of Corollary 5.5. For $k = c \log n$ and $\tau = e^{-2/c}$ the bound (5.30) becomes

$$\begin{aligned} p_{\text{fa}} &\leq \exp \left(\log \frac{\binom{k}{2}}{\tau} + c \log n \left\{ \log \frac{n}{k} + 1 + \frac{c \log n - 1}{2} \left(-\frac{2}{c} \right) \right\} \right) \\ &= \frac{1}{\tau} \exp \left(\log \binom{k}{2} + (\log n)(-c \log k + c + 1) \right) \rightarrow 0. \end{aligned}$$

For $\tau = \left(\frac{k}{ne}\right)^{\frac{2+\epsilon}{k-1}}$ the bound (5.30) becomes:

$$\begin{aligned} p_{\text{fa}} &\leq \exp \left(\frac{2+\epsilon}{k-1} \log \frac{n}{k} + \log \binom{k}{2} + k \left\{ -\frac{\epsilon}{2} \log \frac{n}{k} - \frac{\epsilon}{2} \right\} \right) \\ &= \exp \left(\left[-\frac{k\epsilon}{2} + \frac{2+\epsilon}{k-1} \right] \log \frac{n}{k} + \log \binom{k}{2} - \frac{\epsilon k}{2} \right) \\ &\rightarrow 0 \text{ if } 3/\epsilon < k \leq n. \end{aligned}$$

For $\tau \leq 1 - \frac{2(1+\epsilon)}{k-1} \log \frac{ne}{k}$ and $k = \omega(\log n)$, since

$$\frac{2(1+\epsilon)}{k-1} \log \frac{ne}{k} \rightarrow 0, \quad \text{and} \quad \frac{2(1+\epsilon)}{k-1} \log \frac{ne}{k} = \left(\frac{1+\epsilon}{1+\epsilon/2} \right) \frac{2+\epsilon}{k-1} \log \frac{ne}{k},$$

it follows that $1 - \frac{2(1+\epsilon)}{k-1} \log \frac{ne}{k} \leq \left(\frac{k}{ne}\right)^{\frac{2+\epsilon}{k-1}}$ so $p_{\text{fa}} \rightarrow 0$ by the previous case.

For $\tau = \left(\frac{k}{ne}\right)^{\frac{(1+\epsilon)k}{\binom{k}{2}-1}}$ and $3 \leq k \leq n$ the bound (5.30) becomes:

$$p_{\text{fa}} \leq \exp \left(\log \binom{k}{2} - \epsilon k \log \frac{ne}{k} \right) \rightarrow 0.$$

For $\tau = \tau_\epsilon \triangleq \left(\frac{k}{n}\right)^{\frac{2+\epsilon}{k-1}}$ the bound (5.30) becomes:

$$\begin{aligned} p_{\text{fa}} &\leq \exp \left(\frac{2+\epsilon}{k-1} \log \frac{n}{k} + \log \binom{k}{2} + k \left\{ -\frac{\epsilon}{2} \log \frac{n}{k} + 1 \right\} \right) \\ &= \exp \left(\left[-\frac{k\epsilon}{2} + \frac{2+\epsilon}{k-1} \right] \log \frac{n}{k} + \log \binom{k}{2} + k \right) \\ &\rightarrow 0 \text{ if } k = o(n) \text{ and } k > 3/\epsilon. \end{aligned}$$

□

Corollary 5.6 (Impossibility regimes). *Under the hard-cluster community model, each of the following conditions implies the impossibility of weak detection:*

- $k = c \log n$ for a constant $c > 0$ and a constant $\tau > e^{-2/c}$;
- $k = o(\sqrt{n})$ and $\tau \geq \exp\left(-\frac{(2-\epsilon)\log(n/k^2)}{k-1}\right)$ for a constant $\epsilon > 0$;
- $k = \omega(\log n)$ and $\tau \geq 1 - \frac{2(1-\epsilon)}{k-1} \log\left(1 + \frac{\epsilon_n n}{k^2}\right)$ for positive constants ϵ and ϵ_n with $\epsilon_n \rightarrow 0$;
[These conditions imply $\tau \rightarrow 1$ so $\tau \geq \exp\left\{\frac{2}{k-1} \log\left(1 + \frac{\epsilon_n n}{k^2}\right)\right\}$ for sufficiently large n , where $\exp\left\{\frac{2}{k-1} \log\left(1 + \frac{\epsilon_n n}{k^2}\right)\right\}$ is the value of τ obtained by setting the left hand side of (5.31) to ϵ_n .]

5.4.1 Achievability for hard-cluster community model – interval test

Under H_1 , for the planted pair (C^*, Θ^*) , we have $X_{ij} \in [\Theta^*, \Theta^* + 2\pi\tau]$ for all $i, j \in C^*, i \neq j$. Hence the test succeeds with $\theta = \Theta^*$ and $C = C^*$, so $p_{\text{miss}} = 0$.

We now compute the false alarm probability for the interval test with $\ell = 2\pi\tau$. Under H_0 , the edge labels $(X_{ij})_{1 \leq i < j \leq n}$ are i.i.d. uniform on $[0, 2\pi)$. For a fixed subset $C \subset [n]$ with $|C| = k$, let

$$E_C \triangleq \left\{ \exists \theta \in [0, 2\pi) : X_{ij} \in [\theta, \theta + 2\pi\tau] \quad \forall i, j \in C, i \neq j \right\}.$$

By a union bound over all communities,

$$p_{\text{fa}} = \mathbb{Q}\left(\bigcup_{C: |C|=k} E_C\right) \leq \binom{n}{k} \mathbb{Q}(E_C), \quad (5.32)$$

where $\mathbb{Q}(E_C)$ does not depend on the particular choice of C .

To bound $\mathbb{Q}(E_C)$ while scanning over θ , note that if E_C occurs, then in particular all $\binom{k}{2}$ angles $\{X_{ij} : i < j, i, j \in C\}$ lie in some arc of length $2\pi\tau$. Let e^* denote an edge in C attaining the minimum angle among these m angles. Then E_C implies that all remaining $\binom{k}{2} - 1$ angles lie in the arc $[X_{e^*}, X_{e^*} + 2\pi\tau]$. Taking a union bound over the choice of the minimizing edge yields $\mathbb{Q}(E_C) \leq \binom{k}{2} \tau \binom{k}{2}^{-1}$. Therefore,

$$\begin{aligned} p_{\text{fa}} &\leq \binom{k}{2} \binom{n}{k} \tau \binom{k}{2}^{-1} \leq \binom{k}{2} \left(\frac{ne}{k}\right)^k \tau \binom{k}{2}^{-1} \\ &= \exp\left(\log \frac{\binom{k}{2}}{\tau} + k \left\{ \log \frac{n}{k} + 1 + \frac{k-1}{2} \log \tau \right\}\right). \end{aligned}$$

This proves Theorem 5.3(a).

5.4.2 Converse for hard-cluster community model

We bound $\text{Var}_{\mathbb{Q}}(L)$ above, where $L(X) = \frac{d\mathbb{P}}{d\mathbb{Q}}(X)$ and $X = (X_{ij})_{1 \leq i < j \leq n}$. Since $\mathbb{P}$ and $\mathbb{Q}$ are absolutely continuous, $\text{TV}(\mathbb{P}, \mathbb{Q}) \leq \frac{1}{2} \sqrt{\text{Var}_{\mathbb{Q}}(L)}$, so if $\text{Var}_{\mathbb{Q}}(L) \rightarrow 0$ (equivalently $\mathbb{E}_{\mathbb{Q}}[L^2] \rightarrow 1$), then $\text{TV}(\mathbb{P}, \mathbb{Q}) \rightarrow 0$ and weak detection is impossible.

Averaging over the possible values (C, θ) of (C^*, Θ^*) we find

$$L(X) = \int_0^{2\pi} \frac{1}{\binom{n}{k}} \sum_{C \subset [n]: |C|=k} L_{C, \theta}(X) \frac{d\theta}{2\pi},$$

where

$$L_{C, \theta}(X) = \begin{cases} \tau^{-\binom{k}{2}} & \text{if } X_{ij} \in [\theta, \theta + 2\pi\tau] \text{ for all } i, j \in C, i < j, \\ 0 & \text{otherwise.} \end{cases}$$

For C, θ, C', θ' fixed, let $S = |C \cap C'|$ and

$$\delta \triangleq \frac{1}{2\pi} \text{length}([\theta, \theta + 2\pi\tau] \cap [\theta', \theta' + 2\pi\tau]).$$

Then

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[L_{C, \theta}(X) L_{C', \theta'}(X)] &= \frac{1}{\tau^{2\binom{k}{2}}} \mathbb{Q} \left\{ \begin{array}{l} X_{ij} \in [\theta, \theta + 2\pi\tau] \cap [\theta', \theta' + 2\pi\tau] \text{ for } i, j \in C \cap C', i < j, \\ X_{ij} \in [\theta, \theta + 2\pi\tau] \text{ for } i, j \in C, \{i, j\} \not\subset C \cap C', i < j, \\ X_{ij} \in [\theta', \theta' + 2\pi\tau] \text{ for } i, j \in C', \{i, j\} \not\subset C \cap C', i < j \end{array} \right\} \\ &= \frac{\delta^{\binom{S}{2}} \tau^{k(k-1) - S(S-1)}}{\tau^{k(k-1)}} = \delta^{\binom{S}{2}} \tau^{-2\binom{S}{2}}. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[L^2] &= \mathbb{E}_{C, C', \Theta, \Theta'} [\mathbb{E}_{\mathbb{Q}}[L_{C, \Theta}(X) L_{C', \Theta'}(X)]] \\ &= \mathbb{E}_S [\mathbb{E}_{\Theta, \Theta'} [\delta^{\binom{S}{2}} \tau^{-2\binom{S}{2}}]] = \mathbb{E}_S \left[\mathbb{E}_{\Theta, \Theta'} \left[\left(\frac{\delta}{\tau^2} \right)^{\binom{S}{2}} \right] \right]. \end{aligned}$$

Since the maximum overlap length is $2\pi\tau$, we have $\delta \leq \tau$, hence $\delta/\tau^2 \leq 1/\tau$ and

$$\mathbb{E}_{\mathbb{Q}}[L^2] \leq \mathbb{E}_S \left[\left(\frac{1}{\tau} \right)^{\binom{S}{2}} \right] \stackrel{(a)}{=} \mathbb{E}_S \left[\left(\frac{1}{\tau} \right)^{\frac{k-1}{2} \times S} \right],$$

where (a) used $\binom{S}{2} = \frac{S(S-1)}{2} \leq \frac{k-1}{2} S$ for $0 \leq S \leq k$.

The variable S is distributed as hypergeometric(n, k, k), which is dominated in the convex order by $\text{Binom}(k, p)$ with $p = k/n$. Let $z = (1/\tau)^{\frac{k-1}{2}}$. Using that $s \mapsto z^s$ is convex for $z \geq 0$,

$$\mathbb{E}_S[z^S] \leq (1 - p + pz)^k = \left(1 + \frac{k}{n}(z - 1) \right)^k.$$

Finally, since $1 + x \leq e^x$,

$$\mathbb{E}_{\mathbb{Q}}[L^2] \leq \exp\left(\frac{k^2}{n}(z-1)\right) = \exp\left(\frac{k^2}{n}\left(\left(\frac{1}{\tau}\right)^{\frac{k-1}{2}} - 1\right)\right).$$

If (5.31) holds, the exponent tends to 0, hence $\mathbb{E}_{\mathbb{Q}}[L^2] \rightarrow 1$ and $\text{Var}_{\mathbb{Q}}(L) = \mathbb{E}_{\mathbb{Q}}[L^2] - 1 \rightarrow 0$, which implies $\text{TV}(\mathbb{P}, \mathbb{Q}) \rightarrow 0$ and completes the proof of Theorem 5.3(b).

5.5 The von Mises community model

Recall the functions

$$R(\kappa) = \frac{I_0(2\kappa)}{I_0^2(\kappa)} \quad A(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)},$$

where I_0 and I_1 are the modified Bessel functions of the first kind, of order 0 and 1 respectively. See Appendix C.1 for definitions and properties of these functions.

Theorem 5.4. *Consider the von Mises community model with parameters (n, k, κ) .*

(a) *(Achievability) The interval test with window length $2\pi\tau$ and threshold k satisfies*

$$p_{\text{fa}} \leq \exp\left(\log \frac{\binom{k}{2}}{\tau} + k \cdot \left\{\log \frac{n}{k} + 1 + \frac{k-1}{2} \log \tau\right\}\right), \quad (5.33)$$

$$p_{\text{miss}} \leq \binom{k}{2} \cdot \frac{\exp\{\cos(\pi\tau) \cdot \kappa\}}{2\pi^2 I_0(\kappa) \cdot \kappa \cdot |\sin(\pi\tau)|}. \quad (5.34)$$

Thus, strong detection is achieved by the interval test if the right hand sides of (5.33) and (5.34) converge to 0. Moreover, the coherence test for any integer $B \geq 3$, $\epsilon \in (0, 1)$ and suitable threshold β satisfies

$$p_{\text{fa}} \leq B \cdot \exp\left\{k \left(\log \left(\frac{ne}{k}\right) - (1 - \epsilon/4)^2 \cdot (k-1) \cdot A^2(\kappa) \cdot \cos^2(\pi/B)/2\right)\right\}, \quad (5.35)$$

$$p_{\text{miss}} \leq \exp\left(-\frac{\epsilon^2 K A^2(\kappa)}{32}\right). \quad (5.36)$$

Thus, strong detection is achieved by the coherence test if the right hand sides of (5.35) and (5.36) converge to 0.

(b) *(Converse) Weak detection is impossible if*

$$\frac{k^2}{n} \left(\exp\left(\frac{(k-1) \log R(\kappa)}{2}\right) - 1\right) \xrightarrow{n \rightarrow \infty} 0. \quad (5.37)$$

Corollary 5.7 (Achievability regimes). *Under the von Mises community model, strong detection is possible in each of the following regimes.*

- $3/\epsilon < k = o(\log n)$ for some $\epsilon \in (0, 1)$ and $\kappa \geq \left(\frac{4}{\pi^2} + \epsilon\right) (\log k) \left(\frac{n}{k}\right)^{\frac{4(1+\epsilon)}{k-1}}$. [Use the interval test. Setting $\tau \equiv \tau_\epsilon \triangleq (k/n)^{\frac{2+\epsilon}{k-1}}$ and using the interval test with window length $2\pi\tau$ and subset size k , we have from (5.33) that $p_{\text{fa}} \rightarrow 0$ (see proof of Corollary 5.5). Further, using that $\tau \rightarrow 0$, $\kappa \rightarrow \infty$, $\cos(\pi\tau) - 1 = -\frac{(\pi\tau)^2}{2}(1 + o(1))$, $\sin(\pi\tau) = \pi\tau(1 + o(1))$, and $I_0(\kappa) \sim \frac{e^\kappa}{\sqrt{2\pi\kappa}}$, the equation (5.34) yields

$$\log p_{\text{miss}} \leq (\text{const}) + 2 \log k - \frac{1}{2} \log(\kappa\tau^2) - \frac{\pi^2 \kappa \tau^2 (1 + o(1))}{2}, \quad (5.38)$$

Therefore, since $\kappa\tau^2 \geq (\frac{4}{\pi^2} + \epsilon)(\log k) \left(\frac{n}{k}\right)^{\frac{2\epsilon}{k-1}}$, the last (negative) term in (5.38) dominates and $\log p_{\text{miss}} \rightarrow -\infty$.]

- $k = c \log n$ for some $c > 2$ and fixed κ with $A^2(\kappa)c > 2$. [Use the coherence test. Take B sufficiently large and ϵ sufficiently small.]
- $k = c \log n$ and $\kappa \geq \frac{2 \log k}{1 - \cos(\pi e^{-2/c})}$. [Set $\tau = e^{-2/c}$ and use the interval test with window size $2\pi\tau$. Note $p_{\text{fa}} \rightarrow 0$ for this choice of k and τ (see proof of Corollary 5.5). Using $I_0(\kappa) \sim \frac{e^\kappa}{\sqrt{2\pi\kappa}}$ (because $\kappa \rightarrow \infty$), (5.34) yields $\log p_{\text{miss}} \leq (\text{const}) + 2 \log k - \frac{1}{2} \log(\kappa) + (\cos(\pi\tau) - 1) \kappa \rightarrow -\infty$.]
- $k = \omega(\log n)$ and $\kappa \geq (1 + \epsilon) \sqrt{\frac{8 \log n}{k-1}}$ for some ϵ with $0 < \epsilon < 1$. [Use the coherence test. The righthand sides of (5.36) and (5.35) are monotone decreasing in κ so we may assume $\kappa = (1 + \epsilon) \sqrt{\frac{8 \log n}{k-1}}$. The conditions imply $\kappa \rightarrow 0$ so that $A(\kappa) \sim \frac{\kappa}{2}$ (see the appendix). Since $(1 - \epsilon/4)(1 + \epsilon) \geq 1 + \epsilon/2$ we can select B large enough that $(1 - \epsilon/4)(1 + \epsilon) \cos(\pi/B) \geq 1 + \epsilon/4$. Then the exponent in (5.35) is less than or equal to $k \{ \log(\frac{n\epsilon}{k}) - (1 + \epsilon/4)^2 \log n \}$ so that $p_{\text{fa}} \rightarrow 0$. Also, $K\kappa^2 \rightarrow \infty$ so $p_{\text{miss}} \rightarrow 0$.]

Corollary 5.8 (Impossibility regime). *Under the von Mises community model, each of the following conditions implies the impossibility of weak detection. Note that they require κ to be smaller for larger k and in particular $\kappa \rightarrow 0$ for $k = \omega(\log n)$.*

- $k = o(\log n)$ and $\kappa \leq (n/k^2)^{\frac{4-\epsilon}{k-1}}$ for some $\epsilon > 0$. [These conditions imply $\kappa \rightarrow \infty$ so $R(\kappa) \sim \sqrt{\pi\kappa}$. The conditions also imply $\log \frac{k^2}{n} + \frac{(k-1) \log(\sqrt{\pi\kappa}(1+o(1)))}{2} \rightarrow -\infty$.]
- $k = c \log n$ for some $c > 0$ and $c \log R(\kappa) < 2$. [This implies $\log \frac{k^2}{n} + \frac{(k-1) \log(R(\kappa))}{2} \rightarrow -\infty$.]
- $k = \omega(\log n)$ and $k \leq n^c$ for some c with $0 < c < 0.5$ and $\kappa^2 \leq \frac{4(1-2c-\epsilon) \log n}{k-1}$ for some $\epsilon > 0$. [These conditions imply $\kappa \rightarrow 0$ so that $\log R(\kappa) \sim \frac{\kappa^2}{2}$ and they also imply $\log \frac{k^2}{n} + \frac{(k-1)\kappa^2(1+o(1))}{4} \rightarrow -\infty$.]

Remark 5.1. The first condition of Corollary 5.8 matches the first sufficient condition for strong recovery in Corollary 5.7, up to the $\log k$ factor. Moreover, if we strengthen the condition on k a bit more to $k = o\left(\frac{\log n}{\log \log n}\right)$ then $\log k = o\left(\left(\frac{n}{k^2}\right)^\epsilon\right)$ in which case the factor $\log k$ can be absorbed into

the other term and is thus not meaningful. The second condition of Corollary 5.8 lines up well with the second sufficient condition for strong recovery in Corollary 5.7 – there is a factor of 2 difference for κ^2 for the second condition and large values of c . In particular, as $\kappa \rightarrow 0$, $\log R(\kappa) \sim \frac{\kappa^2}{2}$ and $A^2(\kappa) \sim \frac{\kappa^2}{4}$. Similarly, the third condition of Corollary 5.8 lines up well with the fourth sufficient condition for strong recovery in Corollary 5.7; there is again a factor of 2 difference for the value of κ^2 when c is close to 0.

5.5.1 Achievability for von Mises community model – interval test

Under H_0 , the edge variables are i.i.d. uniform on $[0, 2\pi)$, exactly as in the hard-cluster community model. Therefore, the false-alarm bound (5.33) is the same as (5.30), already proven.

The interval detector returns decision H_1 if the labels of all intracommunity edges fall into the detection window centered at the true value of Θ . So p_{miss} is less than or equal to the probability under $\mathbb{P}$ that at least one of the $\binom{k}{2}$ such labels does not fall into the detection interval centered at the true value of Θ . So by the union bound,

$$p_{\text{miss}} \leq \binom{k}{2} (1 - p_\kappa(\tau)),$$

where p_κ is defined in Theorem 5.2.

Using the fact that if Θ is uniformly distributed over $[0, 2\pi]$ then $\cos \Theta$ has pdf $\frac{1}{\pi\sqrt{1-t^2}}$ for $t \in [-1, 1]$, we find

$$\begin{aligned} 1 - p_\kappa(\tau) &= \frac{1}{2\pi I_0(\kappa)} \int_{\{\theta \in [0, 2\pi] : \cos \theta \leq \cos(\pi\tau)\}} e^{\kappa \cos \theta} d\theta \\ &= \frac{1}{2\pi^2 I_0(\kappa)} \int_{-1}^{\cos(\pi\tau)} \frac{e^{\kappa t}}{\sqrt{1-t^2}} dt \\ &\leq \frac{1}{2\pi^2 I_0(\kappa) \sqrt{1 - \cos^2(\pi\tau)}} \int_{-\infty}^{\cos(\pi\tau)} e^{\kappa t} dt \\ &= \frac{\exp\{(\cos(\pi\tau) - 1)\kappa\}}{2\pi^2 I_0(\kappa) e^{-\kappa} \cdot \kappa \cdot |\sin(\pi\tau)|} \end{aligned}$$

which establishes (5.34).

Remark 5.2. Consider the interval test. If instead of using window size $2\pi\tau_\epsilon$ we were to use window size $2\pi\tau_f$ where

$$\tau_f = \left(\frac{k}{ne} \right)^{\frac{(1+\epsilon)k}{\binom{k}{2}-1}}$$

and if

$$\kappa \geq \tau_f^{-2} \left(\frac{4}{\pi^2} + \epsilon \right) (\log k) \left(\frac{n}{k} \right)^{\frac{2\epsilon}{k-1}}$$

then strong recovery would be achieved for $k = o(\log n)$ (i.e. the condition $k \geq 3/\epsilon$ would not be needed).

5.5.2 Achievability for von Mises community model – coherence test

We bound p_{fa} . By a union bound, we have for any $C \subset [n]$ with $|C| = k$,

$$p_{\text{fa}} \leq \binom{n}{k} \cdot \mathbb{Q}\left(\left|\sum_{e \in E(C)} Z_e\right| > \beta\right). \quad (5.39)$$

Let B be an integer with $B \geq 3$ and consider a B -gon inscribed in the unit circle in the complex plane. The distance of each side to the origin is $\cos(\frac{\pi}{B})$. Thus by symmetry and a union bound,

$$\begin{aligned} p_{\text{fa}} &\leq \binom{n}{k} \cdot B \cdot \mathbb{Q}\left(\Re\left(\sum_{e \in E(C)} Z_e\right) > \beta \cos\left(\frac{\pi}{B}\right)\right) \\ &= \binom{n}{k} \cdot B \cdot \mathbb{Q}\left(S_c > \beta \cos\left(\frac{\pi}{B}\right)\right), \end{aligned} \quad (5.40)$$

where $S_c \triangleq \sum_{e \in E(C)} \cos(X_e)$. For any $s > \mathbb{E}_{\mathbb{Q}}[S_c] = 0$, with $K = \binom{k}{2}$,

$$\mathbb{Q}(S_c \geq s) \stackrel{(A)}{\leq} \inf_{\theta > 0} \exp\left\{-\theta s + K \log \mathbb{E}[e^{\theta \cos(X_{12})}]\right\} \stackrel{(B)}{\leq} \inf_{\theta > 0} e^{-\theta s + K \frac{\theta^2}{4}} \stackrel{(C)}{=} e^{-s^2/K}, \quad (5.41)$$

where (A) uses independence and identical distributions of X_e under $\mathbb{Q}$, (B) uses that

$$\mathbb{E}_{\mathbb{Q}}[e^{\theta \cos(X_i)}] = \frac{1}{2\pi} \int_0^{2\pi} e^{\theta \cos t} dt = I_0(\theta) \leq e^{\frac{1}{4}\theta^2},$$

and (C) uses that $\theta^* = 2s/K$ optimizes the expression. Combining (5.40) and (5.41) yields:

$$p_{\text{fa}} \leq \binom{n}{k} \cdot B \cdot \exp\left\{-\frac{\beta^2 \cos^2(\pi/B)}{K}\right\} \leq B \cdot \exp\left\{k \log\left(\frac{ne}{k}\right) - \frac{\beta^2 \cos^2(\pi/B)}{K}\right\}. \quad (5.42)$$

Setting $\beta = (1 - \frac{\epsilon}{4})K A(\kappa)$ for an arbitrary ϵ with $0 < \epsilon < 1$ yields (5.35).

We turn to bounding the miss probability. For any set E of edges,

$$\left|\sum_{e \in E} Z_e\right| = \max_{\theta} \sum_{e \in E} \cos(X_e - \theta) \geq \sum_{e \in E} \cos(X_e - \Theta^*),$$

where Θ^* is the planted angle. Therefore,

$$p_{\text{miss}} = \mathbb{P}\left(\max_{C: |C|=k} \left|\sum_{e \in E(C)} Z_e\right| \leq \beta\right) \leq \mathbb{P}\left(\left|\sum_{e \in E(C^*)} Z_e\right| \leq \beta\right) \leq \mathbb{P}\left(\sum_{e \in E(C^*)} \cos(X_e - \Theta^*) \leq \beta\right), \quad (5.43)$$

where C^* is the planted set. It is easily verified that the random variables $\{\cos(X_e - \Theta^*)\}_{e \in E(C^*)}$ are i.i.d. and supported on $[-1, 1]$, with mean

$$\int_0^{2\pi} \cos(x - \Theta^*) \cdot \frac{\exp(\kappa \cos(x - \Theta^*))}{2\pi I_0(\kappa)} dx \stackrel{(a)}{=} \frac{1}{I_0(\kappa)} \cdot \left[\frac{1}{2\pi} \int_0^{2\pi} \cos(y) e^{\kappa \cos(y)} dy \right] \stackrel{(b)}{=} \frac{I_1(\kappa)}{I_0(\kappa)} =: A(\kappa),$$

where (a) uses the change of variable $y = x - \Theta^*$ and (b) uses the definition of the modified Bessel function $I_1(\cdot)$. Applying Hoeffding's inequality yields for any $u > 0$,

$$\mathbb{P} \left(\sum_{e \in E(C^*)} \cos(X_e - \Theta^*) \leq K A(\kappa) - u \right) \leq \exp \left(- \frac{u^2}{2K} \right). \quad (5.44)$$

In particular, combining (5.43) and (5.44) and setting $u = \epsilon K A(\kappa)/4$ and $\beta = (1 - \frac{\epsilon}{4})K A(\kappa)$ yields (5.36).

5.5.3 Converse for the von Mises community model

Let $K \equiv \binom{k}{2}$. As in the flat model, the conditional likelihood ratio for a fixed candidate (C, θ) is given by

$$L_{C,\theta}(X) = I_0(\kappa)^{-K} \exp \left\{ \kappa \sum_{e \in E(C)} \cos(X_e - \theta) \right\},$$

where $E(C)$ is the set of $\binom{k}{2}$ unordered pairs inside C . The (unconditional) likelihood ratio is the mixture $L(X) = \mathbb{E}_{C,\Theta}[L_{C,\Theta}(X)]$.

Let (C, Θ) and (C', Θ') be independent draws of the planted parameters. Proceed as in the flat model, using independence of edges under $\mathbb{Q}$:

$$\mathbb{E}_{\mathbb{Q}}[L_{C,\Theta}(X) L_{C',\Theta'}(X) \mid C, \Theta, C', \Theta'] = \left[\frac{I_0(2\kappa \cos \frac{\Theta - \Theta'}{2})}{I_0^2(\kappa)} \right]^{\binom{S}{2}},$$

where $S \triangleq |C \cap C'|$ and we use the fact that $|E(C) \cap E(C')| = \binom{S}{2}$ is the number of overlapping intra community edges. Averaging over (C, Θ) and (C', Θ') thus yields

$$\mathbb{E}_{\mathbb{Q}}[L^2] = \mathbb{E}_{C,C'} \left[\frac{1}{2\pi} \int_0^{2\pi} \left(\frac{I_0(2\kappa \cos \frac{\phi}{2})}{I_0^2(\kappa)} \right)^{\binom{S}{2}} d\phi \right], \quad \phi \equiv \Theta - \Theta'. \quad (5.45)$$

We further upper bound by maximizing over ϕ . Since $I_0(2\kappa \cos \frac{\phi}{2}) \leq I_0(2\kappa)$ for all ϕ ,

$$\mathbb{E}_{\mathbb{Q}}[L^2] \leq \mathbb{E}_{C,C'} \left[R(\kappa)^{\binom{S}{2}} \right],$$

where $R(\kappa) \triangleq \frac{I_0(2\kappa)}{I_0^2(\kappa)}$. The random variable S has the hypergeometric(n, k, k) distribution. By convex ordering, the z -transform of this distribution is dominated by the z -transform of the Binom($k, k/n$)

distribution:

$$\mathbb{E}[z^S] \leq (1 - p + pz)^k, \quad \text{for all } z \geq 0.$$

Using $\binom{s}{2} \leq \frac{k-1}{2} s$ for $0 \leq s \leq k$,

$$R(\kappa)^{\binom{S}{2}} = \exp\left((\log R(\kappa)) \cdot \binom{S}{2}\right) \leq \exp\left(\frac{(k-1)\log R(\kappa)}{2} S\right). \quad (5.46)$$

Setting $t \equiv \frac{(k-1)\log R(\kappa)}{2}$ and $z = e^t$, we obtain

$$\mathbb{E}_{\mathbb{Q}}[L^2] \leq \mathbb{E}[e^{tS}] \leq (1 - p + pe^t)^k \leq \exp\left(k p (e^t - 1)\right) = \exp\left(\frac{k^2}{n} (e^t - 1)\right).$$

This proves the bound

$$\mathbb{E}_{\mathbb{Q}}[L^2] \leq \exp\left\{\frac{k^2}{n} \left(\exp\left(\frac{(k-1)\log R(\kappa)}{2}\right) - 1\right)\right\}. \quad (5.47)$$

Therefore, $\mathbb{E}_{\mathbb{Q}}[L^2] \rightarrow 1$ (and so detection is impossible) whenever the exponent in (5.47) goes to 0. This is precisely the necessary condition (5.37). This completes the proof of Theorem 5.4(b).

Part II

Robust Inference and Consensus on Networks

Chapter 6

Robust graph matching with adversarial corruption

6.1 Introduction

The graph matching problem and its motivating applications — in social networks [NS09, NS08], biological networks [SXB08, KHGPM16], natural language processing [HNM05], and computer vision [SS05] — were introduced in Chapter 1 and developed in further detail in Chapter 3. Over a decade of progress has led to a sound understanding of the fundamental limits of graph matching in the case of correlated Erdős-Rényi graphs; an overview is provided in Section 6.1.1. The bulk of this theory rests on the assumption that the two networks are drawn from a common generative model and differ only through independent edge subsampling. Real networks routinely violate this assumption, and the existing analyses leave open the question of how much corruption a given matching algorithm can tolerate before its guarantees break down.

This chapter takes up that question, with a specific focus on corruption at the *node* level rather than the edge level. A user’s Twitter account may get hacked, causing them to connect and disconnect arbitrarily with other users. A protein in a yeast two-hybrid PPI assay may interact randomly with other proteins due to a variety of factors: it may itself be an unknown transcription factor, leading to false positives, or it may fail to express, be toxic to the cell, or require post-translational modifications that do not take place in yeast cells, leading to false negatives [Fio19, KPA⁺12]. These phenomena are not well captured by edge-corruption models. The number of corrupted node pairs may be quadratic in the size of the graph, but there is a spatial clustering of the noise: each corrupted edge has at least one endpoint in a (typically small) set of misbehaving nodes. In this

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sense, algorithms for graph matching may themselves be viewed as robust algorithms for graph isomorphism, and this chapter argues that robustness towards *node corruptions* is an important axis along which such robustness should be characterized.

Contributions To our knowledge, this chapter provides the first systematic study of fundamental limits for graph matching with node corruption. Two models are introduced:

- (a) *Weakly corrupted graphs (WCG)*: The adversary selects a random set of nodes in each network and resamples all the edges adjacent to the set without observing the graphs. This models random behavior of unknown proteins in a PPI network.
- (b) *Strongly corrupted graphs (SCG)*: The adversary selects an arbitrary set of nodes in each network and rewires all the edges adjacent to the set after observing the graphs. This models malicious behavior of hacked users in social networks.

For the **WCG** model, we show that no estimator correctly matches any positive fraction of corrupted nodes. Conversely, under appropriate conditions, the k -core estimator correctly matches almost all of the uncorrupted nodes and none of the corrupted nodes. Under a further condition that is also necessary, it identifies the corrupted nodes and correctly matches all the uncorrupted nodes. Our simulations suggest that there is a gap between these fundamental limits and the performance of commonly used computationally feasible algorithms.

For the **SCG** model, we show that an analogous detection of corrupted nodes is impossible. Even so, when only one of the networks is corrupted, the maximum overlap estimator outputs a matching that correctly matches a positive fraction of the uncorrupted nodes. An explicit lower bound on the fraction of correctly matched nodes as a function of the fraction of corrupted nodes is also derived.

6.1.1 Related work

The problem of finding necessary and sufficient conditions for matching correlated random graphs was considered in [PG11]. Ever since, a growing line of work has improved these results for exact recovery [CK16, CK17], almost-exact recovery [CKMP19, WXY22] and partial recovery [HM23, GML21, DD22]. A parallel line of work has investigated the performance of computationally feasible algorithms [BCL⁺19, DMWX21, FMWX22, MRT21], culminating in algorithms that run provably well in polynomial time when the graphs are far from isomorphic [MRT23, MWXY23, DL23].

All these works study correlated Erdős-Rényi graphs, for which the fundamental limits of achievability and impossibility are now well understood. Subsequently, an emerging line of work is expanding the scope of the problem. For instance, [RS21] and [GRS22] study the graph matching problem in correlated stochastic block models, and [RS23] studies information theoretic limits of graph matching in inhomogeneous random graphs.

Recently, there is growing interest in studying robust variants of estimation problems in graphs

when a positive fraction of nodes are corrupted. For example, [AJK⁺22] studies the problem of estimating the parameter p of an Erdős-Rényi graph in the presence of such an adversary. More recently, [LM22] and [HDdS23] study the community detection problem when nodes are corrupted. Finally, the model in [MM18] allows for a simpler version of node corruptions in graph matching, but studies worst-case performance when a sublinear fraction of nodes are corrupted. All these results provide insight into robustness of algorithms, and facilitate development of algorithms better suited for real-world networks.

6.2 Preliminaries

Notation Let $[n]$ denote the set $\{1, 2, \dots, n\}$ and let $\binom{[n]}{2}$ denote the set of unordered pairs $\{\{u, v\} : u, v \in [n] \text{ and } u \neq v\}$. For a graph G on n nodes, assume that its node set $V(G)$ is $[n]$, and so its edge set $E(G)$ is a subset of $\binom{[n]}{2}$. In this chapter, graphs are undirected and unweighted, so denote $G\{i, j\} = 1$ if $\{i, j\} \in E(G)$ and 0 otherwise. The graph G is sampled from the Erdős-Rényi distribution, denoted $G \sim \text{ER}(n, p)$, if G has n nodes and each edge in G exists with probability p . Let π be a permutation on $[n]$ and denote by G^π the graph obtained by relabeling nodes in G according to π , so that

$$G\{i, j\} = G^\pi\{\pi(i), \pi(j)\} \quad \forall \{i, j\} \in \binom{[n]}{2}.$$

Standard asymptotic notation ($O(\cdot)$, $o(\cdot)$, $\Theta(\cdot)$, $\dots$) is used throughout, and it is implicit that $n \rightarrow \infty$.

Throughout this chapter, $\text{Bern}(p)$ and $\text{Bin}(n, p)$ denote respectively the Bernoulli and binomial distribution. The hypergeometric distribution is denoted by $\text{HypGeom}(n, k, m)$. A random variable with this distribution counts the number of successes in a sample of k elements drawn without replacement from a population of n individuals, of which m elements are considered successes.

6.2.1 Correlated graphs and corruption models

In all definitions below, n is a positive integer and p, s are in $[0, 1]$. Further, G_1 and G_2 are graphs with $V(G) = [n]$ and $\pi^\star$ is a permutation on $[n]$.

Definition 6.1 (CER model). The tuple $(G_1, G_2, \pi^\star)$ is sampled from the correlated Erdős-Rényi distribution $\text{CER}(n, p, s)$ if two graphs G_1 and G'_2 are obtained by independently subsampling each edge of a parent graph $G \sim \text{ER}(n, p)$ with probability s . Independently, a permutation $\pi^\star$ is sampled uniformly at random, and G_2 is obtained as $G_2 = G'_2 \pi^\star$.

Marginally, G_1 and G_2 each follow the $\text{ER}(n, ps)$ distribution. However, the two graphs are edge-wise correlated according to a latent permutation $\pi^\star$. Next, two models of corruption are presented, motivated respectively by applications in protein-protein interaction and social network de-anonymization.

Definition 6.2 (WCG and SCG Models). Let (G_1, G_2, π^*) be a sample from $\text{CER}(n, p, s)$. Let γ and λ be in $[0, 1]$. Consider an adversary that selects two sets of nodes $\mathcal{B}_1 \subseteq V(G_1)$ and $\mathcal{B}_2 \subseteq V(G_2)$ such that $|\mathcal{B}_1| = \lambda\gamma n$ and $|\mathcal{B}_2| = (1-\lambda)\gamma n$. Let $\mathcal{E}_{\mathcal{B}_1}$ (resp. $\mathcal{E}_{\mathcal{B}_2}$) denote all the node pairs adjacent to $\mathcal{B}_1$ (resp. $\mathcal{B}_2$) in $\binom{[n]}{2}$:

$$\begin{aligned}\mathcal{E}_{\mathcal{B}_1} &= \left\{ \{i, j\} \in \binom{[n]}{2} : i \in \mathcal{B}_1 \text{ or } j \in \mathcal{B}_1 \right\}, \\ \mathcal{E}_{\mathcal{B}_2} &= \left\{ \{i, j\} \in \binom{[n]}{2} : i \in \mathcal{B}_2 \text{ or } j \in \mathcal{B}_2 \right\}.\end{aligned}$$

- **WCG model:** The weak adversary selects $\mathcal{B}_1$ and $\mathcal{B}_2$ uniformly at random and independent of G_1 and G_2 . It then assigns the edge status of each node pair in $\mathcal{E}_{\mathcal{B}_1}$ and $\mathcal{E}_{\mathcal{B}_2}$ independently from the $\text{Bern}(ps)$ distribution. The corrupted graphs are denoted $\tilde{G}_1$ and $\tilde{G}_2$. The tuple $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ is said to be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$.
- **SCG model:** A strong adversary A is any rule to select the sets $\mathcal{B}_1$ and $\mathcal{B}_2$, and the edge status of all node pairs in $\mathcal{E}_{\mathcal{B}_1}$ and $\mathcal{E}_{\mathcal{B}_2}$. The corrupted graphs are denoted $\tilde{G}_1$ and $\tilde{G}_2$, and the tuple $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ is said to be distributed according to $\text{SCG}(n, p, s, \gamma, \lambda, A)$.

In words, the adversary corrupts a total of γn nodes, of which a fraction λ are in G_1 and the rest are in G_2 . It then modifies the edge status of each node pair with at least one corrupted end point. Note that the WCG model defines a joint distribution on $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$. In contrast, for the SCG model, one must explicitly define an adversary A to obtain a distribution on $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$.

6.2.2 Matchings and estimators

Definition 6.3 (Matching). A *matching* μ is an injective function with domain $\text{dom}(\mu) \subseteq [n]$ and codomain $[n]$.

Note that permutations are matchings with domain equal to $[n]$. An *estimator* $\mathcal{E}$ is a mapping that takes in a pair of corrupted graphs $(\tilde{G}_1, \tilde{G}_2)$ and outputs a matching μ . In doing so, it attempts to recover the latent permutation π^* between the uncorrupted graphs G_1 and G_2 . Two estimators that have been studied in the absence of any adversary are the maximum overlap estimator $\hat{\mathcal{E}}_{\text{MO}}$ and the k -core estimator $\hat{\mathcal{E}}_k$. They are presented next using the following definition.

Definition 6.4 (Intersection Graph). Let H_1 and H_2 be two graphs and let μ be a matching. The intersection graph $H_1 \wedge_\mu H_2$ is a graph with node set $\text{dom}(\mu)$, such that for any two nodes $i, j \in \text{dom}(\mu)$, the pair $\{i, j\}$ is an edge in $H_1 \wedge_\mu H_2$ if and only if $\{i, j\}$ is an edge in H_1 and $\{\mu(i), \mu(j)\}$ is an edge in H_2 .

Maximum overlap estimator For two graphs H_1 and H_2 , the maximum overlap estimator $\hat{\mathcal{E}}_{\text{MO}}(H_1, H_2)$ outputs a matching $\hat{\mu}_{\text{MO}}$ that maximizes the number of edges in the corresponding

intersection graph:

$$\hat{\mu}_{\text{MO}} \in \arg \max_{\mu} |E(H_1 \wedge_{\mu} H_2)|.$$

The maximum overlap matching is the maximum likelihood estimator for exact recovery in the absence of the adversary, and is therefore optimal in that setting.

k -core estimator The k -core of a graph G , denoted $\text{core}_k(G)$ is the largest set of vertices A of G such that the induced subgraph on A has minimum degree at least k . For any two graphs H_1 and H_2 and non-negative integer k , a matching μ is said to be a k -core matching of H_1 and H_2 if the minimum degree in $H_1 \wedge_{\mu} H_2$ is at least k .

The k -core estimator $\hat{\mathcal{E}}_k(H_1, H_2)$ selects a k -core matching $\hat{\mu}_k$ such that $|\text{dom}(\hat{\mu}_k)|$ is at least as large as $|\text{dom}(\mu_k)|$, for any other k -core matching μ_k .

6.2.3 Recovery objectives

For a matching μ and a permutation π^* on $[n]$, denote by $\text{ov}(\mu, \pi^*)$ the overlap between μ and π^* , i.e. the number of nodes on which μ and π^* agree:

$$\text{ov}(\mu, \pi^*) \triangleq |\{i \in \text{dom}(\mu) : \mu(i) = \pi^*(i)\}|.$$

Upon observing only the pair of corrupted graphs $(\tilde{G}_1, \tilde{G}_2)$, the objective is to find a matching $\hat{\mu}$ to maximize the overlap between $\hat{\mu}$ and the latent permutation π^* . Definition 6.5 captures this notion.

Definition 6.5 (α -recovery). Let $\alpha \in (0, 1]$. An estimator that outputs a matching $\hat{\mu}$ is said to achieve

- (i) α -recovery, if $\mathbb{P}\left(\frac{\text{ov}(\hat{\mu}, \pi^*)}{n} \geq \alpha\right) = 1 - o(1)$.
- (ii) almost α -recovery, if for every $\varepsilon > 0$,

$$\mathbb{P}\left(\frac{\text{ov}(\hat{\mu}, \pi^*)}{n} \geq \alpha - \varepsilon\right) = 1 - o(1).$$

The terms *exact recovery* and *almost exact recovery* are used frequently in literature, and correspond respectively to 1-recovery and almost 1-recovery. Similarly, *partial recovery* corresponds to the existence of some $\alpha > 0$ for which α -recovery is achievable.

Graph matching is often a precursor to downstream tasks. Subsequently, an estimator is often useful in practice only if it correctly matches all the nodes in its domain. This concept is made rigorous through the notion of *precision*.

Definition 6.6 (Precision). The *precision* ρ of a matching $\hat{\mu}$ is the fraction of the matching that

is correct, i.e.

$$\rho(\hat{\mu}) \triangleq \frac{\text{ov}(\hat{\mu}, \pi^*)}{|\text{dom}(\hat{\mu})|}.$$

For a sequence of graph-pairs $(\tilde{G}_1, \tilde{G}_2)_n$ on n vertices, an estimator $\mathcal{E}(\tilde{G}_1, \tilde{G}_2)$ that outputs a matching $\hat{\pi}$ is *precise* if $\mathbb{P}(\rho(\hat{\pi}) = 1) = 1 - o(1)$. For any $\varepsilon > 0$, it is said to be ε -imprecise if $\mathbb{P}(\rho(\hat{\pi}) \leq 1 - \varepsilon) = 1 - o(1)$.

6.3 Main results

Impossibility and achievability results are presented separately for the WCG and SCG models. In all the results, n is a positive integer and p, s, γ, λ are real numbers such that $p \in (0, 1)$, $s \in (0, 1]$, and $\gamma, \lambda \in [0, 1]$. For $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ from the WCG model, denote by $\mathcal{B}'_2$ the pre-image of $\mathcal{B}_2$ under π^* , i.e.

$$\mathcal{B}'_2 \triangleq \{i \in [n] : \pi^*(i) \in \mathcal{B}_2\}.$$

6.3.1 Results on the WCG model

Our first result is an impossibility result that holds for any estimator.

Theorem 6.1. *Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$. Let $\mathcal{E}(\tilde{G}_1, \tilde{G}_2)$ be any estimator that returns a matching μ . Let $\alpha^* = 1 - \gamma + \lambda(1 - \lambda)\gamma^2$.*

(i) *If $\mathcal{E}$ is precise, then*

$$\mathbb{P}(\text{dom}(\mu) \subseteq (\mathcal{B}_1 \cup \mathcal{B}'_2)^c) = 1 - o(1).$$

(ii) *If $\mathcal{E}$ achieves almost α -recovery, then $\alpha \leq \alpha^*$.*

(iii) *Let $p = C \log(n)/n$ and $\lambda \in \{0, 1\}$. If $\mathcal{E}$ is precise and achieves α^* -recovery, then $C \geq 1/(s^2 \alpha^*)$.*

Part (i) of Theorem 6.1 states that no precise estimator can correctly recover any of the corrupted nodes with high probability. Part (ii) precludes the possibility of almost 1-recovery (and therefore also 1-recovery) when $\gamma > 0$, in stark contrast to known achievability results in the absence of adversary.

Next, we show that when the average degrees of G_1 and G_2 are logarithmic in the number of nodes n (i.e. $p = C \log(n)/n$ for some positive constant C), the k -core estimator performs optimally for an appropriate choice of k . Specifically, we prove that there is a threshold $\tau \equiv \tau(s, \gamma, \lambda)$ such that if $C > \tau$, then the k -core estimator identifies and matches all the uncorrupted nodes. Further, if $C < \tau$, then the estimator matches all but a vanishing fraction of the uncorrupted nodes and none of the corrupted nodes.

Theorem 6.2. Let C be a positive constant and let $p = C \log(n)/n$. Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$, and let $\hat{\mu}_k$ be the matching output by the k -core estimator $\hat{\mathcal{E}}_k(\tilde{G}_1, \tilde{G}_2)$ with $k = \sqrt{\log n}$. Let $\alpha^* = 1 - \gamma + \lambda(1 - \lambda)\gamma^2$.

(i) $\hat{\mathcal{E}}_k$ is precise.

(ii) If $C > 1/(s^2\alpha^*)$, then $\hat{\mathcal{E}}_k(\tilde{G}_1, \tilde{G}_2)$ achieves α^* -recovery if $\lambda \in \{0, 1\}$, and achieves almost α^* -recovery if $\lambda \in (0, 1)$. Further,

$$\mathbb{P}(\text{dom}(\hat{\mu}_k) = (\mathcal{B}_1 \cup \mathcal{B}_2')^c) = 1 - o(1). \quad (6.1)$$

(iii) If $C < 1/(s^2\alpha^*)$, then $\hat{\mathcal{E}}_k(\tilde{G}_1, \tilde{G}_2)$ achieves almost α^* -recovery for all $\lambda \in [0, 1]$. Further,

$$\mathbb{P}(|(\mathcal{B}_1 \cup \mathcal{B}_2')^c \setminus \text{dom}(\hat{\mu}_k)| = o(n)) = 1 - o(1). \quad (6.2)$$

6.3.2 Results on the SCG model

First, we show that in the SCG model, no estimator can be precise.

Theorem 6.3. Suppose $\gamma > 0$. There exists an adversary $\mathbf{A}'$ for which the output $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ of $\text{SCG}(n, p, s, \gamma, \lambda, \mathbf{A}')$ satisfies the following: If an estimator $\mathcal{E}(\tilde{G}_1, \tilde{G}_2)$ returns a matching μ with $|\text{dom}(\mu)| = \Theta(n)$, then $\mathcal{E}$ is ε^* -imprecise, where ε^* is any real number such that $\varepsilon^* < \frac{\max(\lambda, 1-\lambda)\gamma}{2}$.

Thus, no estimator can identify a set $S \subset [n]$ containing a positive fraction of nodes, such that all nodes in S are correctly matched. Despite this, we show that if only one of the networks is compromised, then imprecise recovery of a positive fraction of nodes is possible under appropriate conditions. We state two achievability results below. Theorem 6.4 deals with the case when average degrees are logarithmic in n , whereas Theorem 6.5 deals with the case when p is constant.

Theorem 6.4. Let $\lambda \in \{0, 1\}$ and $\alpha \in [0, 1]$. If

$$\gamma < \frac{s(1 - \alpha^2)}{4}, \quad (6.3)$$

then there exists a constant $C' \equiv C'(\alpha, \gamma)$ such that for all $C > C'$ and $p = C \log(n)/n$, and for all adversaries $\mathbf{A}$ and outputs $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ of $\text{SCG}(n, p, s, \gamma, \lambda, \mathbf{A})$, the maximum overlap estimator achieves α -recovery.

Theorem 6.5. Suppose p is constant, $\lambda \in \{0, 1\}$ and $\alpha \in [0, 1]$. If

$$\gamma < 1 - \sqrt{1 - \frac{s^2 p (1 - p) (1 - \alpha^2)}{2}}, \quad (6.4)$$

then for all adversaries $\mathbf{A}$ and outputs $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ of $\text{SCG}(n, p, s, \gamma, \lambda, \mathbf{A})$, the maximum overlap estimator achieves α -recovery.

6.4 Proof outlines

Proofs for all the main results of this chapter are outlined, with details deferred to the supplementary material.

6.4.1 The WCG model

The performance of the k -core estimator against the weak adversary is analyzed. First, the impossibility result is proved using an indistinguishability argument.

Proof of Theorem 6.1. (i) It suffices to show that no estimator can correctly match any node i in $\mathcal{B}_1 \cup \mathcal{B}'_2$ with high probability. Consider the joint distribution of the collection

$$\left\{ \tilde{G}_1\{i, j\}, \tilde{G}_2\{\pi^*(i), \pi^*(j)\} \right\}_{j \in [n], j \neq i}.$$

These $2(n-1)$ random variables are each distributed as $\text{Bern}(ps^2)$ and mutually independent, since either $G_1\{i, j\}$ or $G_2\{\pi^*(i), \pi^*(j)\}$ is resampled because either $i \in \mathcal{B}_1$ or $\pi^*(i) \in \mathcal{B}_2$. Further, this joint distribution is the same for all nodes i in the set $\mathcal{B}_1 \cup \mathcal{B}'_2$, and so the nodes within it are statistically indistinguishable. Consequently, no estimator can match any subset $\overline{M}$ of nodes in $\mathcal{B}_1 \cup \mathcal{B}'_2$ better than random guessing. Lemma D.1 shows that the random guessing estimator is precise if and only if $\mathbb{P}(\overline{M} = \phi) = 1 - o(1)$, and the desired result follows.

(ii) Lemma D.1 implies that any estimator can at best match correctly the set $(\mathcal{B}_1 \cup \mathcal{B}'_2)^c$ and at most a sublinear number of nodes in $(\mathcal{B}_1 \cup \mathcal{B}'_2)$. However, $|\mathcal{B}_1 \cup \mathcal{B}'_2|/n$ converges in probability to $\gamma - \lambda(1-\lambda)\gamma^2$. This follows from Lemma D.2, where it is shown that $|\mathcal{B}_1 \cap \mathcal{B}'_2|/n$ converges in probability to $\lambda(1-\lambda)\gamma$. Since no more than a sublinear number of nodes in $\mathcal{B}_1 \cup \mathcal{B}'_2$ are correctly matched, it follows that the fraction of correctly matched nodes, α is strictly upper bounded by $|\mathcal{B}_1 \cup \mathcal{B}'_2|^c/n + \varepsilon$ for every $\varepsilon > 0$. We conclude that $\alpha \leq 1 - \gamma + \lambda(1-\lambda)\gamma^2$, as desired.

(iii) Assume $\lambda = 1$ so that $\mathcal{B}'_2 = \phi$, although a similar proof works for $\lambda = 0$. With probability $1 - o(1)$:

$$\alpha^* n \stackrel{(a)}{\leq} \text{ov}(\mu, \pi^*) \leq |\text{dom}(\mu)| \stackrel{(b)}{\leq} |\mathcal{B}_1^c| = \alpha^* n,$$

where (a) is true because $\mathcal{E}$ achieves α^* recovery and (b) uses (i) since $\mathcal{E}$ is a precise estimator. The above string of inequalities are thus equalities. Thus, (i) yields that $\text{dom}(\mu) = \mathcal{B}_1^c$ with probability $1 - o(1)$. Since $\mathcal{E}$ achieves α^* -recovery, it follows that μ has correctly matched all the vertices in $\mathcal{B}_1^c$. We show that this is only possible when $C \geq 1/(s^2\alpha^*)$. For a graph G and vertex subset $X \subseteq V(G)$,

let $G|_X$ denote the induced subgraph of G on X . Then, with probability $1 - o(1)$:

$$\begin{aligned} H_1 &\triangleq \tilde{G}_1|_{\text{dom}(\mu)} = \tilde{G}_1|_{\mathcal{B}_1^c} \stackrel{(c)}{=} G_1|_{\mathcal{B}_1^c}, \\ H_2 &\triangleq \tilde{G}_2|_{\pi^*(\text{dom}(\mu))} = \tilde{G}_2|_{\pi^*(\mathcal{B}_1^c)} \stackrel{(d)}{=} G_2|_{\pi^*(\mathcal{B}_1^c)}, \end{aligned}$$

where (c) is because no node pair in $\tilde{G}_1|_{\mathcal{B}_1^c}$ is influenced by the adversary, and (d) is because $G_2 = \tilde{G}_2$. Thus, $(H_1, H_2, \pi^*|_{\text{dom}(\mu)}) \sim \text{CER}(\alpha^*n, p, s)$. Recovering $\pi^*|_{\text{dom}(\mu)}$ is the exact graph recovery problem between H_1 and H_2 , which is impossible whenever $\alpha^*nps^2 < 1$, i.e. whenever $C < 1/(s^2\alpha^*)$ [CK17, WXY22]. $\square$

Next, a proof sketch for Theorem 6.2 is presented.

Proof of Theorem 6.2. (i) The proof is deferred to Section D.1.1.

(ii) The union bound yields for any $\delta \geq 0$,

$$\mathbb{P}(\hat{\mathcal{E}}_k \text{ achieves almost } \alpha^* \text{-recovery}) \geq 1 - p_1 - p_2 - p_3,$$

where

$$p_1 = \mathbb{P}(\hat{\mu}_k \neq \pi^*|_{\text{core}_k(\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2)}), \quad (6.5)$$

$$p_2 = \mathbb{P}(\text{core}_k(\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2) \neq (\mathcal{B}_1 \cup \mathcal{B}_2')^c), \quad (6.6)$$

$$p_3 = \mathbb{P}(|(\mathcal{B}_1 \cup \mathcal{B}_2')^c| < (1 - \delta)\alpha^*n) \quad (6.7)$$

From Theorem D.1 and Lemma D.5 in the proof of (i), it follows that $p_1 = o(1)$. The bulk of the analysis is to show that $p_2 = o(1)$ whenever $C > 1/(s^2\alpha^*)$. This is shown in Lemma D.6 in the supplementary material. Finally, for any $\delta > 0$, it follows from Lemma D.2 that $p_3 = o(1)$ for all $\lambda \in [0, 1]$.

When $\lambda \in \{0, 1\}$, we have $p_3 = 0$ even when $\delta = 0$. This is because either $\mathcal{B}_1 = \phi$ or $\mathcal{B}_2' = \phi$, and therefore $|\mathcal{B}_1 \cup \mathcal{B}_2'| = \gamma n = \alpha^*n$ in this setting. However, setting $\delta = 0$ corresponds to achieving α^* -recovery.

(iii) Let M^* denote $\text{core}_k(\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2)$. The union bound yields for any $\delta \geq 0$,

$$\mathbb{P}(\hat{\mathcal{E}}_k \text{ achieves almost } \alpha^* \text{-recovery}) \geq 1 - p_1 - p_4,$$

where p_1 is defined in (6.5), and

$$p_4 = \mathbb{P}(|M^*| < (1 - \delta)\alpha^*n). \quad (6.8)$$

Algorithm 3: Adversary A'

Inputs : $G_1, G_2, \gamma, \lambda$
Outputs: $\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2$

- 1 Initialize $\tilde{G}_1 = G_1$ and $\tilde{G}_2 = G_2$
- 2 Select $\mathcal{B}_1 = \{1, 2, \dots, \lambda\gamma n\}$, and $\mathcal{B}_2$ such that $\mathcal{B}'_2 = \{1, 2, \dots, (1 - \lambda)\gamma n\}$
- 3 Partition $\mathcal{B}_1$ into $\mathcal{B}_{1,1} \triangleq \{1, \dots, \lambda\gamma n/2\}$ and $\mathcal{B}_{1,2} \triangleq \{\lambda\gamma n/2 + 1, \dots, \lambda\gamma n\}$
- 4 **for** $\{(i, j) \text{ such that } i \in \mathcal{B}_{1,1} \text{ and } j \in [n]\}$ **do**
- 5 Set $\tilde{G}_1\{i, j\} = G_1\{\lambda\gamma n/2 + i, j\}$
- 6 **end**
- 7 **for** $\{(i, j) \text{ such that } i \in \mathcal{B}_{1,2} \text{ and } j \in [n]\}$ **do**
- 8 Set $\tilde{G}_1\{i, j\} = G_1\{i - \lambda\gamma n/2, j\}$
- 9 **end**
- 10 **return** $\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2$

Lemma D.7 shows that $p_4 = o(1)$ for any $\delta > 0$. From part (i) of this theorem, it also follows that $p_1 = o(1)$. Therefore, $\hat{\mathcal{E}}_k$ achieves almost α^* -recovery. It remains to prove (6.2), i.e. $\hat{\mathcal{E}}_k$ recovers all but a vanishing fraction of the uncorrupted nodes. Since $p_1 = o(1)$, it suffices to instead show that for any $\delta' > 0$

$$p_5 \triangleq \mathbb{P}(|(\mathcal{B}_1 \cup \mathcal{B}'_2)^c \setminus M^*| > \delta' n) = 1 - o(1).$$

Indeed, denoting $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}'_2$, it follows that for any $\delta' > 0$, and $\varepsilon = \delta/\alpha^*$ that

$$\begin{aligned} p_5 &\leq \mathbb{P}(M^* \not\subseteq \mathcal{B}^c) + \mathbb{P}(|\mathcal{B}^c \setminus M^*| > \delta n) \cap \{M^* \subseteq \mathcal{B}^c\} \\ &\leq o(1) + \mathbb{P}(|\mathcal{B}^c| - |M^*| > \delta n) \\ &\leq o(1) + \mathbb{P}(|\mathcal{B}^c| > (1 + \varepsilon/2)\alpha^* n) + \mathbb{P}(|M^*| < (1 - \varepsilon/2)\alpha^* n) \\ &\stackrel{(a)}{=} o(1) + o(1) + o(1), \end{aligned}$$

where (a) uses both Lemma D.2 and Lemma D.7. □

6.4.2 The SCG model

Proof of Theorem 6.3.

Without loss of generality, assume $\lambda \geq 1/2$ (else λ and $1 - \lambda$ may be interchanged in this proof). Consider the adversary A' in Algorithm 3, which effectively swaps nodes in $\mathcal{B}_{1,1}$ and $\mathcal{B}_{1,2}$. Let $\tilde{\pi}$ denote the corresponding permutation, i.e.

$$\tilde{\pi}(i) = \begin{cases} \lambda\gamma n/2 + i, & i \in \mathcal{B}_{1,1} \\ i - \lambda\gamma n/2, & i \in \mathcal{B}_{1,2} \\ i, & \text{otherwise} \end{cases}.$$

It follows that the graphs $G_1^{\tilde{\pi}}$ and $\tilde{G}_1$ are isomorphic and have the same node labeling. Therefore, no estimator will be able to identify any node which is not a fixed point of $\tilde{\pi}$ with probability greater than $1/2$.

Let μ be the matching output by an estimator $\mathcal{E}(\tilde{G}_1, \tilde{G}_2)$ and let M denote $\text{dom}(\mu)$. Since $|M| = \Theta(n)$, there exists a sequence (ε_n) such that $|M| = \varepsilon_n n$ and $\liminf_{n \rightarrow \infty} \varepsilon_n > 0$. Consider the sets

$$\mathcal{X} = \{i \in M \cap \mathcal{B}_{1,1} : \mu(i) = \pi^*(i)\} \quad \text{and} \quad \mathcal{Y} = \{i \in M \cap \mathcal{B}_{1,1}^c : \mu(i) = \pi^*(i)\}.$$

Let $p_1 \triangleq \mathbb{P}(\text{ov}(\mu, \pi^*)/|M| > \varepsilon^*)$. Then, for any $\delta > 0$:

$$p_1 = \mathbb{P}\left(\frac{|\mathcal{X}|}{|M|} + \frac{|\mathcal{Y}|}{|M|} > \varepsilon^*\right) \tag{6.9}$$

$$\leq \mathbb{P}\left(\frac{|\mathcal{X}|}{|M|} > \delta\right) + \mathbb{P}\left(\frac{|\mathcal{Y}|}{|M|} > 1 - (\varepsilon^* + \delta)\right) \tag{6.10}$$

$$\stackrel{(a)}{\leq} \underbrace{2^{-\delta \varepsilon_n n} + \mathbb{P}\left(\frac{|M \cap \mathcal{B}_{1,1}^c|}{|M|} > 1 - (\varepsilon^* + \delta)\right)}_{(*)}. \tag{6.11}$$

Here, (a) is because the pair $(\tilde{G}_1, \tilde{G}_2)$ does not contain the information needed to correctly match node i in $\mathcal{B}_{1,1}$ with probability more than $1/2$, even if an oracle were to correctly match all other nodes in $\mathcal{B}_{1,1}$. Next, $(*)$ is analyzed. Since it is impossible to determine if a node i is in $\mathcal{B}_{1,1}$, it follows that $|M \cap \mathcal{B}_{1,1}^c| \sim \text{HypGeom}(n, \varepsilon_n n, (1 - \lambda\gamma/2)n)$. This is because the set $M \cap \mathcal{B}_{1,1}^c$ may be viewed as being constructed by sampling $\varepsilon_n n$ nodes from $[n]$ without replacement, where a sampled node i is labeled a success if and only if $i \in \mathcal{B}_{1,1}^c$. Using standard formulas:

$$\begin{aligned} \mathbb{E}|M \cap \mathcal{B}_{1,1}^c| &= \left(1 - \frac{\lambda\gamma}{2}\right) \varepsilon_n n, \\ \text{Var}(|M \cap \mathcal{B}_{1,1}^c|) &= \frac{\lambda\gamma(1 - \lambda\gamma)\varepsilon_n(1 - \varepsilon_n)}{4} \times \frac{n^2}{n - 1}. \end{aligned}$$

Choose δ in (6.10) to be sufficiently small so that $\varepsilon^* + \delta < \frac{\lambda\gamma}{2}$. Then,

$$\begin{aligned} (*) &\leq \mathbb{P}\left(\left|\frac{|M \cap \mathcal{B}_{1,1}^c|}{|M|} - \left(1 - \frac{\lambda\gamma}{2}\right)\right| > \frac{\lambda\gamma}{2} - (\varepsilon^* + \delta)\right) \\ &\leq \frac{\lambda\gamma(1 - \lambda\gamma)(1 - \varepsilon_n)}{4\varepsilon_n(\lambda\gamma/2 - (\varepsilon^* + \delta))^2} \times \frac{1}{n - 1} \end{aligned} \tag{6.12}$$

where (6.12) uses Chebyshev's inequality. Finally, since $\liminf_{n \rightarrow \infty} \varepsilon_n > 0$, it follows that (6.12) is $o(1)$. Therefore, $p_1 = o(1)$. This concludes the proof. $\square$

Next, the maximum overlap estimator is analyzed when only one network is compromised. Without loss of generality, assume that $\lambda = 1$, so that $\tilde{G}_2 = G_2$.

Some notation is in order. For any matching μ , let $X(\mu)$ denote the number of edges in $G_1 \wedge_\mu G_2$. Similarly, let $\tilde{X}(\mu)$ denote the number of edges in $\tilde{G}_1 \wedge_\mu \tilde{G}_2$. Recall that the maximum overlap matching is defined as $\hat{\mu}_{\text{MO}} \in \arg \max_\mu \tilde{X}(\mu)$.

Note that it may be assumed without loss of generality that $\text{dom}(\hat{\mu}_{\text{MO}}) = [n]$, since $\tilde{X}(\cdot)$ satisfies a monotonicity property: extending the domain of any matching to $[n]$ does not decrease the number of edges in $\tilde{G}_1 \wedge_\mu \tilde{G}_2$. Assume further, with out loss of generality, that the latent correspondence π^* is the identity permutation id. Recall that a fixed point of a permutation π is an input i such that $\pi(i) = i$. Let $\mathcal{T}^\alpha$ (resp. $\mathcal{T}^{\leq \alpha}$) denote the set of all permutations with exactly (resp. at most) αn fixed points. It is shown below that $\hat{\mu}_{\text{MO}} \notin \mathcal{T}^{\leq \alpha}$. It suffices to prove:

$$\mathbb{P}\left(\tilde{X}(\text{id}) > \tilde{X}(\pi) \text{ for all } \pi \in \mathcal{T}^{\leq \alpha}\right) = 1 - o(1). \quad (6.13)$$

The following lemma is a useful ingredient for the proof of both Theorem 6.4 and Theorem 6.5.

Lemma 6.1. *For any adversary $\mathbf{A}$, any permutation π , and any output $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \text{id})$ of $\text{SCG}(n, p, s, \gamma, 1, \mathbf{A})$:*

$$\tilde{X}(\text{id}) - \tilde{X}(\pi) > X(\text{id}) - X(\pi) - Z,$$

where

$$Z \triangleq \max_{\substack{S, T \subseteq [n] \\ |S|, |T| \leq \gamma n}} \sum_{\substack{i \in S \cup T \\ j \in [n]}} G_2\{i, j\}. \quad (6.14)$$

Proof. For any selection $\mathcal{B}_1$ of γn nodes in G_1 , recall that $\mathcal{E}_{\mathcal{B}_1} \triangleq \left\{ \{i, j\} \in \binom{[n]}{2} : i \in \mathcal{B}_1 \text{ or } j \in \mathcal{B}_1 \right\}$. The adversary selects $\mathcal{B}_1$ and sets the edge status of all node pairs in $\mathcal{E}_{\mathcal{B}_1}$ to either 0 or 1. For any node pair $e = \{i, j\}$ and graph H , let $H(e)$ be a shorthand for $H\{i, j\}$. For any set $\mathcal{B}_1$ and any permutation π :

$$\left(\tilde{X}(\pi) - \tilde{X}(\text{id})\right) - \left(X(\pi) - X(\text{id})\right) = \sum_{e \in \mathcal{E}_{\mathcal{B}_1}} \left(\tilde{G}_1(e) - G_1(e)\right) \left(G_2^\pi(e) - G_2(e)\right) \quad (6.15)$$

$$\begin{aligned} &\stackrel{(a)}{\leq} \sum_{e \in \mathcal{E}_{\mathcal{B}_1}} \mathbb{1}\{G_1(e)=1\} \cdot \mathbb{1}\{G_2(e)=1\} \cdot \mathbb{1}\{G_2^\pi(e)=0\} \\ &\quad + \sum_{e \in \mathcal{E}_{\mathcal{B}_1}} \mathbb{1}\{G_1(e)=0\} \cdot \mathbb{1}\{G_2(e)=0\} \cdot \mathbb{1}\{G_2^\pi(e)=1\} \\ &\leq \sum_{e \in \mathcal{E}_{\mathcal{B}_1}} \left(\mathbb{1}\{G_2(e)=1\} + \mathbb{1}\{G_2^\pi(e)=1\}\right). \end{aligned} \quad (6.16)$$

Here, (a) is because each term of the sum in (6.15) is in the set $\{-1, 0, 1\}$, and equals 1 if and only if the adversary sets $\tilde{G}_1(e) = 1 - G_1(e)$ whenever $(G_1(e), G_2(e), G_2^\pi(e))$ is either $(1, 1, 0)$ or $(0, 0, 1)$.

Note that (6.16) is maximized when $\mathcal{B}_1$ and π are chosen to maximize $|E(G_2) \cap \mathcal{E}_{\mathcal{B}_1}| + |E(G_2^\pi) \cap \mathcal{E}_{\mathcal{B}_1}|$. Therefore,

$$\begin{aligned} \left(\tilde{X}(\pi) - \tilde{X}(\text{id}) \right) - \left(X(\pi) - X(\text{id}) \right) &\leq \max_{\mathcal{B}_1, \pi(\mathcal{B}_1)} \sum_{e \in \mathcal{E}_{\mathcal{B}_1}} (G_2(e) + G_2^\pi(e)) \\ &\stackrel{(b)}{\leq} \max_{\substack{S, T \subseteq [n] \\ |S|, |T| \leq \gamma n}} \sum_{\substack{i \in S \cup T \\ j \in [n]}} G_2\{i, j\}, \end{aligned}$$

as desired. Here, (b) follows from the fact that $|\mathcal{B}_1| = |\pi(\mathcal{B}_1)| \leq \gamma n$. This concludes the proof. $\square$

Proof of Theorem 6.4. Let Z be as in Equation (6.14). Let Δ_2 denote the maximum node degree in the graph G_2 . Since $|S| + |T| \leq 2\gamma n$, it follows that $Z \leq 2\gamma n \Delta_2$. Let $\mathbf{E}$ denote the error event

$$\mathbf{E} = \bigcup_{\pi \in \mathcal{T}^{\leq \alpha}} \left\{ \tilde{X}(\text{id}) - \tilde{X}(\pi) < 0 \right\}. \quad (6.17)$$

Applying Lemma 6.1 yields for any $\varepsilon > 0$:

$$\mathbb{P}(\mathbf{E}) \leq \mathbb{P} \left(\bigcup_{\pi \in \mathcal{T}^{\leq \alpha}} \{X(\text{id}) - X(\pi) < 2\gamma n \Delta_2\} \right) \leq \sum_{i=1}^3 p_i,$$

where

$$p_1 = \mathbb{P} \left(X(\text{id}) \leq (1 - \varepsilon) \binom{n}{2} p s^2 \right), \quad (6.18)$$

$$p_2 = \mathbb{P}(\Delta_2 > (1 + \varepsilon) n p s), \quad (6.19)$$

$$p_3 = \mathbb{P} \left(\bigcup_{\pi \in \mathcal{T}^{\leq \alpha}} \left\{ X(\pi) \geq (1 - \varepsilon) \binom{n}{2} p s^2 - 2\gamma n (1 + \varepsilon) n p s \right\} \right) \quad (6.20)$$

Lemmas D.8 and D.9 show that $p_1 = o(1)$ and $p_2 = o(1)$ for any $\varepsilon > 0$ and sufficiently large C . Furthermore, Lemma D.11 shows that $p_3 = o(1)$ for sufficiently small $\varepsilon > 0$ and sufficiently large $C > 0$, whenever $\gamma < s(1 - \alpha^2)/4$. This requires a Chernoff argument, using bounds on the moment generating function of $X(\pi)$ obtained by analyzing the orbital decomposition of π . Altogether, it follows that $\mathbb{P}(\mathbf{E}) = o(1)$, which concludes the proof. $\square$

Proof of Theorem 6.5. Let Z be as defined in Equation (6.14), and let Γ denote the constant $\Gamma = \binom{\gamma n}{2} + \gamma(1 - \gamma)n^2$. Since for any choice of S and T :

$$|\{\{i, j\} : i \in S \cup T, j \in [n]\}| \leq 2\Gamma,$$

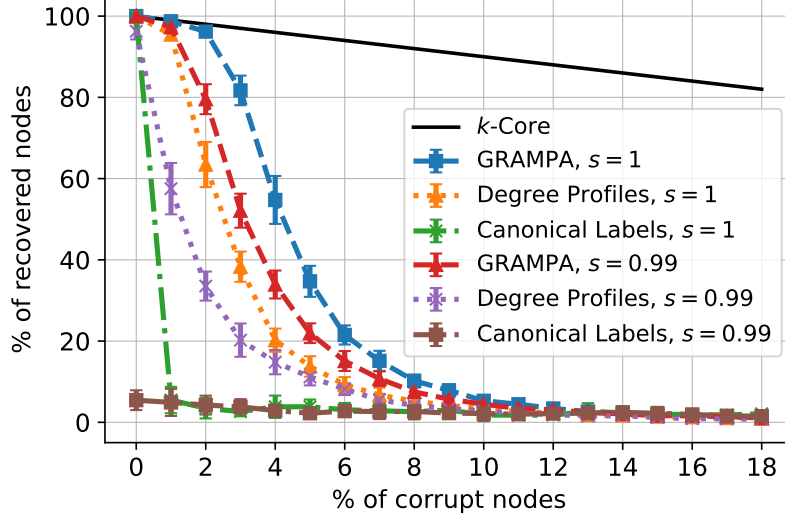


Figure 6.1: WCG model, $(n, p, \lambda) = (10^3, 0.1, 1)$

it follows that $Z \leq 2\Gamma$. Let $\mathbf{E}$ denote the error event in (6.17). Consequently, for any $\varepsilon > 0$,

$$\mathbb{P}(\mathbf{E}) \leq \mathbb{P}\left(\bigcup_{\pi \in \mathcal{T}^{\leq \alpha}} \{X(\text{id}) - X(\pi) < 2\Gamma\}\right) \leq p_1 + p_4,$$

where

$$p_1 = \mathbb{P}\left(X(\text{id}) \leq (1 - \varepsilon) \binom{n}{2} p s^2\right), \quad (6.21)$$

$$p_4 = \mathbb{P}\left(\bigcup_{\pi \in \mathcal{T}^{\leq \alpha}} \left\{X(\pi) \geq (1 - \varepsilon) \binom{n}{2} p s^2 - 2\Gamma\right\}\right). \quad (6.22)$$

Lemma D.8 shows that $p_1 = o(1)$ for any $\varepsilon > 0$. Furthermore, Lemma D.13, shows that $p_4 = o(1)$ whenever the condition (6.4) is satisfied. This uses similar techniques as the proof of Lemma D.11, although the resulting sufficient conditions for recovery are quite different. Combining, it follows that $\mathbb{P}(\mathbf{E}) = o(1)$ as desired. $\square$

6.5 Discussion

Feasible Algorithms The maximum overlap and k -core estimators are useful to serve as a baseline and to establish theoretical guarantees. However, they do not run in polynomial time, and have limited utility in practice. It is an open question to analyze the performance of *computationally feasible* algorithms when an adversary corrupts nodes. Figure 6.1 compares the asymptotic guarantee of the k -core estimator against simulation results for the following estimators.

- (a) GRAMPA [FMWX22] is a spectral algorithm that uses the spectrum of the adjacency matrices

Algorithm 4: Example adversary, $\lambda = 1/2$.

Inputs : $G_1, G_2, \gamma, \lambda$
Outputs: $\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2$

- 1 Initialize $\tilde{G}_1 = G_1$ and $\tilde{G}_2 = G_2$
- 2 Set $\mathcal{B}_1 = \{1, 2, \dots, \frac{\gamma n}{2}\}, \mathcal{B}_2 = \{\frac{\gamma n}{2} + 1, \dots, \gamma n\}$
- 3 Set $\mathcal{G}_1 = \{i \in [n] : i > \gamma n, i \text{ is odd}\}$ and $\mathcal{G}_2 = \{i \in [n] : i > \gamma n, i \text{ is even}\}$
- 4 **for** k **in** $\{1, 2\}$ **do**
- 5 **for** $\{i, j\}$ *such that* $i \in \mathcal{B}_k$ *and* $j \in \mathcal{G}_k$ **do**
- 6 Set $\tilde{G}_k\{i, j\} = 1$
- 7 **end**
- 8 **end**
- 9 **return** $\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2$

to match the two graphs. The code is available in [FMWX20].

- (b) DEGREE PROFILING [DMWX21] associates with each node a signature, specifically the histogram of the degrees of its neighbors. It then matches nodes based on signature proximity. The code is available in [DMWX20].
- (c) CANONICAL LABELING [DCKG19] first matches nodes with outlier degrees, and uses them as seeds to match the remaining nodes.

Clearly, there is large gap between the performance of these algorithms and the asymptotic guarantees of the k -core estimator. These algorithms are not robust to the *random* noise in the setting without the adversary, since they require $s \rightarrow 1$ for good performance [FMWX22, DMWX21, DCKG19]. Perhaps unsurprisingly, they are also not robust to the *spatial* noise induced by the node-based adversary.

SCG Model The SCG model is significantly more complicated when both networks are compromised. For simplicity, assume that $\lambda = 1/2$, and let $p = C \log n/n$ for some positive constant C . We show that there is a simple adversarial action that can cause the maximum overlap estimator to recover none of the nodes correctly. Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \text{id})$ be an output of $\text{SCG}(n, p, s, \gamma, \lambda)$ obtained as described in Algorithm 4. The adversary selects disjoint sets $\mathcal{B}_1$ and $\mathcal{B}_2$ and forces nodes to have high degrees in such a way that the overlap is maximized when every node is wrongly mapped. This is rigorously proved in Appendix D.3. One may study variants that pre-process the graphs, but by Theorem 6.3, precise recovery of nodes would be impossible even in that setting.

6.6 Conclusion

This work studied two models for graph matching when a positive fraction of nodes interact adversarially with their network. The two models are motivated by practical aspects of network alignment: the SCG framework models the malicious behavior of hacked users in a social network, whereas the

WCG framework models the random behavior of stochastic interactors in protein-protein interaction networks.

For the WCG model, our impossibility result states that no positive fraction of the corrupted nodes may be correctly matched. Conversely, under appropriate conditions, the k -core estimator correctly matches almost all of the uncorrupted nodes and none of the corrupted nodes. Under a further condition which is necessary, the k -core estimator also identifies and recovers all the uncorrupted nodes. In contrast, even the simpler problem of detecting corrupted nodes is impossible to solve in the SCG model. Even so, the maximum overlap estimator successfully matches a positive fraction of nodes under appropriate conditions.

Looking ahead, feasible algorithms that approach the performance guarantees of the k -core estimator in the WCG model would be useful in practice. One may also study *seeded* robust graph matching, where the correspondence is known for a small subset of nodes.

Chapter 7

Blockchain security when messages are lost

7.1 Introduction

Blockchain is the data structure used by peers (miners) in a peer-to-peer network to maintain a common ledger in a decentralized manner. The consistency of this ledger is ensured through consensus protocols such as the longest-chain protocol. Following this protocol, an honest miner groups transactions into a block and appends its block to the longest chain in its view, before broadcasting the new blockchain to all other peers. Further, the system may have adversarial users that deviate from the protocol arbitrarily. Despite adversarial users attempting to disrupt the system and peer-to-peer communication incurring message delays, the parties following the protocol must agree on a consistent ledger.

Blockchain security has been studied under various consensus protocols (see [BSAB⁺19, GK20] for a survey). Of these, the longest-chain protocol is of great interest, due its heavy use in modern blockchain implementations. The longest-chain protocol has been modeled under various assumptions: for example, discrete time is used in [GKR20, BKM⁺20], and continuous time dynamics is used in [LGR21, Ren19, DKT⁺20]. Further, the protocol has also been studied for a variety of leader election mechanisms in the consensus protocol. For instance, [PSS17, Ren19, GKL20] assume the proof-of-work mechanism, whereas [PS17, KRDO17, FZ17] assume a proof-of-stake mechanism. All these works establish security of the longest-chain protocol for the synchronous communication model, where communication delays are upper bounded by a known finite constant. A common theme among these results is that in the synchronous delay model, the longest-chain protocol is

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‘secure’ under sufficient honest representation, with high probability.

In this chapter, we analyze the impact of message losses on the security of the longest-chain protocol following proof-of-work leader election, by introducing and analyzing an appropriate communication network model. We motivate this by reviewing some existing communication models in the literature and the known security guarantees associated with them.

7.1.1 Related Work

The underlying communication network can delay the successful delivery of peer-to-peer message broadcasts. Popular blockchains such as Bitcoin use the Internet as their communication network. Since this communication is subject to delay, it is natural to model the delays incurred by each block, and study the impact of delay on the security of the longest-chain protocol.

Let $0 \leq i < j$. Let b_i represent the i -th mined honest block. Let $\text{delay}(b_i \rightarrow b_j)$ denote the time taken for block b_i to reach the miner of block b_j , and let β represent the fraction of adversarial computational power in the system. Finally, let λ be the rate at which blocks are mined in the system. Various descriptions of $\text{delay}(b_i \rightarrow b_j)$ lead to different communication network models:

Instantaneous Model The original white-paper by Satoshi Nakamoto [Nak08] assumes an ideal communication channel, i.e. $\text{delay}(b_i \rightarrow b_j) = 0$. In this model, the longest-chain protocol is provably secure when the honest computational power in the system exceeds the adversarial computational power, i.e. when $\beta < 1 - \beta$, or equivalently, when $\beta < 1/2$.

Synchronous Model The model assumes a deterministic delay for each block that is upper bounded by a known constant Δ , i.e., $\text{delay}(b_i \rightarrow b_j) \leq \Delta < \infty$. This delay effectively reduces the growth rate of the chain held by an honest user. Even so, it has been proved [DKT⁺20, GKR20] that the synchronous model is secure with high probability if and only if

$$\beta < \frac{1}{1 + (1 - \beta)\lambda\Delta} (1 - \beta),$$

where λ is the total mining rate of the honest users.

Partially Synchronous Model The partially synchronous model assumes the existence of some unknown and adversarially chosen ‘Global Stabilization Time (GST)’ such that the delays are unbounded before GST, but bounded after it [DLS88]. Therefore, at any time t , the delay satisfies $\text{delay}(b_i \rightarrow b_j) \leq \Delta + \max(0, \text{GST} - t)$. If certain conditions are met, the partially synchronous model is known to be secure with high probability after the Global Stabilization Time [NTT21].

Sleepy Model The sleepy model considers the setting where miners may either be online or offline, and their participation status may change during the execution of the protocol [PS17]. Let

h_i denote the miner of block b_i . The incurred delay is thus

$$\text{delay}(b_i \rightarrow b_j) = \begin{cases} 0 & h_j \text{ is awake when } b_i \text{ is mined} \\ \infty & h_j \text{ is asleep when } b_i \text{ is mined} \end{cases}.$$

Pass and Shi [PS17] showed that consensus can be achieved in the sleepy model with high probability, if a majority of the awake miners at any point in time are honest.

Random Delay Model The random delay model assumes that the point-to-point delays are independent and identically distributed, i.e. $\text{delay}(b_i \rightarrow b_j) \sim \mathbf{X}$, where $\mathbf{X}$ is some known distribution. The longest-chain protocol is shown to be secure with high probability in the random delay model, if the delay distribution satisfies certain conditions and the adversarial representation in the system is below a certain threshold [SGH21].

Except for the random delay model, none of the above models account for the possibility that point-to-point communication may incur infinite delay, i.e. messages may be lost at random. For instance, the sleepy model allows infinite delay for users that are offline, but does not account for noise in the communication process. In contrast, we introduce and analyze a new communication model to study the impact of lost messages on blockchain security.

7.1.2 Contributions

0- ∞ Model We introduce the 0- ∞ model, where the delays are independent and identically distributed over the set $\{0, \infty\}$. Specifically, for any $i, j \geq 0$ such that $i < j$:

$$\text{delay}(b_i \rightarrow b_j) = \begin{cases} 0 & \text{with probability } 1 - d \\ \infty & \text{with probability } d \end{cases}.$$

This simple model postulates that a message sent point-to-point is either immediately received or permanently lost. This delay is independent for each user, and for each block. The modeling choice aligns with our objective of studying the effect of message losses.

We remark that the 0- ∞ model is a special case of the i.i.d. random delay model introduced in [SGH21], which identifies a region of tolerable adversarial power as a function of the delay distribution. Specifically, if d is the probability of message loss and β is the fraction of computational power in the system that is adversarial, it is shown that the 0- ∞ model is secure with high probability when $\beta < \frac{1-2d}{2(1-d)}$. However, this characterization is not tight for the 0- ∞ model, and the analysis in [SGH21] breaks down in the high-noise regime. For example, security of the model cannot be established when $d > 1/2$, i.e. more than half the messages are lost on average.

It is reasonable to wonder if adversarial computational power can at all be tolerated in the high-noise regime, for instance, when almost all messages are lost. Our work answers this question in

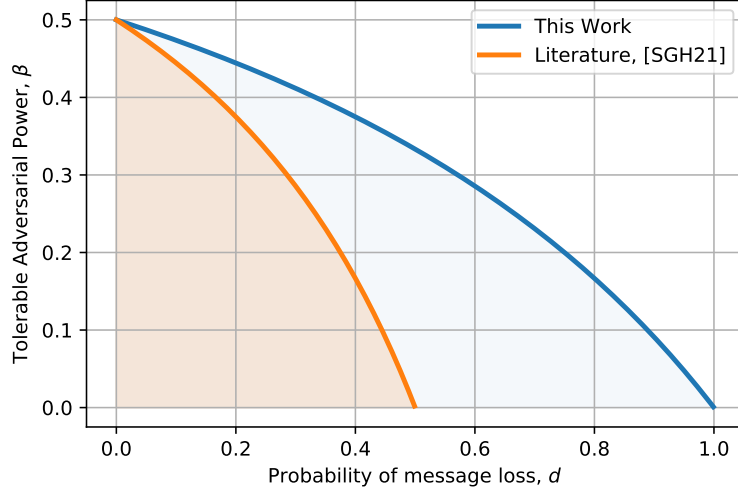


Figure 7.1: Improvement in the region of tolerable adversarial power

the affirmative, by expanding the known security threshold for the $0\text{-}\infty$ model. In particular, our sufficient condition for security is $\frac{\beta}{1-\beta} < 1 - d$. Figure 7.1 shows this improvement.

Our method of analysis is significantly different from that in [SGH21]: we introduce a transmission-graph that captures the history of communication delays between blocks, and identify special paths in the graph that are linked to random variables which are amenable to analysis. Specifically, we identify special objects such as forward-special and backward-special blocks, and associate with them random variables such as forward-unheard and backward-unheard. Our technique also presents a new approach to infer the inclusion of special blocks in the chain held by an honest user through the concept of user-unheard-criterion. The method of analysis is inspired from [DKT⁺20], where security of the synchronous model is established by considering races between honest and adversarial chains. However, our approach does not rely on message delays being finite, and we hope that the tools we introduce are of utility in the study of consensus mechanisms in more general settings, as well as of independent mathematical interest. Our contributions are summarized as:

- We introduce the $0\text{-}\infty$ model as a playground for studying the impact of message losses. This model provides a starting step for more complex models involving message losses.
- We introduce combinatorial objects of independent interest such as the transmission-graph. We also identify random variables (forward-unheard, backward-unheard) associated with this graph that are amenable to analysis, and introduce the user-unheard-criterion. These concepts may be utilized in security analysis of blockchain protocols in more general settings.
- We prove that the longest-chain protocol is secure in the $0\text{-}\infty$ model if certain conditions are met. These conditions are fairly general, and considerably extend the known threshold of tolerable adversarial power. In this regime, we show that the probability of security violation decays almost exponentially in the security parameter.

7.2 Model

In this section, we describe our system model. We consider the setting where infinitely many miners participate in the longest-chain protocol for an infinite duration, and use proof-of-work as the leader election mechanism.

Ledgers, Transactions, Miners, and Blocks Blockchain is the data structure at the heart of the decentralized mechanism to maintain and update a ledger. The ledger is simply an ordered list of transactions. Transactions are assumed to be available to all the miners as soon as they are made. Miners verify the validity of transactions, and update the ledger by grouping the transactions into blocks and linking blocks to form a blockchain. A block is an abstract data structure that contains a hash pointer to a parent block, a cryptographic signature of the block's miner, transactions, and other metadata. The first block in the system is called the genesis block.

Longest-Chain Protocol and Proof-of-Work Miners follow the proof-of-work longest-chain protocol for consensus. Following this protocol, a miner groups any and all transactions that are not included in this longest chain into a block, and attempts to append the block to the longest chain in its view. To do so, it must solve a hash puzzle and include the solution as proof-of-work. If the miner is successful, it broadcasts its chain as a message to other miners over a peer-to-peer network, subject to a communication delay. Upon receiving this message, an honest miner adopts the new chain if the received chain is longer than the chain in the miner's memory. Ties are broken using any deterministic rule, for example, by choosing the chain that terminates in the block that hashes to a lower value. The process continues indefinitely. We assume there are infinitely many miners, and at any finite time, a miner who successfully solves the hash puzzle is doing so for the first time, almost surely.

Parties in the Protocol We refer to parties in the protocol as users. Users that contribute to modifying the ledger through appending blocks to the blockchain are called miners. A miner is either honest or corrupt. Honest miners follow the longest-chain protocol, whereas corrupt miners may deviate from the protocol. For simplicity, all corrupt users are grouped into a single entity called the adversary. This allows corrupt miners to communicate instantaneously, and captures the strong setting of perfect coordination between corrupt miners. The adversary can mine on any previously mined block, but its block must contain the proof-of-work to be valid. It can reveal its chain to any subset of honest users, and can delay its message by arbitrary amounts of time. It can also not include all the transactions it knows about that were not in ancestor blocks. We use β to represent the fraction of computational power in the system that is adversarial.

Mining Process The mining process is abstracted as follows. Let λ denote the total mining rate of the system. We consider a continuous time model where blocks are mined as a Poisson process with rate λ . Since β denotes the fraction of power that is adversarial and since successive mining

instances are independent, adversarial block arrivals follow a Poisson process with rate $\beta\lambda$.

Blockchains and Blocktrees From any block, a unique sequence of blocks leading back to the genesis block can be identified via the hash pointers. We call this sequence a blockchain, or simply a chain. The convention is that the genesis block is the first block of a chain, and the terminating block is called the tip. At any given slot, honest users store a single chain in their memory.

Communication Delays We consider the setting where messages are either instantaneously delivered or permanently lost in an independent and identically distributed manner. Let $\text{delay}(b_i \rightarrow b_j)$ denote the delay incurred by block b_i to reach the miner of block b_j . It is assumed that:

$$\text{delay}(b_i \rightarrow b_j) = \begin{cases} 0, & \text{with probability } 1 - d \\ \infty, & \text{with probability } d \end{cases}.$$

Here, d is the probability of message loss in an instance of point-to-point communication.

7.3 Main results

In this section, we outline the desired security properties and present our main result. We define security on the level of transactions. It is desirable that a transaction eventually makes it to the ledger, and stays permanently at the same position in the ledger. This notion is formalized in Definition 7.1.

Definition 7.1 (Security). Let $\tau > 0$. Let $\mathcal{H}$ be any set of honest users. For any $h \in \mathcal{H}$, let $\mathcal{C}_h(t)$ denote the chain held by user h at time t . We say that a transaction tx made at some time s is $(\tau, \mathcal{H})$ -secure if for any $h_1, h_2 \in \mathcal{H}$ and any $s_1, s_2 > s + \tau$, it holds that tx is included in a block b that is at the same position in $\mathcal{C}_{h_1}(s_1)$ and $\mathcal{C}_{h_2}(s_2)$.

In the literature, security of a transaction is often defined as the confluence of *persistence* and *liveness*. A transaction satisfies liveness if it is eventually added to the ledger, and it satisfies persistence if it remains in the same position in the ledger for all future time, after a confirmation time. We remark that our definition of security implies these notions of persistence and liveness, and is consistent with existing definitions of security, such as in [GKL15, DKT⁺20]. Specifically, if a transaction tx satisfies $(\tau, \mathcal{H})$ -security, then it is part of the chain held by all users in $\mathcal{H}$ before a confirmation time τ time has elapsed. Furthermore, once this confirmation time elapses, the transaction remains at the same position in the ledger for all future time. Our main result shows that if certain conditions are satisfied, then any transaction tx satisfies $(\tau, \mathcal{H})$ security except with a probability that decays almost exponentially in the confirmation time and scales linearly in the size of $\mathcal{H}$. It is stated as Theorem 7.1.

Theorem 7.1 (Main Result). *Let β be the fraction of computational power in the system that is*

adversarial, and d be the probability of message loss. If $\frac{\beta}{1-\beta} < (1-d)$, then for every $\varepsilon > 0$, there exist positive constants a and b such that for all $\tau \geq 0$ and for any honest transaction $\mathbf{tx}$ and any finite set of honest users $\mathcal{H}$:

$$\mathbb{P}(\mathbf{tx} \text{ violates } (\tau, \mathcal{H})\text{-security}) \leq \exp(-a\tau^{1-\varepsilon}) + |\mathcal{H}| \exp(-b\tau).$$

The result states that under a certain threshold of tolerable adversarial power, the probability of security violation for any transaction and any finite set of users decays (almost) exponentially in the confirmation time. Hence, this violation probability can be made arbitrarily small by appropriately selecting the confirmation time. The sufficient condition $\frac{\beta}{1-\beta} < 1-d$ significantly improves the known threshold of tolerable adversarial power for the $0-\infty$ model (Figure 7.1). We also remark that our bound for the probability violation comprises of two terms, the latter of which scales linearly in $|\mathcal{H}|$. This linear scaling is expected, because no single message is successfully transmitted to all users in the model. Therefore, requiring a larger set of users to permanently adopt a transaction in their ledger requires a larger waiting time.

7.4 Definitions and preliminaries

This section introduces key quantities that are used extensively in the analysis. In Section 7.4.1, we introduce the combinatorial objects on which the analysis is performed, such as the main-blocktree and the transmission-graph. Properties of these objects are presented alongside to motivate their purpose. In Section 7.4.2, key random variables that are amenable to analysis, and associated with special paths in the transmission-graph are identified. These quantities are illustrated through an example in Section 7.4.3. In Section 7.4.4, these random variables are used to define ‘catch-up events’, and the notion of ‘ η -Nakamoto blocks’. Finally, Section 7.4.5 introduces the ‘user-unheard-criterion’, which is used to infer useful information about the blockchain held by a user.

7.4.1 Graphs and trees

Three combinatorial objects at the core of our analysis are the main-blocktree and the transmission-graph.

Main-blocktree Any block can be uniquely traced back to the first block in the system (called genesis block). The set of all blocks generated (mined) up till time t forms a directed tree, which we refer to as the main-blocktree and denote it by $\mathbf{MB}(t)$. Here, $\mathbf{MB}(t) = (V_t, E_t)$, where the vertex set V_t is the set of all blocks mined up till time t and the set of directed edges E_t comprises all parent-to-child block pairs. $\mathbf{MB}(t)$ represents the global information about the system, and both honest and adversarial blocks are included in it. Figure 7.2(a) shows an example of the vicinity of the j -th honest block, b_j in some $\mathbf{MB}(t)$.

Definition 7.2 (Heights). Let b_j be the j -th honest block. The height of b_j in a blocktree is the

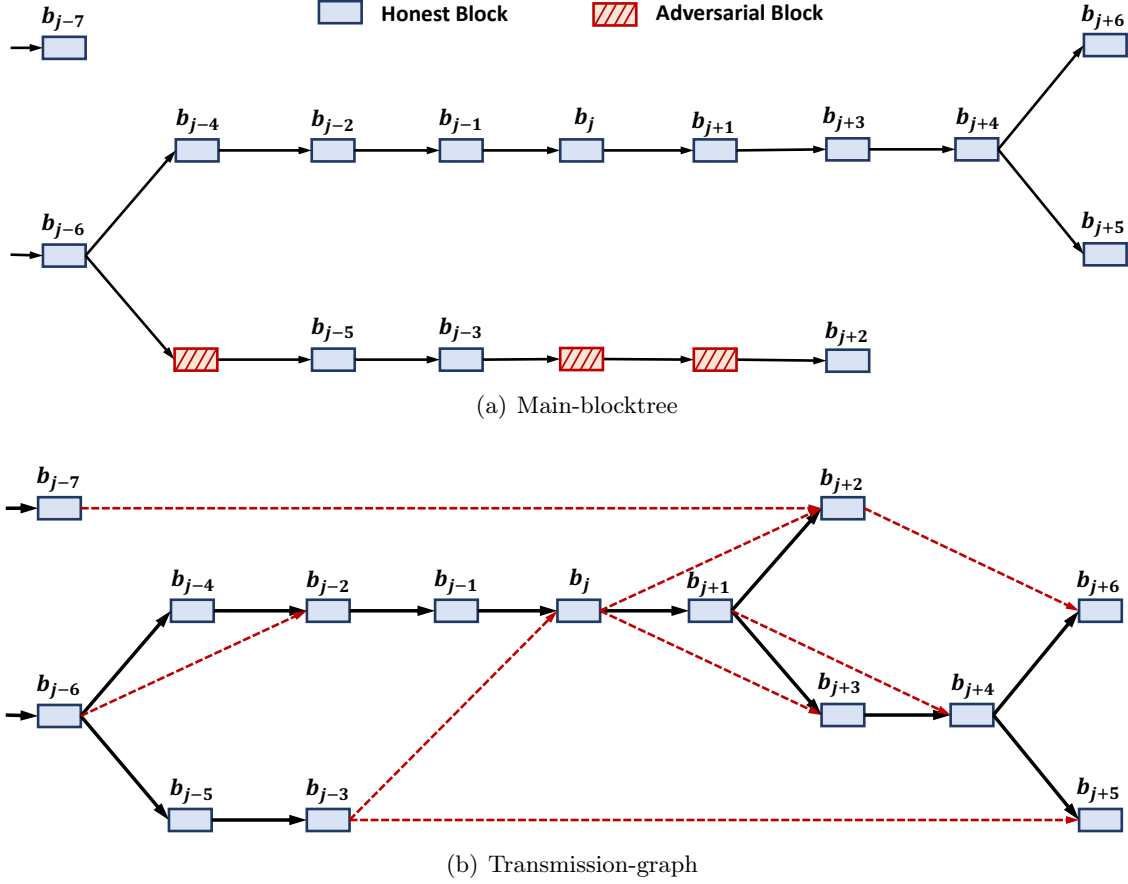


Figure 7.2: Graphs and trees used in analysis

length of the directed path (counting edges) from the genesis block to b_j . We denote the height of b_j in the main-blocktree by $\text{height}_{\text{MB}}(b_j)$.

Transmission-graph At any time t , we associate with the main-blocktree $\text{MB}(t)$, a graph consisting of only honest blocks that we call the transmission-graph. The transmission-graph at time t , denoted $\text{TG}(t)$ is a directed acyclic graph that represents the history of network delays among the honest miners. Here, $\text{TG}(t) = (V_t, E_t)$, where the vertex set V_t is the set of honestly mined blocks up till time t . An edge between b_i and b_j is present if $\text{delay}(b_i \rightarrow b_j) = 0$. A useful observation about $\text{TG}(T)$ is presented in Lemma 7.1.

Lemma 7.1. *Let b_i and b_k be the i -th and k -th honest blocks such that $i < k$. At any time t , suppose there exists a path $\mathcal{A}_{i,k}$ of length n from b_i to b_k in $\text{TG}(t)$, i.e.*

$$\mathcal{A}_{i,k}: b_i = v_0 - v_1 - \dots - v_n = b_k.$$

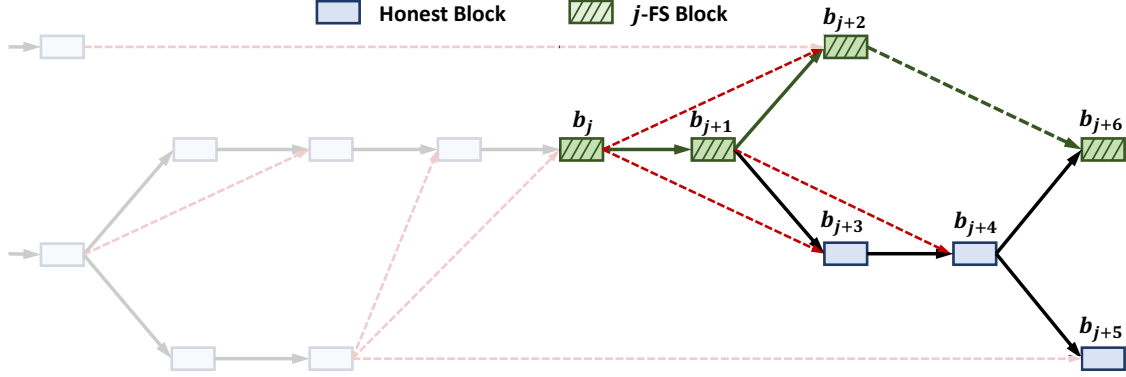


Figure 7.3: j -FS Sequence

Then,

$$\text{height}_{\text{MB}}(b_k) - \text{height}_{\text{MB}}(b_i) \geq n$$

Proof. If v_{i-1} and v_i are two blocks in $\text{TG}(t)$ such that there is an edge from v_{i-1} to v_i , then the miner of block v_i has heard of block v_{i-1} . Therefore, it adds its block at a height greater than that of v_{i-1} , so we have

$$\text{height}_{\text{MB}}(v_i) \geq \text{height}_{\text{MB}}(v_{i-1}) + 1$$

Repeatedly applying this inequality over the path from b_i to b_k yields the desired result. $\square$

7.4.2 Special sequences of honest blocks

Relative to the j -th honest block b_j , we define sequences of special blocks that correspond to forward and backward paths in $\text{TG}(t)$. We also define notions of ‘forward unheard’ and ‘backward unheard’.

Forward Special Blocks

Relative to the j -th honest block b_j , we define a sequence of ‘forward special (FS)’ blocks as follows.

Definition 7.3 (j -FS Sequence). Let $j \geq 0$ and let b_j be the j -th honest block. The j -FS sequence is a sequence of blocks $(b_j^0, b_j^1, b_j^2, \dots)$ such that $b_j^0 = b_j$, and for all $k \geq 1$, b_j^k is the first block to hear of b_j^{k-1} .

We refer to blocks in the j -FS sequence as j -FS blocks. For $0 \leq j < k$, denote by $\text{FS}_j(b_k)$ the number of j -FS blocks mined between b_j and b_k (inclusive). Note that $\text{FS}_j(b_{k-1}) \geq 1$, because b_j is always a j -FS block. The j -FS sequence associated with the transmission-graph in Figure 7.2(b) is shown in Figure 7.3. Notice that $\text{FS}_j(b_{j+5}) = 3$ and $\text{FS}_j(b_{j+6}) = 4$. This example is explored in more detail in Section 7.4.3.

Remark 7.1. Let $0 \leq j \leq k$. Let d be the probability of message loss. The random variable $\text{FS}_j(b_k)$ has the same distribution as $1 + \sum_{i=1}^{k-j} \text{Bern}_i(1-d)$, where $\text{Bern}_i(1-d)$ are i.i.d. Bernoulli random variables with success probability equal to $1-d$. This is because b_j is a j -FS block, and every subsequent block is independently j -FS with probability $1-d$.

Lemma 7.2. *Let $j \geq 0$. The j -FS sequence $(b_j^0, b_j^1, b_j^2, \dots)$ is a forward directed path in the transmission-graph. Further, if $k > i \geq 0$, then the heights of the j -FS blocks b_j^i and b_j^k satisfy:*

$$\text{height}_{\text{MB}}(b_j^k) - \text{height}_{\text{MB}}(b_j^i) \geq k - i$$

Proof. Let $k \geq 0$. Since the miner of b_j^{k+1} has heard of b_j^k , there is an edge from b_j^k to b_j^{k+1} . The conclusion follows from Lemma 7.1. $\square$

Definition 7.4 (Forward Unheard). Let $k \geq j \geq 0$. The Forward Unheard for block b_k with respect to block b_j is denoted $\text{FU}_j(b_k)$ and defined as the number of consecutive j -FS blocks that the miner of b_k has not heard of, going backwards along the j -FS sequence from the last such block mined before b_k . If the miner of b_k has not heard of any j -FS block, then we toss independent biased coins (with failure probability equal to the probability of message loss) and continue to increment the count until a success is encountered.

Remark 7.2. Let $0 \leq j \leq k$. Let d be the probability of message loss. The random variable $\text{FU}_j(b_k)$ has the same distribution as $\text{Geom}(1-d) - 1$, where $\text{Geom}(1-d)$ is a geometric random variable, with minimum value 1. Further, if $k' \geq 0$ such that $k' \neq k$, then $\text{FU}_j(b_k)$ and $\text{FU}_j(b_{k'})$ are independent.

The intuition for defining $\text{FU}_j(b_k)$ as above is illustrated through an example in Section 7.4.3.

Backward Relative Special Honest

Relative to the j -th honest block b_j , we define a sequence of ‘backward special (BS) blocks as follows.

Definition 7.5 (j -BS Sequence). Let $j \geq 0$, and let b_j be the j -th honest block. The j -BS sequence is a sequence of blocks $(b_j^0, b_j^{-1}, b_j^{-2}, \dots)$ such that $b_j^0 = b_j$ and b_j^{-i} is the most recently mined block heard by $b_j^{-(i-1)}$ for all $i \geq 1$.

For $0 \leq i \leq j$, denote by $\text{BS}_j(b_i)$ the number of j -BS blocks mined between b_i and b_j (inclusive). Note that $\text{BS}_j(b_i) \geq 1$, since b_j is always a j -BS block. The j -BS sequence associated with the transmission-graph in Figure 7.2(b) is shown in Figure 7.4. Notice that $\text{BS}_j(b_{j-5}) = 4$ and $\text{BS}_j(b_{j-6}) = 5$. This example is explored in more detail in Section 7.4.3.

Remark 7.3. Let $0 \leq i \leq j$. Let d be the probability of message loss. The random variable $\text{BS}_i(b_j)$ has the same distribution as $1 + \sum_{k=1}^{j-i} \text{Bern}_k(1-d)$, where $\text{Bern}_k(1-d)$ are i.i.d. Bernoulli random variables with success probability equal to $1-d$. This is because b_j is a j -BS block, and every previous block is independently j -BS with probability $1-d$.

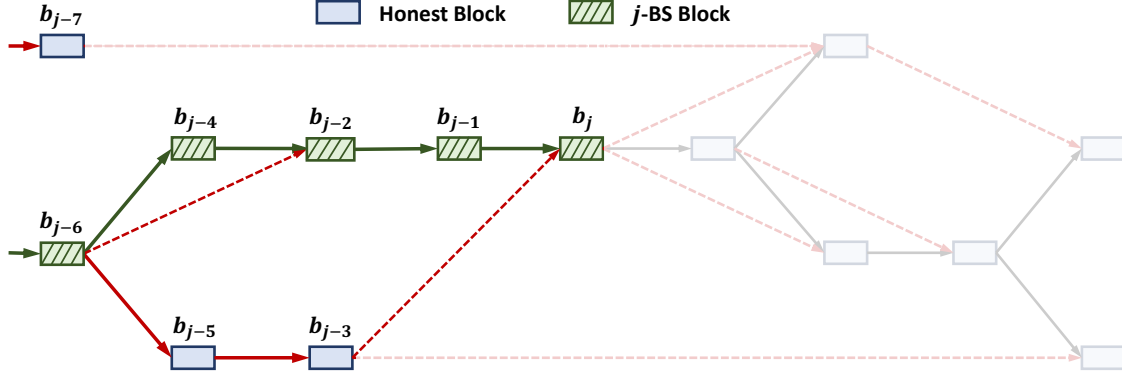


Figure 7.4: j -BS Sequence

Lemma 7.3. Let $j \geq 0$. The j -BS sequence $(b_j^0, b_j^{-1}, b_j^{-2}, \dots)$ is a backward directed path in transmission-graph. Further, if $k > i \geq 0$, then the heights of the j -BS blocks b_j^{-i} and b_j^{-k} satisfy:

$$\text{height}_{\text{MB}}(b_j^{-k}) - \text{height}_{\text{MB}}(b_j^{-i}) \geq k - i$$

Proof. Let $i \geq 0$. Since the miner of $b_j^{-(i-1)}$ has heard of b_j^{-i} , there is an edge from b_j^{-i} to $b_j^{-(i-1)}$. The conclusion follows from Lemma 7.1. $\square$

Definition 7.6 (Backward Unheard). Let $j \geq i \geq 0$. The Backward Unheard for block b_i with respect to block b_j is denoted $\text{BU}_j(b_i)$ and defined as the number of consecutive j -BS blocks whose miners have not heard of b_i , going backwards along the j -BS sequence from the first such block mined after b_i . If none of the miners of j -BS blocks have heard of b_i , then we toss independent biased cones (with failure probability equal to the probability of message loss) and continue to increment the count until a success is encountered.

Remark 7.4. Let $0 \leq i \leq j$. Let d be the probability of message loss. The random variable $\text{BU}_j(b_i)$ has the same distribution as $\text{Geom}(1 - d) - 1$, where $\text{Geom}(1 - d)$ is a geometric random variable, with minimum value 1. Further, if $i' \geq 0$ such that $i' \neq i$, then $\text{BU}_j(b_i)$ and $\text{BU}_j(b_{i'})$ are independent.

The intuition for defining $\text{BU}_j(b_i)$ as above is illustrated through an example in Section 7.4.3. The usefulness of these quantities is evident from Lemma 7.4.

Lemma 7.4. Let $i < j < k$.

- (i) If $\text{FS}_j(b_{k-1}) > \text{FU}_j(b_k)$, then $\text{height}_{\text{MB}}(b_k) - \text{height}_{\text{MB}}(b_j) \geq \text{FS}_j(b_{k-1}) - \text{FU}_j(b_k)$.
- (ii) If $\text{BS}_j(b_{i+1}) > \text{BU}_j(b_i)$, then $\text{height}_{\text{MB}}(b_j) - \text{height}_{\text{MB}}(b_i) \geq \text{BS}_j(b_{i+1}) - \text{BU}_j(b_i)$.

Proof. We prove statement (i). Let $m = \text{FS}_j(b_{k-1}) - \text{FU}_j(b_k) > 0$. Consider the j -FS sequence

$(b_j^0, b_j^1, \dots, b_j^m, \dots)$. Note that $\text{FS}_j(b_{k-1})$ is the number of j -FS blocks mined before b_k , and $\text{FU}_j(b_k)$ is the number of consecutive blocks from this sequence that were not heard by b_k going backward. Therefore, b_k has heard of b_j^m , and will mine at a greater height. Since the j -FS blocks are all mined at different heights, we have

$$\text{height}_{\text{MB}}(b_k) - \text{height}_{\text{MB}}(b_j) > \text{height}_{\text{MB}}(b_j^m) - \text{height}_{\text{MB}}(b_j^0) \geq m = \text{FS}_j(b_{k-1}) - \text{FU}_j(b_k),$$

as desired. The proof of statement (ii) is essentially the same, because the forward sequences FS_j and FU_j map to the backward sequences BS_j and BU_j under reversing the directions of the edges in the transmission-graph. $\square$

7.4.3 An example

Some of the concepts introduced above are best understood through an example. Consider the main-blocktree in Figure 7.2(a) and its associated transmission-graph in Figure 7.2(b). The same transmission-graph is shown again in Figure 7.5, where the j -FS and j -BS blocks are highlighted. Recall that in the transmission-graph, a directed edge from b_j to b_k indicates that the miner of b_k has heard of the block b_j .

j -FS Sequence By definition, b_j is a j -FS block. The miner of the next block b_{j+1} has heard of the most recent j -FS block b_j , so $b_j^1 = b_{j+1}$ is a j -FS block. The miner of b_{j+2} has heard of b_j^1 , so we have that $b_{j+2} = b_j^2$ is also a j -FS block. However, the miners of b_{j+3} , b_{j+4} and b_{j+5} have not heard of b_j^2 , so these blocks are not j -FS. Finally, the miner of b_{j+6} has heard of b_j^2 , so we have that $b_j^3 = b_{j+6}$. Notice that no two j -FS blocks can share the same height.

Forward Unheard Let us consider $\text{FU}_j(b_k)$ with $k = j + 4$. Starting from b_k , we count the number of consecutive j -FS blocks that the miner of b_k has not heard of, going backwards in the j -FS sequence. The j -FS sequence at the mining time of b_k is (b_j, b_{j+1}, b_{j+2}) . Going backwards in this sequence, we see that the miner of b_k has not heard of b_{j+2} , but has heard of b_{j+1} . Therefore, we stop the count and have $\text{FU}_j(b_{j+4}) = 1$.

j -BS Sequence By definition, b_j is a j -BS block. Since the miner of the most recent j -BS block (b_j) has heard of the previous block b_{j+1} , we have that $b_{j-1} = b_j^{-1}$ is a j -BS block. Similarly, the miner of b_j^{-1} has heard of b_{j-2} , so we have that $b_{j-2} = b_j^{-2}$ is a j -BS block. However, the miner of b_j^{-2} has not heard of b_{j-3} , and so b_{j-3} is not a j -BS block. Notice that no two j -BS blocks can share the same height.

Backward Unheard Let us consider $\text{BU}_j(b_i)$ with $i = j - 3$. Starting from the first j -BS block mined after b_i , we count the number of consecutive j -BS blocks that have not heard of b_i , going backwards in the j -BS sequence. The j -BS sequence at the mining time of b_i is b_j, b_{j-1}, b_{j-2} . Going

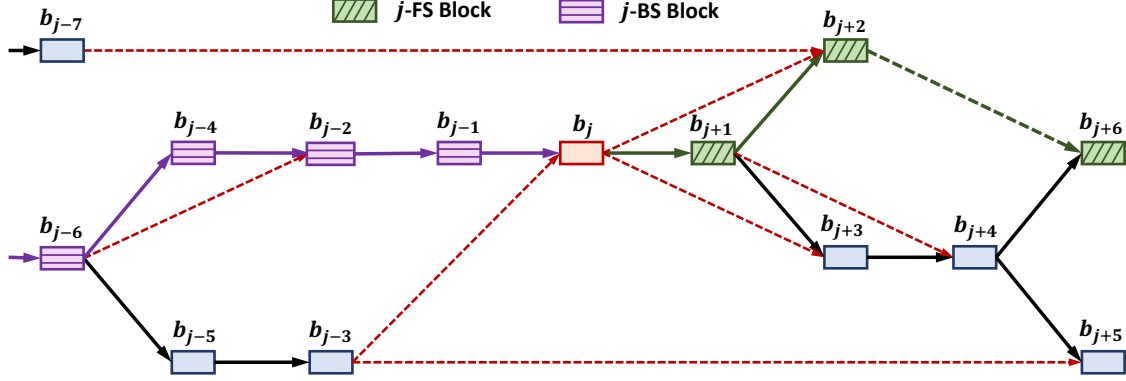


Figure 7.5: Example of j -FS and j -BS sequence

backwards in this sequence, we see that the miners of b_{j-2} and b_{j-1} have not heard of b_i , but the miner of b_j has. Therefore, we stop the count and have $\text{BU}_j(b_{j-3}) = 2$.

7.4.4 Catch-up events and η -Nakamoto blocks

In this section, we define catch-up events and η -Nakamoto blocks.

Adversarial arrivals Let $a(b_i, b_j)$ denote the number of adversarial blocks mined between the mining times of the i -th and j -th honest blocks. Similarly, let $a(b_j^i, b_j^k)$ denote the number of adversarial blocks mined between the i -th and k -th j -FS blocks.

Definition 7.7 (Catch-up Events). Let $0 \leq i < j < k$, and let $0 < \eta \leq 1$. The forward and backward catch-up events are respectively defined as:

$$\vec{\text{B}}_{j,k}^{(\eta)}: a(b_j, b_k) \geq \eta \cdot \text{FS}_j(b_{k-1}) - \text{FU}_j(b_k) \quad (7.1)$$

$$\overleftarrow{\text{B}}_{i,j}^{(\eta)}: a(b_i, b_j) \geq \eta \cdot \text{BS}_j(b_{i+1}) - \text{BU}_j(b_i) \quad (7.2)$$

These events are catch-up events in the following sense. If the event $\vec{\text{B}}_{j,k}^{(\eta)}$ occurs for some $j < k$ and $0 < \eta \leq 1$, then more adversarial blocks have been mined in $[\tau_j, \tau_k]$ than *effective* j -FS blocks. For instance, when $\eta = 1$, more adversarial blocks have been mined in $[\tau_j, \tau_k]$ than the number of blocks in the j -FS sequence from b_j up to the last j -FS block heard by the miner of b_k . If these adversarial blocks were to form a side chain rooted at b_j , then there is a possibility that b_k mines on this side chain. However, such an attack would fail if the catch-up events did not occur, because there would necessarily be a chain longer than the adversarial side chain that the miner of b_k is aware of. Here, η is a robustness measure: If the catch-up event does not occur for a small value of η , then the honest blocks have a considerable lead over adversarial side chains.

Longest Chains of the main-blocktree A longest chain of the main-blocktree at time t is a path in $\text{MB}(t)$ whose length is no shorter than any path in $\text{MB}(t)$. Notice that $\text{MB}(t)$ can have

multiple longest chains.

η -Nakamoto Blocks A desirable property for honest block b_j is the existence of some $0 < \eta \leq 1$, such that none of the events $\vec{B}_{j,k}^{(\eta)}$ occur for any $i < j$ and none of the events $\overleftarrow{B}_{i,j}^{(\eta)}$ occur for any $k > j$. Intuitively, this ensures that adversarial chains rooted at any block b_i with $i < j$ are never long enough for any block b_k with $k > j$ to extend them. In turn, this suggests that block b_j would be in every longest chain of the main-blocktree right from being mined. We formalize this property using the notion of η -Nakamoto blocks below, and make this intuition rigorous in Section 7.3.

Definition 7.8 (η -Nakamoto Block). Let $j \geq 0$ and $0 < \eta \leq 1$. The j -th honest block b_j is said to be an η -Nakamoto block if the event

$$N_j^{(\eta)} : \left[\bigcap_{i: i < j} [\overleftarrow{B}_{i,j}^{(\eta)}]^c \right] \cap \left[\bigcap_{k: k > j} [\vec{B}_{j,k}^{(\eta)}]^c \right] \quad (7.3)$$

occurs.

We remark that an η -Nakamoto block is also an η' -Nakamoto block, for any $\eta' \in [\eta, 1]$.

7.4.5 User-unheard-criterion

η -Nakamoto blocks are useful because they belong to every longest chain of the main-blocktree. However, this is not equivalent to η -Nakamoto blocks belonging to any *user's* chain for all future time. This is because at any given time, a given user has not necessarily heard of all the blocks in $MB(t)$. In this section, we introduce the tools that are relevant to analyzing the state of a user's chain with respect to the main-blocktree. We begin by introducing ‘Unheard’ with respect to a user.

Definition 7.9 (User-unheard, $\text{Unheard}_h(b_j^k)$). Let $j \geq 0$ and let b_j be the j -th honest block. Let $(b_j^0, b_j^1, \dots)$ denote the j -FS sequence. Let h be an honest user. For any $k \geq 0$, we define $\text{Unheard}_h(b_j^k)$ as the number of consecutive j -FS blocks that the user h has not heard of, going backwards along the j -FS sequence from b_j^k . If h has not heard of any j -FS block, then we toss independent biased coins (with failure probability equal to the probability of message loss) and continue to increment the count until a success is encountered.

Remark 7.5. Let $j, k \geq 0$. Let h be any honest user and d be the probability of message loss. Let $\text{Geom}(1 - d)$ be a geometric random variable, with minimum value 1. The distribution of the random variable $\text{Unheard}_h(b_j^k)$ depends on h :

- If h has not mined any block after the mining time of b_j , then $\text{Unheard}_h(b_j^k)$ has the same distribution as $\text{Geom}(1 - d) - 1$.

- If h is a miner of a j -FS block, then h has heard of its own block, as well as the j -FS block that immediately preceded it. However, all other delays from the miners of j -FS blocks to h are still independent and identically distributed. In this case, $\text{Unheard}_h(b_j^k)$ is stochastically dominated by $\text{Geom}(1 - d)$.
- If h is a miner of a block b_ℓ with $\ell > j$, such that b_ℓ is not a j -FS block, then the delay from the most recent j -FS block before b_ℓ to h is infinity. However, all other delays from the miners of j -FS blocks to h are still independent and identically distributed. In this case, $\text{Unheard}_h(b_j^k)$ is stochastically dominated by $\text{Geom}(1 - d)$.

In all cases, we have $\text{Unheard}_h(b_j^k) \leq \text{Geom}(1 - d)$, where the $\leq$ sign indicates stochastic domination.

We compare ‘forward-unheard’ (denoted $\text{FU}_j(b_k)$) and ‘user-unheard’ (denoted $\text{Unheard}_h(b_j^k)$). Although similar in spirit, the quantity $\text{FU}_j(b_k)$ counts the number of consecutive j -FS blocks not heard by the *miner* of block b_k , going backwards in the j -FS sequence from the last such block before b_k , whereas the quantity $\text{Unheard}_h(b_j^k)$ fixes an honest user h and similarly counts the number of consecutive such blocks not heard by h . Since b_k and b_ℓ are mined by two different miners, then $\text{FU}_j(b_k)$ and $\text{FU}_j(b_\ell)$ are independent random variables. However, $\text{Unheard}_h(b_j^k)$ and $\text{Unheard}_h(b_j^\ell)$ need not be independent.

Next, we introduce the user-unheard-criterion, which will later allow us to infer useful information about the state of a user’s chain from the main-blocktree.

Definition 7.10 (User-unheard-criterion). Let h be an honest user. Let $j \geq 1$, $k_0 \geq 1$, and $0 < \eta < 1$. We say that the (h, j, η, k_0) -user-unheard-criterion is satisfied if

$$\text{Unheard}_h(b_j^k) < \left(\frac{1 - \eta}{2}\right) k, \quad \forall k \geq k_0, \quad (7.4)$$

7.5 Proof outline

In this section, we present an outline of the proof of Theorem 7.1. First, deterministic and probabilistic results are stated. These results are used as building blocks in the proof sketch of our main result, Theorem 7.1 which is presented in Section 7.5.3. Rigorous proofs of all the results, including the main result, are relegated to the appendix.

7.5.1 Deterministic results

Theorem 7.2. Let $j \geq 1$ and $0 < \eta \leq 1$. Let b_j be the j -th honest block, and τ_j be the mining time of b_j . Suppose b_j is an η -Nakamoto block. Then:

- (i) b_j is the unique honest block at its height in $\text{MB}(t)$ for all $t \geq \tau_j$.

(ii) b_j is in every longest chain of $\text{MB}(t)$ for all $t \geq \tau_j$.

(iii) b_j is the unique block at its height in $\text{MB}(\tau_j)$.

(iv) For any k such that $k > j$, b_k is a descendant of b_j in the main-blocktree.

Theorem 7.3. *Let h be an honest user. Let $j \geq 1$ and $0 < \eta < 1$. Let $k_0 = \lceil \frac{2\eta}{1-\eta} \rceil$. Let $\mathcal{C}_h(t)$ denote the chain held by h at time t . If b_j is an η -Nakamoto block, and if the (h, j, η, k_0) -user-unheard-criterion is satisfied, then $b_j \in \mathcal{C}_h(t)$ for all $t \geq \tau_j^{k_0}$, where $\tau_j^{k_0}$ is the mining time of the k_0 -th j -FS block.*

7.5.2 Probabilistic results

Lemma 7.5. *Let $0 \leq i < j \leq j' < k$, and $0 \leq \eta \leq 1$. The events:*

- $\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}$ and $\overrightarrow{\mathbf{B}}_{j',k}^{(\eta)}$ are independent.
- $\overrightarrow{\mathbf{B}}_{i,j}^{(\eta)}$ and $\overleftarrow{\mathbf{B}}_{j',k}^{(\eta)}$ are independent.

Theorem 7.4. *Let $j \geq 1$ and $0 \leq \eta \leq 1$. Recall that b_j is the j -th honest block and $\mathbf{N}_j^{(\eta)}$ is the event that b_j is an η -Nakamoto block. If $\frac{\beta}{1-\beta} < \eta \cdot (1-d)$, then there exists a positive constant $p_0 > 0$ such that*

$$\mathbb{P}\left(\mathbf{N}_j^{(\eta)}\right) \geq p_0 > 0$$

Lemma 7.6. *Let $0 < \eta \leq 1$ and $i < j < k$. Let $\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}$ and $\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}$ be the catch-up events defined in (7.1), (7.2). If $\frac{\beta}{1-\beta} < \eta \cdot (1-d)$, there exists a constant $c > 0$ such that*

$$\begin{aligned} \mathbb{P}\left(\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}\right) &\leq e^{-c(k-j)} \\ \mathbb{P}\left(\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}\right) &\leq e^{-c(j-i)} \end{aligned}$$

Theorem 7.5. *Let $0 < \eta \leq 1$. Let β be the fraction of computational power in the system that is adversarial and d be the probability of message loss. Let $B_{s,s+t}^{(\eta)}$ be the event that there are no η -Nakamoto blocks in $[s, s+t]$. If $\frac{\beta}{1-\beta} < \eta \cdot (1-d)$, then there exists a constant $c_0 > 0$ such that for any $s, t \geq 0$,*

$$\mathbb{P}\left(B_{s,s+t}^{(\eta)}\right) \leq e^{-c_0\sqrt{t}}$$

Theorem 7.6. *Let $0 < \eta \leq 1$. Let β be the fraction of computational power in the system that is adversarial and d be the probability of message loss. Let $B_{s,s+t}^{(\eta)}$ be the event that there are no η -Nakamoto blocks in $[s, s+t]$. If $\frac{\beta}{1-\beta} < \eta \cdot (1-d)$, then for every $\varepsilon > 0$, there exist positive*

constants A, a such that for any $s, t > 0$,

$$\mathbb{P}\left(B_{s,s+t}^{(\eta)}\right) \leq A \exp\left(-at^{1-\varepsilon}\right).$$

Theorem 7.7. *Let $0 < \eta < 1$. Suppose the fraction β of computational power in the system that is adversarial, and the probability d of message loss satisfies $\frac{\beta}{1-\beta} < \eta \cdot (1-d)$. Given $s \geq 0$, let b_J be the first η -Nakamoto block mined after time s . There exist constants $C, c > 0$ such that for any honest user h and for all $k' \geq 1$,*

$$\mathbb{P}\left(\left(h, J, \eta, k'\right)\text{-user-unheard-criterion fails}\right) \leq Ce^{-ck'}$$

7.5.3 Proof sketch

η -Nakamoto blocks are special blocks that are part of every longest chain in the main-blocktree, for all time after they are mined (Theorem 7.2). Therefore, if a transaction is included in an η -Nakamoto block or in any of its ancestors, then it will be included in every longest chain of the main-blocktree. Furthermore, if an honest user is up-to-date with the main-blocktree (specifically, by satisfying a relevant user-unheard-criterion, and therefore having heard of more forward special blocks with respect to the η -Nakamoto block, than adversarial blocks), then it is reasonable to expect that the honest user will also include the transaction in its chain. This idea is formalized in the user-unheard-criterion, and made rigorous in Theorem 7.3.

The core idea behind the proof of the main result (Theorem 7.1) is to show that η -Nakamoto blocks occur frequently with high probability (Theorem 7.6). In turn, this implies that a transaction is highly likely to be included in an η -Nakamoto block or its ancestor, soon after it is made. Theorem 7.7 then shows that it is also highly likely that a relevant user-unheard-criterion will hold for the honest user, once a waiting time has elapsed.

To prove Theorem 7.6, we bootstrap from a milder version of it, which is stated as Theorem 7.5. Inspired by the proof strategy in [DKT⁺20], we prove Theorem 7.5 by separating catch-up events into long and short-term catch-up events. We show that long term catch-ups occur rarely as a consequence of Lemma 7.6, and short-term catch-up probabilities are bounded using a lower bound on the probability that the j -th block is an η -Nakamoto block (Theorem 7.4), and the fact that non-overlapping catch-up events are independent (Lemma 7.5). This establishes that a transaction is highly likely to be included in the main-blocktree for all future time, once a waiting period has elapsed. This proves Theorem 7.6.

Next, we show that an honest user is likely to always be up-to-date with the main-blocktree, forever after a waiting time. This is done by explicitly bounding the probability that a relevant user-unheard-criterion is violated using tools from stochastic analysis. This proves Theorem 7.7.

Theorems 7.6 and 7.7 together imply the main result. The details of this proof are presented in

Appendix [E.2.7](#).

7.6 Conclusion

In this chapter, we introduced the $0-\infty$ model: a framework to study the impact of random message losses on the security of the proof-of-work longest-chain protocol. We investigated the security of this protocol by analyzing the transmission-graph, a dynamically evolving graph that captures the delays incurred by the blocks mined by honest miners. Specifically, we studied special sequences of blocks and identified random variables associated with them that are amenable to analysis. These random variables were used to define useful objects and desirable events, such as η -Nakamoto blocks and user-unheard-criterion respectively. These ideas allowed us to generalize analysis techniques from the synchronous delay model to a setting where delays are possibly infinite. We showed that the condition $\frac{\beta}{1-\beta} < 1 - d$ is sufficient for a transaction to satisfy desired security properties except with a probability that decays almost exponentially in the security parameter. This greatly improved the known threshold of the fraction of adversarial power that is tolerable for a given probability of message loss in an instance of point-to-point communication.

Chapter 8

Conclusions

A recurring question in this thesis is: given networks drawn from a structured model, what can be inferred about their latent structure, and where does that inferential capability break down? The chapters answered this question in distinct settings: multi-graph matching, multi-graph correlation detection, planted phase-coherent structure in circular data, matching under node corruption, and longest-chain consensus under message loss. A handful of observations connect them.

A takeaway from Part I of the thesis is that multiple correlated views of the same latent structure carry information that no pair of views contains, and joint analysis can succeed where pairwise analysis cannot. In Chapter 3, exact graph matching across m correlated Erdős–Rényi graphs at edge probability $p = C \log n/n$ is shown to be feasible exactly when $Cs(1 - (1 - s)^{m-1}) > 1$. For $m = 2$ this collapses to the classical condition $Cs^2 > 1$ [CK17]; for $m \geq 3$, the threshold is strictly weaker, and a parameter regime opens in which no pair of graphs admits exact matching but the m -graph problem does. A parallel phenomenon shows up for correlation detection on Gaussian-weighted graphs in Chapter 4, where the detection threshold scales as $\rho^2 \gtrsim (\log n)/(mn)$, an improvement over the two-graph case that may turn an infeasible problem into a feasible one.

A takeaway from Part II of the thesis is that deviations from an idealized model are reflected in performance guarantees, both through the *magnitude* of the deviation (which typically enters the guarantee as a continuously varying bound), and through its *precise form*, which can determine qualitative behavior. In Chapter 6, recovery guarantees for graph matching depend continuously on the fraction γ of corrupted nodes, but the qualitative behavior is highly sensitive to the type of corruption: under random corruption, the corrupted nodes are themselves identifiable and the uncorrupted subgraph is exactly recoverable; under adversarial corruption at the same value of γ , identification is information-theoretically impossible, though a positive fraction of nodes can still be matched. In Chapter 7, the security of the longest-chain protocol traces a continuous boundary in the message-loss rate d and the adversarial computational fraction β , with the explicit threshold $\beta/(1 - \beta) < 1 - d$ improving on the prior state of the art for the same model. The precise form of

deviation here is a lossy communication model, where message is either instantaneously delivered or permanently lost.

8.1 Directions for future research

The individual chapters outline specific problems stemming from each of them that make for meaningful future research. Presently, we identify three broader research questions, not specific to any particular chapter, that are raised by the overall conclusions of this thesis.

Partial recovery with multiple graphs. The thesis studied multi-graph effects through the exact recovery problem in the Bernoulli (Erdős–Rényi) case and the detection problem in the Gaussian case. The problem of *partial recovery*, in which only a positive fraction of nodes need be matched correctly across all m graphs, is a complementary problem in the multi-graph regime. Vassaux and Massoulié [VM25] recently established an impossibility result for partial recovery with multiple Bernoulli graphs, and an achievability result for the analogous detection problem in the Bernoulli setting was given in [Och26]; matching converse and achievability bounds, however, remain to be established. The unifying question is how the partial recovery threshold scales with the number of graphs m , and in particular whether it coincides with the detection threshold, as is the case in the two-graph setting. Settling this would clarify whether multi-graph data carry information that partial-recovery algorithms can convert into matched nodes, or whether there is a real gap between knowing that the graphs are correlated and being able to recover any positive fraction of the underlying correspondence. The answer would also bear directly on the design of multi-graph matching algorithms, since most efficient algorithms in the two-graph case are partial-recovery algorithms by construction.

Information–computation gaps in multi-graph matching. The achievability results of Chapter 3 rest on the k -core estimator, which is not known to admit a polynomial-time implementation. In the two-graph case, the gap between the information-theoretic threshold and what is achievable in polynomial time has been substantially closed by recent work. The chandelier counting algorithm of [MWXY23] and related efficient algorithms [DL23] attain the information-theoretic threshold for two-graph matching in polynomial time, under appropriate conditions. Some of these algorithms also have the property of being *precise*: they make no errors on the matching they commit to. Precision is exactly the structural property that the transitive closure step of Chapter 3 relies on to combine pairwise matchings into a global matching without compounding errors. No analogous polynomial-time algorithm is known for $m \geq 3$ correlated graphs. A natural and apparently approachable question is whether the transitive closure framework can be instantiated with one of these precise polynomial-time pairwise estimators in place of the k -core estimator, and whether the resulting composition reaches the information-theoretic threshold in appropriate parameter regimes. A positive answer would close the practical-versus-fundamental gap for multi-graph matching. A similar question is relevant for the related detection problem: the generalized likelihood ratio test

in Chapter 4 requires looping over all permutations and is not efficient. Can its information-theoretic guarantees be realized with a computationally efficient algorithm?

Towards algorithms that work on real networks. Random graph models are idealized models; the Erdős–Rényi model is the simplest. The stochastic block model and the inhomogeneous random graph add a layer of structure that brings the model closer to real networks but is still highly idealized. The robustness considerations of Chapters 6 and 7 add another layer, by studying how guarantees survive when a fraction of the graph or the channel deviates from these idealizations. Real networks, however, deviate from the models in further ways that the thesis does not address. Degree distributions are heavy-tailed; edges and nodes carry attributes and weights; the underlying network evolves in time; and side information, such as seed correspondences or partial labels, is often available. Each of these directions has been studied in isolation in various places in the literature, but a complete picture of how they interact is missing. The broader question is to what extent the analytical insights collected here actually inform the design of algorithms that perform well on empirical network data, where the simplifying assumptions break in several directions simultaneously. The long-term motivation for pushing theory in any of the directions above is to close this gap.

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Appendix A

Supplementary material for Chapter 3

A.1 Maximum likelihood estimator for exactly matching m homogeneous graphs

We show that in the homogeneous setting, the maximum likelihood estimator is given by the permutation profile that minimizes the number of edges in the corresponding union graph, i.e.

$$(\hat{\pi}_{12}^{\text{MLE}}, \dots, \hat{\pi}_{1m}^{\text{MLE}}) \in \arg \min_{\pi_{12}, \dots, \pi_{1m}} |E(G_1 \vee G_2^{\pi_{12}} \vee \dots \vee G_m^{\pi_{1m}})|.$$

Lemma A.1. *Let $G_1, \dots, G_m$ be correlated Erdős-Rényi graphs obtained from the subsampling model with parameters p and s . Further, let $\pi_{12}, \dots, \pi_{1m}$ denote a collection of permutations on $[n]$. Then*

$$\log \mathbb{P}(G_1, \dots, G_m \mid \pi_{12}^* = \pi_{12}, \dots, \pi_{1m}^* = \pi_{1m}) \propto \text{const.} - |E(G_1 \vee G_2^{\pi_{12}} \vee \dots \vee G_m^{\pi_{1m}})|,$$

where const. depends only on p , s and $G_1, \dots, G_m$.

Proof. Consider the setting where $p_{ij} = p$ for all $i \neq j$. Notice that

$$\begin{aligned} \mathbb{P}(G_1, \dots, G_m \mid \pi_{12}^*, \dots, \pi_{1m}^*) &= \prod_{e \in \binom{[n]}{2}} \mathbb{P}(G_1(e), G_2(\pi_{12}^*(e)) \dots, G_m(\pi_{1m}^*(e)) \mid \pi_{12}^*, \dots, \pi_{1m}^*) \\ &= \prod_{e \in \binom{[n]}{2}} \mathbb{P}(G_1(e), G'_2(e) \dots, G'_m(e)) \end{aligned} \tag{A.1}$$

where for a node pair $e = \{u, v\}$, the shorthand $\pi(e)$ denotes $\{\pi(u), \pi(v)\}$. The edge status of any node pair e in the graph tuple $(G_1, G'_2, \dots, G'_m)$ can be any of the 2^m bit strings of length m , but the corresponding probability in (A.1) depends only on the number of ones and zeros in the bit string. For $i \in [m]$, let α_i denote the number of node pairs e whose corresponding tuple

$(G_1(e), G'_2(e), \dots, G'_m(e))$ has exactly i 1's:

$$\alpha_i \triangleq \sum_{e \in \binom{[n]}{2}} \mathbf{1} \{ (G_1(e), G'_2(e), \dots, G'_m(e)) \text{ has exactly } i \text{ 1's} \}.$$

Two key observations are in order. First, it follows by definition that $\alpha_0 + \alpha_1 + \dots + \alpha_m = \binom{n}{2}$. Second, by definition of α_i , it follows that

$$\sum_{i=0}^m i\alpha_i = \sum_{e \in \binom{[n]}{2}} \sum_{j=1}^m G_j(e) = \sum_{e \in \binom{[n]}{2}} \sum_{j=1}^m G'_j(e) \quad (\text{A.2})$$

is constant, independent of $\pi_{12}^*, \dots, \pi_{1m}^*$. It follows then that

$$\begin{aligned} (\text{A.1}) &= (1 - p + p(1 - s)^m)^{\alpha_0} \times \prod_{i=1}^m (ps^i(1 - s)^{m-i})^{\alpha_i} \\ &= (1 - p + p(1 - s)^m)^{\alpha_0} \times p^{\sum_{i=1}^m \alpha_i} \times \prod_{i=1}^m (s^i(1 - s)^{m-i})^{\alpha_i} \\ &= (1 - p + p(1 - s)^m)^{\alpha_0} \times p^{\binom{n}{2} - \alpha_0} \times \left(\frac{s}{1 - s} \right)^{\sum_{i=1}^m i\alpha_i} \times (1 - s)^{m \sum_{i=1}^m \alpha_i} \\ &= \left(\frac{1 - p + p(1 - s)^m}{p(1 - s)^m} \right)^{\alpha_0} \times (p(1 - s)^m)^{\binom{n}{2}} \times \left(\frac{s}{1 - s} \right)^{\sum_{i=1}^m i\alpha_i} \\ &\propto \left(1 + \frac{1 - p}{p(1 - s)^m} \right)^{\alpha_0}, \end{aligned}$$

where the last step uses (A.2). Finally, since $\frac{1-p}{p(1-s)^m} > 0$, it follows that the log-likelihood satisfies

$$\log(\mathbb{P}(G_1, \dots, G_m \mid \pi_{12}^*, \dots, \pi_{1m}^*)) \propto \text{const.} + \alpha_0,$$

i.e. maximizing the likelihood corresponds to selecting $\pi_{12}, \dots, \pi_{1m}$ to maximize α_0 - the number of node pairs e for which $G_1(e) = G_2(\pi_{12}(e)) = \dots = G_m(\pi_{1m}(e)) = 0$. This is equivalent to minimizing the number of edges in the union graph $G_1 \vee G_2^{\pi_{12}} \vee \dots \vee G_m^{\pi_{1m}}$, as desired. $\square$

Remark A.1. In the case of two graphs, minimizing the number of edges in the union graph $G_1 \vee_{\pi} G_2$ is equivalent to maximizing the number of edges in the intersection graph $G_1 \wedge_{\pi} G_2$. This is consistent with existing literature on two graphs [CK16, WXY22].

A.2 On negative results for exact recovery with two graphs

We build up to the proof of Lemma 3.1 by presenting first a useful lemma about inhomogeneous random graphs.

Lemma A.2. *Let $G \sim \text{RG}(n, \mathbf{Q})$ be an inhomogeneous random graph, and let $N \triangleq \binom{K}{2}$, where K*

is the number of isolated nodes in G . Then,

$$\mathbb{P}(N \geq 1) \geq \exp(-(2n-4)p_{\max}^2 - 6p_{\max}) \cdot \frac{\mu^2 - 2\mu \max_{1 \leq i \leq n} e^{-d_i}}{4(\mu^2 + \mu + 1)}, \quad (\text{A.3})$$

where $\mu \triangleq \sum_{i=1}^n e^{-d_i}$.

Proof. Let Z_i denote the indicator random variable that node i is isolated in G . Further, let $\tilde{Z}_i \sim \text{Bern}(e^{-d_i})$ denote a collection of independent random variables and define $\tilde{N} \triangleq \sum_{1 \leq i < j \leq n} \tilde{Z}_i \cdot \tilde{Z}_j$. We use the second moment method by relating the moments of N and $\tilde{N}$.

Bounding the second moment. Notice that

$$\mathbb{E}[N^2] = \sum_{1 \leq i < j \leq n} \sum_{1 \leq k < \ell \leq n} \mathbb{E}[Z_i Z_j Z_k Z_\ell], \quad (\text{A.4})$$

$$\mathbb{E}[\tilde{N}^2] = \sum_{1 \leq i < j \leq n} \sum_{1 \leq k < \ell \leq n} \mathbb{E}[\tilde{Z}_i \tilde{Z}_j \tilde{Z}_k \tilde{Z}_\ell]. \quad (\text{A.5})$$

Consider the following three exhaustive cases.

– *Case 1:* $i = k$ and $j = \ell$. Notice that

$$\mathbb{E}[\tilde{Z}_i \tilde{Z}_j \tilde{Z}_k \tilde{Z}_\ell] = \mathbb{E}[\tilde{Z}_i^2] \cdot \mathbb{E}[\tilde{Z}_j^2] = e^{-d_i - d_j}.$$

Therefore,

$$\mathbb{E}[Z_i Z_j Z_k Z_\ell] = (1 - p_{ij}) \prod_{\substack{(i,u) \\ u \neq j}} (1 - p_{iu}) \prod_{\substack{(j,v) \\ v \neq i}} (1 - p_{jv}) \stackrel{(a)}{\leq} e^{-d_i - d_j + p_{ij}} \stackrel{(b)}{\leq} e^{6p_{\max}} \cdot \mathbb{E}[\tilde{Z}_i \tilde{Z}_j \tilde{Z}_k \tilde{Z}_\ell], \quad (\text{A.6})$$

where (a) uses that $1 - x \leq e^{-x}$ for all x , and (b) uses that $p_{ij} \leq p_{\max} \leq 6p_{\max}$.

– *Case 2:* $j = k$. Notice that

$$\mathbb{E}[\tilde{Z}_i \tilde{Z}_j \tilde{Z}_k \tilde{Z}_\ell] = \mathbb{E}[\tilde{Z}_i] \cdot \mathbb{E}[\tilde{Z}_j^2] \cdot \mathbb{E}[\tilde{Z}_\ell] = e^{-d_i - d_j - d_\ell}.$$

Therefore, similarly to case 1, we have

$$\begin{aligned} \mathbb{E}[Z_i Z_j Z_k Z_\ell] &= \prod_{f \in \binom{\{i,j,\ell\}}{2}} (1 - p_f) \prod_{\substack{(i,u) \\ u \neq j,\ell}} (1 - p_{iu}) \prod_{\substack{(j,v) \\ v \neq i,\ell}} (1 - p_{jv}) \prod_{\substack{(\ell,w) \\ w \neq i,j}} (1 - p_{\ell w}) \\ &\leq e^{-d_i - d_j - d_\ell + p_{ij} + p_{i\ell} + p_{j\ell}} \leq e^{6p_{\max}} \cdot \mathbb{E}[\tilde{Z}_i \tilde{Z}_j \tilde{Z}_k \tilde{Z}_\ell] \end{aligned} \quad (\text{A.7})$$

– *Case 3: i, j, k, ℓ distinct.* Notice that

$$\mathbb{E}[\tilde{Z}_i \tilde{Z}_j \tilde{Z}_k \tilde{Z}_\ell] = \mathbb{E}[\tilde{Z}_i] \cdot \mathbb{E}[\tilde{Z}_j] \cdot \mathbb{E}[\tilde{Z}_k] \cdot \mathbb{E}[\tilde{Z}_\ell] = e^{-d_i - d_j - d_k - d_\ell}.$$

Therefore, similarly to case 1, we have

$$\begin{aligned} \mathbb{E}[Z_i Z_j Z_k Z_\ell] &= \prod_{f \in \binom{\{i,j,k,\ell\}}{2}} (1 - p_f) \prod_{\substack{(i,u) \\ u \neq j,k,\ell}} (1 - p_{iu}) \prod_{\substack{(j,v) \\ v \neq i,k,\ell}} (1 - p_{jv}) \prod_{\substack{(k,w) \\ w \neq i,j,\ell}} (1 - p_{kw}) \prod_{\substack{(\ell,x) \\ x \neq i,j,k}} (1 - p_{\ell x}) \\ &\leq e^{-d_i - d_j - d_k - d_\ell + p_{ij} + p_{ik} + p_{i\ell} + p_{jk} + p_{j\ell} + p_{k\ell}} \leq e^{6p_{\max}} \cdot \mathbb{E}[\tilde{Z}_i \tilde{Z}_j \tilde{Z}_k \tilde{Z}_\ell] \end{aligned} \quad (\text{A.8})$$

Consequently,

$$\begin{aligned} \mathbb{E}[\tilde{N}^2] &= \sum_{1 \leq i < j \leq n} \sum_{1 \leq k < \ell \leq n} \left[e^{-d_i - d_j} \mathbf{1}_{\{i=k, j=\ell\}} + e^{-d_i - d_j - d_\ell} \mathbf{1}_{\{j=k\}} + e^{-d_i - d_j - d_k - d_\ell} \mathbf{1}_{\{i \neq j \neq k \neq \ell\}} \right] \\ &\leq \sum_{1 \leq i < j \leq n} \sum_{1 \leq k < \ell \leq n} \left[e^{-d_i - d_j} + e^{-d_i - d_j - d_\ell} + e^{-d_i - d_j - d_k - d_\ell} \right] \leq \mu^2 + \mu^3 + \mu^4, \end{aligned} \quad (\text{A.9})$$

where $\mu \triangleq \sum_{1 \leq i \leq n} e^{-d_i}$. Combining equations (A.4) and (A.5) with (A.6), (A.7), (A.8) and (A.9) yields that

$$\mathbb{E}[N^2] \leq e^{6p_{\max}} \cdot \mathbb{E}[\tilde{N}^2] \leq e^{6p_{\max}} \mu^2 (\mu^2 + \mu + 1). \quad (\text{A.10})$$

Bounding the first moment. Notice that

$$\mathbb{E}[\tilde{N}] = \sum_{1 \leq i < j \leq n} \mathbb{E}[\tilde{Z}_i \tilde{Z}_j] = \sum_{1 \leq i < j \leq n} e^{-d_i - d_j} = \frac{1}{2} \left(\mu^2 - \sum_{i=1}^n e^{-2d_i} \right) \geq \frac{\mu}{2} \left(\mu - \max_{1 \leq i \leq n} e^{-d_i} \right), \quad (\text{A.11})$$

where the last step uses that $e^{-2d_k} \leq e^{-d_k} \cdot \max_{1 \leq i \leq n} e^{-d_i}$ for each k . Furthermore,

$$\begin{aligned} \mathbb{E}[N] &= \sum_{1 \leq i < j \leq n} \mathbb{E}Z_i Z_j = \sum_{1 \leq i < j \leq n} \left[(1 - p_{ij}) \prod_{\substack{(i,u) \\ u \neq j}} (1 - p_{iu}) \prod_{\substack{(j,v) \\ v \neq i}} (1 - p_{jv}) \right] \\ &\stackrel{(a)}{\geq} \sum_{1 \leq i < j \leq n} \left[e^{-p_{ij} - p_{ij}^2} \prod_{\substack{(i,u) \\ u \neq j}} e^{-p_{iu} - p_{iu}^2} \prod_{\substack{(j,v) \\ v \neq i}} e^{-p_{jv} - p_{jv}^2} \right] \\ &\stackrel{(b)}{\geq} e^{-(2n-1)p_{\max}^2} \cdot \sum_{1 \leq i < j \leq n} e^{-d_i - d_j + p_{ij}} \stackrel{(c)}{\geq} e^{-(2n-1)p_{\max}^2} \cdot \mathbb{E}[\tilde{N}], \end{aligned} \quad (\text{A.12})$$

where (a) uses that $1 - x \geq e^{-x-x^2}$ for all $x \in [0, 1/2]$, (b) uses that $e^{-p_{ij}^2} \geq e^{-p_{\max}^2}$, and (c) uses

that $e^{p_{ij}} \geq 1$ for all i and j . Finally, combining equations (A.10), (A.11) and (A.12), we have that

$$\frac{\mathbb{E}[N]^2}{\mathbb{E}[N^2]} \geq e^{-(4n-2)p_{\max}^2 - 6p_{\max}} \frac{\mathbb{E}[\tilde{N}]^2}{\mathbb{E}[\tilde{N}^2]} \geq \exp\left(-(4n-2)p_{\max}^2 - 6p_{\max}\right) \frac{\mu^2 - 2\mu \max_{1 \leq i \leq n} e^{-d_i}}{4(\mu^2 + \mu + 1)},$$

where the last step uses that

$$\mathbb{E}[\tilde{N}]^2 \geq \frac{\mu^2}{4} \left(\mu^2 - 2\mu \cdot \max_{1 \leq i \leq n} e^{-d_i} + \max_{1 \leq i \leq n} e^{-2d_i} \right) \geq \frac{\mu^2}{4} \left(\mu^2 - 2\mu \max_{1 \leq i \leq n} e^{-d_i} \right).$$

Since $\mathbb{P}(N \geq 1) \geq \frac{\mathbb{E}N^2}{\mathbb{E}N^2}$, the proof is complete. $\square$

A.2.1 Proof of Lemma 3.1

Let N be the number of pairs of isolated nodes in H . It follows from Lemma A.2 that

$$\begin{aligned} \mathbb{P}(N \geq 1) &\geq \exp\left(-(2n-4)q_{\max}^2 - 6q_{\max}\right) \cdot \frac{\lambda^2 - 2\lambda \max_{1 \leq i \leq n} e^{-r_i}}{4(\lambda^2 + \lambda + 1)} \\ &\geq (1 - o(1)) \cdot \frac{\lambda^2 - 2\lambda \max_{1 \leq i \leq n} e^{-r_i}}{4(\lambda^2 + \lambda + 1)}, \end{aligned} \tag{A.13}$$

where $\lambda \triangleq \sum_{i=1}^n e^{-r_i}$, and the last step in (A.13) uses that $q_{\max} = o(1/\sqrt{n})$. Let $r_{[1]} \leq r_{[2]} \leq \dots \leq r_{[n]}$ denote a sorted copy of the sequence $r_1, \dots, r_n$. Consider the following three cases.

- *Case 1.* $r_{[1]} = \omega(1)$. In this setting, it follows that

$$\lambda \max_{1 \leq i \leq n} e^{-r_i s(1-(1-s)^{m-1})} = \lambda e^{-r_{[1]} s(1-(1-s)^{m-1})} = o(1),$$

since $\lambda = \Omega(1)$. Thus,

$$\mathbb{P}(N \geq 1) \geq (1 - o(1)) \cdot \frac{\lambda^2 - o(1)}{4(\lambda^2 + \lambda + 1)} = \Omega(1).$$

- *Case 2.* $r_{[2]} = O(1)$. In this setting, there exists a constant $C > 0$ such that $C \leq e^{-r_{[2]}} \leq e^{-r_{[1]}}$. Let A denote the adjacency matrix of G and let $R_{[1]}$ and $R_{[2]}$ denote the degree of two nodes

with mean degree $r_{[1]}$ and $r_{[2]}$ respectively. Then,

$$\begin{aligned}
\mathbb{P}(R_{[1]} = 0, R_{[2]} = 0) &= \mathbb{P}(R_{[1]} = 0) \cdot \mathbb{P}(R_{[2]} = 0 \mid R_{[1]} = 0) \geq \mathbb{P}(R_{[1]} = 0) \mathbb{P}(R_{[2]} = 0) \\
&= \prod_{i \neq [1]} (1 - q_{[1]i}) \prod_{j \neq [2]} (1 - q_{[2]j}) \stackrel{(a)}{\geq} \prod_{i \neq [1]} e^{-q_{[1]i} - q_{[1]i}^2} \prod_{j \neq [2]} e^{-q_{[2]j} - q_{[2]j}^2} \\
&= e^{-r_{[1]}} e^{-r_{[2]}} \cdot \exp \left(- \sum_{i \neq [1]} q_{[1]i}^2 - \sum_{j \neq [2]} q_{[2]j}^2 \right) \\
&\stackrel{(b)}{\geq} \exp(-2(n-1)q_{\max}^2) \cdot C^2 \stackrel{(c)}{=} \Omega(1).
\end{aligned}$$

Here, (a) uses the inequality $1 - x \geq e^{-x - x^2}$ whenever $x \in [0, 1/2]$, (b) uses that $e^{-q_{ij}^2} \geq e^{-q_{\max}^2}$ for all i, j and also that $C \leq e^{-r_{[2]}} \leq e^{-r_{[1]}}$. Finally, (c) uses that $\exp(-2(n-1)q_{\max}^2) = \Omega(1)$ whenever $q_{\max} = o(1/\sqrt{n})$.

- *Case 3.* $r_{[2]} = \omega(1)$. Consider the induced subgraph G' of G on the vertex set $V = V - \{[1]\}$. The mean degrees of G' satisfy $r'_{[i]} \leq r_{[i]}$, and so it follows that

$$\lambda' \triangleq \sum_{i=2}^n e^{-d'_{[i]}} \leq \sum_{i=2}^n e^{-r_{[i]}} = \Omega(1),$$

and the smallest mean degree in G' diverges, i.e. $r_{[2]} = \omega(1)$. Equivalently, the graph G' satisfies the conditions for Case 1 above, and so there is a constant lower bound C on the probability that the number of *pairs* of isolated nodes N' in G' is at least one. If node $[1]$ does not connect to an isolated pair of nodes in G' , then the pair remains isolated in G as well. Therefore,

$$\mathbb{P}(N \geq 1) \geq (1 - q_{\max})^2 \cdot \mathbb{P}(N' \geq 1) \geq (1 - q_{\max})^2 \cdot C = \Omega(1).$$

In all cases, we have that $\mathbb{P}(N \geq 1) = \Omega(1)$, i.e. there is a non-vanishing probability that two isolated nodes exist in H . This concludes the proof.

A.3 Stochastic domination of crossing edges in $\tilde{\mathcal{H}}(v)$

The objective of this section is to build up to a proof of Lemma 3.11. We start by making a simple observation about products of Binomial random variables.

Lemma A.3. *Let $X_1, \dots, X_m \sim \text{Bern}(s)$ be i.i.d. random variables, and let $B = X_1 + \dots + X_m$ denote their sum. For each ℓ in $\{1, 2, \dots, \lfloor m/2 \rfloor\}$, define*

$$T_\ell = (X_1 + \dots + X_\ell)(X_{\ell+1} + \dots + X_m).$$

For any $\ell_1, \ell_2 \in \{1, 2, \dots, \lfloor m/2 \rfloor\}$ such that $\ell_1 < \ell_2$, and for any $t \in \mathbb{R}$ and any $b \in \{0, 1, \dots, m\}$,

$$\mathbb{P}(T_{\ell_1} > t \mid B = b) \leq \mathbb{P}(T_{\ell_2} > t \mid B = b). \quad (\text{A.14})$$

Proof. Consider overlapping but exhaustive cases:

Case 1: $t < 0$. Since $T_\ell \geq 0$ almost surely for all ℓ , the inequality (A.14) holds.

Case 2: $t \geq b - 1$. Note that conditioned on $B = b$, it follows that $T_1 \in \{0, b - 1\}$. Therefore, the left hand side of (A.14) equals zero, and the inequality holds.

Case 3: $b = 0$ or $b = 1$. In this case, T_ℓ is identically zero for all ℓ , so (A.14) holds.

Case 4: $b \geq 2$ and $0 \leq t < b - 1$. For any $\ell \in \{1, 2, \dots, \lfloor m/2 \rfloor\}$,

$$\begin{aligned} \mathbb{P}(T_\ell > t \mid B = b) &= \frac{\mathbb{P}(\{(X_1 + \dots + X_\ell)(X_{\ell+1} + \dots + X_m) > t\} \cap \{X_1 + \dots + X_m = b\})}{\mathbb{P}(X_1 + \dots + X_m = b)} \\ &= \frac{\sum_{i: i(b-i) > t} \mathbb{P}(\{X_1 + \dots + X_\ell = i\} \cap \{X_{\ell+1} + \dots + X_m = b - i\})}{\mathbb{P}(X_1 + \dots + X_m = b)} \\ &\stackrel{(a)}{=} \frac{\sum_{i=1}^{b-1} \mathbb{P}(X_1 + \dots + X_\ell = i) \mathbb{P}(X_{\ell+1} + \dots + X_m = b - i)}{\mathbb{P}(X_1 + \dots + X_m = b)} \\ &\stackrel{(b)}{=} \frac{\sum_{i=1}^{b-1} \binom{\ell}{i} \binom{m-\ell}{b-i}}{\binom{m}{b}} \\ &= \frac{\sum_{i=0}^b \binom{\ell}{i} \binom{m-\ell}{b-i} - \binom{m-\ell}{b} - \binom{\ell}{b}}{\binom{m}{b}} \\ &= \frac{\binom{m}{b} - \binom{m-\ell}{b} - \binom{\ell}{b}}{\binom{m}{b}}, \end{aligned} \quad (\text{A.15})$$

where (a) used the fact that for any t such that $0 \leq t < b - 1$, it is true that

$$\{i : i(b - i) > t\} = \{1, 2, \dots, b - 1\}.$$

Here, the notation for binomial coefficients in (b) involves setting $\binom{n}{k} = 0$ whenever $k < 0$ or $k > n$. Let $f_{m,b}(\ell)$ denote the numerator of (A.15), i.e.

$$f_{m,b}(\ell) \triangleq \binom{m}{b} - \binom{m-\ell}{b} - \binom{\ell}{b}$$

It suffices to show that $f_{m,b}(\ell) - f_{m,b}(\ell - 1) \geq 0$ for all $\ell \in \{2, \dots, \lfloor m/2 \rfloor\}$. Indeed,

$$\begin{aligned} f_{m,b}(\ell) - f_{m,b}(\ell - 1) &= \binom{m - \ell + 1}{b} - \binom{m - \ell}{b} - \left(\binom{\ell}{b} - \binom{\ell - 1}{b} \right) \\ &\stackrel{(c)}{=} \binom{m - \ell}{b - 1} - \binom{\ell - 1}{b - 1} \geq 0, \end{aligned}$$

whenever $m - \ell \geq \ell - 1$, i.e. $\ell \leq \lfloor m/2 \rfloor$. Here, (c) uses the identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and the fact that $\binom{n_1}{k} \geq \binom{n_2}{k}$ whenever $n_1 \geq n_2$. This concludes the proof. $\square$

Corollary A.1. *Let F be a collection of edges in the parent graph G . For any edge $e_r \in F$, let X_i^r denote the indicator random variable $G'_i(e_r) \sim \text{Bern}(ps)$. For each ℓ in $\{1, \dots, \lfloor m/2 \rfloor\}$, define*

$$T_\ell^r = (X_1^r + \dots + X_\ell^r)(X_{\ell+1}^r + \dots + X_m^r).$$

Then, for any $\ell_1, \ell_2 \in \{1, \dots, \lfloor m/2 \rfloor\}$ such that $\ell_1 < \ell_2$, the following stochastic ordering holds

$$\sum_{r=1}^{|F|} T_{\ell_1}^r \preceq \sum_{r=1}^{|F|} T_{\ell_2}^r.$$

Proof. It suffices to show that $T_{\ell_1}^r \preceq T_{\ell_2}^r$ for each r , since the edges are independent. For any t ,

$$\mathbb{P}(T_{\ell_1}^r > t) = \sum_{b=0}^m \mathbb{P}(B = b) \mathbb{P}(T_{\ell_1}^r > t | B = b) \leq \sum_{b=0}^m \mathbb{P}(B = b) \mathbb{P}(T_{\ell_2}^r > t | B = b) = \mathbb{P}(T_{\ell_2}^r > t),$$

which concludes the proof. $\square$

With this, we are ready to prove Lemma 3.11.

Proof of Lemma 3.11. Let $\ell_1, \ell_2 \in \{1, \dots, \lfloor m/2 \rfloor\}$ such that $\ell_1 < \ell_2$. Let $t \in \mathbb{R}$. Consider the parent graph G and label the set of incident edges on v as $\{e_1, \dots, e_{\deg_G(v)}\}$. Denote by X_i^r the

indicator random variable $G'_i(e_r) \sim \text{Bern}(ps)$. It follows that

$$\begin{aligned}
\mathbb{P}(\tilde{c}_v(Z_{\ell_2}) > t) &= \mathbb{P}\left(\sum_{i=1}^{\ell_2} \sum_{j=\ell_2+1}^m \deg_{G'_i \wedge G'_j}(v) \geq t\right) \\
&= \mathbb{P}\left(\sum_{i=1}^{\ell_2} \sum_{j=\ell_2+1}^m \sum_{r=1}^{\deg_G(v)} X_i^r X_j^r > t\right) \\
&= \sum_{d=0}^n \mathbb{P}(\deg_G(v) = d) \mathbb{P}\left(\sum_{r=1}^d ((X_1^r + \cdots + X_{\ell_2}^r)(X_{\ell_2+1}^r + \cdots + X_m^r)) > t\right) \\
&\stackrel{(a)}{\geq} \sum_{d=0}^n \mathbb{P}(\deg_G(v) = d) \mathbb{P}\left(\sum_{r=1}^d ((X_1^r + \cdots + X_{\ell_1}^r)(X_{\ell_1+1}^r + \cdots + X_m^r)) > t\right) \\
&= \mathbb{P}\left(\sum_{i=1}^{\ell_1} \sum_{j=\ell_1+1}^m \sum_{r=1}^{\deg_G(v)} X_i^r X_j^r > t\right) \\
&= \mathbb{P}\left(\sum_{i=1}^{\ell_1} \sum_{j=\ell_1+1}^m \deg_{G'_i \wedge G'_j}(v) \geq t\right) \\
&= \mathbb{P}(\tilde{c}_v(Z_{\ell_1}) > t),
\end{aligned}$$

as desired. Here, (a) uses Corollary [A.1](#). □

Appendix B

Supplementary material for Chapter 4

B.1 Proof of Lemma 4.7

(a) The equality $K'_\pi = K'_\pi$ follows from the fact that if e is an edge with $e \in \binom{O_{k\ell}}{2}$ then $e = \{i, j\}$ such that $\pi_{k\ell}(i) = \tilde{\pi}_{k\ell}(i)$ and $\pi_{k\ell}(j) = \tilde{\pi}_{k\ell}(j)$ so $\pi_{k\ell}(e) = \tilde{\pi}_{k\ell}(e)$.

(b) The first inequality follows directly from the definition and the second from part (a).

(c) Using the definition (4.10),

$$(K_\pi)_{(k,e),(\ell,f)} = \mathbf{1}_{\{k \neq \ell\}} \mathbf{1}_{\{\pi_{k\ell}(e)=f\}} = \mathbf{1}_{\{\ell \neq k\}} \mathbf{1}_{\{\pi_{\ell k}(f)=e\}} = (K_\pi)_{(\ell,f),(k,e)},$$

so K_π is symmetric. Similarly,

$$\begin{aligned} (K'_\pi)_{(k,e),(\ell,f)} &= \mathbf{1}_{\{k \neq \ell\}} \mathbf{1}_{\{\pi_{k\ell}(e)=f\}} \mathbf{1}_{\{e \in \binom{O_{k\ell}}{2}\}} \\ &= \mathbf{1}_{\{\ell \neq k\}} \mathbf{1}_{\{\pi_{\ell k}(f)=e\}} \mathbf{1}_{\{f \in \binom{O_{\ell k}}{2}\}} = (K'_\pi)_{(\ell,f),(k,e)} \end{aligned}$$

so K'_π is also symmetric.

(d) Observe that

$$\begin{aligned} (K_\pi K_{\tilde{\pi}})_{(k,e),(k,e)} &= \sum_{\ell=1}^m \sum_{f \in E_n} (K_\pi)_{(k,e),(\ell,f)} (K_{\tilde{\pi}})_{(\ell,f),(k,e)} \\ &= \sum_{\ell \neq k} \sum_{f \in E_n} \mathbf{1}_{\{\pi_{k\ell}(e)=f\}} \mathbf{1}_{\{\tilde{\pi}_{\ell k}(f)=e\}} \\ &= \sum_{\ell \neq k} \mathbf{1}_{\{\pi_{k\ell}(e)=\tilde{\pi}_{k\ell}(e)\}}. \end{aligned}$$

Taking the trace gives

$$\begin{aligned}
\text{Tr}(K_\pi K_{\tilde{\pi}}) &= \sum_{k=1}^m \sum_{e \in E_n} (K_\pi K_{\tilde{\pi}})_{(k,e),(k,e)} \\
&= \sum_{k=1}^m \sum_{\ell \neq k} \sum_{e \in E_n} \mathbf{1}_{\{\pi_{k\ell}(e) = \tilde{\pi}_{k\ell}(e)\}} \\
&= \sum_{k=1}^m \sum_{\ell \neq k} D_{k\ell}(\pi, \tilde{\pi}).
\end{aligned}$$

Finally, $D_{k\ell} = D_{\ell k}$ and $D_{k\ell}(\pi, \tilde{\pi}) = D_{k\ell}(\sigma)$, so the last sum equals $2 \sum_{1 \leq k < \ell \leq m} D_{k\ell}(\sigma)$.

(e) Observe that

$$\begin{aligned}
(K'_\pi K_{\tilde{\pi}})_{(k,e),(k,e)} &= \sum_{\ell=1}^m \sum_{f \in E_n} (K'_\pi)_{(k,e),(\ell,f)} (K_{\tilde{\pi}})_{(\ell,f),(k,e)} \\
&= \sum_{\ell \neq k} \sum_{f \in E_n} \mathbf{1}_{\{\pi_{k\ell}(e) = f\}} \mathbf{1}_{\{e \in \binom{O_{k\ell}}{2}\}} \mathbf{1}_{\{\tilde{\pi}_{\ell k}(f) = e\}} \\
&= \sum_{\ell \neq k} \mathbf{1}_{\{e \in \binom{O_{k\ell}}{2}\}} \mathbf{1}_{\{\pi_{k\ell}(e) = \tilde{\pi}_{k\ell}(e)\}}.
\end{aligned}$$

Taking the trace gives

$$\begin{aligned}
\text{Tr}(K'_\pi K_{\tilde{\pi}}) &= \sum_{k=1}^m \sum_{e \in E_n} (K'_\pi K_{\tilde{\pi}})_{(k,e),(k,e)} \\
&= \sum_{k=1}^m \sum_{\ell \neq k} \sum_{e \in E_n \cap \binom{O_{k\ell}}{2}} \mathbf{1}_{\{\pi_{k\ell}(e) = \tilde{\pi}_{k\ell}(e)\}} \\
&= \sum_{k=1}^m \sum_{\ell \neq k} \left| E_n \cap \binom{O_{k\ell}}{2} \right| = \sum_{k=1}^m \sum_{\ell \neq k} \binom{d_{k\ell}(\sigma)}{2} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}}.
\end{aligned}$$

Finally, $d_{k\ell}(\sigma) = d_{\ell k}(\sigma)$, so the last sum equals $2 \sum_{1 \leq k < \ell \leq m} \binom{d_{k\ell}(\sigma)}{2} \mathbf{1}_{\{d_{k\ell}(\sigma) \geq \frac{n}{2}\}}$. Proof of the other equalities is similar (and also $\text{Tr}(AB) = \text{Tr}(BA)$ for matrices of appropriate dimensions in general).

B.2 Proof of Lemma 4.8

For part (a), expanding the diagonal entries of $K_\pi^2 K_{\tilde{\pi}}$ gives

$$(K_\pi^2 K_{\tilde{\pi}})_{(k,e),(k,e)} = \sum_{\ell, k' \in [m]} \sum_{f, e', f' \in E_n} (K_\pi)_{(k,e),(\ell,f)} (K_\pi)_{(\ell,f),(k',e')} (K_{\tilde{\pi}})_{(k',e'),(k,e)}.$$

Using (4.10), this becomes

$$(K_\pi^2 K_{\tilde{\pi}})_{(k,e),(k,e)} = \sum_{\ell,k' \in [m]} \sum_{f,e' \in E_n} \mathbf{1}_{\{k \neq \ell, \ell \neq k', k' \neq k\}} \cdot \mathbf{1}_{\{\pi_{k\ell}(e)=f\}} \cdot \mathbf{1}_{\{\pi_{\ell k'}(f)=e'\}} \cdot \mathbf{1}_{\{\tilde{\pi}_{k'k}(e')=e\}}.$$

For fixed k, ℓ, k' and e , the first two edge indicators uniquely determine $f = \pi_{k\ell}(e)$ and $e' = \tilde{\pi}_{\ell k'}(f)$. Hence the final condition is precisely

$$(\tilde{\pi}_{k'k} \circ \pi_{\ell k'} \circ \pi_{k\ell})(e) = e$$

for ℓ, k' such that k, ℓ, k' are distinct. Summing over all diagonal coordinates (k, e) gives part (a).

Part (b) is similar, expanding the diagonal of $K_\pi K_{\tilde{\pi}} K_\pi K_{\tilde{\pi}}$; the first three indicators determine the intermediate edge variables, and the final indicator becomes $(\tilde{\pi}_{\ell'k} \circ \pi_{k'\ell'} \circ \tilde{\pi}_{\ell k'} \circ \pi_{k\ell})(e) = e$ where the sum is over k, ℓ, k', ℓ' such that $\{k, k'\} \cap \{\ell, \ell'\} = \emptyset$.

Part (c) is identical to part (b). Expanding the diagonal of $K_\pi K_\pi K_\pi K_{\tilde{\pi}}$; the final indicator becomes $(\tilde{\pi}_{\ell'k} \circ \pi_{k'\ell'} \circ \pi_{\ell k'} \circ \pi_{k\ell})(e) = e$.

B.3 Proof of Lemma 4.11

A classical bound on the tail of the distribution of C_j is [ABT03]

$$\mathbb{P}(C_j \geq k) \leq \frac{1}{j^k k!}.$$

(It follows from $\mathbb{P}(C_j \geq k) = \mathbb{P}(\binom{C_j}{k} \geq 1)$, Markov's inequality, and the fact $\mathbb{E}[\binom{C_j}{k}] \leq \frac{1}{j^k k!}$ with equality for $jk \leq n$ [ABT03, Fel57]). Also $C_j \leq n/j$. We shall use the following summation by parts identity: if Z is a nonnegative integer-valued random variable and $a_0, a_1, \dots$ is increasing, then

$$\mathbb{E}[a_Z] = a_0 + \sum_{k \geq 1} (a_k - a_{k-1}) \mathbb{P}(Z \geq k).$$

For part (a), first assume $\theta_n \geq 0$. Then

$$\begin{aligned} \mathbb{E}[e^{\theta_n C_j}] - 1 &\leq \sum_{k \geq 1} (e^{\theta_n k} - e^{\theta_n (k-1)}) \frac{1}{k!} \\ &= (e^{\theta_n} - 1) \sum_{k \geq 1} \frac{e^{\theta_n (k-1)}}{k!} = O(\theta_n) = o(1). \end{aligned}$$

If $\theta_n < 0$, then

$$0 \leq 1 - \mathbb{E}[e^{\theta_n C_j}] \leq \mathbb{E}[e^{|\theta_n| C_j}] - 1 = o(1),$$

so part (a) follows.

To complete the proof of the lemma, the same reduction lets us consider only $\theta_n \geq 0$. Since $\theta_n = o(\log n/n)$, by monotonicity it suffices to take $\theta_n = \log n/(2n)$ and show that $\mathbb{E}[e^{\theta_n C_j^2}] = 1 + o(1)$. Using the same summation identity,

$$\mathbb{E}[e^{\theta_n C_j^2}] - 1 \leq \sum_{k=1}^n \left(e^{\theta_n k^2} - e^{\theta_n (k-1)^2} \right) \frac{1}{k!}.$$

Let $M_n := \left\lfloor \frac{n}{(\log n)^2} \right\rfloor$ and split the last sum into

$$S_{n,1} := \sum_{k=1}^{M_n} \left(e^{\theta_n k^2} - e^{\theta_n (k-1)^2} \right) \frac{1}{k!}, \quad S_{n,2} := \sum_{k=M_n+1}^n \left(e^{\theta_n k^2} - e^{\theta_n (k-1)^2} \right) \frac{1}{k!}.$$

Small k part. For $k \leq M_n$ we have $k^2 \leq M_n k$ and for $x \geq 0$ we have $1 - e^{-x} \leq x$, so that

$$e^{\theta_n k^2} - e^{\theta_n (k-1)^2} = (1 - e^{-\theta_n (2k-1)}) e^{\theta_n M_n k} \leq 2\theta_n k e^{\theta_n M_n k}.$$

Therefore

$$S_{n,1} \leq 2\theta_n \sum_{k \geq 1} \frac{k e^{\theta_n M_n k}}{k!} = O(\theta_n) = o(1).$$

Large k part. For $S_{n,2}$, we use $e^{\theta_n k^2} - e^{\theta_n (k-1)^2} \leq e^{\theta_n k^2}$. Using $k! > (k/e)^k$,

$$\frac{e^{\theta_n k^2}}{k!} < \exp(\theta_n k^2 + k - k \log k) =: e^{f_n(k)}.$$

A direct computation shows:

- $f_n''(k) = 2\theta_n - 1/k$ has exactly one zero at $k_0 = n/\log n$;
- $M_n < k_0 < n$ for large n ;
- $f_n'(M_n) < 0$, $f_n'(k_0) < 0$, and $f_n'(n) = 0$.

Thus f_n' decreases on $[M_n, k_0]$ and increases on $[k_0, n]$, remaining ≤ 0 throughout. Hence f_n is strictly decreasing on $[M_n, n]$, so

$$f_n(k) \leq f_n(M_n) \quad \text{for all } k \in [M_n, n].$$

A direct asymptotic expansion gives

$$f_n(M_n) = -\frac{n}{\log n} + O\left(\frac{n \log \log n}{(\log n)^2}\right) \leq -\frac{1}{2} \frac{n}{\log n}$$

for all large n . Therefore

$$S_{n,2} \leq n \exp\left(-\frac{1}{2} \frac{n}{\log n}\right) \rightarrow 0.$$

Combining the two estimates gives

$$\mathbb{E}[e^{\theta_n C_j^2}] - 1 \leq S_{n,1} + S_{n,2} = o(1),$$

which proves part (b).

B.4 On the test statistic

In this appendix we show that the statistic T defined in (4.3) is approximately a generalized likelihood ratio statistic. By definition, the generalized likelihood ratio is $\max_{\pi} L_{\pi}(X)$ and thresholding any (strictly) increasing function of the generalized likelihood ratio is a generalized likelihood ratio test (GLRT). By (4.11), $L_{\pi}(X)$ is a monotone increasing function of $X^{\top}(I_{mN} - \Sigma_{\pi}^{-1})X$ where $\Sigma_{\pi} = I_{mN} + \rho K_{\pi}$. Calculating as in the proof of Lemma 4.7 it can be shown that

$$K_{\pi}^2 = (m-1)I_{mN} + (m-2)K_{\pi}$$

from which it can be readily checked that

$$\Sigma_{\pi}^{-1} = \frac{1 + (m-2)\rho}{1 + (m-2)\rho - (m-1)\rho^2} I_{mN} - \frac{\rho}{1 + (m-2)\rho - (m-1)\rho^2} K_{\pi}. \quad (\text{B.1})$$

(This may look complicated but it can be derived by considering the case $n = 1$ because up to a permutation of rows and columns depending on π , the matrix Σ_{π} is block diagonal with $\binom{n}{2}$ blocks of the form Σ defined in (4.1).) Therefore, using the fact $X^{\top} K_{\pi} X = 2T_{\pi}(X)$, we get

$$X^{\top}(I_{mN} - \Sigma_{\pi}^{-1})X = \frac{2\rho T_{\pi}(X) - (m-1)\rho^2 \|X\|_2^2}{1 + (m-2)\rho - (m-1)\rho^2}.$$

Since the coefficient of $\|X\|_2^2$ includes ρ^2 which is much smaller than the coefficient 2ρ of $T_{\pi}(X)$, we conclude that $L_{\pi}(X)$ is approximately a monotone increasing function of $T_{\pi}(X)$. Thus, T is approximately a generalized likelihood ratio statistic. (In addition, selecting π to maximize $T_{\pi}(X)$ is the maximum likelihood estimator of the true alignment in case it is known that H_1 is the true hypothesis.)

Appendix C

Supplementary material for Chapter 5

C.1 Bounds and approximations of Bessel functions

This appendix collects bounds and asymptotic values related to the modified Bessel functions of the first kind, $I_0(\kappa)$.

- The power series expansion of I_0 and comparison with the power series expansion of e^x evaluated at $x = \kappa^2/4$ gives, for all κ ,

$$I_0(\kappa) = \sum_{k=0}^{\infty} \frac{1}{(k!)^2} \left(\frac{\kappa}{2}\right)^{2k} \leq e^{\frac{1}{4}\kappa^2}$$

- Let $\kappa > 1/2$. For Θ uniformly distributed over $[0, 2\pi]$, the probability density of $\cos \Theta$ is $\frac{1}{\pi\sqrt{1-t^2}}$ for $-1 < t < 1$. This gives the integral relation (known classically [DLMF]) and bound:

$$I_0(\kappa) = \frac{1}{\pi} \int_{-1}^1 \frac{e^{\kappa t}}{\sqrt{1-t^2}} dt \geq \frac{1}{\pi} \int_{1-\frac{1}{\kappa}}^1 \frac{e^{\kappa t}}{\sqrt{1-t^2}} dt$$

Using $1-t^2 \leq 2(1-t)$ and $e^{\kappa t} \geq e^{\kappa(1-(1-t))}$ over the interval of integration and integrating yields

$$I_0(\kappa) \geq \frac{2\sqrt{2}}{3\pi} \frac{e^{\kappa}}{\sqrt{\kappa}} \geq (0.3) \frac{e^{\kappa}}{\sqrt{\kappa}}.$$

This lower bound is close to the large κ asymptotics:

$$I_0(\kappa) \sim \frac{e^{\kappa}}{\sqrt{2\pi\kappa}} \approx (0.39894) \frac{e^{\kappa}}{\sqrt{2\pi\kappa}}.$$

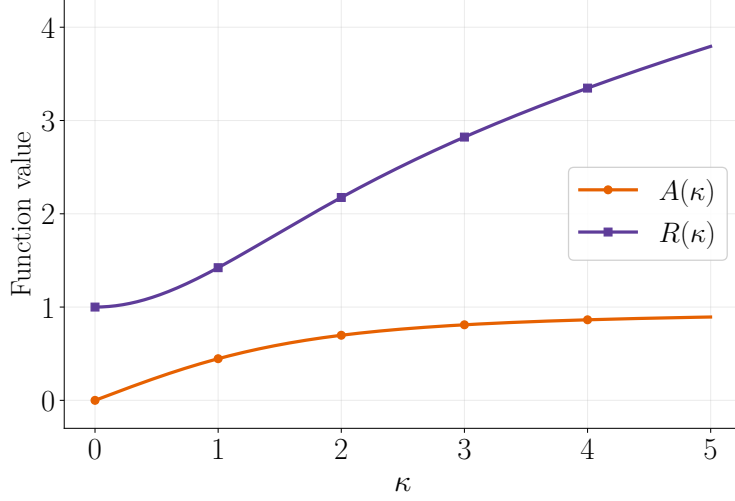


Figure C.1: The functions $A(\kappa)$ and $R(\kappa)$

Accounting for the remainder term, it holds that as $\kappa \rightarrow \infty$,

$$I_0(\kappa) = \frac{e^\kappa}{\sqrt{2\pi\kappa}} (1 + O(\kappa^{-1})) . \quad (\text{C.1})$$

- Let

$$A(\kappa) = \frac{1}{I_0(\kappa)} \cdot \left[\frac{1}{2\pi} \int_0^{2\pi} \cos(y) e^{\kappa \cos(y)} dy \right] = \frac{I_1(\kappa)}{I_0(\kappa)}$$

A Taylor expansion yields (see also Figure C.1)

- As $\kappa \rightarrow 0$, $A(\kappa) = \kappa/2 + O(\kappa^3)$.
- As $\kappa \rightarrow \infty$, $A(\kappa) \rightarrow 1$.

- Let $\rho_\kappa(\theta) \triangleq I_0(2\kappa \cos \theta)/I_0^2(\kappa)$. Lemma 5.1 shows that for every $\kappa > 0$, $\frac{1}{2\pi} \int_0^{2\pi} \rho_\kappa(\theta) d\theta = 1$.
- Let $R(\kappa) \triangleq I_0(2\kappa)/I_0^2(\kappa)$. The maximum of $\rho_\kappa(\theta)$ over θ is $R(\kappa) = \rho_\kappa(0)$. The asymptotic behavior of I_0 described above implies the following asymptotic behavior of R .

- As $\kappa \rightarrow 0$, $R(\kappa) = 1 + \kappa^2/2 + O(\kappa^4)$.
- As $\kappa \rightarrow \infty$, $R(\kappa) = \sqrt{\pi\kappa} (1 + O(\kappa^{-1}))$.

- As $\kappa \rightarrow 0$,

$$\begin{aligned}
D_{\text{KL}}(\text{vonMises}(0, \kappa) \parallel \text{Unif}([0, 2\pi])) &= \kappa \frac{I_1(\kappa)}{I_0(\kappa)} - \log I_0(\kappa) \\
&= \frac{\kappa(\kappa/2 + o(\kappa))}{I_0(\kappa)} - \log I_0(\kappa) \\
&= \frac{\kappa^2}{2} - \frac{\kappa^2}{4} + o(\kappa^2) = \frac{\kappa^2}{4} + o(\kappa^2)
\end{aligned}$$

C.2 Generalized likelihood ratio tests

C.2.1 The interval test for the hard-cluster flat model

For $\theta \in [0, 1]$, define the count

$$N(\theta) \triangleq \sum_{i=1}^N \mathbf{1}\{X_i \in [\theta, \theta + 2\pi\tau] \pmod{1}\}, \quad M_\tau(X) \triangleq \sup_{\theta \in [0, 2\pi]} N(\theta),$$

i.e. $M_\tau(X)$ is the largest number of points that can be captured in an interval of length $2\pi\tau$. Recall that the interval test above with window $2\pi\tau$ and threshold γ is

$$\phi_{\tau, \gamma}(X) \triangleq \mathbf{1}\{M_\tau(X) \geq \gamma\}, \quad \gamma \in \{1, \dots, n\}.$$

Likelihood ratios. Recall the conditional likelihood ratio given (S, θ) ,

$$L_{S, \theta}(x) = \prod_{i \in S} \frac{\mathbf{1}\{x_i \in I_\theta\}}{\tau} = \tau^{-K} \mathbf{1}\{x_i \in I_\theta \forall i \in S\}. \quad (\text{C.2})$$

Averaging over S and Θ gives the (unconditional) likelihood ratio,

$$\begin{aligned}
L(x) &\triangleq \frac{dP}{dQ}(x) = \frac{1}{\binom{N}{K} \tau^K} \int_0^{2\pi} \sum_{\substack{S \subset [N] \\ |S|=K}} \mathbf{1}\{S \subset \{i : x_i \in I_\theta\}\} \frac{d\theta}{2\pi} \\
&= \frac{1}{\binom{n}{K} \tau^K} \int_0^{2\pi} \binom{N(\theta)}{K} \frac{d\theta}{2\pi}.
\end{aligned} \quad (\text{C.3})$$

The last step used that the number of K -subsets contained in a set of size $N(\theta)$ is $\binom{N(\theta)}{K}$. We consider two GLRTs, depending on how we eliminate the parameters (S, θ) , as follows:

(A) *Maximize over both S and θ .* In this case,

$$T_A(x) \triangleq \sup_{\theta \in [0, 1]} \sup_{\substack{S \subset [N] \\ |S|=K}} L_{S, \theta}(x) \stackrel{(a)}{=} \sup_{\theta} \tau^{-K} \mathbf{1}\{N(\theta) \geq K\} \stackrel{(b)}{=} \tau^{-K} \mathbf{1}\{M_\tau(x) \geq K\}, \quad (\text{C.4})$$

where (a) used that given θ , there exists some index set S of points in $[\theta, \theta + 2\pi\tau]$ iff $N(\theta) \geq K$, and (b) uses the definition of $M_\tau(x)$.

Therefore, the decision rule in this case is precisely to compare T_A to any $t \in (0, \tau^{-K}]$:

$$\text{reject } H_0 \iff M_\tau(X) \geq K,$$

i.e. the interval test with threshold $\gamma = K$.

(B) *Average over S , then maximize over θ .* Following (C.3), we have

$$T_B(x) \triangleq \sup_{\theta \in [0, 1]} L_\theta(x) = \frac{1}{\binom{N}{K} \tau^K} \sup_{\theta} \binom{N(\theta)}{K}. \quad (\text{C.5})$$

Since $m \mapsto \binom{m}{K}$ is strictly increasing for $m \geq K$, thresholding T_B is equivalent to thresholding $M_\tau(X)$:

$$T_B(x) > t \iff M_\tau(X) \geq \gamma \quad \text{for some integer } \gamma \geq K \text{ depending on } t, \quad (\text{C.6})$$

i.e. the interval test with tunable $\gamma \geq K$.

C.2.2 The coherence test for the von Mises community model

Consider the detection problem for the von Mises community model with parameters n, k, κ . The log likelihood ratio given $\Theta^* = \theta$ and $C^* = C$ is given by:

$$\begin{aligned} \log L_{\theta, C}(X) &= \kappa \sum_{e \in E(C)} \cos(X_e - \theta) - K \log(I_0(\kappa)) \\ &= \kappa \cos(\theta) \sum_{e \in E(C)} \cos(X_e) + \kappa \sin(\theta) \sum_{e \in E(C)} \sin(X_e) - K \log(I_0(\kappa)) \end{aligned}$$

For each edge e , let $Z_e = e^{iX_e} = \cos(X_e) + i \sin(X_e)$. Maximizing over θ yields:

$$\max_{\theta \in [0, 2\pi]} \log L_{\theta, S}(X) = \kappa \left| \sum_{e \in E(C)} Z_e \right| - K \log(I_0(\kappa))$$

If we further maximize over C and select a threshold β we arrive at the GLRT test

$$\psi_n(\beta) \triangleq \mathbf{1} \left\{ \max_{\substack{C \subseteq [n] \\ |C|=k}} \left| \sum_{e \in E(C)} Z_e \right| \geq \beta \right\}.$$

This is our *coherence test* for the von Mises community model.

C.3 On polynomial-time tests

The *Rayleigh test*, defined below, has been successfully used in literature for detecting that a sample of observations is drawn from a von Mises distribution versus a uniform distribution. We show that it is not as effective in the planted setting, for the von Mises community model.

Let $Z_{ij} = e^{iX_{ij}}$, and consider the test statistic

$$T_{\text{Rayleigh}} \triangleq \left| \sum_{1 \leq i < j \leq n} Z_{ij} \right|.$$

We analyze for threshold β the test $\mathbf{1}\{T_{\text{Rayleigh}} \geq \beta\}$. Let $N_E = \binom{n}{2}$ be the total number of edges.

Bounding false alarm probability. Under H_0 , the Z_{ij} are i.i.d. uniform on the unit circle. Using the concentration bound (5.40) and (5.41):

$$p_{\text{fa}} = \mathbb{Q}(T_{\text{Rayleigh}} \geq \beta) \leq 4 \exp\left(-\frac{\beta^2}{2N_E}\right). \quad (\text{C.7})$$

Bounding the miss probability. Under H_1 , let Θ be the planted angle and let $m = \binom{k}{2}$. Using the same idea as (5.43) and (5.36), we have that the expected value of the sum $S = \sum_{i < j} Z_{ij}$ given Θ is $\mathbb{E}_P[S|\Theta] = mA(\kappa)e^{i\Theta}$. Let $\mu_1 = mA(\kappa)$. We lower bound the statistic by projecting onto the true direction Θ :

$$T_{\text{Rayleigh}} = |S| \geq \Re(Se^{-i\Theta}) = \sum_{i < j} \cos(X_{ij} - \Theta) \triangleq S_{\Theta}.$$

S_{Θ} is a sum of N_E independent real random variables bounded in $[-1, 1]$, with mean μ_1 . By Hoeffding's inequality:

$$p_{\text{miss}} \leq P(S_{\Theta} < \beta) \leq \exp\left(-\frac{(\mu_1 - \beta)^2}{2N_E}\right),$$

provided $\beta < \mu_1$.

To ensure both error probabilities vanish, we must be able to select β such that $\beta^2 = \omega(N_E)$ (for p_{fa}) and $\beta < \mu_1$ (for p_{miss}). This requires $\mu_1^2 = \omega(N_E)$. Choose $\beta = \mu_1/2$, so the total error probability

$$p_{\text{fa}} + p_{\text{miss}} \leq 5 \exp\left(-\frac{\mu_1^2}{8N_E}\right) = 5 \exp\left(-\frac{m^2 A^2(\kappa)}{8N_E}\right).$$

Substituting $m \approx k^2/2$ and $N_E \approx n^2/2$, the sufficient condition for this test to succeed is

$$k^4 A^2(\kappa) = \omega(n^2) \quad \text{or equivalently} \quad k^2 A(\kappa) = \omega(n).$$

This test is computationally efficient but requires a larger community size (e.g., $k = \omega(\sqrt{n})$ if $A(\kappa) = \Theta(1)$), compared to the (computationally hard) coherence test.

C.4 On the relationship between interval and coherence test

We comment on the relationship between the interval test and the coherence test for the community models. In Theorem 5.4, sufficient conditions for strong detection for the von Mises community model are given for $k = o(\log n)$ using the interval test, and for $k = c \log n$ and $k = \omega(\log n)$ using the coherence test. While those sections are complementary, the estimators themselves are rather similar in the regime of small k (and so large κ) as we explain in this section.

The statistic used in the coherence test for a given $C \subset [n]$ with $|C| = k$ satisfies

$$\left| \sum_{e \in E(C)} Z_e \right| = \sum_{e \in E(C)} \cos(X_e - \hat{\Theta}_C) \quad \text{where} \quad \hat{\Theta}_C = \arg \max_{\theta} \sum_{e \in E(C)} \cos(X_e - \theta).$$

If the threshold β has the form $K - \epsilon$ for some $\epsilon \in (0, 1)$ and if the threshold is exceeded:

$$\sum_{e \in E(C)} \cos(X_e - \hat{\Theta}_C) \geq K - \epsilon \tag{C.8}$$

then $\cos(X_e - \hat{\Theta}_C) \geq 1 - \epsilon$ for each e . If ϵ is close to zero then by the Taylor approximation of $\cos$ we have

$$\begin{aligned} \left| \sum_{e \in E(C)} Z_e \right| &\approx K - \frac{1}{2} \sum_{e \in E(C)} |X_e - \hat{\Theta}_C|^2 \\ \hat{\Theta}_C &\approx \arg \min_{\theta} \sum_{e \in E(C)} |X_e - \theta|^2 \end{aligned}$$

Therefore, if we define $V_C = \min_{\theta} \frac{1}{K-1} \sum_{e \in E(C)} |X_e - \theta|^2$, the event (C.8) is approximately the same as $V_C \leq \frac{2\epsilon}{K-1}$. If the values X_e for $e \in E(C)$ are all on the same half of the unit circle then V_C is the usual unbiased estimator of variance for real valued observations. With this motivation we define the *variance test* to be

$$\psi_n(\tau) \triangleq \mathbf{1} \left\{ \min_{\substack{C \subseteq [n] \\ |C|=k}} V_C(X) \leq \sigma^2 \right\}.$$

for some threshold σ^2 .

In summary, the coherence test with threshold $\beta = K - \epsilon$ is approximately the same as the variance test with threshold $\sigma^2 = \frac{2\epsilon}{K-1}$. In turn, the variance test is rather similar to the interval test with threshold K and interval width $\tau = \sigma$ in the sense that the square root of sample variance and the width of the range of $(X_e : e \in E(C))$ are two similar measures of the spread of those random variables.

C.5 On knowing Θ^*

For the community models, our achievability results do not rely on knowledge of the planted phase Θ^* , while our converse bounds continue to hold even if Θ^* is revealed to the decision maker. For the flat models, we can characterize how access to Θ^* can improve performance by reducing the optimal error probabilities, as quantified in the following proposition.

Proposition C.1. *Consider the detection problem for the flat uniform model such that Θ^* is known to the decision maker. If $K^2 \leq N$ and $\tau = o(K^2/N)$ then strong detection is possible. If $K^2 \leq N$ and $\tau = \omega(K^2/N)$ then weak detection is impossible.*

Remark C.1. Comparing Corollary 5.1 and Proposition C.1 shows that the value of knowing Θ is that τ can be larger by a factor $N^{\frac{1}{K-1}}$ for constant K and by a factor $\log N$ for $K = N^\alpha$ for $0 < \alpha < 1/2$.

Proof. Without loss of generality, by symmetry, assume it is known that $\Theta^* = 0$.

(Achievability) Suppose $K^2 \leq N$ and $\tau = o(K^2/N)$. Let Y denote the number of observations that fall into the interval $[0, \tau]$ and consider the test with threshold γ based on Y . The error probabilities are given by:

$$p_{\text{miss}} = P(K + \text{binom}(N - K, \tau) < \gamma) \quad \text{and} \quad p_{\text{fa}} = P(\text{binom}(N, \tau) \geq \gamma).$$

Note p_{miss} here is equal to the upper bound on p_{miss} used in Section 5.2.1. Hence, we can apply the same upper bound on p_{miss} as in that section. Also, p_{fa} is less than $\mathbb{Q}(B_1)$ which is less than the right hand side of (5.8) (for $\gamma \geq 1 + (N - 1)\tau$) and the right hand side of (5.7), both with the leading factor of N removed. If K is constant then $n\tau \rightarrow 0$ and we take $\gamma = K$ leading to $p_{\text{miss}} = 0$ and $p_{\text{fa}} \rightarrow 0$ by (5.7) with the factor of N removed. If $K \rightarrow \infty$ such that $K^2 \leq N$ and $\tau = o(K^2/N)$ then (5.4) and (5.5) with the term $-\log(N)$ removed both hold and together are sufficient to prove $p_{\text{miss}} \rightarrow 0$ and $p_{\text{fa}} \rightarrow 0$.

(Converse) Suppose that $K^2 \leq N$ and $\tau = \omega\left(\frac{K^2}{N}\right)$. We follow the moment method of the proof of Theorem 5.1(b). With it known that $\Theta^* \equiv 0$, only the case $\theta = \theta' = 0$ is relevant so the variable δ is equal to τ . We thus find:

$$\begin{aligned} E_{\mathbb{Q}}[L^2] &= \sum_{j=0}^K P\{|S \cap S'| = j\} \tau^{-j} \leq \sum_{j=0}^K P\left\{\text{binom}\left(K, \frac{K}{N}\right) = j\right\} \tau^{-j} = \left(1 - \frac{K}{N} + \frac{K}{N\tau}\right)^K \\ &\leq \exp\left(\frac{K^2}{N\tau}\right) \rightarrow 1. \end{aligned}$$

Therefore, weak detection is impossible. □

Remark C.2. The hypothesis test used in the above proof of Proposition C.1 is the maximum likelihood ratio test and is hence Pareto optimal. So another method of proof of the converse part of the Proposition would be to show $\min_{\gamma} p_{\text{fa}} + p_{\text{miss}} \rightarrow 1/2$ for that test.

C.6 Community recovery and connection to stochastic block models

The community models in this chapter differ from stochastic block models for a planted dense community in that the marginal distributions of the edge labels is the same under both distributions, $\mathbb{P}$ and $\mathbb{Q}$. However, if a genie were to reveal the value of the parameter Θ then the community models in this chapter fall into the realm of stochastic block models with general edge label distributions P and Q as in [WX21, HWX17]. Under that model for given n, k, P, Q it is assumed there is a subset C^* of cardinality k drawn uniformly at random from among subsets of $[n]$ of cardinality k . A symmetric $n \times n$ matrix A with zero diagonal is observed such that for all $1 \leq i < j \leq n$, A_{ij} are independent and $A_{ij} \sim P$ if $i, j \in C^*$ and $A_{ij} \sim Q$ otherwise. Necessary and sufficient information theoretic conditions for weak and strong detection are given in [WX21] and for weak and exact recovery are given in [HWX17]. The necessary conditions there translate to necessary conditions for the model here because the genie giving Θ can be ignored. For simplicity we assume without loss of generality that $\Theta = 0$.

To apply [HWX17] we let P denote the von Mises distribution with parameters $\kappa, 0$ and Q be the uniform distribution on $[0, 2\pi]$. The likelihood ratio is given by $\frac{dP}{dQ}(x) = \frac{e^{\kappa \cos(x)}}{I_0(\kappa)}$ which is bounded if κ is bounded in which case Assumption 1 of [HWX17] holds by Lemma 1 in [HWX17]. Thus, by Theorem 1 in [HWX17] in a regime with κ bounded a necessary condition on κ for weak recovery is $K \cdot D_{\text{KL}}(P||Q) \rightarrow \infty$ and $\liminf \frac{(K-1)D_{\text{KL}}(P||Q)}{\log(n/K)} \geq 2$. Note that

$$D_{\text{KL}}(P||Q) = \mathbb{E}_P \left[\log \frac{dP}{dQ} \right] = \mathbb{E}_P[\kappa \cos(\Theta)] - \log I_0(\kappa) = \kappa A(\kappa) - \log I_0(\kappa)$$

For $\kappa \rightarrow 0$, $D_{\text{KL}}(P||Q) \sim \frac{\kappa^2}{4}$. Thus, if we consider k varying with n such that $\Omega(\log n) \leq k \leq o(n)$ we find that a necessary condition for weak recovery is $\liminf \frac{(k-1)\kappa^2}{\log(n/k)} \geq 2$, which is satisfied for example if $\kappa^2 \geq \frac{2 \log(n/k)}{k}$. [We can also try using Theorem 2 of [HWX17] to get a necessary condition for exact recovery.]

If Θ were known we could also apply results from [HWX18] (related to [DM15]) to yield a polynomial time message passing algorithm for weak recovery and, after a cleanup procedure, exact recovery, if $\Omega(\sqrt{n}) \leq k \leq o(n)$ under a signal-to-noise ratio condition (that is not information theoretically tight for weak recovery but might be for exact recovery). The analysis in those papers is based on the moments of the observations so we change variables and work with the observations W_{ij} given by $W_{ij} = \sqrt{2} \cos(X_{ij} - \Theta)$. The new observations form a sufficient statistic (assuming Θ is known) so there is no information theoretic loss. We have that if $\{i, j\} \not\subset C^*$ then $\mathbb{E}[W_{ij}] = 0$ and

$\mathbb{E}[W_{i,j}^2] = 1$ and if $\{i, j\} \subset C^*$ then $\mu = \mathbb{E}[W_{ij}] = \sqrt{2}A(\kappa) = \kappa/\sqrt{2} + O(\kappa^3)$ so the message passing algorithm succeeds if $\lambda = \frac{\kappa^2 K^2}{2n} > \frac{1}{e}$.

The message passing algorithm can be adapted to work without knowing Θ as follows. Given a large integer B we could run B copies of the message passing algorithm in parallel, where the b^{th} version is applied to the observation matrix $W^{(b)} = \sqrt{2} \cos(X_{ij} + \frac{b}{2\pi})$. Then for each b , $\mu_b = E[W_{ij}|\Theta^* = \theta] = \cos(\theta - \frac{b}{2\pi})A(\kappa)$ Therefore, $\max_b \mu_b \geq \cos(\frac{1}{4\pi B}) A(\kappa)$.

Thus, if λ is fixed with $\lambda > \frac{1}{e}$ then we select a positive integer B large enough that $\lambda \cos(\frac{1}{4\pi B}) > \frac{1}{e}$. Then if κ is chosen so that $\lambda = \frac{\kappa^2 K^2}{2n}$ the combined message passing algorithm can achieve both error probabilities converging to zero – that is, strong detection in polynomial time.

Appendix D

Supplementary material for Chapter 6

D.1 Proofs for the WCG model

This section presents the proofs pertaining to the WCG model. First, the lemmas used in the proof of the impossibility result, Theorem 6.1 are presented. Lemma D.1 bounds the performance of the estimator that matches vertices in $\mathcal{B}_1 \cup \mathcal{B}'_2$ by random guessing, and Lemma D.2 uses a simple concentration argument to bound the size of $\mathcal{B}_1 \cup \mathcal{B}'_2$. Armed with these, we proceed to establish that the k -core estimator is precise in Section D.1.1. It is followed by Section D.1.2, where supporting lemmas for Theorem 6.2(ii) and (iii) are respectively presented.

Lemma D.1. *Let n be a positive integer, and let p, s, γ, λ be in $[0, 1]$. Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^\star)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$, and let $\mathcal{B}'_2$ denote the pre-image of $\mathcal{B}_2$ under $\pi^\star$. Let $\bar{\mu}$ be a matching output by an estimator $\bar{\mathcal{E}}$ such that its domain $\text{dom}(\bar{\mu}) \subseteq \mathcal{B}_1 \cup \mathcal{B}'_2$ and its codomain is the set $\pi^\star(\mathcal{B}_1 \cup \mathcal{B}'_2)$. Conditioned on the domain and codomain, suppose that $\bar{\mathcal{E}}$ matches nodes in its domain by random guessing. Let $\delta > 0$.*

(i) *If $\bar{\mathcal{E}}$ is precise, then*

$$\mathbb{P}(\text{dom}(\bar{\mu}) = \phi) = 1 - o(1).$$

(ii) *If $\bar{\mathcal{E}}$ is δ -imprecise, then for any $\varepsilon > 0$:*

$$\mathbb{P}(\text{ov}(\bar{\mu}, \pi^\star) > \varepsilon n) = o(1) \tag{D.1}$$

Proof. (i) Note that $\text{dom}(\bar{\mu}) \subseteq \mathcal{B}_1 \cup \mathcal{B}'_2$ by definition. Since each element in $\text{dom}(\bar{\mu})$ is mapped randomly to an element in $\pi^\star(\mathcal{B}_1 \cup \mathcal{B}'_2)$, and since the mapping is injective, it follows that the

probability that all nodes in $\text{dom}(\bar{\mu})$ are correctly matched is given by

$$\begin{aligned} \mathbb{P}\left(\bigcap_{i \in \text{dom}(\bar{\mu})} \{\bar{\mu}(i) = \pi^*(i)\}\right) &= \frac{1}{|\mathcal{B}_1 \cup \mathcal{B}'_2|} \times \frac{1}{|\mathcal{B}_1 \cup \mathcal{B}'_2| - 1} \times \cdots \times \frac{1}{|\mathcal{B}_1 \cup \mathcal{B}'_2| - |\text{dom}(\bar{\mu})| + 1} \\ &= \frac{(|\mathcal{B}_1 \cup \mathcal{B}'_2| - |\text{dom}(\bar{\mu})|)!}{|\mathcal{B}_1 \cup \mathcal{B}'_2|!}. \end{aligned}$$

Since $|\mathcal{B}_1 \cup \mathcal{B}'_2| \geq \max(\lambda, 1 - \lambda) \gamma n = \Omega(n)$, it follows that the above probability is $o(1)$ if $|\text{dom}(\bar{\mu})| \geq 1$, and equals 1 if and only if $|\text{dom}(\bar{\mu})| = 0$. Since $\bar{\mathcal{E}}$ achieves precise recovery, the above probability must be $1 - o(1)$, which then implies $\mathbb{P}(\text{dom}(\bar{\mu}) = \emptyset) = 1 - o(1)$.

(ii) For any node $m \in \text{dom}(\bar{\mu})$, let X_m denote the indicator event $\mathbb{1}\{\bar{\mu}(m) = \pi^*(m)\}$. Notice that $\text{ov}(\bar{\mu}, \pi^*) = \sum_{m \in \text{dom}(\bar{\mu})} X_m$, and so it follows that

$$\begin{aligned} \mathbb{E}\text{ov}(\bar{\mu}, \pi^*) &= \sum_{m \in \text{dom}(\bar{\mu})} \mathbb{E}X_m = \sum_{m \in \text{dom}(\bar{\mu})} \mathbb{P}(\bar{\mu}(m) = \pi^*(m)) = \sum_{m \in \text{dom}(\bar{\mu})} \frac{1}{|\text{dom}(\bar{\mu})|} = 1, \\ \text{Var}(\text{ov}(\bar{\mu}, \pi^*)) &= \mathbb{E}Y^2 - 1 = \left(\sum_{m_1 \in \text{dom}(\bar{\mu})} \sum_{m_2 \in \text{dom}(\bar{\mu})} \mathbb{E}X_{m_1} X_{m_2} \right) - 1 \stackrel{(a)}{=} 1, \end{aligned}$$

where (a) uses the fact that

$$\mathbb{E}X_{m_1} X_{m_2} = \mathbb{P}(\{\bar{\mu}(m_1) = \pi^*(m_1)\} \cap \{\bar{\mu}(m_2) = \pi^*(m_2)\}) = \begin{cases} \frac{1}{|\text{dom}(\bar{\mu})|} \times \frac{1}{|\text{dom}(\bar{\mu})| - 1}, & \text{if } m_1 \neq m_2 \\ \frac{1}{|\text{dom}(\bar{\mu})|}, & \text{if } m_1 = m_2. \end{cases}$$

Thus, Chebyshev's inequality yields

$$\mathbb{P}(\text{ov}(\bar{\mu}, \pi^*) \geq \varepsilon n) \leq \mathbb{P}(|\text{ov}(\bar{\mu}, \pi^*) - 1| \geq \varepsilon n - 1) \leq \frac{1}{(\varepsilon n - 1)^2} = o(1),$$

which concludes the proof. $\square$

Lemma D.2. *Let n be a positive integer, and let p, s, γ, λ be in $[0, 1]$. Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$, and let $\mathcal{B}'_2$ denote the pre-image of $\mathcal{B}_2$ under π^* . Then, for any $\varepsilon > 0$*

$$\mathbb{P}\left(\left|\frac{|\mathcal{B}_1 \cap \mathcal{B}'_2|}{n} - \lambda(1 - \lambda)\gamma^2\right| > \varepsilon\right) = o(1).$$

Proof. Without loss of generality, assume that $\lambda \geq 1/2$, and that $\mathcal{B}_1 = \{1, 2, \dots, \lambda\gamma n\}$, since the number of elements in $\mathcal{B}_1 \cap \mathcal{B}'_2$ is independent of the elements in $\mathcal{B}_1$. Since π^* is independent of the sets $\mathcal{B}_1$ and $\mathcal{B}_2$, selecting $\mathcal{B}_2$ uniformly at random is equivalent to selecting $\mathcal{B}'_2$ uniformly at random. View $\mathcal{B}'_2$ as being constructed by sampling $(1 - \lambda)\gamma n$ nodes from $[n]$ without replacement. A node i sampled this way is labeled a success if $i \in \mathcal{B}_1$ and a failure otherwise. The number

of successes after $(1 - \lambda)\gamma n$ trials is exactly $|\mathcal{B}_1 \cap \mathcal{B}'_2|$, and is described by the hypergeometric distribution $\text{HypGeom}(n, (1 - \lambda)\gamma n, \lambda\gamma n)$. Using standard formulas for the mean and variance of the hypergeometric distribution yields

$$\begin{aligned}\mathbb{E}|\mathcal{B}_1 \cap \mathcal{B}'_2| &= \lambda(1 - \lambda)\gamma^2 n, \\ \text{Var}(|\mathcal{B}_1 \cap \mathcal{B}'_2|) &= \lambda(1 - \lambda)\gamma^2(1 - \lambda\gamma)(1 - (1 - \lambda)\gamma) \times \frac{n^2}{n - 1}.\end{aligned}$$

Therefore, applying Chebyshev's inequality to $|\mathcal{B}_1 \cap \mathcal{B}'_2|/n$ yields that for any constant $\varepsilon > 0$:

$$\mathbb{P}\left(\left|\frac{|\mathcal{B}_1 \cap \mathcal{B}'_2|}{n} - \lambda(1 - \lambda)\gamma\right| \geq \varepsilon\right) \leq \lambda(1 - \lambda)\gamma^2(1 - \lambda\gamma)(1 - (1 - \lambda)\gamma) \times \frac{1}{\varepsilon^2(n - 1)} = o(1),$$

as desired. $\square$

D.1.1 Proof of Theorem 6.2(i)

This subsection analyzes the k -core estimator. To show that the k -core estimator is precise, it suffices to establish that with high probability, the output of the k -core estimator $\hat{\mu}_k$ is such that $\text{dom}(\hat{\mu}_k)$ is exactly the k -core of the true intersection graph $\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2$, and furthermore that the mapping $\hat{\mu}_k$ agrees with π^* on its domain. Theorem D.1 and Lemma D.5 together establish this. A similar result is proved in [CKMP19] for Erdős-Rényi graphs when no adversary is present. The techniques introduced there were extended to analyze the k -core estimator for graph matching in correlated stochastic block models [GRS22] and inhomogeneous random graphs [RS23]. These techniques are adapted to the WCG model below.

Definition D.1 (Weak k -core matching). Let H_1 and H_2 be two graphs, and let μ^* and μ be matchings. We say that μ is a weak k -core matching of H_1 and H_2 with respect to μ^* if the average degree in $H_1 \wedge_{\mu} H_2$ of all the nodes $i \in M$ such that $\mu(i) \neq \mu^*(i)$ is at least k , i.e.

$$f(\mu; \mu^*, H_1, H_2, k) \triangleq \sum_{i \in \text{dom}(\mu): \mu(i) \neq \mu^*(i)} \deg_{H_1 \wedge_{\mu} H_2}(i) \geq k \times |\{i \in \text{dom}(\mu): \mu(i) \neq \mu^*(i)\}|.$$

When the context is clear, we omit the parameters from the notation and simply use $f(\mu)$.

Definition D.2 (Maximal matching). Let μ^* and μ be matchings. A matching μ is a maximal matching with respect to μ^* (or simply, a μ^* -maximal matching) if for every $i \in \text{dom}(\mu^*)$, either $i \in \text{dom}(\mu)$ or $\mu^*(i) \in \text{range}(\mu)$.

Remark D.1. Every matching can be uniquely extended to a μ^* -maximal matching.

Denote by $\mathcal{M}(\mu^*, d)$ the set of all matchings μ which are μ^* -maximal and such that there are exactly d nodes in M for which the images under μ and μ^* disagree. Let $\mathcal{M}(\mu^*) \triangleq \bigcup_{d=0}^n \mathcal{M}(\mu^*, d)$.

Lemma D.3. Let μ^* be a matching, and suppose that μ is a k -core matching of two graphs H_1 and

H_2 . Then, there exists a matching $\mu' \in \mathcal{M}(\mu^*)$ such that μ' is a weak k -core matching.

Proof. Since μ is a k -core matching, it is also a weak k -core matching with respect to μ^* . Let μ' denote the unique extension of μ to a μ^* -maximal matching. Since the extension only involves adding elements from μ^* , it follows that

$$\begin{aligned} k |\{i : \mu'(i) \neq \mu^*(i)\}| &= k |\{i : \mu(i) \neq \mu^*(i)\}| \\ &\leq \sum_{j \in \{i : \mu(i) \neq \mu^*(i)\}} \deg_{H_1 \wedge_{\mu} H_2}(j) \\ &\leq \sum_{j \in \{i : \mu'(i) \neq \mu^*(i)\}} \deg_{H_1 \wedge_{\mu'} H_2}(j), \end{aligned}$$

and so the average degree in $H_1 \wedge_{\mu'} H_2$ of all the nodes $i \in \text{dom}(\mu')$ such that $\mu'(i) \neq \mu^*(i)$ is at least k . Since μ' is a μ^* -maximal matching by construction, the result follows. $\square$

Lemma D.3 establishes that if no weak k -core matchings exist in $\mathcal{M}(\mu^*, d)$ for any $d > 0$, then any k -core matching must agree with μ^* . The main advantage of restricting the search to $\mathcal{M}(\mu^*)$ is that there are much fewer μ^* -maximal matchings than matchings.

Lemma D.4. $|\mathcal{M}(\mu^*, d)| \leq n^{2d}/d!$.

Proof. Any μ^* -maximal matching can be identified by the set $\{(i, \mu(i)) : \mu(i) \neq \mu^*(i)\}$. Since the cardinality of this set is d , there are $\binom{n}{d}$ choices of i and at most $\binom{n}{d}$ choices of $\mu(i)$ that preserve the injectivity of μ . Further, there are at most $d!$ ways to match up the d nodes according to μ that preserve the injectivity of μ . It follows that

$$|\mathcal{M}(\mu^*, d)| \leq d! \left(\binom{n}{d} \right)^2 \leq d! \left(\frac{n^d}{d!} \right) = \frac{n^{2d}}{d!}.$$

$\square$

Theorem D.1. Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$. Let $\hat{\mu}_k$ denote the matching output by the k -core estimator $\hat{\mathcal{E}}_k(\tilde{G}_1, \tilde{G}_2)$. Then,

$$\mathbb{P} \left(\text{dom}(\hat{\mu}_k) = \text{core}_k \left(\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2 \right) \text{ and } \hat{\mu}_k = \pi^*|_{\text{dom}(\hat{\mu}_k)} \right) \geq 2 - \exp(-n^2 \xi),$$

where

$$\xi \triangleq \max_{1 \leq d \leq n} \max_{(M, \mu) \in \mathcal{M}(d)} \mathbb{P}(f(\mu) \geq kd)^{1/d}.$$

Proof. Let $\mathcal{K}$ denote the set of all k -core matchings of $\tilde{G}_1$ and $\tilde{G}_2$, and let $\mathcal{H}$ denote the event

$$\mathcal{H} \triangleq \bigcap_{\mu \in \mathcal{K}} \bigcap_{i \in \text{dom}(\mu)} \{\mu(i) = \pi^*(i)\}$$

First, since $\hat{\mu}_k$ itself is a k -core matching, the event $\mathcal{H}$ implies that $\hat{\mu}_k(i) = \pi^*(i)$ for all $i \in \text{dom}(\hat{\mu}_k)$. Let $M^* = \text{core}_k(\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2)$. We now show the event $\mathcal{H}$ implies that $\text{dom}(\hat{\mu}_k) = M^*$.

First, note that $\pi^*|_{M^*}$ is a k -core matching. Therefore, by the maximality property in the definition of the k -core estimator, we have that $|\text{dom}(\hat{\mu}_k)| \geq |M^*|$. Therefore, to show that $\text{dom}(\hat{\mu}_k) = M^*$, it suffices to simply show that $\text{dom}(\hat{\mu}_k) \subseteq M^*$ whenever $\mathcal{H}$ occurs. Assume to the contrary that $L \triangleq \text{dom}(\hat{\mu}_k) \setminus M^*$ is non-empty. Then, on the event $\mathcal{H}$, the subgraph of $\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2$ that is induced on $M^* \cup L$ also has minimum degree k , contradicting the maximality of M^* . Therefore, $\text{dom}(\hat{\mu}_k) = M^*$ if $\mathcal{H}$ occurs. It follows that

$$\mathbb{P}\left(\text{dom}(\hat{\mu}_k) = \text{core}_k(\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2) \text{ and } \hat{\mu}_k = \pi^*|_{\text{dom}(\hat{\mu}_k)}\right) \geq \mathbb{P}(\mathcal{H}).$$

To prove the theorem, it suffices to show that $\mathbb{P}(\mathcal{H}^c) \leq \exp(n^2\xi) - 1$. For any graph G , let $d_{\min}(G)$ denote its minimum degree. Indeed,

$$\begin{aligned} \mathbb{P}(\mathcal{H}^c) &= \mathbb{P}\left(\bigcup_{\mu \in \mathcal{K}} \bigcup_{i \in \text{dom}(\mu)} \{\mu(i) \neq \pi^*(i)\}\right) \\ &= \mathbb{P}\left(\bigcup_{\substack{\mu \text{ such that} \\ \exists i \in \text{dom}(\mu): \mu(i) \neq \pi^*(i)}} d_{\min}(\tilde{G}_1 \wedge_{\mu} \tilde{G}_2) \geq k\right) \\ &\stackrel{(a)}{\leq} \sum_{d=1}^n \mathbb{P}\left(\bigcup_{\substack{\mu \text{ such that} \\ |\{i \in \text{dom}(\mu): \mu(i) \neq \pi^*(i)\}| = d}} f(\mu) \geq kd\right) \\ &\stackrel{(b)}{\leq} \sum_{d=1}^n \mathbb{P}\left(\bigcup_{\mu \in \mathcal{M}(\pi^*, d)} f(\mu) \geq kd\right) \\ &\stackrel{(c)}{\leq} \sum_{d=1}^n |\mathcal{M}(d)| \left(\max_{1 \leq d \leq n} \max_{\mu \in \mathcal{M}(\pi^*, d)} \mathbb{P}(f(\mu) \geq kd)\right) \\ &\stackrel{(d)}{\leq} \sum_{d=1}^n \frac{(n^2\xi)^d}{d!} \\ &\leq \exp(n^2\xi) - 1. \end{aligned}$$

where (a) partitions the set of all matchings based on the number of disagreements of the matching with π^* and uses a union bound, (b) follows from Lemma D.3, (c) follows from a union bound, and (d) is from the definition of ξ and an application of Lemma D.4. $\square$

Next, we show that ξ decays rather quickly under appropriate conditions. The proof below follows [GRS22] and [RS23], although their focus is the stochastic block model and the inhomogeneous random graph model respectively. Below, we adapt the argument to the WCG model.

Lemma D.5. *Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$. Let $\mathcal{M}(\pi^*, d)$ be the set of all π^* -maximal matchings μ such that $|\{i \in \text{dom}(\mu) : \mu(i) \neq \pi^*(i)\}| = d$.*

$$\xi = \max_{1 \leq d \leq n} \max_{\mu \in \mathcal{M}(\pi^*, d)} \mathbb{P}(f(\mu) \geq kd)^{1/d}.$$

Suppose that $p = \frac{C \log n}{n}$ for some $C > 0$. If $k \geq 13$, then $\xi = o(n^{-2})$.

Proof. For any matching $\mu \in \mathcal{M}(\pi^*, d)$, define the following sets:

$$\begin{aligned} \mathcal{A}(\mu) &\triangleq \{(i, j) \in \text{dom}(\mu) \times \text{dom}(\mu) : \mu(i) \neq \pi^*(i)\}, \\ \mathcal{T}(\mu) &\triangleq \{(i, j) \in \mathcal{A}(\mu) : \mu(i) = \pi^*(j) \text{ and } \mu(j) = \pi^*(i)\}, \\ \mathcal{N}(\mu) &\triangleq \mathcal{A}(\mu) \setminus \mathcal{T}(\mu). \end{aligned}$$

For a graph G , let $G(i, j)$ denote the (i, j) -th entry of its adjacency matrix. First, note that $|\mathcal{A}(\mu)| \leq d|\text{dom}(\mu)| \leq dn$, and that $|\mathcal{T}(\mu)| \leq d$. This is because μ makes d errors by definition. It follows that

$$\begin{aligned} f(\mu) &= \sum_{\substack{i \in \text{dom}(\mu) \\ \mu(i) \neq \pi^*(i)}} \deg_{\tilde{G}_1 \wedge_\mu \tilde{G}_2}(i) \\ &= \sum_{(i, j) \in \mathcal{A}(\mu)} \tilde{G}_1(i, j) \tilde{G}_2(\mu(i), \mu(j)) \\ &= 2 \sum_{\substack{(i, j) \in \mathcal{T}(\mu) \\ i < j}} \tilde{G}_1(i, j) \tilde{G}_2(\mu(i), \mu(j)) + \sum_{(i, j) \in \mathcal{N}(\mu)} \tilde{G}_1(i, j) \tilde{G}_2(\mu(i), \mu(j)) \\ &=: 2X_{\mathcal{T}} + X_{\mathcal{N}}. \end{aligned}$$

It is easy to see that $X_{\mathcal{T}}$ and $X_{\mathcal{N}}$ are independent, since they involve disjoint node pairs. Furthermore, for the same reason, the individual terms in $X_{\mathcal{T}}$ are also independent. More importantly,

$$\tilde{G}_1(i, j) \tilde{G}_2(\mu(i), \mu(j)) \sim \begin{cases} \text{Bern}(p^2 s^2), & i \in \mathcal{B}_1 \cup \mathcal{B}'_2 \text{ or } j \in \mathcal{B}_1 \cup \mathcal{B}'_2 \\ \text{Bern}(ps^2), & \text{otherwise} \end{cases}.$$

Therefore, it follows that

$$X_{\mathcal{T}} \preceq \text{Bin}(|\mathcal{T}(\mu)|, ps^2) \preceq \text{Bin}(d, ps^2), \quad (\text{D.2})$$

where $\preceq$ denotes stochastic domination of the RHS. Next, the $X_{\mathcal{N}}$ is analyzed, which is slightly

more complicated because the summands may be correlated. To circumvent this, partition $\mathcal{N}(\mu)$ into $\mathcal{N}_1(\mu)$, $\mathcal{N}_2(\mu)$, and $\mathcal{N}_3(\mu)$ and define

$$X_{\mathcal{N}_j} \triangleq \sum_{\substack{(i,j) \in \mathcal{N}_j(\mu) \\ i < j}} \tilde{G}_1(i, j) \tilde{G}_2(\mu(i), \mu(j)), \quad j \in \{1, 2, 3\}.$$

It follows that $X_{\mathcal{N}} \leq 2(X_{\mathcal{N}_1} + X_{\mathcal{N}_2} + X_{\mathcal{N}_3})$. Here, the factor of 2 accounts for the restriction that $i < j$. Furthermore, it is possible to partition $\mathcal{N}$ in such a way that $X_{\mathcal{N}_1}$, $X_{\mathcal{N}_2}$ and $X_{\mathcal{N}_3}$ are mutually independent. To see this, consider two node pairs (i, j) and (a, b) in $\binom{M}{2}$. The random variables $\tilde{G}_1(i, j) \tilde{G}_2(\mu(i), \mu(j))$ and $\tilde{G}_1(a, b) \tilde{G}_2(\mu(a), \mu(b))$ are dependent if and only if one of the following two conditions hold:

$$\{\mu(i), \mu(j)\} = \{\pi^*(a), \pi^*(b)\} \quad (\text{D.3})$$

$$\{\mu(a), \mu(b)\} = \{\pi^*(i), \pi^*(j)\}. \quad (\text{D.4})$$

Consider then the dependency graph H on the node set $V(H) \triangleq \{\{i, j\} : (i, j) \in \mathcal{N}(\mu)\}$ such that two nodes in this graph $\{i, j\}$ and $\{a, b\}$ have an edge between them if and only if they satisfy (D.3) or (D.4). Since each node in H has at most 2 neighbors, it follows that H is 3-colorable. Letting $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}_3$ be the partition corresponding to the 3 colors, it can be seen that $X_{\mathcal{N}_1}$, $X_{\mathcal{N}_2}$ and $X_{\mathcal{N}_3}$ are independent sums of Binomial random variables. Specifically, for each $(i, j) \in \mathcal{N}(\mu)$, we have that $\tilde{G}_1(i, j) \tilde{G}_2(\mu(i), \mu(j)) \sim \text{Bern}(p^2 s^2)$. It follows that for each $m \in \{1, 2, 3\}$:

$$X_{\mathcal{N}_m} \preceq \text{Bin}(|\mathcal{N}_m(\mu)|, p^2 s^2) \preceq \text{Bin}(dn, p^2 s^2),$$

where we have used the fact that $|\mathcal{N}_m(\mu)| \leq |\mathcal{N}(\mu)| \leq |\mathcal{A}(\mu)| \leq dn$. Finally,

$$\begin{aligned} \mathbb{P}(f(\mu) \geq kd) &\leq \mathbb{P}(2X_{\mathcal{T}} + X_{\mathcal{N}} \geq kd) \\ &\leq \mathbb{P}(2(X_{\mathcal{T}} + X_{\mathcal{N}_1} + X_{\mathcal{N}_2} + X_{\mathcal{N}_3}) \geq kd) \\ &\leq \sum_{m=1}^3 \mathbb{P}(2X_{\mathcal{T}} + 6X_{\mathcal{N}_m} \geq kd). \end{aligned}$$

The Chernoff bound can then be applied to the binomial distribution. It follows that for any $\theta > 0$:

$$\begin{aligned} \mathbb{P}(f(\mu) \geq kd) &\leq \sum_{m=1}^3 e^{-\theta kd} \mathbb{E}[e^{2\theta X_{\mathcal{T}}}] \mathbb{E}[e^{6\theta X_{\mathcal{N}_m}}] \\ &\leq 3e^{-\theta kd} \left(1 + ps^2(e^{2\theta} - 1)\right)^d \left(1 + p^2 s^2 e^{6\theta} - 1\right)^{dn} \\ &\stackrel{(a)}{\leq} 3 \exp\left(-d\left(\theta k - e^{2\theta} ps^2 - ne^{6\theta} p^2 s^2\right)\right), \end{aligned}$$

where (a) follows from the fact that $(1+x)^t \leq e^{tx}$. Finally, set $\theta = c' \log n$, and observe that when

$c < 1/6$:

$$\begin{aligned} e^{2\theta} p s^2 &= n^{2c'-1} C s^2 \log n = o(1), \\ n e^{6\theta} p^2 s^2 &= n^{6c-1} C^2 s^2 (\log n)^2 = o(1). \end{aligned}$$

Finally, since $k \geq 13$, it can be ensured that $ck > 2$ by choosing $c = \frac{1}{6} - \varepsilon$ for sufficiently small ε . This yields

$$\xi \leq \mathbb{P}(f(\mu) \geq kd)^{1/d} \leq (3 \exp(-d(ck \log n - o(1))))^{1/d} = o(n^{-2}),$$

as desired. $\square$

D.1.2 Supporting lemmas for Theorem 6.2

Lemma D.6. *Let n and k be positive integers, and let p, s, γ, λ be real numbers such that $0 \leq p, s, \gamma, \lambda \leq 1$. Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$. Suppose that $p = \frac{C \log(n)}{n}$ for some positive constant C . Let $\alpha^* = 1 - \gamma + \lambda(1 - \lambda)\gamma^2$. If $C > \frac{1}{s^2 \alpha^*}$, then the k -core M^* of $\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2$ with $k = \sqrt{\log n}$ satisfies*

$$\mathbb{P}(M^* = (\mathcal{B}_1 \cup \mathcal{B}_2')^c) = 1 - o(1).$$

Proof. Let the matching $\tilde{\pi}^*$ with $\text{dom}(\tilde{\pi}^*) = (\mathcal{B}_1 \cup \mathcal{B}_2')^c$, denote the restriction of the true permutation π^* to the node set $(\mathcal{B}_1 \cup \mathcal{B}_2')^c$. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ denote the events

$$\begin{aligned} \mathcal{H}_1: \quad & \bigcap_{i \in \mathcal{B}_1 \cup \mathcal{B}_2'} \left\{ \deg_{\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2}(i) < k \right\} \\ \mathcal{H}_2: \quad & \bigcap_{j \notin \mathcal{B}_1 \cup \mathcal{B}_2'} \left\{ \deg_{\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2}(j) > k \right\}. \end{aligned}$$

Note that the intersection graph in $\mathcal{H}_1$ is with respect to π^* whereas the intersection graph in $\mathcal{H}_2$ is with respect to $\tilde{\pi}^*$. Let $\mathcal{H} = \mathcal{H}_1 \cap \mathcal{H}_2$. It suffices to show that $\mathbb{P}(\mathcal{H}^c) = o(1)$. This implies that with high probability, no node in $\mathcal{B}_1 \cup \mathcal{B}_2'$ has degree greater than k in the intersection graph $\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2$, and therefore cannot belong to its k -core. Furthermore, it also implies that with high probability, the subgraph of $\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2$ induced on the node set $(\mathcal{B}_1 \cup \mathcal{B}_2')^c$ has minimum degree at least k . In turn, this implies that $\mathbb{P}(M^* = (\mathcal{B}_1 \cup \mathcal{B}_2')^c) = 1 - o(1)$, as desired.

Bounding $\mathbb{P}(\mathcal{H}_1^c)$ Since the sets $\mathcal{B}_1$ and $\mathcal{B}_2'$ are selected independent of G_1 and G_2 , it follows for any $i \in \mathcal{B}_1 \cup \mathcal{B}_2'$ that $\deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) \sim \text{Bin}(n-1, p^2 s^2)$. This is because for any $i \in \mathcal{B}_1 \cup \mathcal{B}_2'$ and $j \in [n]$ such that $j \neq i$, Therefore, $\tilde{G}_1(i, j)$ and $\tilde{G}_2(\pi^*(i), \pi^*(j))$ are independent Bernoulli random

variables with mean ps . Therefore, for any $\delta > 0$:

$$\begin{aligned}
\mathbb{P}(\mathcal{H}_1^c) &= \mathbb{P}\left(\bigcup_{i \in \mathcal{B}_1 \cup \mathcal{B}'_2} \left\{ \deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) > (1+\delta)np^2s^2 \right\}\right) \\
&\leq \mathbb{P}\left(|\mathcal{B}_1 \cup \mathcal{B}'_2| > 1.01(1-\alpha^*)n\right) \\
&+ \mathbb{P}\left(\bigcup_{i \in \mathcal{B}_1 \cup \mathcal{B}'_2} \left\{ \deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) > (1+\delta)np^2s^2 \right\} \cap \left\{ |\mathcal{B}_1 \cup \mathcal{B}'_2| \leq 1.01(1-\alpha^*)n \right\}\right) \\
&\leq \mathbb{P}\left(|\mathcal{B}_1 \cup \mathcal{B}'_2| > 1.01(1-\alpha^*)n\right) + 1.01(1-\alpha^*)n \times \mathbb{P}\left(\text{Bin}(n-1, p^2s^2) > (1+\delta)np^2s^2\right) \\
&\stackrel{(a)}{\leq} o(1) + 1.01(1-\alpha^*)n \times \mathbb{P}\left(\text{Bin}(n-1, p^2s^2) > (1+\delta)np^2s^2\right) \\
&\stackrel{(b)}{\leq} o(1) + 2n \times \mathbb{P}\left(\text{Bin}(n, p^2s^2) > (1+\delta)np^2s^2\right) \\
&\stackrel{(c)}{\leq} o(1) + 2n \times \left(\frac{\exp(\delta)}{(1+\delta)^{1+\delta}}\right)^{np^2s^2}, \tag{D.5}
\end{aligned}$$

where (a) follows from Lemma D.2 and (b) follows from $\alpha^* \geq 0$ and the stochastic domination $\text{Bin}(n-1, q) \preceq \text{Bin}(n, q)$ for any $q \in [0, 1]$. Finally, (c) follows from Lemma 3.6. Setting $\delta = \frac{n}{(\log n)^2}$ and recalling that $p = \frac{C \log n}{n}$, it follows that $(1+\delta)np^2s^2 = C^2s^2 + o(1)$. Therefore, Equation (D.5) can be written as

$$\begin{aligned}
\mathbb{P}(\mathcal{H}_1^c) &\leq \mathbb{P}\left(\bigcup_{i \in \mathcal{B}_1 \cup \mathcal{B}'_2} \left\{ \deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) > C^2s^2 + o(1) \right\}\right) \\
&\leq o(1) + 2n \times \left(\frac{\exp\left(\frac{n}{(\log n)^2}\right)}{\left(1 + \frac{n}{(\log n)^2}\right)^{1 + \frac{n}{(\log n)^2}}}\right)^{C^2s^2 \frac{(\log n)^2}{n}} \\
&= \begin{cases} o(1), & Cs > 1 \\ \omega(1), & Cs < 1 \end{cases}.
\end{aligned}$$

Since $C > 1/(s^2\alpha^*)$, it follows that $Cs > \frac{1}{s\alpha^*} \geq 1$. This is because $s\alpha^* \leq 1$. Therefore, for any $\varepsilon > 0$ and any k' such that $k' > C^2s^2 + \varepsilon$, it is true that

$$\mathbb{P}(\mathcal{H}_1^c) \leq \mathbb{P}\left(\bigcup_{i \in \mathcal{B}_1 \cup \mathcal{B}'_2} \left\{ \deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) > k' \right\}\right) = o(1). \tag{D.6}$$

Indeed, our choice of $k = \sqrt{\log n}$ satisfies $k > C^2s^2 + \varepsilon$ for all sufficiently large n . Therefore, $\mathbb{P}(\mathcal{H}_1^c) = o(1)$ as desired.

Bounding $\mathbb{P}(\mathcal{H}_2^c)$ The probability that a node in $\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2$ has degree lesser than k is computed. For any $i \in (\mathcal{B}_1 \cup \mathcal{B}_2')^c$,

$$\deg_{\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2}(i) = \sum_{\substack{j \in (\mathcal{B}_1 \cup \mathcal{B}_2')^c \\ j \neq i}} \tilde{G}_1(i, j) \tilde{G}_2(\pi^*(i), \pi^*(j)).$$

Since i and j are both in $(\mathcal{B}_1 \cup \mathcal{B}_2')^c$, it follows that

$$\tilde{G}_1(i, j) \tilde{G}_2(\pi^*(i), \pi^*(j)) = G_1(i, j) G_2(\pi^*(i), \pi^*(j)) \sim \text{Bern}(ps^2).$$

Further, for any $j_1, j_2 \in (\mathcal{B}_1 \cup \mathcal{B}_2')^c$ such that $j_1 \neq j_2 \neq i$, it follows by independence across edges that $G_1(i, j_1) G_2(\pi^*(i), \pi^*(j_1))$ and $G_1(i, j_2) G_2(\pi^*(i), \pi^*(j_2))$ are independent random variables. Therefore, it follows that $\deg_{\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2}(i) \sim \text{Bin}((1 - \gamma)n - 1, ps^2)$. We have from a union bound that for any $\delta' \in (0, 1)$:

$$\begin{aligned} \mathbb{P}(\mathcal{H}_2^c) &= \mathbb{P}\left(\bigcup_{j \in (\mathcal{B}_1 \cup \mathcal{B}_2')^c} \left\{ \deg_{\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2}(i) < (1 - \delta')(\alpha^*n - 1)ps^2 \right\}\right) \\ &\leq \mathbb{P}\left(|(\mathcal{B}_1 \cup \mathcal{B}_2')| > 1.01(1 - \alpha^*)n\right) \\ &\quad + \mathbb{P}\left(\bigcup_{j \in (\mathcal{B}_1 \cup \mathcal{B}_2')^c} \left\{ \deg_{\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2}(i) < (1 - \delta')(\alpha^*n - 1)ps^2 \right\} \cap |(\mathcal{B}_1 \cup \mathcal{B}_2')| \leq 1.01(1 - \alpha^*)n\right) \\ &\stackrel{(d)}{\leq} o(1) + 2n \times \mathbb{P}\left(\text{Bin}(\alpha^*n - 1, ps^2) < (1 - \delta')(\alpha^*n - 1)ps^2\right) \\ &\stackrel{(e)}{\leq} o(1) + 2n \times \left(\frac{e^{-\delta'}}{(1 - \delta')^{1 - \delta'}}\right)^{ps^2(\alpha^*n - 1)} \\ &\stackrel{(f)}{=} o(1) + 2n \times \left(\frac{e^{-\delta'}}{(1 - \delta')^{1 - \delta'}}\right)^{\left(Cs^2\alpha^* - \frac{Cs^2}{n}\right)\log n} \\ &\stackrel{(g)}{=} \begin{cases} o(1), & Cs^2\alpha^* > 1 \\ \omega(1), & \text{otherwise} \end{cases}. \end{aligned} \tag{D.7}$$

Here, in (d), Lemma D.2 is used along with the fact that $\alpha^* \geq 0$. Further, in (e), Lemma 3.6 is used, and in (f) we have set $p = C \log(n)/n$.

To see why (g) is true, consider the function

$$g(a, x, n) = nx^{a \log n}.$$

Notice that if $a \log(x) < -1$, then $\lim_{n \rightarrow \infty} g(a, x, n) = 0$. Equivalently, this sufficient condition requires

$x < \exp(-1/a)$. Setting $a = Cs^2\alpha^* - \frac{Cs^2}{n}$, this condition reduces to

$$x < \exp\left(-\frac{1}{Cs^2\alpha^* - \frac{Cs^2}{n}}\right). \quad (\text{D.8})$$

When $Cs^2\alpha^* > 1$, there is a sufficiently small constant $\varepsilon > 0$ such that for all n sufficiently large, it is true that $Cs^2\alpha^* - \frac{Cs^2}{n} > 1 + \varepsilon$. Therefore, (D.8) holds if $x < \exp\left(-\frac{1}{1+\varepsilon}\right)$. For our purpose, x is a function of δ' :

$$x(\delta') = \frac{e^{-\delta'}}{(1 - \delta')^{1-\delta'}}.$$

Clearly, $x(0) = 1$ and the right-hand side above is continuous and decreasing on $(0, 1)$. Therefore, there exists $\delta'' > 0$ such that for all sufficiently small ε :

$$x(\delta'') = \exp\left(-\frac{1}{1 + \varepsilon/2}\right) < \exp\left(-\frac{1}{1 + \varepsilon}\right),$$

which satisfies (D.8) and hence implies (g). Selecting $\delta' = \delta''$, it follows from (D.7) that for any k' such that $k' < (1 - \delta'')Cs^2\alpha^* \log(n) - o(1)$:

$$\mathbb{P}(\mathcal{H}_2^c) \leq \mathbb{P}\left(\bigcup_{j \in (\mathcal{B}_1 \cup \mathcal{B}_2')^c} \left\{\deg_{\tilde{G}_1 \wedge_{\tilde{\pi}^*} \tilde{G}_2} < k'\right\}\right) = o(1). \quad (\text{D.9})$$

Indeed, our choice of $k = \sqrt{\log n}$ satisfies this condition for sufficiently large n . Combining Equation (D.6) and Equation (D.9) via a union bound, it follows that $\mathbb{P}(\mathcal{H}^c) = o(1)$ as desired. This concludes the proof. $\square$

Lemma D.7. *Let n and k be positive integers, and let p, s, γ, λ be real numbers such that $0 \leq p, s, \gamma, \lambda \leq 1$. Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{WCG}(n, p, s, \gamma, \lambda)$, and let $\mathcal{B}_2'$ denote the pre-image of $\mathcal{B}_2$ under π^* . Suppose that $p = \frac{C \log(n)}{n}$ for some positive constant C . Let $\alpha^* = 1 - \gamma + \lambda(1 - \lambda)\gamma^2$. If $C < \frac{1}{s^2\alpha^*}$, then the k -core M^* of $\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2$ with $k = \sqrt{\log n}$ satisfies*

$$\mathbb{P}\left(|(\mathcal{B}_1 \cup \mathcal{B}_2')^c \setminus M^*| = o(n)\right) = 1 - o(1).$$

Proof. From Theorem 6.2 (i) and Theorem 6.1 (i), it follows that $\mathbb{P}(M^* \subseteq (\mathcal{B}_1 \cup \mathcal{B}_2')^c) = 1 - o(1)$. It remains to show that all but a vanishing fraction of the nodes in $(\mathcal{B}_1 \cup \mathcal{B}_2')^c$ are also in M^* . Let $\tilde{\pi}^*$ with $\text{dom}(\tilde{\pi}^*) = (\mathcal{B}_1 \cup \mathcal{B}_2')^c$ denote the restriction of the permutation π^* to the node set $(\mathcal{B}_1 \cup \mathcal{B}_2')^c$. Let $\text{core}_k(G)$ denote the k -core of a graph G . First, it is shown that none of the nodes in $\mathcal{B}_1 \cup \mathcal{B}_2'$ belong to the k -core with high probability. Since for any $i \in \mathcal{B}_1 \cup \mathcal{B}_2'$, it holds that $\deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) \sim \text{Bin}(n - 1, p^2 s^2) \preceq \text{Bin}(n, p^2 s^2)$, and since $|\mathcal{B}_1 \cup \mathcal{B}_2'| \leq |\mathcal{B}_1| + |\mathcal{B}_2'| \leq \gamma n$, it follows

by the union bound that for any $\delta > 0$

$$\begin{aligned} \mathbb{P} \left(\bigcup_{i \in \mathcal{B}_1 \cup \mathcal{B}'_2} \left\{ \deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) \geq (1 + \delta)np^2s^2 \right\} \right) &\leq \gamma n \times \mathbb{P}(\text{Bin}(n, p^2s^2) > (1 + \delta)np^2s^2) \\ &\leq \gamma n \times \left(\frac{\exp(\delta)}{(1 + \delta)^{1+\delta}} \right)^{(n-1)p^2s^2}. \end{aligned}$$

Setting $p = C \log(n)/n$ yields

$$\mathbb{P} \left(\bigcup_{i \in \mathcal{B}_1 \cup \mathcal{B}'_2} \left\{ \deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) \geq (1 + \delta)C^2s^2 \frac{\log(n)^2}{n} \right\} \right) \leq \gamma n \times \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^{C^2s^2 \frac{\log(n)^2}{n}}.$$

Setting $\delta = \frac{n}{C^2s^2 \log(n)^{3/2}} - 1$ yields

$$\mathbb{P} \left(\bigcup_{i \in \mathcal{B}_1 \cup \mathcal{B}'_2} \left\{ \deg_{\tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2}(i) \geq \sqrt{\log(n)} \right\} \right) \leq \gamma n \left(\frac{\exp \left(\frac{n}{C^2s^2 \log(n)^{3/2}} - 1 \right)}{\left(\frac{n}{C^2s^2 \log(n)^{3/2}} \right)^{\frac{n}{C^2s^2 \log(n)^{3/2}}}} \right)^{C^2s^2 \frac{\log(n)^2}{n}} = o(1), \quad (\text{D.10})$$

whenever $Cs > 0$. Therefore, setting $k = \sqrt{\log(n)}$ yields $\mathbb{P}(\mathcal{E}) = o(1)$, where $\mathcal{E}$ denotes the event that there exists a node in $\mathcal{B}_1 \cup \mathcal{B}'_2$ with degree larger than k . Therefore, on the event $\mathcal{E}^c$, it follows that no node in $\mathcal{B}_1 \cup \mathcal{B}'_2$ belongs to the k -core. It remains to show that all but a vanishing fraction of nodes in $(\mathcal{B}_1 \cup \mathcal{B}'_2)^c$ form the k -core. Consider then the trimmed graph $H \triangleq \tilde{G}_1 \wedge_{\pi^*} \tilde{G}_2$ on the node set $(\mathcal{B}_1 \cup \mathcal{B}'_2)^c$. Thus, on the event $\mathcal{E}^c$, it is true that

$$\text{core}_k(H) \subseteq M^* \subseteq (\mathcal{B}_1 \cup \mathcal{B}'_2)^c. \quad (\text{D.11})$$

On the other hand, since $\mathcal{B}_1$ and $\mathcal{B}'_2$ are selected independent of the graphs G_1 and G_2 , it follows that the graph H itself is an Erdős-Rényi graph on $|(\mathcal{B}_1 \cup \mathcal{B}'_2)^c|$ nodes, where each edge is present with probability $Cs \frac{\log n}{n}$. The k -core of an Erdős-Rényi graph is a well studied problem. We invoke Theorem 2 of [Luc91], restated below for convenience.

Proposition D.1. *Let $c(n) = (n - 1)p$ denote the average degree in an Erdős-Rényi graph G on n nodes with edge probability p . For every $\varepsilon > 0$, there is a constant d , such that for $c = c(n) > d$ and $k = k(n) \leq c - c^{0.5+\varepsilon}$, it is true that*

$$\mathbb{P}(|\text{core}_k(G)| \geq n(1 - \exp(-c^\varepsilon))) = 1 - o(1).$$

First, some intuition is presented. Observe that the average degree c in H is given by

$$(|(\mathcal{B}_1 \cup \mathcal{B}'_2)^c| - 1) \times \frac{Cs \log(n)}{n} = \Theta(\log(n))$$

with high probability. Further, $k = \sqrt{\log(n)} \leq c - c^{0.5+\varepsilon}$ whenever $\varepsilon < 0.5$. Therefore, the graph H with $k = \sqrt{\log n}$ satisfies the conditions of Proposition D.1 with high probability, and so the k -core contains almost all the nodes of H with high probability. Let $\alpha^* = 1 - \gamma + \lambda(1 - \lambda)\gamma^2$. Formally, for any $\delta > 0$:

$$\begin{aligned} \mathbb{P}(|(\mathcal{B}_1 \cup \mathcal{B}'_2)^c \setminus M^*| > \delta n) &\leq \mathbb{P}(\mathcal{E}) + \mathbb{P}(|(\mathcal{B}_1 \cup \mathcal{B}'_2)^c| < (1 - \delta)(1 - \alpha^*)n) \\ &\quad + \mathbb{P}\left(|\text{core}_{\sqrt{\log n}}(H)| < (1 - \delta)(1 - \alpha^*)n (1 - \exp(-c^{0.1}))\right) \end{aligned} \quad (\text{D.12})$$

$$\leq o(1) + o(1) + o(1), \quad (\text{D.13})$$

where we have showed in (D.10) that the first term is $o(1)$. The second term is $o(1)$ due to Lemma D.2, and the third term is $o(1)$ due to Proposition D.1. It is emphasized that (D.12) holds because $\text{core}_k(H) \subseteq M^* \subseteq (\mathcal{B}_1 \cup \mathcal{B}'_2)^c$ under the event $\mathcal{E}^c$. This concludes the proof. $\square$

D.2 Proofs for the SCG model

This section analyzes the maximum overlap estimator in the context of the SCG model. First, a simple concentration inequality is shown for the number of edges in $G_1 \wedge_{\pi^*} G_2$. Then, the moment generating function of the random variable $X(\pi)$ is bounded using techniques introduced in [CK17] and refined in [WXY22], among others.

Lemma D.8. *Let n be a positive integer and s be a real number such that $s \in (0, 1]$. Let $C > 0$ be constant and let $p \geq C \log(n)/n$. Let $(G_1, G_2, \pi^*) \sim \text{CER}(n, p, s)$. Let $X(\text{id})$ denote the number of edges in the intersection graph $G_1 \wedge_{\pi^*} G_2$. Then, for any $\varepsilon \in (0, 1)$:*

$$\mathbb{P}\left(X(\text{id}) \leq (1 - \varepsilon) \binom{n}{2} ps^2\right) = o(1).$$

Proof. First, notice that $G_1 \wedge_{\pi^*} G_2$ is also an Erdős-Rényi graph on n nodes, where each edge is present with probability ps^2 . It follows that the number of edges in the graph $X(\text{id}) \sim \text{Bin}\left(\binom{n}{2}, ps^2\right)$. Thus,

$$\begin{aligned} \mathbb{P}\left(X(\text{id}) \leq (1 - \varepsilon) \binom{n}{2} ps^2\right) &\stackrel{(a)}{\leq} \left(\frac{e^{-\varepsilon}}{(1 - \varepsilon)^{1-\varepsilon}}\right)^{\binom{n}{2} ps^2} \\ &\stackrel{(b)}{\leq} \left(\frac{e^{-\varepsilon}}{(1 - \varepsilon)^{1-\varepsilon}}\right)^{\frac{Cs^2}{2}(n-1)\log(n)} \\ &= o(1), \end{aligned}$$

where (a) follows from Lemma 3.6 and (b) is true because $e^{-\varepsilon} < (1 - \varepsilon)^{1-\varepsilon}$ for all $\varepsilon \in (0, 1)$. $\square$

Lemma D.9. *Let $s \in (0, 1]$ and $\varepsilon > 0$ be fixed, and $C > 0$ be constant. Let $(G_1, G_2) \sim \text{CER}(n, p, s)$ be a correlated pair of Erdős-Rényi graphs, where $p = C \log(n)/n$ and let Δ_2 denote the maximum node degree in G_2 . If $C > \frac{3}{s\varepsilon^2}$, then*

$$\mathbb{P}(\Delta_2 > (1 + \varepsilon)nps) = o(1).$$

Proof. Let $v_i \in [n]$ denote a node and let $\deg(v_i)$ denote its degree in G_2 . By the union bound and the fact that $p = C \log(n)/n$,

$$\begin{aligned} \mathbb{P}(\Delta_2 > (1 + \varepsilon)nps) &\leq \sum_{i=1}^n \mathbb{P}(\deg(v_i) > (1 + \varepsilon) \cdot Cs \log(n)) \\ &= n \times \mathbb{P}(\text{Bin}(n-1, C \log(n)/n) > (1 + \varepsilon) \cdot Cs \log(n)) \\ &\leq n \times \mathbb{P}(\text{Bin}(n, C \log(n)/n) > (1 + \varepsilon) \cdot Cs \log(n)) \\ &\stackrel{(a)}{\leq} n \times \exp\left(-\frac{\varepsilon^2 Cs \log(n)}{3}\right) \\ &= n^{1 - \frac{\varepsilon^2 Cs}{3}} \\ &\stackrel{(b)}{=} o(1), \end{aligned}$$

where (a) uses Theorem 4.4 of [MU17], and (b) is because $C > \frac{3}{s\varepsilon^2}$. $\square$

Lemma D.10. *Let π be a permutation on $[n]$, and let $(G_1, G_2, \pi^*) \sim \text{CER}(n, p, s)$. Let $X(\pi)$ denote the number of edges in $G_1 \wedge_{\pi} G_2$. The moment generating function of $X(\pi)$ is given by*

$$\mathbb{E}e^{tX(\pi)} = \prod_{k=1}^{\binom{n}{2}} L_k^{N_k}$$

where N_k is the number of k -orbits in the edge decomposition of π and

$$L_k \triangleq \text{Tr}(L^k), \tag{D.14}$$

where the 2×2 matrix L is given as

$$\begin{bmatrix} 1 - ps & \frac{ps}{1-ps} (1 - p + p(1-s)^2 + ps(1-s)e^t) \\ 1 - ps & ps(1-s + se^t) \end{bmatrix}$$

Moreover, $L_k \leq L_2^{k/2}$.

Proof. For a permutation π , Let $\mathcal{O}^\pi = \bigcup_{k=1}^{\binom{n}{2}} \mathcal{O}_k^\pi$ denote the edge orbit decomposition of π , where

$\mathcal{O}_k^\pi$ is the set of all k length edge orbits of π . By independence across edge orbits:

$$\begin{aligned}\mathbb{E}\exp(tX(\pi)) &= \mathbb{E}\exp\left(t \sum_{\{i,j\} \in \binom{[n]}{2}} G_1\{i,j\} G_2\{\pi(i), \pi(j)\}\right) \\ &= \mathbb{E}\exp\left(t \sum_{O \in \mathcal{O}^\pi} \sum_{(i,j) \in O} G_1\{i,j\} G_2\{\pi(i), \pi(j)\}\right) \\ &= \prod_{O \in \mathcal{O}^\pi} \mathbb{E}\exp(tX_O(\pi)),\end{aligned}$$

where $X_O(\pi) = \sum_{i,j \in O} G_1\{i,j\} G_2\{\pi(i), \pi(j)\}$. Let $(a_i, b_i)_{i=1}^{k+1}$ with $(a_{k+1}, b_{k+1}) = (a_1, b_1)$ denote mutually independent random variables so that $a_i, b_i \sim \text{Bern}(ps)$ and $\mathbb{P}(b_i = 1 \mid a_i = 1) = s$. Let $L_k \triangleq \mathbb{E}e^{tX_O}$ for any edge orbit O such that $|O| = k$. Then,

$$\begin{aligned}L_k &= \mathbb{E}\exp\left(t \sum_{j=1}^k a_j b_{j+1}\right) = \mathbb{E} \prod_{j=1}^k \exp(t \cdot a_j b_{j+1}) \\ &= \mathbb{E} \mathbb{E} \prod_{j=1}^k \exp(t \cdot a_j b_{j+1}) \mid b_1, \dots, b_k \\ &= \mathbb{E} \mathbb{E} \prod_{j=1}^k \exp(t \cdot a_j b_{j+1}) \mid b_j, b_{j+1} \\ &\stackrel{(a)}{=} \mathbb{E} \prod_{j=1}^k \mathbb{E}\exp(t \cdot a_j b_{j+1}) \mid b_j, b_{j+1} \\ &= \sum_{\substack{(\theta_1, \dots, \theta_k) \in \{0,1\}^k \\ \theta_{k+1} = \theta_1}} \mathbb{P}((b_1, \dots, b_k) = (\theta_1, \dots, \theta_k)) \prod_{j=1}^k \mathbb{E}\exp(t \cdot a_j b_{j+1} \mid b_j = \theta_j, b_{j+1} = \theta_{j+1})\end{aligned}\tag{D.15}$$

where (a) follows from conditional independence of edges in G_1 given G_2 . On the other hand, Equation (D.15) is exactly equal to

$$(D.15) = \text{Tr}(L^k),$$

where the 2×2 matrix L is given by

$$L(\ell, m) = \mathbb{E}\exp(t \cdot a_1 b_2) \mid b_1 = \ell, b_2 = m \times \mathbb{P}(b_2 = m), \quad \ell, m \in \{0, 1\} \tag{D.16}$$

Computing the conditional expectations gives as desired,

$$\begin{bmatrix} 1 - ps & \frac{ps}{1-ps} (1 - p + p(1-s)^2 + ps(1-s)e^t) \\ 1 - ps & ps(1-s + se^t) \end{bmatrix}.$$

Finally, we show that $L_k \leq L_2^{k/2}$. Since the eigenvalues of L are given by $\frac{T \pm \sqrt{T^2 - 4D}}{2}$, where T and D are the trace and determinant of L respectively, it follows that

$$L_k = \text{Tr}(L^k) = \left(\frac{T + \sqrt{T^2 - 4D}}{2} \right)^k + \left(\frac{T - \sqrt{T^2 - 4D}}{2} \right)^k,$$

and it follows that $L_k \leq L_2^{k/2}$. □

Corollary D.1. *Evaluating the trace of L and L^2 ,*

$$\begin{aligned} L_1 &= 1 - ps + ps(1 - s + se^t) \\ L_2 &= (1 - ps)^2 + 2ps(1 - p + p(1-s)^2 + ps(1-s)e^t) + (ps(1-s + se^t))^2. \end{aligned}$$

Proof. This follows from a direct computation using Equation (D.14). □

Lemma D.11. *Let n be a positive integer and s, γ, α be reals such that $s \in (0, 1]$ and $\gamma, \alpha \in [0, 1]$. Let $\mathbf{A}$ be any adversary and let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed according to $\text{SCG}(n, p, s, \gamma, 1, \mathbf{A})$. Let p_3 be the event defined in (6.20). If $\gamma < s(1 - \alpha^2)/4$, then $p_3 = o(1)$.*

Proof. By definition of p_3 and the union bound,

$$p_3 \leq \sum_{k=0}^{\alpha n} \mathbb{P} \left(\bigcup_{\pi \in \mathcal{T}^{k/n}} \left\{ X(\pi) \geq (1 - \varepsilon) \binom{n}{2} ps^2 - 2\gamma n(1 + \varepsilon) nps \right\} \right).$$

The Chernoff bound then yields for any $t > 0$:

$$\mathbb{P} \left(X(\pi) \geq (1 - \varepsilon) \binom{n}{2} ps^2 - \gamma n(1 + \varepsilon) nps \right) \leq \exp \left(-t \left[(1 - \varepsilon) \binom{n}{2} ps^2 - 2\gamma n(1 + \varepsilon) nps \right] \right) \cdot \mathbb{E} e^{tX(\pi)}. \quad (\text{D.17})$$

For each π , let n_k^π (resp. N_k^π) denote the number of k -orbits in the node (resp. edge) decomposition of π . The MGF of $X(\pi)$ is handled using Lemma D.10:

$$\begin{aligned} \mathbb{E} e^{tX(\pi)} &= \prod_{\ell=1}^{\binom{n}{2}} L_\ell^{N_\ell^\pi} \leq L_1^{\binom{n_1^\pi}{2} + n_2^\pi} \cdot \prod_{\ell=2}^{\binom{n}{2}} L_2^{\ell/2} \\ &= L_1^{\binom{n_1^\pi}{2} + n_2^\pi} \cdot L_2^{\frac{1}{2} \left(\binom{n}{2} - \binom{n_1^\pi}{2} - n_2^\pi \right)} \end{aligned} \quad (\text{D.18})$$

Substituting Equation (D.18) in Equation (D.17) yields

$$\mathbb{P} \left(X(\pi) \geq (1 - \varepsilon) \binom{n}{2} ps^2 - 2(1 + \varepsilon) \gamma n^2 ps \right) \leq \exp(\zeta(\pi)),$$

where

$$\zeta(\pi) = -t \left((1 - \varepsilon) \binom{n}{2} ps^2 - 2(1 + \varepsilon) \gamma n^2 ps \right) + \frac{1}{2} \binom{n}{2} \log(L_2) + \frac{1}{2} \left(\binom{n_1^\pi}{2} + n_2^\pi \right) \log \left(\frac{L_1^2}{L_2} \right) \quad (\text{D.19})$$

and

$$\begin{aligned} L_1 &= 1 - ps + ps(1 - s + se^t) \\ L_2 &= (1 - ps)^2 + 2ps(1 - p + p(1 - s)^2 + ps(1 - s)e^t) + (ps(1 - s + se^t))^2. \end{aligned}$$

It is easy to verify that for all $p, s \in [0, 1]$ and $t > 0$, the quantity $\frac{L_1^2}{L_2} \geq 1$, and so an upper bound on $\zeta(\pi)$ can be obtained by using the bound $n_2^\pi \leq n - n_1^\pi$. Let β_π denote the fraction of fixed points in the permutation π . By definition, $n_1^\pi = \beta_\pi n$. Using $p = \frac{C \log(n)}{n}$ yields

$$\begin{aligned} L_1 &= 1 + \frac{C \cdot (e^t - 1) s^2 \log(n)}{n}, \\ L_2 &= 1 + \frac{C^2 (e^t - 1) s^2 (2 + (e^t - 1) s^2) (\log n)^2}{n^2}. \end{aligned}$$

Substituting these in Equation (D.19) yields

$$\zeta(\pi) \leq T_1(\pi) + T_2(\pi) + T_3(\pi) + T_4(\pi) + T_5(\pi) + T_6(\pi),$$

where

$$\begin{aligned}
T_1(\pi) &= -t \left(\frac{(1-\varepsilon)s^2}{2} - 2(1+\varepsilon)\gamma s \right) \times \frac{C \log n}{n} \times n^2 = \Theta(n \log n) \\
T_2(\pi) &= \frac{\beta_\pi^2 n^2}{2} \log(L_1) = \frac{\beta_\pi^2 n^2}{2} \log \left(1 + \frac{C \cdot (e^t - 1)s^2 \log(n)}{n} \right) \\
&\leq \frac{\beta_\pi^2 n^2}{2} \times \frac{C(e^t - 1)s^2 \log n}{n} = O(n \log n) \\
T_3(\pi) &= \frac{(1 - \beta_\pi^2)n^2}{4} \log(L_2) = \frac{(1 - \beta_\pi^2)n^2}{4} \log \left(1 + \frac{C^2 (e^t - 1)s^2 (2 + (e^t - 1)s^2) (\log n)^2}{n^2} \right) \\
&\leq \frac{(1 - \beta_\pi^2)n^2}{4} \times \frac{C^2 (e^t - 1)s^2 (2 + (e^t - 1)s^2) (\log n)^2}{n^2} = O((\log n)^2) \\
T_4(\pi) &= \frac{t(1-\varepsilon)ps^2}{2} \times n = \frac{C \cdot t(1-\varepsilon)s^2}{2} \log n = \Theta(\log n) \\
T_5(\pi) &= \frac{(2 - 3\beta_\pi)n}{2} \log(L_1) = \frac{(2 - 3\beta_\pi)n}{2} \log \left(1 + \frac{C(e^t - 1)s^2 \log(n)}{n} \right) \\
&\leq \frac{(2 - 3\beta_\pi)n}{2} \times \frac{C(e^t - 1)s^2 \log(n)}{n} = O(\log n) \\
T_6(\pi) &= -\frac{3(1 - \beta_\pi)n}{4} \log(L_2) = -\frac{3(1 - \beta_\pi)n}{4} \log \left(1 + \frac{C^2 (e^t - 1)s^2 (2 + (e^t - 1)s^2) (\log n)^2}{n^2} \right) \\
&\leq -\frac{3(1 - \beta_\pi)n}{4} \times \frac{C^2 (e^t - 1)s^2 (2 + (e^t - 1)s^2) (\log n)^2}{n^2} = O\left(\frac{(\log n)^2}{n}\right).
\end{aligned}$$

Since the dominant terms are $T_1(\pi)$ and $T_2(\pi)$, it follows that for sufficiently large n and any $t > 0$:

$$\sum_{i=1}^6 T_i(\pi) \leq (1 + \varepsilon) \left(\frac{-t(1-\varepsilon)s^2}{2} + 2t(1+\varepsilon)\gamma s + \frac{\beta_\pi^2 s^2}{2} (e^t - 1) \right) \times Cn \log n. \quad (\text{D.20})$$

Next, the condition in the hypothesis of the theorem is invoked, i.e. $\gamma < s(1 - \alpha^2)/4$. Since $\beta_\pi \leq \alpha < 1$ for all $\pi \in \mathcal{T}^{\leq \alpha}$, it follows also that $\gamma < s(1 - \beta_\pi^2)/4$. Let

$$t^* = \log \left(\frac{1}{\beta_\pi^2} \left(1 - \frac{4\gamma}{s} \right) \right). \quad (\text{D.21})$$

Note that $t^* > 0$. Also note that

$$-\frac{t^* s^2}{2} + 2t^* \gamma s + \frac{\beta_\pi^2}{2} s^2 (e^{t^*} - 1) = -\frac{s}{2} (s\beta_\pi^2 + 4\gamma - s + (s - 4\gamma) \log(s - 4\gamma) - (s - 4\gamma) \log(s\beta_\pi^2)), \quad (\text{D.22})$$

and that the RHS of Equation (D.22) is strictly negative whenever the condition $\gamma < s(1 - \beta_\pi^2)/4$ is satisfied (see Lemma D.12). Therefore, there exists a sufficient small $\varepsilon > 0$ and a sufficiently large

$C > 0$ such that

$$\sum_{i=1}^6 T_i(\pi) \leq (1+\varepsilon) \left(\frac{-t^*(1-\varepsilon)s^2}{2} + 2t^*(1+\varepsilon)\gamma s + \frac{\beta_\pi^2 s^2}{2}(e^{t^*} - 1) \right) \cdot Cn \log n < -\frac{\beta_\pi + 1}{2} n \log(n).$$

This yields that for sufficiently large n :

$$\zeta(\pi) \leq -\frac{\beta_\pi + 1}{2} n \log n.$$

Finally, this yields

$$\begin{aligned} p_3 &\leq \sum_{k=0}^{\alpha n} \sum_{\pi \in \mathcal{T}^{k/n}} \mathbb{P} \left(X(\pi) \geq (1-\varepsilon) \binom{n}{2} p s^2 - 2(1+\varepsilon) \gamma n^2 p s \right) \\ &= \sum_{k=0}^{\alpha n} n^k \exp \left(-\frac{1}{2} \left(\frac{k}{n} + 1 \right) n \log(n) \right) \\ &= \sum_{k=0}^{\alpha n} \exp \left(-\frac{1}{2} (n-k) \log n \right) \\ &= \sum_{k=(1-\alpha)n}^n \exp \left(-\frac{1}{2} k \log n \right) \\ &\leq \frac{\exp \left(-\frac{1-\alpha}{2} n \log n \right)}{1 - \exp \left(-\frac{1}{2} \log n \right)} \\ &= o(1) \end{aligned}$$

This concludes the proof. $\square$

Lemma D.12. Suppose β_π, γ are constants such that $\beta_\pi > 0$ and $\gamma < s(1 - \beta_\pi^2)/4$. Then,

$$s\beta_\pi^2 + 4\gamma - s + (s - 4\gamma) \log(s - 4\gamma) - (s - 4\gamma) \log(s\beta_\pi^2) > 0$$

Proof. Let $x = s\beta_\pi^2$ and $y = s - 4\gamma$. Further, let $z = y/x$. Then,

$$s\beta_\pi^2 + 4\gamma - s + (s - 4\gamma) \log(s - 4\gamma) - (s - 4\gamma) \log(s\beta_\pi^2) = x(1 - z + z \log(z)).$$

First, note that $x > 0$. Furthermore, $z > 1$ since $\gamma < \frac{s(1-\beta_\pi^2)}{4}$ implies $x < y$. Finally, note that the function $z \mapsto 1 - z + z \log z$ is convex and minimized at $z = 1$, where it equals 0. It follows that the desired quantity is positive whenever $z > 1$. $\square$

Lemma D.13. Let n be a positive integer and p, s, γ, α be real numbers such that $p \in (0, 1)$, $s \in (0, 1]$, and $\gamma, \alpha \in [0, 1]$. Let $\mathbf{A}$ be any adversary and let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \pi^*)$ be distributed

according to $\text{SCG}(n, p, s, \gamma, 1, \mathbf{A})$. Let $\varepsilon \in (0, 1)$ and let p_4 be the event defined in (6.22). If

$$\gamma < 1 - \sqrt{1 - \frac{s^2 p(1-p)(1-\alpha^2)}{2}},$$

then $p_4 = o(1)$.

Proof. By definition of p_4 and the union bound,

$$\begin{aligned} p_4 &= \mathbb{P} \left(\bigcup_{k=0}^{\alpha n} \bigcup_{\pi \in \mathcal{T}^{k/n}} \left\{ X(\pi) \geq (1-\varepsilon) \binom{n}{2} p s^2 - 2 \left[\binom{\gamma n}{2} - \gamma(1-\gamma)n^2 \right] \right\} \right) \\ &\leq \sum_{k=0}^{\alpha n} \mathbb{P} \left(\bigcup_{\pi \in \mathcal{T}^{k/n}} \left\{ X(\pi) \geq (1-\varepsilon) \binom{n}{2} p s^2 - 2 \left[\binom{\gamma n}{2} - \gamma(1-\gamma)n^2 \right] \right\} \right) \end{aligned}$$

The Chernoff bound is used below to bound these terms. It follows for any $t > 0$:

$$\begin{aligned} \mathbb{P} \left(X(\pi) \geq (1-\varepsilon) \binom{n}{2} p s^2 - 2 \left[\binom{\gamma n}{2} - \gamma(1-\gamma)n^2 \right] \right) \\ \leq \exp \left(-t \left((1-\varepsilon) \binom{n}{2} p s^2 - 2 \left[\binom{\gamma n}{2} - \gamma(1-\gamma)n^2 \right] \right) \right) \cdot \mathbb{E} e^{tX(\pi)} \end{aligned} \quad (\text{D.23})$$

For each π , let n_k^π (resp. N_k^π) denote the number of k -orbits in the node (resp. edge) decomposition of π . The MGF of $X(\pi)$ is handled using Lemma D.10:

$$\mathbb{E} e^{tX(\pi)} = \prod_{\ell=1}^{\binom{n}{2}} L_\ell^{N_\ell^\pi} \leq L_1^{\binom{n_1^\pi}{2} + n_2^\pi} \cdot \prod_{\ell=2}^{\binom{n}{2}} L_2^{\ell/2} = L_1^{\binom{n_1^\pi}{2} + n_2^\pi} \cdot L_2^{\frac{1}{2} \left(\binom{n}{2} - \binom{n_1^\pi}{2} - n_2^\pi \right)} \quad (\text{D.24})$$

Substituting Equation (D.24) in Equation (D.23) yields

$$\mathbb{P} \left(X(\pi) \geq (1-\varepsilon) \binom{n}{2} p s^2 - 2 \left[\binom{\gamma n}{2} - \gamma(1-\gamma)n^2 \right] \right) \leq \exp(\zeta(\pi)),$$

where

$$\zeta(\pi) = -t \left((1-\varepsilon) \binom{n}{2} p s^2 - 2 \left[\binom{\gamma n}{2} - \gamma(1-\gamma)n^2 \right] \right) + \frac{1}{2} \binom{n}{2} \log(L_2) + \frac{1}{2} \left(\binom{n_1^\pi}{2} + n_2^\pi \right) \log \frac{L_1^2}{L_2}$$

Let β_π denote the fraction of fixed points in the permutation π . By definition, $n_1^\pi = \beta_\pi n$. Letting $\tilde{p} = p(1-\varepsilon)$ and using the fact that $n_2^\pi \leq n - n_1^\pi$, it follows that

$$\zeta(\pi) \leq \left(-\frac{t}{2} (\tilde{p}s^2 + 2\gamma^2 - 4\gamma) + \frac{1}{4} \log(L_2) + \frac{\beta_\pi^2}{4} \log \left(\frac{L_1^2}{L_2} \right) \right) n^2 \quad (\text{D.25})$$

$$+ \left(-\frac{t}{2} (2\gamma - \tilde{p}s^2) - \frac{1}{4} \log(L_2) + \frac{2-3\beta_\pi}{4} \log \left(\frac{L_1^2}{L_2} \right) \right) n \quad (\text{D.26})$$

Next, it is shown that if (6.4) is satisfied, then there exists $\delta > 0$ such that coefficient of the n^2 term in (D.25) is less than or equal to $-\delta$. To that end, notice that this coefficient is strictly negative when

$$\tilde{p}s^2 + 2\gamma^2 - 4\gamma > \inf_{t>0} \left\{ \frac{1}{t} \left(\frac{1}{2} \log(L_2) + \frac{\beta_\pi^2}{2} \log\left(\frac{L_1^2}{L_2}\right) \right) \right\} = \inf_{t>0} \left\{ \log\left(L_1^{\frac{\beta_\pi^2}{t}} \cdot L_2^{\frac{1-\beta_\pi^2}{2t}} \right) \right\} \quad (\text{D.27})$$

Therefore, (D.27) holds if

$$\exp(\tilde{p}s^2 + 2\gamma^2 - 4\gamma) > \exp\left(\inf_{t>0} \log\left(L_1^{\frac{\beta_\pi^2}{t}} \cdot L_2^{\frac{1-\beta_\pi^2}{2t}} \right) \right) \quad (\text{D.28})$$

$$= \inf_{t>0} L_1^{\frac{\beta_\pi^2}{t}} \cdot L_2^{\frac{1-\beta_\pi^2}{2t}} \quad (\text{D.29})$$

since $L_1, L_2 \geq 1$ and therefore the objective function being infimized in Equation (D.28) is always non-negative. Observing that the objective function for the infimization in (D.29) is monotonically increasing in t , and substituting for L_1 and L_2 from Corollary D.1, it follows that

$$\begin{aligned} (\text{D.29}) &= \lim_{t \rightarrow 0} \left\{ (1-ps+ps(1-s+se^t))^{\beta_\pi^2/t} \right. \\ &\quad \times \left. \left((1-ps)^2 + 2ps(1-p+p(1-s)^2 + ps(1-s)e^t) + (ps(1-s+se^t))^2 \right)^{\frac{1-\beta_\pi^2}{2t}} \right\} \\ &= \exp\left(ps^2(p + \beta_\pi^2 - p\beta_\pi^2) \right), \end{aligned}$$

i.e. the condition (D.27) holds if $\exp(\tilde{p}s^2 + 2\gamma^2 - 4\gamma) > \exp(ps^2(p + \beta_\pi^2 - p\beta_\pi^2))$. Rewriting this as a condition on γ yields that Equation (D.27) holds whenever

$$\gamma < 1 - \sqrt{1 - \frac{ps^2((1-p)(1-\beta_\pi^2) - \varepsilon)}{2}}. \quad (\text{D.30})$$

Indeed, since $\beta_\pi \leq \alpha$ for all $\pi \in \mathcal{T}^{\leq \alpha}$, it follows that whenever (6.4) holds, ε can be chosen sufficiently small so that (D.30) is satisfied for all $\pi \in \mathcal{T}^{\leq \alpha}$. It follows that there exists $\delta > 0$ such that the coefficient of the n^2 term in (D.25) is less than or equal to $-\delta$. Therefore, for sufficiently large n :

$$\zeta(\pi) \leq -\frac{\delta}{2}n^2 \leq -\frac{1}{2}(\beta_\pi + 1)n \log(n).$$

Combining all the above,

$$\begin{aligned}
p_4 &\leq \sum_{k=0}^{\alpha n} \sum_{\pi \in \mathcal{T}^{k/n}} \mathbb{P} \left(X(\pi) \geq \binom{n}{2} \tilde{p} s^2 - 2 \left[\binom{\gamma n}{2} - \gamma(1-\gamma)n^2 \right] \right) \\
&\leq \sum_{k=0}^{\alpha n} n^k \exp \left(-\frac{1}{2} \left(\frac{k}{n} + 1 \right) n \log(n) \right) \\
&= \sum_{k=0}^{\alpha n} \exp \left(-\frac{1}{2} (n-k) \log n \right) \\
&= \sum_{k=(1-\alpha)n}^n \exp \left(-\frac{1}{2} k \log n \right) \\
&\leq \frac{\exp \left(-\frac{1-\alpha}{2} n \log n \right)}{1 - \exp \left(-\frac{1}{2} \log n \right)} \\
&= o(1)
\end{aligned}$$

as desired. This concludes the proof. $\square$

D.3 The SCG model when both networks are compromised

In this section, the maximum overlap estimator is analyzed for the case when the adversary corrupts the same number of nodes in both graphs, i.e. $\lambda = 1/2$. The action of the adversary is described by Algorithm 4. A similar adversary can cause the maximum overlap estimator to fail horribly when $\lambda \in (0, 1)$ but $\lambda \neq 1/2$. However, the setting of $\lambda = 1/2$ conveys the main ideas without complicating notation and will be the focus of the analysis.

Theorem D.2. *Let $(\mathcal{B}_1, \mathcal{B}_2, \tilde{G}_1, \tilde{G}_2, \text{id})$ be distributed according to $\text{SCG}(n, p, s, \gamma, 1/2, \mathbf{A})$, where the adversary $\mathbf{A}$ is described in Algorithm 4. Let $\hat{\pi}^{\text{MO}}$ denote the matching output by the maximum overlap estimator $\hat{\mathcal{E}}_{\text{MO}}(\tilde{G}_1, \tilde{G}_2)$. Then, for any $\varepsilon > 0$,*

$$\mathbb{P}(\text{ov}(\hat{\pi}^{\text{MO}}, \text{id}) \geq \varepsilon n) = o(1).$$

Proof. Assume without loss of generality that $\gamma \leq 1/2$, since the same argument works by appropriately interchanging the role of γ and $1 - \gamma$. For any permutation π , let $X(\pi)$ denote the number of edges in the intersection graph $G_1 \wedge_{\pi} G_2$. Similarly, let $\tilde{X}(\pi)$ denote the number of edges in the intersection graph $\tilde{G}_1 \wedge_{\pi} \tilde{G}_2$. For $j \in \{1, 2\}$, let $E(\tilde{G}_j)$ denote the set of edges present in $\tilde{G}_j$, and furthermore, let $\tilde{E}(\tilde{G}_j) \subseteq E(\tilde{G}_j)$ denote the edges added by the adversary in Algorithm 4.

Let $\tilde{\pi}$ be a matching with $\text{dom}(\tilde{\pi}) = [n]$ such that

$$\tilde{\pi}(i) \in \begin{cases} \mathcal{B}_2, & i \in \mathcal{B}_1 \\ \mathcal{B}_1, & i \in \mathcal{B}_2 \\ \mathcal{G}_2, & i \in \mathcal{G}_1 \\ \mathcal{G}_1, & i \in \mathcal{G}_2 \end{cases},$$

where $\mathcal{G}_1, \mathcal{G}_2$ are as defined in Algorithm 4. Since $|\mathcal{B}_1| = |\mathcal{B}_2|$ and $|\mathcal{G}_1| = |\mathcal{G}_2|$, such a matching exists. Note that $\text{ov}(\tilde{\pi}, \text{id}) = 0$. First, a lower bound on $\tilde{X}(\tilde{\pi})$ is computed.

$$\tilde{X}(\tilde{\pi}) = \left| E \left(\tilde{G}_1 \wedge_{\tilde{\pi}} \tilde{G}_2 \right) \right| \geq \left| \tilde{E} \left(\tilde{G}_1 \wedge_{\tilde{\pi}} \tilde{G}_2 \right) \right| \stackrel{(a)}{=} |\mathcal{B}_1| \times |\mathcal{G}_1| = \frac{\gamma n}{2} \times \frac{(1-\gamma)n}{2} = \frac{\gamma(1-\gamma)}{4} \times n^2,$$

where (a) follows from the fact that each node in $\mathcal{B}_1$ connects to every node in $\mathcal{G}_1$ in the graph $\tilde{G}_1 \wedge_{\tilde{\pi}} \tilde{G}_2$.

Thus, in order to complete the proof, it suffices to show that

$$\mathbb{P} \left(\bigcup_{\varepsilon \in (0,1]} \bigcup_{\substack{\pi \\ \text{ov}(\pi, \text{id}) = \varepsilon n}} \tilde{X}(\pi) > \frac{\gamma(1-\gamma)}{4} \times n^2 \right) = o(1). \quad (\text{D.31})$$

A union bound argument is presented below to show (D.31). Before proceeding, let us collect some observations about the degrees of nodes in $\tilde{G}_1$ and $\tilde{G}_2$.

Remark D.2. Let $\mathcal{B}_1, \mathcal{B}_2, \mathcal{G}_1, \mathcal{G}_2$ be as defined in Algorithm 4. Since the adversary only adds edges and does not remove any edges,

$$\deg_{\tilde{G}_1}(i) \leq \begin{cases} \deg_{G_1}(i) + \frac{(1-\gamma)n}{2}, & i \in \mathcal{B}_1 \\ \deg_{G_1}(i) + \frac{\gamma n}{2}, & i \in \mathcal{G}_1 \\ \deg_{G_1}(i), & i \in \mathcal{B}_2 \cup \mathcal{G}_2 \end{cases}$$

and

$$\deg_{\tilde{G}_2}(j) \leq \begin{cases} \deg_{G_2}(j) + \frac{(1-\gamma)n}{2}, & j \in \mathcal{B}_2 \\ \deg_{G_2}(j) + \frac{\gamma n}{2}, & j \in \mathcal{G}_2 \\ \deg_{G_2}(j), & j \in \mathcal{B}_1 \cup \mathcal{G}_1 \end{cases} \quad (\text{D.32})$$

Continuing, let $([n], \pi')$ be a matching such that $\text{ov}(\pi', \text{id}) = \varepsilon n$. Then,

$$\tilde{X}(\pi') = \frac{1}{2} \sum_{i \in [n]} \deg_{\tilde{G}_1 \wedge_{\pi'} \tilde{G}_2}(i) \leq \frac{1}{2} \sum_{i \in [n]} \min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(\pi'(i)) \right) = \frac{1}{2} (Z_1 + Z_2), \quad (\text{D.33})$$

where

$$Z_1 \triangleq \sum_{\substack{i \in [n] \\ \pi'(i)=i}} \min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(i) \right), \quad Z_2 \triangleq \sum_{\substack{i \in [n] \\ \pi'(i) \neq i}} \min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(\pi'(i)) \right)$$

First, the term Z_1 is analyzed. Clearly, for any $i \in [n]$, Remark D.2 yields

$$\min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(i) \right) \leq \min \left(\deg_{G_1}(i), \deg_{G_2}(i) \right),$$

and so it follows that

$$Z_1 \leq \varepsilon n \times \max_{i \in [n]} \left(\min \left(\deg_{G_1}(i), \deg_{G_2}(i) \right) \right) \leq n \times \max_{i \in [n]} \deg_{G_1}(i). \quad (\text{D.34})$$

Next, the term Z_2 is analyzed. Notice that it can be upper bounded as

$$Z_2 \leq Z_{2,1} + Z_{2,2} + Z_{2,3} + Z_{2,4}, \quad (\text{D.35})$$

where each term is defined and then bounded using Remark D.2 as

$$Z_{2,1} \triangleq \sum_{\substack{i \in [n] \\ i \in \mathcal{B}_2 \cup \mathcal{G}_2}} \min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(\pi'(i)) \right) \leq |\mathcal{B}_2 \cup \mathcal{G}_2| \max_{i \in [n]} \left(\deg_{G_1}(i) \right) \leq n \max_{i \in [n]} \left(\deg_{G_1}(i) \right), \quad (\text{D.36})$$

$$Z_{2,2} \triangleq \sum_{\substack{i \in [n] \\ \pi'(i) \in \mathcal{B}_1 \cup \mathcal{G}_1}} \min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(\pi'(i)) \right) \leq |\mathcal{B}_1 \cup \mathcal{G}_1| \max_{\pi'(i) \in [n]} \left(\deg_{G_2}(\pi'(i)) \right) \leq n \max_{i \in [n]} \left(\deg_{G_2}(i) \right), \quad (\text{D.37})$$

$$\begin{aligned} Z_{2,3} &\triangleq \sum_{\substack{i \in [n] \\ i \in \mathcal{B}_1 \\ \pi'(i) \in \mathcal{B}_2 \cup \mathcal{G}_2}} \min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(\pi'(i)) \right) \\ &\leq \left| \{i : i \in \mathcal{B}_1, \pi'(i) \in \mathcal{B}_2 \cup \mathcal{G}_2\} \right| \left[\frac{(1-\gamma)n}{2} + \max_{i \in [n]} \left(\deg_{G_1}(i) \right) \right], \end{aligned} \quad (\text{D.38})$$

$$\begin{aligned} Z_{2,4} &\triangleq \sum_{\substack{i \in [n] \\ i \in \mathcal{G}_1 \\ \pi'(i) \in \mathcal{B}_2 \cup \mathcal{G}_2}} \min \left(\deg_{\tilde{G}_1}(i), \deg_{\tilde{G}_2}(\pi'(i)) \right) \\ &\leq \left| \{i : i \in \mathcal{G}_1, \pi'(i) \in \mathcal{B}_2 \cup \mathcal{G}_2\} \right| \left[\frac{\gamma n}{2} + \max_{i \in [n]} \left(\deg_{G_1}(i) \right) \right]. \end{aligned} \quad (\text{D.39})$$

Let $\varepsilon_1(\pi')$ denote the fraction of nodes i in $\mathcal{B}_1$ such that $\pi'(i) \in \mathcal{B}_2 \cup \mathcal{G}_2$. Similarly, let $\delta_1(\pi')$ denote the fraction of nodes j in $\mathcal{G}_1$ such that $\pi'(j) \in \mathcal{B}_2 \cup \mathcal{G}_2$. Therefore, (D.38) and (D.39) can be written as

$$Z_{2,3} \leq \varepsilon_1(\pi')|\mathcal{B}_1| \left(\frac{(1-\gamma)n}{2} + \max_{i \in [n]} (\deg_{G_1}(i)) \right) \leq \varepsilon_1(\pi') \frac{\gamma(1-\gamma)}{4} n^2 + n \max_{i \in [n]} (\deg_{G_1}(i)), \quad (\text{D.40})$$

$$Z_{2,4} \leq \delta_1(\pi')|\mathcal{G}_1| \left(\frac{\gamma n}{2} + \max_{i \in [n]} (\deg_{G_1}(i)) \right) \leq \delta_1(\pi') \frac{\gamma(1-\gamma)}{4} n^2 + n \max_{i \in [n]} (\deg_{G_2}(i)). \quad (\text{D.41})$$

Therefore, combining (D.35)-(D.41) yields

$$Z_2 \leq 2n \max_{i \in [n]} (\deg_{G_1}(i)) + 2n \max_{i \in [n]} (\deg_{G_2}(i)) + (\varepsilon_1(\pi') + \delta_1(\pi')) \frac{\gamma(1-\gamma)}{4} \times n^2 \quad (\text{D.42})$$

Substituting (D.34) and (D.42) in (D.33),

$$\tilde{X}(\pi') \leq \frac{Z_1 + Z_2}{2} \leq \frac{1}{2} \left(3n \max_{i \in [n]} \deg_{G_1}(i) + 2n \max_{i \in [n]} (\deg_{G_2}(i)) + (\varepsilon_1(\pi') + \delta_1(\pi')) \frac{\gamma(1-\gamma)}{4} \times n^2 \right). \quad (\text{D.43})$$

The following claim is made next.

Claim D.1. Let $\varepsilon > 0$ and π' be any permutation such that $\text{ov}(\pi', \text{id}) \geq \varepsilon n$. Let ε_1 and δ_1 be as defined above. Then, $\varepsilon_1(\pi') + \delta_1(\pi') < 2$.

Proof. The proof of the claim relies on a simple observation: If node i is not a fixed point of π' , then neither is the node $\pi'(i)$. Specifically, for each node $i \in \mathcal{B}_1 \cup \mathcal{G}_1$ such that $\pi'(i) \in \mathcal{B}_2 \cup \mathcal{G}_2$, there exists a unique node $k = \pi'(i)$ in $\mathcal{B}_2 \cup \mathcal{G}_2$ such that $\pi'(k) \neq k$. This is true simply because π' is a permutation and therefore bijective.

Assume to the contrary that $\varepsilon_1(\pi') + \delta_1(\pi') = 2$. Since $\varepsilon_1(\pi')$ and $\delta_1(\pi')$ are fractions, our assumption also implies that $\varepsilon_1(\pi') = 1$ and $\delta_1(\pi') = 1$. Since the number of wrongly matched nodes in $\mathcal{B}_1$ (resp. $\mathcal{G}_1$) is at least as large as the number of nodes in $\mathcal{B}_1$ (resp. $\mathcal{G}_1$) that are mapped to $\mathcal{B}_2 \cup \mathcal{G}_2$, it follows that

$$\begin{aligned} |\{i \in [n] : \pi'(i) \neq i\}| &\geq 2(\varepsilon_1(\pi')|\mathcal{B}_1| + \delta_1(\pi')|\mathcal{G}_1|) \\ &= 2(|\mathcal{B}_1| + |\mathcal{G}_1|) \\ &= n, \end{aligned}$$

which contradicts the fact that $\text{ov}(\pi', \text{id}) \geq \varepsilon n$. Here, the factor of 2 in (D.44) is due to the aforementioned observation. It is concluded that $\varepsilon_1(\pi') + \delta_1(\pi') < 2$. $\square$

Claim D.1 confirms the existence of a small positive constant ε' such that $\varepsilon_1(\pi') + \delta_1(\pi') \leq 2 - \varepsilon'$.

Combining this with (D.43) and using the fact that $2n \leq 3n$ for all n yields

$$\tilde{X}(\pi') \leq \frac{3n}{2} \left(\max_{i \in [n]} \deg_{G_1}(i) + \max_{i \in [n]} \deg_{G_2}(i) \right) + (1 - \varepsilon'/2) \times \left(\frac{\gamma(1-\gamma)}{4} \times n^2 \right). \quad (\text{D.44})$$

Therefore,

$$\begin{aligned} \mathbb{P} \left(\tilde{X}(\pi') > \frac{\gamma(1-\gamma)}{4} \times n^2 \right) &\leq \mathbb{P} \left(\frac{3n}{2} \left(\max_{i \in [n]} \deg_{G_1}(i) + \max_{i \in [n]} \deg_{G_2}(i) \right) > \frac{\varepsilon'}{2} \times \frac{\gamma(1-\gamma)}{4} \times n^2 \right) \\ &= \mathbb{P} \left(\max_{i \in [n]} \deg_{G_1}(i) + \max_{i \in [n]} \deg_{G_2}(i) > \frac{\gamma(1-\gamma)}{12} \times \varepsilon' n \right) \\ &\leq \mathbb{P} \left(\max_{i \in [n]} \deg_{G_1}(i) > \frac{\gamma(1-\gamma)}{24} \times \varepsilon' n \right) \\ &\quad + \mathbb{P} \left(\max_{i \in [n]} \deg_{G_2}(i) > \frac{\gamma(1-\gamma)}{24} \times \varepsilon' n \right) \\ &\stackrel{(a)}{\leq} 2n \times \mathbb{P} \left(\text{Bin}(n, ps^2) > \frac{\gamma(1-\gamma)}{24} \times \varepsilon' n \right), \\ &\stackrel{(b)}{\leq} 2n \times \mathbb{P} (\text{Bin}(n, ps^2) > n) \end{aligned}$$

where (a) uses a union bound, the stochastic dominance of $\text{Bin}(n, ps^2)$ over $\text{Bin}(n-1, ps^2)$, and (b) uses the fact that $\varepsilon'\gamma(1-\gamma)/24 < 1$. We use the tail bound on the Binomial distribution [MU17]

$$\mathbb{P} (\text{Bin}(n, ps^2) > (1 + \delta)nps^2) \leq \exp(-nps^2\delta^2/3),$$

with $\delta = \frac{n}{Cs^2 \log(n)} - 1$ to get

$$\mathbb{P} \left(\tilde{X}(\pi') > \frac{\gamma(1-\gamma)}{4} \times n^2 \right) \leq 2n \times \mathbb{P} (\text{Bin}(n, ps^2) > n) \leq 2n \times \exp \left(-\frac{n^2}{3Cs^2 \log(n)} \right).$$

It remains to perform a union bound over all π' such that $\text{ov}(\pi', \text{id}) > \varepsilon n$. However, the number of such permutations is trivially upper bounded by $n! < \exp(n^{1.1} \log(n))$ for all sufficiently large n . Therefore,

$$\mathbb{P} \left(\bigcup_{\varepsilon \in (0,1]} \bigcup_{\text{ov}(\pi, \text{id}) = \varepsilon n} \tilde{X}(\pi) > \frac{\gamma(1-\gamma)}{4} \times n^2 \right) \leq 2n \exp(n^{1.1} \log(n)) \times \exp \left(-\frac{n^2}{3Cs^2 \log(n)} \right) = o(1),$$

as desired. This completes the proof. $\square$

Appendix E

Supplementary material for Chapter 7

E.1 Proofs of deterministic results

E.1.1 Proof of Theorem 7.2

Let $N_j^{(\eta)}$ be the event that b_j is an η -Nakamoto block. Recall that

$$N_j^{(\eta)} = \left[\bigcap_{i: i < j} [\overleftarrow{B}_{i,j}^{(\eta)}]^c \right] \cap \left[\bigcap_{k: k > j} [\overrightarrow{B}_{j,k}^{(\eta)}]^c \right]$$

Suppose $N_j^{(\eta)}$ occurs, so that b_j is an η -Nakamoto block.

- (i) Let $t \geq \tau_j$. We show that b_j is the unique honest block at its height in $MB(t)$, i.e. for any $i, k \geq 0$ such that $i < j < k$, we show that

$$\text{height}_{MB}(b_i) < \text{height}_{MB}(b_j) < \text{height}_{MB}(b_k). \quad (\text{E.1})$$

We begin by proving the first inequality in (E.1). Since $N_j^{(\eta)}$ occurs, it follows that the event $[\overleftarrow{B}_{i,j}^{(\eta)}]^c$ occurs. Therefore, we have

$$\begin{aligned} \text{height}_{MB}(b_j) - \text{height}_{MB}(b_i) &\stackrel{(a)}{\geq} BS_j(b_{i+1}) - BU_j(b_i) \\ &\stackrel{(b)}{\geq} \eta \cdot BS_j(b_{i+1}) - BU_j(b_i) \\ &\stackrel{(c)}{>} a(b_i, b_j), \end{aligned}$$

where (a) is from Lemma 7.4, (b) follows from the fact that $\eta \leq 1$, and (c) is the definition of $[\overleftarrow{B}_{i,j}^{(\eta)}]^c$. Since $a(b_i, b_j) \geq 0$, it follows that $\text{height}_{MB}(b_i) < \text{height}_{MB}(b_j)$. Similarly, the occurrence of $[\overrightarrow{B}_{j,k}^{(\eta)}]^c$ implies that $\text{height}_{MB}(b_j) < \text{height}_{MB}(b_k)$. Thus, b_j is the unique

honest block at its height in $\text{MB}(t)$.

- (ii) We show that b_j is in every longest chain of $\text{MB}(t)$. Let $\mathcal{V}$ be any longest chain in $\text{MB}(t)$, i.e.

$$\mathcal{V} = v_0 - v_1 - \cdots - v_{\text{height}_{\text{MB}}(b_j)} - \cdots - v_{m-1} - v_m,$$

where v_0 is the genesis block, $v_{\text{height}_{\text{MB}}(b_j)} =: v$ is the block at the height of b_j , and m is the height of $\text{MB}(t)$. It suffices to show that $v = b_j$. In fact, since b_j is the unique honest block at its height, it suffices to only show that v must be an honest block. Starting from v and traversing blocks in $\mathcal{V}$ in the backward direction, let b_i denote the first honest block encountered in $\mathcal{V}$, not including v itself. Since the genesis block is honest, such a b_i exists. Similarly, we also want to traverse blocks in $\mathcal{V}$ along the forward direction. Here, we have two cases:

Case 1: There is an honest block in $\mathcal{V}$ after v . In this case, let b_k be the first honest block in $\mathcal{V}$ after v . Let H denote $\text{height}_{\text{MB}}(b_k)$, and $\mathcal{V}_1$ be the sequence $b_i - \cdots - v - \cdots - b_k$.

Case 2: There is no honest block in $\mathcal{V}$ after v . In this case, let b_k be the first honest block mined after time t . Let $H = m$ denote the height of the main-blocktree, and $\mathcal{V}_1$ be the sequence $b_i - \cdots - v - \cdots - v_m$.

In both cases, let N denote the number of blocks in $\mathcal{V}_1$, not including b_i . Notice that the blocks are at consecutive heights. We have:

$$\begin{aligned} N &= H - \text{height}_{\text{MB}}(b_i) \\ &\stackrel{(a)}{\geq} \text{height}_{\text{MB}}(b_j) + \text{FS}_j(b_{k-1}) - \text{FU}_j(b_k) - \text{height}_{\text{MB}}(b_i) \\ &\stackrel{(b)}{\geq} (\text{FS}_j(b_{k-1}) - \text{FU}_j(b_k)) + (\text{BS}_j(b_{i+1}) - \text{BU}_j(b_i)) \\ &\stackrel{(c)}{\geq} (a(b_i, b_j) + 1) + (a(b_j, b_k) + 1) \\ &> a(b_i, b_k) + 1, \end{aligned}$$

where (a) follows from Lemma 7.4 in case 1, and the fact that j -FS blocks are mined at different heights in case 2. Further, (b) is due to Lemma 7.4 and the fact that $\eta \leq 1$, and (c) follows because the events $\left[\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}\right]^c$ and $\left[\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}\right]^c$ occur. However, by definition of b_i and b_k , all blocks in $\mathcal{V}_1$ after b_i except for b_k and possibly v are adversarial. Therefore, there must be at least one honest block in $\mathcal{V}_1$ strictly between b_i and b_k . Since the only possibility for this is v , it follows that v must be an honest block. Since b_j is the unique honest block at the height of v , we conclude that b_j is in every longest chain of $\text{MB}(t)$, as desired.

- (iii) Let v be a block in $\text{MB}(\tau_j)$ such that $\text{height}_{\text{MB}}(v) = \text{height}_{\text{MB}}(b_j)$. Let $\mathcal{T}$ be the tline of ancestors of v . Traversing blocks along the backward direction starting from v , let b_i denote the first honest block encountered. The portion of $\mathcal{T}$ between b_i and v consists of only

adversarial blocks. However,

$$\begin{aligned}
\text{height}_{\text{MB}}(v) - \text{height}_{\text{MB}}(b_i) &= \text{height}_{\text{MB}}(b_j) - \text{height}_{\text{MB}}(b_i) \\
&\stackrel{(a)}{\geq} \eta \cdot \text{BS}_j(b_{i+1}) - \text{BU}_j(b_i) \\
&\stackrel{(b)}{>} a(b_i, b_j),
\end{aligned}$$

where (a) is from Lemma 7.4, and (b) is true because b_j is an η -Nakamoto block. Since v is the only possibility for an honest block in the portion of $\mathcal{T}$ after b_i , it follows that v must be an honest block. From (i), we conclude that $v = b_j$. This concludes the proof.

- (iv) Let $k > j$. Since b_j is an η -Nakamoto block, we have from the proof of (i) that $\text{height}_{\text{MB}}(b_k) > \text{height}_{\text{MB}}(b_j)$. Let v denote the ancestor of b_k at the height of b_j in the main-blocktree. Starting from v and traversing blocks along the ancestors of v , let b_i denote the first honest block encountered. This is exactly case 1 in the proof of (ii), and it follows that $v = b_j$.

E.1.2 Proof of Theorem 7.3

Let $|\mathcal{C}_h(t)|$ denote the number of blocks in $\mathcal{C}_h(t)$. Let τ_j denote the mining time of b_j , and let τ_j^k denote the mining time of b_j^k . Since b_j is an η -Nakamoto block, we know that the event $\left[\vec{\mathbf{B}}_{j,k}^{(\eta)}\right]^c$ occurs for all $k > j$. In turn, this implies

$$a(b_j^0, b_j^{k+1}) < \eta \cdot (k+1) \quad \forall k > 0.$$

However, $\eta(k+1) \leq \left(\frac{1+\eta}{2}\right)k$ whenever $k \geq k_0$. Further, since (h, j, η, k_0) -user-unheard-criterion holds, we have that $\text{Unheard}_h(b_j^k) < \left(\frac{1-\eta}{2}\right)k$ whenever $k \geq k_0$. These facts imply:

$$a(b_j^0, b_j^{k+1}) < k - \text{Unheard}_h(b_j^k) \quad \forall k \geq k_0. \quad (\text{E.2})$$

Fix $k \geq k_0$. Let t be such that $\tau_j^k \leq t < \tau_j^{k+1}$. We show that h includes b_j in its chain at time t .

Since $a(b_j^0, b_j^{k+1}) \geq 0$, it follows from (E.2) that $\text{Unheard}_h(b_j^k) < k$. Therefore, h has heard of at least one j -FS block. Therefore, $|\mathcal{C}_h(t)| \geq \text{height}_{\text{MB}}(b_j)$. Let $v \in \mathcal{C}_h(t)$ be the block at the height of b_j . To show that $b_j \in \mathcal{C}_h(t)$, it suffices by statement (iv) in Theorem 7.2 to prove that $\mathcal{C}_h(t)$ contains at least one honest block mined in $[\tau_j^k, \infty)$.

Let $\mathcal{C}_h(t)^{\lceil v}$ denote the sub-chain of $\mathcal{C}_h(t)$ starting from v , i.e. $\mathcal{C}_h(t)^{\lceil v}$ contains blocks of $\mathcal{C}_h(t)$ that are at height no less than that of v . We show that (E.2) implies that $\mathcal{C}_h(t)^{\lceil v}$ cannot contain all adversarial blocks. We know from statement (iii) of Theorem 7.2 that b_j is the unique block at its height in $\text{MB}(\tau_j)$. Therefore, all blocks in $\mathcal{C}_h(t)^{\lceil v}$ are mined at or after time τ_j . Since $t < \tau_j^{k+1}$, we

have

$$a(b_j^0, b_j^{k+1}) \stackrel{(a)}{<} k - \text{Unheard}_h(b_j^k) \stackrel{(b)}{\leq} |\mathcal{C}_h(t)^{\uparrow v}|,$$

where (a) is the same as (E.2) and (b) follows from the fact that h has heard of b_j^m , where $m = k - \text{Unheard}_h(b_j^k)$. Therefore, h adopts a chain that has length at least $\text{height}_{\text{MB}}(v) + m$. We conclude that there exists an honest block $b_k \in \mathcal{C}_h(t)^{\uparrow v}$. From statement (iv) in Theorem 7.2, it follows that $\mathcal{C}_h(t)$ contains b_j .

The above argument is true for all $t \in [\tau_j^k, \tau_j^{k+1})$, so it follows that $h \in \mathcal{C}_h(t)$ for all $t \in [\tau_j^k, \tau_j^{k+1})$. Since this is true for all $k \geq k_0$, it follows that $h \in \mathcal{C}_h(t)$ for all $t \geq \tau_j^{k_0}$, as desired.

E.2 Proofs of probabilistic results

E.2.1 Proof of Lemma 7.5

We prove only the first statement, since the second uses a similar argument. For any $m \in \{i, j, j', k\}$, let τ_m denote the mining time of b_m . The LHS of the event

$$\vec{\mathbf{B}}_{j',k}^{(\eta)}: a(b_{j'}, b_k) \geq \eta \cdot \text{FS}_{j'}(b_{k-1}) - \text{FU}_{j'}(b_k)$$

depends on the number of adversarial arrivals in $[\tau_{j'}, \tau_k]$. Further, the RHS depends on the delays from $b_{j'}$ to all the honest blocks mined in $[\tau_{j'}, \tau_k]$.

In contrast, the LHS of the event

$$\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}: a(b_i, b_j) \geq \eta \cdot \text{BS}_j(b_{i+1}) - \text{BU}_j(b_i)$$

depends on the number of adversarial arrivals in $[\tau_i, \tau_j]$. Further, the RHS depends on the delays from all the honest blocks mined in $[\tau_i, \tau_j]$ to b_j .

Since honest and adversarial arrivals are independent Poisson processes, and since the delay associated with any two blocks in $[\tau_i, \tau_j]$ is independent of the delay associated with any two blocks in $[\tau_{j'}, \tau_k]$, and since the two intervals do not overlap, it follows that $\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}$ and $\vec{\mathbf{B}}_{j',k}^{(\eta)}$ are independent.

E.2.2 Proof of Theorem 7.4

Fix j . Recall that the event $\mathbf{N}_j^{(\eta)} = \overleftarrow{\mathbf{E}}_j^{(\eta)} \cap \vec{\mathbf{E}}_j^{(\eta)}$, where $\overleftarrow{\mathbf{E}}_j^{(\eta)} = \bigcap_{i: i < j} [\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}]^\complement$ and $\vec{\mathbf{E}}_j^{(\eta)} = \bigcap_{k: k > j} [\vec{\mathbf{B}}_{j,k}^{(\eta)}]^\complement$. From Lemma 7.5, it follows that $\overleftarrow{\mathbf{E}}_j^{(\eta)}$ and $\vec{\mathbf{E}}_j^{(\eta)}$ are independent events. Therefore,

$$\mathbb{P}(\mathbf{N}_j^{(\eta)}) = \mathbb{P}(\overleftarrow{\mathbf{E}}_j^{(\eta)} \cap \vec{\mathbf{E}}_j^{(\eta)}) = \mathbb{P}(\overleftarrow{\mathbf{E}}_j^{(\eta)}) \mathbb{P}(\vec{\mathbf{E}}_j^{(\eta)}).$$

Thus, it suffices to show the existence of $p > 0$ such that $\mathbb{P}(\vec{\mathbf{E}}_j^{(\eta)}) \geq p$ and $\mathbb{P}(\vec{\mathbf{E}}_j^{(\eta)}) \geq p$.

Notation: If an i.i.d. random process is a sequence of random variables with known distribution, say Geometric or Bernoulli with parameter q , we refer to the i -th random variable in the sequence as $\text{Geom}_i(q)$ or $\text{Bern}_i(q)$ respectively.

Showing $\mathbb{P}(\vec{\mathbf{E}}_j^{(\eta)}) \geq p > 0$: By definition, the event $\vec{\mathbf{E}}_j^{(\eta)}$ occurs if

$$a(b_j, b_k) - \eta \cdot \text{FS}_j(b_{k-1}) < -\text{FU}_j(b_k) \quad \forall k > j. \quad (\text{E.3})$$

Notice that the LHS and RHS are independent random variables. The random variables in the LHS are:

$$a(b_j, b_k) = \sum_{i=1}^{k-j} (\text{Geom}_i(1 - \beta) - 1), \quad (\text{E.4})$$

$$\text{FS}_j(b_{k-1}) = 1 + \sum_{i=1}^{k-j-1} \text{Bern}_i(1 - d), \quad (\text{E.5})$$

where equation (E.4) follows from the fact that there are $\text{Geom}(1 - \beta) - 1$ adversarial arrivals between two successive honest arrivals, and the fact that the number of adversarial arrivals in disjoint intervals is independent. Equation (E.5) follows from Remark 7.1.

Let $(\mathbf{X}_i : i \geq 1)$ be an i.i.d. random process, with $\mathbf{X}_i \sim \text{Geom}_i(1 - \beta) - 1 - \eta \cdot \text{Bern}_i(1 - d)$. Let $\mathbf{W}_j = \sum_{i=1}^j \mathbf{X}_i$ denote the sum of the first j terms of the process $(\mathbf{X}_i : i \geq 1)$. Finally, it follows from Remark 7.2 that $(\text{FU}_j(b_k) : k > j)$ is identical to the i.i.d. random process $(\mathbf{Y}_i : i > 0)$, with $\mathbf{Y}_i \sim \text{Geom}_i(1 - d) - 1$. Using these random variables, equation (E.3) can be equivalently stated as

$$\sum_{i=1}^{k-j} \mathbf{X}_i < -\mathbf{Y}_{k-j} \quad \forall k > j. \quad (\text{E.6})$$

Let γ be a constant to be determined later, such that $0 < \gamma < \eta$. The inequality in (E.6) holds if $\mathbf{E}_A \cap \mathbf{E}_B$ occur, where

$$\begin{aligned} \mathbf{E}_A : \sum_{j=1}^i \mathbf{X}_j &< -\gamma i \quad \forall i \geq 1 \\ \mathbf{E}_B : -\gamma i &< -\mathbf{Y}_i \quad \forall i \geq 1. \end{aligned}$$

Here, $\mathbf{E}_A$ and $\mathbf{E}_B$ are independent events. Thus, we have $\mathbb{P}(\vec{\mathbf{E}}_j^{(\eta)}) \geq \mathbb{P}(\mathbf{E}_A \cap \mathbf{E}_B) = \mathbb{P}(\mathbf{E}_A) \mathbb{P}(\mathbf{E}_B)$. It suffices to show the existence of $p_1, p_2 > 0$ such that $\mathbb{P}(\mathbf{E}_A) \geq p_1$ and $\mathbb{P}(\mathbf{E}_B) \geq p_2$. First, we

bound $\mathbb{P}(\mathbf{E}_B)$.

$$\mathbb{P}(\mathbf{E}_B) = \mathbb{P}\left(\bigcap_{i \geq 1} \{\mathbf{Y}_i \leq \gamma i\}\right) = \prod_{i=1}^{\infty} (1 - d^{1+\gamma i}) =: p_2 > 0,$$

since $(\mathbf{Y}_i: i \geq 1)$ is an i.i.d process with $1 + \mathbf{Y}_i \sim \text{Geom}_i(1 - d)$. Notice that $p_2 > 0$ for all $0 \leq d < 1$ and for all $\gamma > 0$.

Next, we bound $\mathbb{P}(\mathbf{E}_A)$. Fix some $\ell \in \mathbb{N}$, and let c be a positive constant given by $c = (\eta - \gamma)\ell$. Consider the following two desirable events:

$$\begin{aligned} \mathcal{G}_1: \mathbf{W}_\ell &\leq -\gamma\ell - c \\ \mathcal{G}_2: \max_{m \geq 0} (\mathbf{W}_{\ell+m} - \mathbf{W}_\ell + \gamma m) &< c \end{aligned}$$

It is clear that $\mathcal{G}_1 \cap \mathcal{G}_2 \implies \mathbf{E}_A$, and that $\mathcal{G}_1$ and $\mathcal{G}_2$ are independent events, since Poisson arrivals over disjoint intervals are independent. Therefore, it suffices to find constants $p_{11}, p_{12} > 0$ such that $\mathbb{P}(\mathcal{G}_1) \geq p_{11}$ and $\mathbb{P}(\mathcal{G}_2) \geq p_{12}$.

$$\mathbb{P}(\mathcal{G}_1) = \mathbb{P}\left(\sum_{i=1}^{\ell} \mathbf{X}_i \leq -\gamma\ell - c\right) \tag{E.7}$$

$$= \mathbb{P}\left(\sum_{i=1}^{\ell} (\text{Geom}_i(1 - \beta) - 1 - \eta \cdot \text{Bern}_i(1 - d)) \leq -\gamma\ell - c\right) \tag{E.8}$$

$$= \mathbb{P}\left(\sum_{i=1}^{\ell} (\text{Geom}_i(1 - \beta) - \eta \cdot \text{Bern}_i(1 - d)) \leq (1 - \gamma)\ell - c\right) \tag{E.9}$$

$$\geq \mathbb{P}\left(\bigcap_{i=1}^{\ell} [\{\text{Geom}_i(1 - \beta) = 1\} \cap \{\text{Bern}_i(1 - d) = 1\}]\right) \tag{E.10}$$

$$= p_{11} > 0, \tag{E.11}$$

for fixed ℓ , since $\gamma < \eta$ and since c is chosen to be sufficiently small. Here, the inequality in (E.10) holds because one way to satisfy the inequality in the event in (E.9) is when both LHS and RHS of the inequality equal $(1 - \eta)\ell$.

It remains to show $p_{12} > 0$. Consider the random process $(\mathbf{Z}_i: i \geq 1)$, where $\mathbf{Z}_i = \mathbf{X}_i + \gamma$. Then, $(\sum_{i=1}^m \mathbf{Z}_i: m \geq 1)$ and $(\mathbf{W}_{\ell+m} - \mathbf{W}_\ell + \gamma m: m \geq 1)$ follow the same distribution. The Kingman bound [Kin64] yields,

$$\mathbb{P}(\mathcal{G}_2) = \mathbb{P}\left(\max_{m \geq 0} \sum_{i=1}^m \mathbf{Z}_i < c\right) \geq 1 - e^{-\theta^* c} =: p_{12},$$

where

$$\theta^* = \sup \left\{ \theta > 0 : \mathbb{E} \left[e^{\theta Z_1} \leq 1 \right] \right\}.$$

Since $c > 0$, we see that $p_{12} > 0$ if $\theta^* > 0$. We show that $\theta^* > 0$ if $\frac{\beta}{1-\beta} < \eta(1-d)$. A simple computation yields

$$\mathbb{E}[Z_1] = \frac{\beta}{1-\beta} - \eta \cdot (1-d) + \gamma,$$

If $\frac{\beta}{1-\beta} < \eta(1-d)$, then there exists $0 < \gamma < \eta$ such that $\mathbb{E}[Z_1] < 0$. Since $\mathbb{E}[e^{\theta Z_1}]_{\theta=0} = 1$, and $\frac{d}{d\theta} \mathbb{E}[e^{\theta Z_1}]_{\theta=0} = \mathbb{E}[Z_1] < 0$, we know that there exists $\theta_1 > 0$ such that $\mathbb{E}[e^{\theta Z_1}]_{\theta=\theta_1} < 1$, which then implies $\theta^* > \theta_1 > 0$.

The above argument is summarized as

$$\mathbb{P} \left(\vec{\mathcal{E}}_j^{(\eta)} \right) \geq \mathbb{P}(\mathcal{E}_A) \mathbb{P}(\mathcal{E}_B) \geq \mathbb{P}(\mathcal{G}_1) \cdot \mathbb{P}(\mathcal{G}_2) \cdot \mathbb{P}(\mathcal{E}_B) \geq p_{11} \cdot p_{12} \cdot p_2 =: p > 0. \quad (\text{E.12})$$

Showing $\mathbb{P} \left(\overleftarrow{\mathcal{E}}_j^{(\eta)} \right) \geq p > 0$: For any $i, k \geq 0$ such that $j - i = k - j$, it follows from Remarks 7.1, 7.2, 7.3 and 7.4 that $\mathbb{P} \left(\overleftarrow{\mathcal{B}}_{i,j}^{(\eta)} \right) = \mathbb{P} \left(\overrightarrow{\mathcal{B}}_{j,k}^{(\eta)} \right)$. By symmetry, we have

$$\begin{aligned} \mathbb{P} \left(\overleftarrow{\mathcal{E}}_j^{(\eta)} \right) &= \mathbb{P} \left(\bigcap_{i=0}^j \left[\overleftarrow{\mathcal{B}}_{i,j}^{(\eta)} \right]^c \right) \\ &= \mathbb{P} \left(\bigcap_{k=j+1}^{2j+1} \left[\overrightarrow{\mathcal{B}}_{j,k}^{(\eta)} \right]^c \right) \\ &\geq \mathbb{P} \left(\bigcap_{k:k>j} \left[\overrightarrow{\mathcal{B}}_{j,k}^{(\eta)} \right]^c \right) \\ &= \mathbb{P} \left(\vec{\mathcal{E}}_j^{(\eta)} \right) \\ &\geq p, \end{aligned}$$

where the constant $p > 0$ is the same as in (E.12). This concludes the proof.

E.2.3 Proof of Lemma 7.6

First, we show the existence of $c > 0$ such that $\mathbb{P} \left(\overrightarrow{\mathcal{B}}_{j,k}^{(\eta)} \right) \leq e^{-c(k-j)}$. Recall that the event $\overrightarrow{\mathcal{B}}_{j,k}^{(\eta)}$ is defined as

$$\overrightarrow{\mathcal{B}}_{j,k}^{(\eta)} : a(b_j, b_k) \geq \eta \cdot \text{FS}_j(b_{k-1}) - \text{FU}_j(b_k).$$

From Remarks 7.1, 7.2, 7.3 and 7.4, these random variables are characterized as:

$$\begin{aligned} a(b_j, b_k) &= \sum_{i=1}^{k-j} (\text{Geom}_i(1-\beta) - 1), \\ \text{FS}_j(b_{k-1}) &= 1 + \sum_{i=1}^{k-j-1} \text{Bern}_i(1-d), \\ \text{FU}_j(b_k) &= \text{Geom}(1-d) - 1. \end{aligned}$$

Further, the three random variables $a(b_j, b_k)$, $\text{FS}_j(b_{k-1})$, $\text{FU}_j(b_k)$ are mutually independent. Consider the following desirable events associated with them.

$$\begin{aligned} \mathcal{G}_1: a(b_j, b_k) &< \frac{\beta}{1-\beta} (k-j) (1+\varepsilon) \\ \mathcal{G}_2: \text{FS}_j(b_{k-1}) &> (1-d) (k-j) (1-\delta) \\ \mathcal{G}_3: \text{FU}_j(b_k) &\leq (k-j) \left[(1-\delta) \eta \cdot (1-d) - \frac{\beta}{1-\beta} (1+\varepsilon) \right] \end{aligned}$$

Clearly, $\mathcal{G}_1 \cap \mathcal{G}_2 \cap \mathcal{G}_3 \implies \left[\vec{\text{B}}_{j,k}^{(\eta)} \right]^c$. Therefore, we have

$$\mathbb{P} \left(\left[\vec{\text{B}}_{j,k}^{(\eta)} \right]^c \right) \geq \mathbb{P}(\mathcal{G}_1 \cap \mathcal{G}_2 \cap \mathcal{G}_3) = \mathbb{P}(\mathcal{G}_1) \mathbb{P}(\mathcal{G}_2) \mathbb{P}(\mathcal{G}_3). \quad (\text{E.13})$$

It suffices to find bounds for each term separately in the RHS of (E.13).

Bounding $\mathcal{G}_1$: Since $\mathbb{E}[a(b_j, b_k)] = \frac{\beta}{1-\beta} (k-j)$, it follows from the Chernoff bound that

$$\begin{aligned} \mathbb{P}(\mathcal{G}_1^c) &= \mathbb{P} \left(a(b_j, b_k) \geq \frac{\beta}{1-\beta} (k-j) (1+\varepsilon) \right) \leq e^{-c_1(k-j)}, \\ \text{where } c_1 &= \frac{\beta}{1-\beta} (1+\varepsilon) \log(1+\varepsilon) + \frac{1}{1-\beta} (1+\beta\varepsilon) \log(1+\beta\varepsilon) > 0 \end{aligned}$$

Bounding $\mathcal{G}_2$: It follows from the Hoeffding bound that

$$\begin{aligned} \mathbb{P}(\mathcal{G}_2^c) &= \mathbb{P} \left(1 + \sum_{i=1}^{k-j-1} \text{Bern}_i(1-d) \leq (1-d) (k-j) (1-\delta) \right) \\ &\leq \mathbb{P} \left(\sum_{i=1}^{k-j} \text{Bern}_i(1-d) \leq (1-d) (k-j) (1-\delta) \right) \\ &\leq e^{-c_2(k-j)}, \end{aligned}$$

where $c_2 = \frac{\delta^2(1-d)}{2} > 0$.

Bounding $\mathcal{G}_3$: Consider the quantity $c_4 = \left\lceil (1 - \delta) \eta \cdot (1 - d) - \frac{\beta}{1 - \beta} (1 + \varepsilon) \right\rceil$. If $\frac{\beta}{1 - \beta} < \eta (1 - d)$, then there exist $\varepsilon, \delta > 0$ such that $c_4 > 0$. Since $1 + \text{FU}_j(b_k)$ follows a geometric distribution, we have

$$\mathbb{P}(\mathcal{G}_3^c) = \mathbb{P}(\text{Geom}(1 - d) - 1 > c_4(k - j)) = 1 - d^{c_4(k - j)} \geq 1 - e^{c_3(k - j)},$$

for some $c_3 > 0$.

Combining these facts together, we revisit (E.13). We have

$$\mathbb{P}\left(\left[\vec{\mathbf{B}}_{j,k}^{(\eta)}\right]^c\right) \geq \mathbb{P}(\mathcal{G}_1) \mathbb{P}(\mathcal{G}_2) \mathbb{P}(\mathcal{G}_3) \quad (\text{E.14})$$

$$= (1 - \mathbb{P}(\mathcal{G}_1^c)) (1 - \mathbb{P}(\mathcal{G}_2^c)) (1 - \mathbb{P}(\mathcal{G}_3^c)) \quad (\text{E.15})$$

$$\geq \left(1 - e^{-c_1(k-j)}\right) \left(1 - e^{-c_2(k-j)}\right) \left(1 - e^{-c_3(k-j)}\right) \quad (\text{E.16})$$

$$\geq \left(1 - e^{-c_0(k-j)}\right)^3 \quad (\text{E.17})$$

$$\geq 1 - e^{-c(k-j)}, \quad (\text{E.18})$$

where $c_0 = \max\{c_1, c_2, c_3\} > 0$, and subsequently $c > 0$. We conclude the existence of $c > 0$ for which $\mathbb{P}\left(\vec{\mathbf{B}}_{j,k}^{(\eta)}\right) \leq e^{-c(k-j)}$.

It remains to show that $\mathbb{P}\left(\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}\right) \leq e^{-c(j-i)}$. The proof is very similar, so the details are omitted. Recall that

$$\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}: \quad a(b_i, b_j) \geq \eta \cdot \text{BS}_j(b_{i+1}) - \text{BU}_j(b_i),$$

where the random variables involved may be written as

$$\begin{aligned} a(b_i, b_j) &= \sum_{k=1}^{j-i} (\text{Geom}_k(1 - \beta) - 1), \\ \text{BS}_j(b_{i+1}) &= 1 + \sum_{k=1}^{j-i-1} \text{Bern}_k(1 - d), \\ \text{BU}_j(b_i) &= \text{Geom}(1 - d) - 1, \end{aligned}$$

Thus, if $k - j = j - i$, we see that

- $a(b_i, b_j)$ and $a(b_j, b_k)$ follow the same distribution.
- $\text{BS}_j(b_{i+1})$ and $\text{FS}_j(b_{k-1})$ follow the same distribution.
- $\text{BU}_j(b_i)$ and $\text{FU}_j(b_k)$ follow the same distribution.

Therefore, the same concentration inequalities apply, and we conclude that for the same constant c

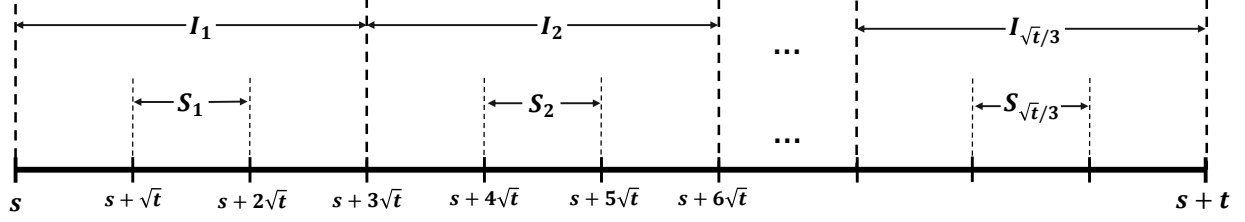


Figure E.1: Partitioning $[s, s+t]$ into sub-intervals

in (E.18), we have $\mathbb{P}\left(\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}\right) \leq e^{-c(j-i)}$.

E.2.4 Proof of Theorem 7.5

For any $i \geq 0$, let τ_i denote the mining time of block b_i . Partition the interval $[s, s+t]$ into $\sqrt{t}$ intervals of length $\sqrt{t}$ each. Group these sub-intervals into threes, so that there are $\sqrt{t}/3$ groups of sub-intervals, namely $I_1, I_2, \dots, I_{\sqrt{t}/3}$. Thus, $I_\ell = [s + 3(\ell - 1)\sqrt{t}, s + 3\ell\sqrt{t}]$. Further, let S_ℓ represent the middle sub-interval of I_ℓ , so that $S_\ell = [s + (3\ell - 2)\sqrt{t}, s + (3\ell - 1)\sqrt{t}]$, as shown in Figure E.1.

Consider the following desirable events.

$$\mathcal{G}_1: \bigcap_{j: \tau_j \in [s+\sqrt{t}, s+t-\sqrt{t}]} \left[\left(\bigcap_{\substack{i < j \\ \tau_i < \tau_j - \sqrt{t}}} [\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}]^c \right) \cap \left(\bigcap_{\substack{k > j \\ \tau_k > \tau_j + \sqrt{t}}} [\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}]^c \right) \right]$$

$$\mathcal{G}_2: \bigcup_{\ell=1}^{\sqrt{t}/3} \mathcal{H}_\ell, \text{ where } \mathcal{H}_\ell: \bigcup_{j: \tau_j \in S_\ell} \left[\left(\bigcap_{\substack{i < j \\ \tau_i \geq \tau_j - \sqrt{t}}} [\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}]^c \right) \cap \left(\bigcap_{\substack{k > j \\ \tau_k \leq \tau_j + \sqrt{t}}} [\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}]^c \right) \right]$$

First, observe that $\mathcal{G}_1 \cap \mathcal{G}_2 \implies [B_{s,s+t}^{(\eta)}]^c$. This is because $\mathcal{G}_1$ ensures that no catch-up events (neither $\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}$, nor $\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}$) occur when b_i and b_k are separated in time by more than $\sqrt{t}$ from b_j , for any b_j mined in $[s + \sqrt{t}, s + t - \sqrt{t}]$. This means that the existence of a block b_j in this interval for which no catch-up event occurs whenever i and k are within $\sqrt{t}$ of τ_j is sufficient to ensure that b_j is a Nakamoto block. This is exactly the event $\mathcal{G}_2$. Since $\bigcup_{\ell} S_\ell \subset [s + \sqrt{t}, s + t - \sqrt{t}]$, it follows that $\mathcal{G}_1 \cap \mathcal{G}_2 \implies [B_{s,s+t}^{(\eta)}]^c$. Thus,

$$\mathbb{P}\left(B_{s,s+t}^{(\eta)}\right) \leq \mathbb{P}(\mathcal{G}_1^c) + \mathbb{P}(\mathcal{G}_2^c).$$

Next, we bound the probability of each term in the RHS separately.

Bounding $\mathbb{P}(G_1^c)$: Fix $\delta > 0$. Consider the following events:

$$\begin{aligned} D_1 &: \{ \# \{i: \tau_i \in [s, s+t]\} > 2\lambda_h t \}, \\ D_2 &: \left\{ \exists \tau_i, \tau_k \in [s, s+t]: (k-i) < (1-\delta)\lambda_h\sqrt{t}, \tau_k - \tau_i > \sqrt{t} \right\}. \end{aligned}$$

By the tail bound for Poisson random variables, we know that $\mathbb{P}(D_1) \leq e^{-c_0 t}$ for some $c_0 > 0$. We now show that $\mathbb{P}(D_2) \leq e^{-c_1\sqrt{t}}$ for some $c_1 > 0$. Let $T_{i,k} \triangleq \tau_k - \tau_i$ be the random variable denoting the time between the i -th and k -th mining. Let $M(t) = (1-\delta)\lambda_h\sqrt{t}$. Notice that

$$\bigcap_{i \in [s, s+t]} \left\{ T_{i, i+M(t)} < \sqrt{t} \right\} \implies D_2^c.$$

For any honest arrival time τ_i , we have

$$\mathbb{E}[T_{i, i+M(t)}] = \mathbb{E}\left[\sum_{j=i}^{i+M(t)} (\tau_j - \tau_{j-1})\right] = \frac{M(t)}{\lambda_h} = (1-\delta)\sqrt{t}.$$

Applying the Chernoff bound, we see that there exists $c > 0$ such that

$$\mathbb{P}\left(T_{i, i+M(t)} > \sqrt{t}\right) \leq \mathbb{P}\left(T_{i, i+M(t)} > (1-\delta)\sqrt{t} + \delta\sqrt{t}\right) \leq e^{-c\sqrt{t}}.$$

Thus, we get

$$\begin{aligned} \mathbb{P}(D_2) &= \mathbb{P}\left(\bigcup_{i: \tau_i \in [s, s+t]} \left\{ T_{i, i+M(t)} > \sqrt{t} \right\}\right) \\ &\leq \mathbb{P}\left(\left(\bigcup_{i: \tau_i \in [s, s+t]} \left\{ T_{i, i+M(t)} > \sqrt{t} \right\}\right) \cap D_1^c\right) + \mathbb{P}(D_1) \\ &\leq \left(\sum_{i=1}^{2\lambda_h t} \mathbb{P}\left(T_{i, i+M(t)} > \sqrt{t}\right)\right) + e^{-c_0 t} \\ &\leq e^{-c_1\sqrt{t}}, \end{aligned}$$

for some $c_1 > 0$. Thus, $\mathbb{P}(\mathbf{D}_2) \leq e^{-c_1\sqrt{t}}$. We may therefore bound $\mathbb{P}(\mathbf{G}_1^c)$ as

$$\begin{aligned}
\mathbb{P}(\mathbf{G}_1^c) &\leq \mathbb{P}(\mathbf{D}_1 \cup \mathbf{D}_2) + \mathbb{P}(\mathbf{G}_1^c \cap \mathbf{D}_1^c \cap \mathbf{D}_2^c) \\
&\leq \mathbb{P}(\mathbf{D}_1) + \mathbb{P}(\mathbf{D}_2) + \mathbb{P}(\mathbf{G}_1^c \cap \mathbf{D}_1^c \cap \mathbf{D}_2^c) \\
&\leq e^{-c_0 t} + e^{-c_1\sqrt{t}} + \sum_{j=1}^{2\lambda_h t} \left[\left(\sum_{i=0}^{j-M(t)} \mathbb{P}(\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}) \right) + \left(\sum_{k=j+M(t)}^{\infty} \mathbb{P}(\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}) \right) \right] \\
&\leq e^{-c_0 t} + e^{-c_1\sqrt{t}} + \sum_{j=1}^{2\lambda_h t} \left[\left(\sum_{i=0}^{j-M(t)} e^{-c(j-i)} \right) + \left(\sum_{k=j+M(t)}^{\infty} e^{-c(k-j)} \right) \right] \\
&\leq e^{-c_0 t} + e^{-c_1\sqrt{t}} + \sum_{j=1}^{2\lambda_h t} \left(2 \sum_{m=M(t)}^{\infty} e^{-cm} \right) \\
&= e^{-c_0 t} + e^{-c_1\sqrt{t}} + \frac{4\lambda_h t}{1 - e^{-c}} e^{-cM(t)} \\
&= e^{-c_0 t} + e^{-c_1\sqrt{t}} + \frac{4\lambda_h t}{1 - e^{-c}} e^{-c(1-\delta)\lambda_h\sqrt{t}} \\
&\leq e^{-c_3\sqrt{t}},
\end{aligned}$$

for some $c_3 > 0$.

Bounding $\mathbb{P}(\mathbf{G}_2^c)$: We have $\mathbf{G}_2^c = \bigcap_{\ell=1}^{\sqrt{t}/3} \mathbf{H}_\ell^c$. Notice that $\mathbf{H}_\ell^c$ are mutually independent for distinct ℓ by Lemma 7.5. Recall that

$$\mathbf{H}_\ell = \bigcup_{j: \tau_j \in S_\ell} \mathbf{R}_j^\ell, \text{ where } \mathbf{R}_j^\ell: \left(\bigcap_{\substack{i < j \\ \tau_i \geq \tau_j - \sqrt{t}}} [\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}]^c \right) \cap \left(\bigcap_{\substack{k > j \\ \tau_k \leq \tau_j + \sqrt{t}}} [\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}]^c \right).$$

Let M_ℓ be the number of honest blocks mined in S_ℓ , and N_ℓ be the number of η -Nakamoto blocks mined in S_ℓ . Since $\mathbf{H}_\ell$ is contained in the event $M_\ell \geq 1$, we have for each $\ell \in \{1, 2, \dots, \sqrt{t}/3\}$:

$$\mathbb{P}(\mathbf{H}_\ell) = \mathbb{P}\left(\bigcup_{j: \tau_j \in S_\ell} \mathbf{R}_j^\ell\right) = \mathbb{P}\left(\left(\bigcup_{j: \tau_j \in S_\ell} \mathbf{R}_j^\ell\right) \cap \{M_\ell \geq 1\}\right) \geq \mathbb{P}\left(\bigcup_{j: \tau_j \in S_\ell} \mathbf{N}_j^{(\eta)}\right) \geq \frac{p_0^2}{2}, \quad (\text{E.19})$$

where p_0 is the lower bound on the probability that b_j is an η -Nakamoto block, obtained in Theorem 7.4. The inequality in (E.19) deserves some elaboration: Since M_ℓ is a non-negative integer valued random variable, we have from the second moment method that

$$\mathbb{P}(N_\ell \geq 1) = \mathbb{P}(N_\ell > 0) \geq \frac{(\mathbb{E}[N_\ell])^2}{\mathbb{E}[N_\ell^2]} \geq \frac{(p_0 \lambda_h \sqrt{t})^2 t}{\lambda_h \sqrt{t} + (\lambda_h \sqrt{t})^2} \geq \frac{\lambda_h^2 p_0^2 t}{2\lambda_h^2 t} \geq \frac{p_0^2}{2},$$

for sufficiently large t . Here, we used the fact that $\mathbb{E}[N_\ell] = p_0 \lambda_h \sqrt{t}$, and $\mathbb{E}[N_\ell^2] \leq \mathbb{E}[M_\ell^2] = \lambda_h \sqrt{t} + (\lambda_h \sqrt{t})^2$.

Thus, $\mathbb{P}(\mathbf{H}_\ell^c) \leq 1 - \frac{p_0^2}{2} < 1$, which yields

$$\mathbb{P}(\mathbf{G}_2^c) = \mathbb{P}\left(\bigcap_{\ell=1}^{\sqrt{t}/3} \mathbf{H}_\ell^c\right) = \prod_{\ell=1}^{\sqrt{t}/3} \mathbb{P}(\mathbf{H}_\ell^c) \leq \left(1 - \frac{p_0^2}{2}\right)^{\sqrt{t}/3} \leq e^{-c_4 \sqrt{t}},$$

for some $c_4 > 0$, since the $\mathbf{H}_\ell$'s are mutually independent events. Therefore, we conclude that

$$\mathbb{P}\left(B_{s,s+t}^{(\eta)}\right) \leq \mathbb{P}(\mathbf{G}_1^c) + \mathbb{P}(\mathbf{G}_2^c) \leq e^{-c_3 \sqrt{t}} + e^{-c_4 \sqrt{t}} \leq e^{-c_0 \sqrt{t}},$$

for some $c_0 > 0$, as desired.

E.2.5 Proof of Theorem 7.6

Fix $m > 1$. Consider the following statement for m :

$$\mathbf{S}[m] : \forall \theta \geq m, \exists a_\theta > 0, A_\theta > 0 \text{ such that } \mathbb{P}\left(B_{s,s+t}^{(\eta)}\right) \leq A_\theta \exp\left(-a_\theta t^{1/\theta}\right)$$

In Theorem 7.5, we proved that $\mathbf{S}[2]$ is true. Next, we show the following:

$$\mathbf{S}[m] \implies \mathbf{S}\left[\frac{2m-1}{m}\right].$$

Assume $\mathbf{S}[m]$ is true. For any $i \geq 0$, let τ_i denote the mining time of block b_i . Partition the interval $[s, s+t]$ into $t^{\frac{m-1}{2m-1}}$ intervals of length $t^{\frac{m}{2m-1}}$ each. Group these sub-intervals into threes, so that there are $\frac{t^{\frac{m-1}{2m-1}}}{3}$ groups of sub-intervals, namely $I_1, I_2, \dots, I_{\frac{m-1}{t^{\frac{m-1}{2m-1}}/3}}$. Thus,

$$\begin{aligned} I_\ell &= \left[s + 3(\ell-1)t^{\frac{m}{2m-1}}, s + 3\ell t^{\frac{m}{2m-1}}\right], \\ S_\ell &= \left[s + (3\ell-2)t^{\frac{m}{2m-1}}, s + (3\ell-1)t^{\frac{m}{2m-1}}\right]. \end{aligned}$$

Consider the following desirable events.

$$\begin{aligned} \mathcal{G}_1 &: \bigcap_{j: \tau_j \in \left[s+t^{\frac{m}{2m-1}}, s+t-t^{\frac{m}{2m-1}}\right]} \left[\left(\bigcap_{\substack{i < j \\ \tau_i < \tau_j - t^{\frac{m}{2m-1}}}} [\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}]^c \right) \cap \left(\bigcap_{\substack{k > j \\ \tau_k > \tau_j + t^{\frac{m}{2m-1}}}} [\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}]^c \right) \right] \\ \mathcal{G}_2 &: \bigcup_{\ell=1}^{\frac{m-1}{t^{\frac{m-1}{2m-1}}/3}} \mathbf{H}_\ell, \text{ where } \mathbf{H}_\ell: \bigcup_{j: \tau_j \in S_\ell} \left[\left(\bigcap_{\substack{i < j \\ \tau_i \geq \tau_j - t^{\frac{m}{2m-1}}}} [\overleftarrow{\mathbf{B}}_{i,j}^{(\eta)}]^c \right) \cap \left(\bigcap_{\substack{k > j \\ \tau_k \leq \tau_j + t^{\frac{m}{2m-1}}}} [\overrightarrow{\mathbf{B}}_{j,k}^{(\eta)}]^c \right) \right] \end{aligned}$$

Observe that $\mathcal{G}_1 \cap \mathcal{G}_2 \implies \left[B_{s,s+t}^{(\eta)} \right]^c$.

Bounding $\mathbb{P}(\mathcal{G}_1^c)$: We show that $\mathbb{P}(\mathcal{G}_1^c) \leq e^{-c_3 t^{\frac{m}{2m-1}}}$ for some $c_3 > 0$. This is done following the steps in Theorem 7.5. Fix $\delta > 0$ and consider the following events:

$$\begin{aligned} \mathcal{D}_1 &: \{ \# \{i: \tau_i \in [s, s+t]\} > 2\lambda_h t \}, \\ \mathcal{D}_2 &: \left\{ \exists \tau_i, \tau_k \in [s, s+t]: (k-i) < (1-\delta) \lambda_h t^{\frac{m}{2m-1}}, \tau_k - \tau_i > t^{\frac{m}{2m-1}} \right\}. \end{aligned}$$

By the tail bound for Poisson random variables, we know that $\mathbb{P}(\mathcal{D}_1) \leq e^{-c_0 t}$ for some $c_0 > 0$. We now show that $\mathbb{P}(\mathcal{D}_2) \leq e^{-c_1 t^{\frac{m}{2m-1}}}$ for some $c_1 > 0$. Let $T_{i,k} := \tau_k - \tau_i$ be the random variable denoting the time between the i -th and k -th mining. Let $M(t) = (1-\delta) \lambda_h t^{\frac{m}{2m-1}}$. Notice that

$$\bigcap_{i \in [s, s+t]} \left\{ T_{i, i+M(t)} < \sqrt{t} \right\} \implies \mathcal{D}_2^c.$$

For any honest arrival time τ_i , we have

$$\mathbb{E}[T_{i, i+M(t)}] = \mathbb{E} \left[\sum_{j=i}^{i+M(t)} (\tau_j - \tau_{j-1}) \right] = \frac{M(t)}{\lambda_h} = (1-\delta) t^{\frac{m}{2m-1}}.$$

Applying the Chernoff bound, we see that there exists $c > 0$ such that

$$\mathbb{P} \left(T_{i, i+M(t)} > t^{\frac{m}{2m-1}} \right) \leq \mathbb{P} \left(T_{i, i+M(t)} > (1-\delta) t^{\frac{m}{2m-1}} + \delta t^{\frac{m}{2m-1}} \right) \leq e^{-c t^{\frac{m}{2m-1}}}.$$

Thus, we get

$$\begin{aligned} \mathbb{P}(\mathcal{D}_2) &= \mathbb{P} \left(\bigcup_{i: \tau_i \in [s, s+t]} \left\{ T_{i, i+M(t)} > t^{\frac{m}{2m-1}} \right\} \right) \\ &\leq \mathbb{P} \left(\left(\bigcup_{i: \tau_i \in [s, s+t]} \left\{ T_{i, i+M(t)} > t^{\frac{m}{2m-1}} \right\} \right) \cap \mathcal{D}_1^c \right) + \mathbb{P}(\mathcal{D}_1) \\ &\leq \left(\sum_{i=1}^{2\lambda_h t} \mathbb{P} \left(T_{i, i+M(t)} > t^{\frac{m}{2m-1}} \right) \right) + e^{-c_0 t} \\ &\leq e^{-c_1 t^{\frac{m}{2m-1}}}, \end{aligned}$$

for some $c_1 > 0$. Thus, $\mathbb{P}(D_2) \leq e^{-c_1 t^{\frac{m}{2m-1}}}$. We may therefore bound $\mathbb{P}(G_1^c)$ as

$$\begin{aligned}
\mathbb{P}(G_1^c) &\leq \mathbb{P}(D_1 \cup D_2) + \mathbb{P}(G_1^c \cap D_1^c \cap D_2^c) \\
&\leq \mathbb{P}(D_1) + \mathbb{P}(D_2) + \mathbb{P}(G_1^c \cap D_1^c \cap D_2^c) \\
&\leq e^{-c_0 t} + e^{-c_1 t^{\frac{m}{2m-1}}} + \sum_{j=1}^{2\lambda_h t} \left[\left(\sum_{i=0}^{j-M(t)} \mathbb{P}(\overleftarrow{B}_{i,j}^{(\eta)}) \right) + \left(\sum_{k=j+M(t)}^{\infty} \mathbb{P}(\overrightarrow{B}_{j,k}^{(\eta)}) \right) \right] \\
&\leq e^{-c_0 t} + e^{-c_1 t^{\frac{m}{2m-1}}} + \sum_{j=1}^{2\lambda_h t} \left[\left(\sum_{i=0}^{j-M(t)} e^{-c(j-i)} \right) + \left(\sum_{k=j+M(t)}^{\infty} e^{-c(k-j)} \right) \right] \\
&\leq e^{-c_0 t} + e^{-c_1 t^{\frac{m}{2m-1}}} + \sum_{j=1}^{2\lambda_h t} \left(2 \sum_{m=M(t)}^{\infty} e^{-cm} \right) \\
&= e^{-c_0 t} + e^{-c_1 t^{\frac{m}{2m-1}}} + \frac{4\lambda_h t}{1-e^{-c}} e^{-cM(t)} \\
&= e^{-c_0 t} + e^{-c_1 t^{\frac{m}{2m-1}}} + \frac{4\lambda_h t}{1-e^{-c}} e^{-c(1-\delta)\lambda_h t^{\frac{m}{2m-1}}} \\
&\leq e^{-c_3 t^{\frac{m}{2m-1}}},
\end{aligned}$$

for some $c_3 > 0$.

Bounding $\mathbb{P}(G_2^c)$: We have $G_2^c = \bigcap_{\ell=1}^{t^{\frac{m-1}{2m-1}}/3} H_\ell^c$. Notice that H_ℓ^c are mutually independent for distinct ℓ by Lemma 7.5. Since $S[m]$ is true, we have

$$\mathbb{P}(H_\ell^c) \leq \mathbb{P}\left(B_{s+(3\ell-2)t^{\frac{m}{2m-1}}, s+(3\ell-1)t^{\frac{m}{2m-1}}}^{(\eta)}\right) \leq A_m \exp\left(-a_m \left(t^{\frac{m}{2m-1}}\right)^{\frac{1}{m}}\right) \leq A_m \exp\left(-a_m t^{1/(2m-1)}\right)$$

Therefore, it follows that

$$\mathbb{P}(G_2^c) = \mathbb{P}\left(\bigcap_{\ell=1}^{t^{\frac{m-1}{2m-1}}/3} H_\ell^c\right) = \prod_{\ell=1}^{t^{\frac{m-1}{2m-1}}/3} \mathbb{P}(H_\ell^c) \leq \left(A_m \exp\left(-a_m t^{1/(2m-1)}\right)\right)^{t^{\frac{m-1}{2m-1}}/3} \leq e^{-c_4 t^{\frac{m}{2m-1}}},$$

for some $c_4 > 0$, since the H_ℓ 's are mutually independent events. Therefore, we have

$$\mathbb{P}\left(B_{s,s+t}^{(\eta)}\right) \leq \mathbb{P}(G_1^c) + \mathbb{P}(G_2^c) \leq e^{-c_3 t^{\frac{m}{2m-1}}} + e^{-c_4 t^{\frac{m}{2m-1}}} \leq A_{m'} e^{-a_{m'} t^{1/m'}},$$

where $m' = \frac{m}{2m-1}$, and $a_{m'}, A_{m'} > 0$. In turn, this implies that $S\left[\frac{2m-1}{m}\right]$ is true.

Finally, consider the recursion given by $m_{k+1} = \frac{2m_k-1}{m_k}$, and the initial condition $m_1 = 2$. We have proved that $S[m_1]$ is true and that $S[m_{k+1}]$ is true whenever $S[m_k]$ is true. By induction, it follows that $S[m_k]$ is true for all $k \in \mathbb{N}$. Since $m_k = \frac{k+1}{k}$ and $\lim_{k \rightarrow \infty} m_k = 1$, we conclude that for every $\varepsilon > 0$, there exist positive constants A, a such that $\mathbb{P}\left(B_{s,s+t}^{(\eta)}\right) \leq A \exp(-at^{1-\varepsilon})$. This concludes

the proof.

E.2.6 Proof of Theorem 7.7

By the union bound, we have

$$\mathbb{P} \left(\bigcup_{k=k'}^{\infty} \left\{ \text{Unheard}_h(b_J^k) > \left(\frac{1-\eta}{2} \right) k \right\} \right) \leq \sum_{k=k'}^{\infty} \mathbb{P} \left(\text{Unheard}_h(b_J^k) > \left(\frac{1-\eta}{2} \right) k \right). \quad (\text{E.20})$$

From remark 7.5, it follows that $\text{Unheard}_h(b_J^k) \leq \text{Geom}(1-d)$. Thus, for all $k \geq 1$, we have

$$\begin{aligned} \mathbb{P} \left(\text{Unheard}_h(b_J^k) > \left(\frac{1-\eta}{2} \right) k \right) &\leq \mathbb{P} \left(\text{Geom}(1-d) > \left(\frac{1-\eta}{2} \right) k \right) \\ &\leq C_0 \cdot e^{-ck}, \end{aligned}$$

for some positive constants C_0 and c . Combining this with (E.20) yields

$$\mathbb{P} \left(\bigcup_{k=k'}^{\infty} \left\{ \text{Unheard}_h(b_J^k) > \left(\frac{1-\eta}{2} \right) k \right\} \right) \leq \sum_{k=k'}^{\infty} C_0 \cdot e^{-ck} = C \cdot e^{-ck'},$$

where $C = \frac{C_0}{1-e^{-c}}$ is a positive constant. This concludes the proof.

E.2.7 Proof of Theorem 7.1

Before we prove our main result, we recall a useful lemma about Poisson random variables.

Lemma E.1. *Let X be a Poisson random variable with mean μ . Then*

$$(i) \quad \mathbb{P}(X \geq 2\mu) \leq e^{-\frac{4}{3}\mu}.$$

$$(ii) \quad \mathbb{P}(X \leq \frac{1}{2}\mu) \leq e^{-\frac{1}{8}\mu}.$$

Proof. The proof follows from Theorem 4.5 in [MU17]. □

We now prove our main result. Let η be such that $\frac{\beta}{1-\beta} < \eta \cdot (1-d)$. Let $k_0 = \left\lceil \frac{2\eta}{1-\eta} \right\rceil$, and fix an honest subset of users $\mathcal{H}$. The idea of the proof is as follows: if an η -Nakamoto block b_J is mined in the time interval $(s, s+t_1)$, and k_0 number of J -FS blocks are mined before time $s+t_1+t_2$, and if all users $h \in \mathcal{H}$ satisfy the (h, J, η, k_0) -user-unheard-criterion, then Theorems 7.2 and 7.3 together imply that the η -Nakamoto block b_J is included in $\mathcal{C}_h(t)$ for all $t \geq s+t_1+t_2$. Since tx must be included in either b_J or its ancestors, tx satisfies $(t_1+t_2, \mathcal{H})$ -security.

Let b_J be the first η -Nakamoto block mined after time s . Let $T_J = \tau_J - s$ denote the time between s and the mining time of the first η -Nakamoto block. For the r -th honest block b_r and any time t ,

let $N_r(t)$ denote the number of r -FS blocks mined until time t . Consider the following events:

$$\begin{aligned} \mathbf{E}_1 &: T_J > t_1 \\ \mathbf{E}_2^h &: \text{The } (h, J, \eta, k_0)\text{-user-unheard-criterion is violated} \\ \mathbf{E}_3 &: \bigcup_{r: \tau_r \in [s, s+t_1]} \{N_r(s+t_1+t_2) < k_0\}. \end{aligned}$$

By the first paragraph of the proof, the union bound gives

$$\mathbb{P}(\text{tx violates } (t_1 + t_2, \mathcal{H})\text{-security}) \leq \mathbb{P}(\mathbf{E}_1) + \sum_{h \in \mathcal{H}} \mathbb{P}(\mathbf{E}_2^h) + \mathbb{P}(\mathbf{E}_3). \quad (\text{E.21})$$

From Theorem 7.6, we have that for any $\varepsilon > 0$, there exist positive constants A', a' such that $\mathbb{P}(\mathbf{E}_1) \leq A' (\exp -a' t_1^{1-\varepsilon})$.

From Theorem 7.7, we have that there exist positive constants C, c' such that $\mathbb{P}(\mathbf{E}_2^h) \leq C \exp(-c' k_0)$ for all $h \in \mathcal{H}$.

It remains to bound $\mathbb{P}(\mathbf{E}_3)$. Let λ_h denote the aggregate mining rate of the honest users. Let $t_2 > \frac{2k_0}{(1-d)\lambda_h}$. Let M be the number of honest miners in $[s, s+t_1]$, so that M has the Poisson distribution with mean $\lambda_h(1-d)t_1$. Therefore, we have from Lemma E.1 that

$$\mathbb{P}(M > 2\lambda_h t_1) \leq \exp\left(\frac{-4\lambda_h t_1}{3}\right).$$

For $r \geq 1$, consider the r -th honest miner b_r after time s and consider the r -FS sequence. Let U_r denote the number of r -FS blocks mined in $[\tau_r, \tau_r + t_2]$. Then, U_r has the Poisson probability distribution with mean $\lambda_h(1-d)t_2$. Applying Lemma E.1, we get

$$\mathbb{P}(U_r \leq k_0) \leq \mathbb{P}\left(U_r \leq \frac{1}{2}\lambda_h(1-d)t_2\right) \leq \exp\left(\frac{-\lambda_h(1-d)t_2}{8}\right).$$

Therefore, we have

$$\begin{aligned} \mathbb{P}(\mathbf{E}_3) &= \mathbb{P}(\mathbf{E}_3 \cap \{M > 2\lambda_h t_1\}) + \mathbb{P}(\mathbf{E}_3 \cap \{M \leq 2\lambda_h t_1\}) \\ &\leq \mathbb{P}(M > 2\lambda_h t_1) + \sum_{r=1}^{2\lambda_h t_1} \mathbb{P}(U_r \leq k_0) \\ &\leq \exp\left(\frac{-4\lambda_h t_1}{3}\right) + 2\lambda_h t_1 \cdot \exp\left(\frac{-\lambda_h(1-d)t_2}{8}\right). \end{aligned}$$

Therefore, combining (E.21) and the bounds on $\mathbb{P}(\mathbf{E}_1)$, $\mathbb{P}(\mathbf{E}_2^h)$, and $\mathbb{P}(\mathbf{E}_3)$ yields

$$\begin{aligned}
\mathbb{P}(\text{tx violates } (t_1 + t_2, \mathcal{H})\text{-security}) &\leq \mathbb{P}(\mathbf{E}_1) + \sum_{h \in \mathcal{H}} \mathbb{P}(\mathbf{E}_2^h) + \mathbb{P}(\mathbf{E}_3) \\
&\leq A' \exp(-a't_1^{1-\varepsilon}) + \sum_{h \in \mathcal{H}} C \exp(-ct_2) + \exp\left(\frac{-4\lambda_h t_1}{3}\right) \\
&\quad + 2\lambda_h t_1 \exp\left(\frac{-\lambda_h(1-d)t_2}{8}\right) \\
&\leq A \exp(-at_1^{1-\varepsilon}) + B(|\mathcal{H}| + t_1) \exp(-bt_2)
\end{aligned}$$

for some positive constants A , a , B , and b . The following lemma therefore completes the proof of Theorem 7.1.

Lemma E.2. *Let $\varepsilon > 0$. Suppose there exist positive constants A , a , B , b such that for all $t_1 > 0$, $t_2 > 0$ and for any honest transaction tx and any finite set of honest users $\mathcal{H}$:*

$$\mathbb{P}(\text{tx violates } (t_1 + t_2, \mathcal{H})\text{-security}) \leq A \exp(-at_1^{1-\varepsilon}) + B(|\mathcal{H}| + t_1) \exp(-bt_2). \quad (\text{E.22})$$

Then there exist positive constants a''' and b''' such that for all $\tau \geq 0$ and for any honest transaction tx and any finite set of honest users $\mathcal{H}$:

$$\mathbb{P}(\text{tx violates } (\tau, \mathcal{H})\text{-security}) \leq \exp(-a'''\tau^{1-\varepsilon}) + |\mathcal{H}| \exp(-b'''\tau). \quad (\text{E.23})$$

Proof. The lefthand side of (E.22) is zero if $\mathcal{H} = \emptyset$ so assume without loss of generality that $|\mathcal{H}| \geq 1$. Given $\tau \geq 0$, let $t_1 = t_2 = \tau/2$. Then (E.22) yields

$$\mathbb{P}(\text{tx violates } (\tau, \mathcal{H})\text{-security}) \leq A \exp(-(a/2^{1-\varepsilon})\tau^{1-\varepsilon}) + 2B(|\mathcal{H}| + \tau) \exp(-(b/2)\tau). \quad (\text{E.24})$$

Let a' and b' be positive constants such that $a' < a/2^{1-\varepsilon}$ and $b' < b/2$. Let $\bar{\tau}$ be so large that $A \exp(-[(a/2^{1-\varepsilon}) - a']\tau^{1-\varepsilon}) \leq 1$ and $2B \exp(-[(b/2) - b']\tau) \leq 1$ for all $\tau \geq \bar{\tau}$. Then for $\tau \geq \bar{\tau}$

$$\mathbb{P}(\text{tx violates } (\tau, \mathcal{H})\text{-security}) \leq \exp(-a'\tau^{1-\varepsilon}) + (|\mathcal{H}| + \tau) \exp(-b'\tau) \quad (\text{E.25})$$

Let b'' be a positive constant with $b'' < b'$. Then, using the assumption $|\mathcal{H}| \geq 1$,

$$\begin{aligned}
(|\mathcal{H}| + \tau) \exp(-b'\tau) &= |\mathcal{H}| \exp(-b''\tau) + \tau \exp(-b'\tau) - |\mathcal{H}|(\exp(-b''\tau) - \exp(-b'\tau)) \\
&\leq |\mathcal{H}| \exp(-b''\tau) + \tau \exp(-b'\tau) - (\exp(-b''\tau) - \exp(-b'\tau)) \\
&= |\mathcal{H}| \exp(-b''\tau) - (\exp(-b''\tau) - (1 + \tau) \exp(-b'\tau)) \\
&\leq |\mathcal{H}| \exp(-b''\tau) \quad \text{for all } \tau \text{ sufficiently large}
\end{aligned} \quad (\text{E.26})$$

Combining (E.25) and (E.26) implies that there exists $\bar{\tau}'$ such that

$$\mathbb{P}(\text{tx violates } (\tau, \mathcal{H})\text{-security}) \leq \exp(-a'\tau^{1-\varepsilon}) + \tau \exp(-b''\tau) \quad \text{for } \tau \geq \bar{\tau}' \quad (\text{E.27})$$

Select positive constants a''' and b''' such that $a''' < a'$ and $b''' < b''$ and

$$1 \leq \exp(-a'''\tau^{1-\varepsilon}) + \tau \exp(-b'''\tau) \quad \text{for } 0 \leq \tau \leq \bar{\tau}' \quad (\text{E.28})$$

Combining (E.27) and (E.28) yields (E.23) for all $\tau \geq 0$. □

This concludes the proof of Theorem 7.1.